The Emptiness of Everything

Contemplating modern sciences
through Buddhist principles

NGUYEN HOANG HAI

The Emptiness of Everything

Contemplating modern sciences through Buddhist principles

(Second edition: revised and expanded)

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Preface

Many people believe that science and religion cannot coexist due to the conflict between reason and faith. However, Buddhism is very close to science. The principles of Buddhism are both philosophical and scientific. Buddhism and modern science not only share commonalities but can also complement and support each other. This book emerged after many years of the author teaching the courses *Contemporary Science and Buddhism* and *Philosophy of Natural Science* at Vietnam National University, Hanoi. The book is not only a textbook serving students but can also be intended for those interested in and passionate about three fields: Buddhism, philosophy, and science. To follow this book, readers need general knowledge of science in the fields of mathematics, physics, chemistry, biology, and neuroscience. Many contents relate to scientific and technological knowledge at the university level, so if readers have graduated from university or higher, especially in science and technology fields, they will find it easier to follow. However, this does not limit readers because all scientific knowledge is presented at the most basic level. Moreover, with the information explosion on the internet and the assistance of artificial intelligence, this book can reach a wide audience.

The Buddhist scriptures are extremely vast with many different schools. This book only addresses three principles of Buddhism that, according to the author, are the most fundamental: dependent origination, suffering, and the middle way. These three principles are

reflected in modern science about the material world, life, the body, and consciousness. Dependent origination is the source and basis for explaining all things and phenomena in the universe, holding that they do not exist independently but always depend on laws and conditions. Dependent origination in life, especially in human life, leads to no-self—the absence of an independent, unchanging, and self-existent self. The illusion that there always exists a self arises from dependence on the five aggregates and external factors from the natural and social environment, leading to suffering. To be liberated from suffering, Buddhism holds that one must follow the middle way, which avoids extremes and is actualized through the Noble Eightfold Path.

The dialogue between science and religion, especially Buddhism, has long become a topic of interest in many research works by scholars both domestic and international. Among the sciences, physics and cognitive science are often seen as having the most points of intersection with Buddhist thought. In his work *The Tao of Physics*, Fritjof Capra pointed out the profound similarities between modern physics and Eastern systems of thought such as Hinduism, Buddhism, and Taoism [1]. Additionally, Alan Wallace [2] also has many works discussing the relationship between contemporary fields of science and Buddhist studies. In Vietnam, the number of authors interested in this topic remains modest; the most notable is Trinh Xuan Thuan, who has connected the thoughts of physics, astronomy, and cosmology with Buddhist conceptions of the universe and humanity [3, 4]. Not only do scientists turn to Buddhism as a source of philosophical inspiration, but many Buddhist practitioners also actively engage in dialogue with science to explore similar points in understanding reality. Among them, His Holiness the 14th Dalai Lama is a representative figure, with many academic exchanges on Buddhist principles in relation to natural science and cognitive science [5]. Other works approach Buddhism from the perspective of cognitive science, such as *Why Buddhism is True* by Robert Wright

[6], contributing to expanding this interdisciplinary knowledge base. However, Buddhism does not only illuminate a single field of science but also aims at a comprehensive view of the entire structure of reality—from the material world, life, to the spiritual. Based on that perspective, this book seeks to clarify the universality of Buddhist principles in relation to the levels of the universe: physics, biology, and consciousness.

Chapter 1 presents a brief overview of Indian civilization before Buddhism emerged, the reform of Buddhism based on Hinduism, the development of Buddhism in India and Vietnam, and the principles of Buddhism. Dependent origination, the most universal principle appearing in all things and phenomena in the universe, indicates that all things do not exist independently but depend on each other in space and over time. In space, when examining any thing, we must examine it in relation to other things, and as a whole in relation to the components that make up that whole. Over time, all these relationships continuously transform, leading to chains of cause and effect. Each thing is both the result of previous causes and the cause of later results. This is the continuous change of causal chains. Dependent origination leads to emptiness: all things and phenomena depend on other causes (hetu) and conditions (pratitya) to exist, so no phenomenon has a self-existent, independent essence. Because causes and conditions continuously change, things have no fixed essence. The complete absence of a self-existent, independent, and unchanging essence is **emptiness** of all things and phenomena, considered by the Madhyamaka school as the absolute truth, hence taken as the title of this book. The Madhyamaka school equates dependent origination and emptiness: dependent origination is emptiness, emptiness is dependent origination. Therefore, proving that things and phenomena follow the principle of dependent origination means containing emptiness within them. To prove emptiness, things must exhibit two characteristics: (1) compositional nature shows that a thing as a whole is made

up of smaller components; (2) impermanence shows that relationships between components continuously change. Any thing or phenomenon with the above two characteristics has emptiness. This book presents emptiness in four worlds in the following chapters: matter, life, humans, and consciousness.

Chapter 2 presents the emptiness of matter through theories of atoms and subatomic particles. The theory describing the microscopic world is quantum mechanics, requiring scientists to abandon everyday experiences and accept hypotheses contradictory to those experiences. The principles of quantum mechanics such as wave-particle duality, superposition principle, indistinguishability principle, exclusion principle, and uncertainty principle show that microscopic particles are closely related, one depending on and determining the other. Matter is composed of atoms. Atoms are formed by smaller particles such as electrons, protons, and neutrons. According to the standard model of physics, electrons, quarks, and neutrinos are fundamental particles. They are not unchanging and independent but continuously appear and disappear. A distant observer sees them as particles with a self-existent nature. But according to quantum mechanics, they are merely remnants of a continuous process of appearance and disappearance of countless particles in the vacuum. According to string theory, fundamental particles are in turn composed of extremely small strings living in 10 or 11-dimensional space, realities completely different from human perception. The same applies to concepts like space and time. They do not exist independently of occurring events but can expand, contract, and bend by matter within them. Space is considered to emerge from quantum entanglement in a world with fewer dimensions than the world we live in. Meanwhile, the arrow of time is attributed to the increase of entropy in thermodynamics. Time, space, and matter coexist because they depend on each other inseparably. Where there is space and time, there is matter, and vice versa.

Chapter 3 presents emptiness in life in general. If life is consid-

ered a self-sustaining chemical system capable of following Darwin's theory of evolution, then each gene segment within each organism is a living entity, self-replicating units that act as subjects of the evolutionary process. Self-replicators use individual organisms, living machines, to fulfill their purposes. Self-replicators create complex machines, even creating the brain with hundreds of billions of neurons to reward the living machine with consciousness of itself and consciousness of free will. The origin of life relates to almost all events in the universe. From the Big Bang, the formation of elements in stars, supernova explosions when stars die, to conditions on ancient Earth,... all have decisive influence on the formation and development of life on Earth.

Chapter 4 shows emptiness in the human body at three levels: the body, cells, and DNA. The human body is a symbiosis with tens of trillions of other microorganisms, mainly living in the intestinal system. Microorganisms symbiotic with the human body digest food, strengthen the immune system, and assist the nervous system. Human cells are also not independent. Most cells contain foreign organisms with their own genes and protein encoding, now becoming an organelle with an important role in energy production, time control, and determining cell death programming. That is the mitochondrion. Mitochondria are the energy factories of animal cells, keeping time for cells, and they also regulate the speed of human embryo development. Chloroplasts in plant cells have a role similar to mitochondria in animal cells; they are foreign bacteria living symbiotically in plant cells. Recently, people discovered that cyanelles are bacteria in the process of becoming an organelle of sea algae. Human DNA is also an ecosystem where self-replicators (genes) cooperate and compete to survive and develop. Genes encoding and controlling protein encoding in humans occupy only a very small proportion of DNA; the remainder consists of genes of unknown function. They may be self-replicators, using the human body as a living machine.

Chapter 5 shows the emptiness of human consciousness. Consciousness develops from basic perceptions, collective intelligence at different structural levels. The senses only receive a very thin slice of reality to form information transmitted to the brain. The brain enclosed in the skull only receives external information through electrical signals, from which it decodes by space and time to reconstruct a model of external reality with the purpose of predicting the future. Human consciousness is the brain's model of reality. The reflection of consciousness need not be truthful, need not be complete, only needs to be appropriate for the brain to predict many, quickly, and far into the past and future. This chapter presents the mechanism by which the body creates physical pain when tissue is damaged, dissatisfaction after achieving desires, unhappiness when not knowing that ignorance leads to all suffering. That is the emptiness of all things.

The book's content addresses many fields with a very large number of terms, so for convenience in following the book's content, readers use the appendices printed at the end of the book to know the content of terms. For example, the concept *Uncertainty Principle* is mentioned many times in different sections, but readers may not know which section of the book presents detailed content of this concept, so they can look up the term in the *Index*. Abbreviations such as AI, NASA, GDP,... are explained in the *Glossary*. References are numbered according to the order of appearance in the book. In the referencing process, the author strives to use the most recent publications (November 2025) in reputable journals.

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Although the author has spent much time and effort in compiling this book, due to its coverage of many different issues, some information may not be complete or accurate. The author very much looks forward to receiving feedback from readers. Feedback can be sent to the email address: nhhai@outlook.com.

Hanoi, November 2025

Nguyen Hoang Hai

Chapter 1

Principles

Summary. (1) Indian civilization comes from the blending of Aryans and indigenous peoples from approximately 4 thousand years before the Common Era (BCE) with the foundation of the Vedas. An important characteristic of this civilization is caste distinction and deity worship. (2) Buddhism emerged, breaking down caste distinctions and eliminating deity worship to build a religion that is more philosophical than superstitious. (3) The principles of Buddhism include: the principle of dependent origination, suffering, and the principle of the middle way. The principle of dependent origination holds that all things and phenomena have no self-nature, are not independent, unchanging, and exist in relation to each other. Dependent origination is the nature of all things in the universe; dependent origination leading to suffering is an attribute of life. The principle of the middle way is the path to end suffering by not going to extremes and transcending extremes. From the Buddha's concept of no-self, Buddhist thinkers developed the concept of emptiness, considered the supreme truth. Nagarjuna equated dependent origination with emptiness. (4) The emptiness of Buddhism manifests through compositional nature—things exist because they are created from smaller constituents, so things have no self-nature and are not independent; impermanence—the constituents interact with each other and that interaction continuously changes over time, so things are not permanent, unchanging but always in a state of continuous change over time. Proving that

things and phenomena have compositional nature and impermanence means that things and phenomena have emptiness.

1.1 Indian civilization

According to the White Paper *Religion and Religious Policy in Vietnam* published in 2022, Buddhism is the most widespread religion with over 14 million followers, accounting for 53 % of religious followers nationwide [7]. If including those who do not officially identify as Buddhists but have faith in this religion, the number is certainly much larger. Although Buddhism is widespread, it can be seen that Buddhist scriptures are very numerous and diverse, causing confusion for newcomers when they must choose scriptures suitable for themselves. Therefore, understanding the basic principles of Buddhism and the values Buddhism brings to humanity is necessary. This chapter presents the background of Hinduism when Buddhism emerged and the development of Buddhism worldwide and in Vietnam. Subsequently, we study three important principles: dependent origination, suffering, and the middle way.

1.1.1 Indian x

According to plate tectonics theory[1] (plate tectonics), the Earth's surface is formed from different lithospheric plates. They do not stand still but move continuously due to convection of the mantle layer within the Earth. Convection arises from heat escaping from radioactive elements within the core, causing hot material to move upward and cold material to move downward. On the lithospheric plates is the crust layer: oceanic crust with high material density,

[1] Plate tectonics is the geological theory about the structure of the Earth's crust, holding that the Earth's surface is formed from many continuously moving tectonic plates. Initially Alfred Wegener proposed continental drift from the 1920s. This later developed into plate tectonics theory.

approximately 10 km thick, and continental crust with lower material density and thickness up to 70 km. When the crusts move, they can collide with each other to create mountain ranges, deep trenches, and earthquakes, volcanoes, tsunamis. Approximately 50 million years ago, the Indian continental plate separated from the ancient Gondwana continental plate to move northward and collide with the Asian continental plate [8]. When they crashed into each other, they created the boundary of the world's highest mountain range—the Himalayas, reaching heights of over 8 thousand meters, a natural boundary separating the Indian subcontinent from other parts of the world.

Approximately 65 thousand years ago, Homo sapiens migrated from Africa and set foot on the Indian subcontinent, establishing human presence here. Studying the time when humans arrived in India is very important because these are the first ancestors of Indians, genetically contributing approximately 50 %–60 % of genes to today [9]. The remaining genes come from later interbreeding with other peoples in West Asia, Central Asia, and East Asia. At least 8.6 % of India's current population retains almost intact genes from the first settlers of the Indian subcontinent [10].

There are two hypotheses about when humans first set foot on the Indian subcontinent. The first hypothesis holds that sapiens from Africa, crossing the Arabian Peninsula to arrive here, was approximately 100–120 thousand years ago [11]. The second hypothesis is much more recent, approximately 50–65 thousand years ago. The reason for the large difference between the two hypotheses is the eruption of the Toba supervolcano, Indonesia, occurring approximately 74 thousand years ago [12], preventing human settlement in India or, if there was settlement before, survival from that terrible catastrophe was difficult.

This volcanic eruption is the largest recorded by humans and has been studied very carefully. It ejected 2800 km^3 of rock and soil, including 800 km^3 of ash, creating a dust layer approximately 15 cm

thick covering all of Southeast Asia, including the East Sea. This was a terrible catastrophe creating a population bottleneck in human evolution. This catastrophe created a global volcanic winter, strongly affecting Southeast Asia and South Asia. Studies in 2021 led by scientists from Peking University, China, showed that the eruption came from continuous and steady accumulation of magma flows underground, making measurements of surface deformation or underground gas emission of little use. Therefore, determining whether a similar eruption will occur in the future cannot use ordinary observation methods. If a similar explosion occurs, all of Southeast Asia could be wiped out [13].

Scientists from Macquarie University, Australia, used interdisciplinary techniques, combining tooth enamel studies, uranium isotope dating by electron spin resonance, biostratigraphic age estimation to study two human teeth in Lida Ajer Cave on Sumatra Island, Indonesia, showing they are 63–73 thousand years old, about 20 thousand years older than previous studies [14]. Thus, if humans set foot in Southeast Asia no later than approximately 65 thousand years ago, they must have arrived in India before that, because the shortest path from Africa to Southeast Asia must pass through India. Studies analyzing human genes according to haplogroups on DNA of the Y chromosome (Y-DNA) for males and on mitochondrial DNA (mitochondrial DNA – mtDNA) for females[2] show that all humanity originated from Africa approximately 200 thousand years ago and arrived in South Asia approximately 60 thousand years ago [15]. The difference in the two methods above does not create contradiction but can accept that humans settled in India earliest approximately 65 thousand years ago, after the Toba volcanic eruption event.

[2]The Y chromosome is a sex chromosome inherited through the paternal line. Conversely, mitochondrial DNA is a DNA independent of human DNA but inherited through the maternal line. Mitochondria are considered foreign microorganisms that have been symbiotic with human cells for billions of years.

1.1.2 Indus Valley civilization

India subcontinent has two famous rivers originating from the Himalayas and flowing in opposite directions: the Indus River (Indus river) and the Ganges River (Ganges river). The Indus River is over 3180 km long, flowing southwest to the Gulf of Oman. In Sanskrit, *Sindhu* means *river*. This word was used to refer to the Indus River. The Persians, when entering Indian lands, called this river Hindu and the northern region of India where this river flows is Hindustan, meaning the land of rivers. Indians today call themselves Hindus.[3] The Ganges is 2510 km long, flowing southeast, crossing northern India and Bangladesh to the Bay of Bengal. The river is named after a Hindu deity, Ganga. Because it originates from the Himalayas, the river has an endless source of snow water. The Ganges basin has the highest population density in the world. According to Indian legend, the Ganges continues to flow down to the underworld. Indians consider the Ganges their most sacred river, the daughter of the Himalayas. In the heaven it is the Mandakini, on earth it is Ganga, in the underworld it is Patalganga. Because it flows through all three worlds, it is also called Tripathaga (flowing through the three paths).

Like other ancient civilizations that are all linked to water sources, usually rivers,[4] Indian civilization is linked with the Indus and Ganges rivers. Among them, the first civilization is the Indus Valley civilization (Indus valley civilization), one of the world's largest and oldest civilizations, approximately 3–1 thousand years BCE. In 2016, an archaeological study by the Indian Institute of Technology

[3]Before 1947, the region now comprising India, Pakistan, Bangladesh, Myanmar was called British India, where the Indus was the second largest river after the Ganges. Currently, the Indus does not belong to India but is the main river in Pakistan.

[4]Egyptian civilization dates from the fourth millennium BCE, linked with the Nile. Mesopotamian civilization dates from the 3rd millennium BCE, linked with the Tigris and Euphrates rivers. Chinese civilization is linked with the Yellow River and Yangtze.

showed that the Indus Valley civilization may date from approximately 8 thousand years ago, much older than previous publications of approximately 6 thousand years. If the above study is correct, then the Indus civilization, together with the Nile civilization and Mesopotamian civilization, are the three oldest civilizations in the world [16].

The Indus civilization, centered on the two ancient cities of Harappa and Mohenjo-daro (now in Pakistan), had the densest population in the world at that time. Archaeological excavations discovered dense houses built of fired brick, developed agriculture, and an ordered society. Many public baths were hypothesized for purifying the body before participating in worship ceremonies. The people had animistic beliefs and fertility beliefs worshipping male (linga) and female (yoni) reproductive symbols. Here, people find the close connection between purity, sexual symbols, and worship ceremonies.

However, the Indus civilization mysteriously disappeared around 1 thousand years BCE. The common hypothesis holds that climate change altered the entire monsoon climate, leading to drought over 5 thousand years ago. By analyzing oxygen-18 isotopes in animal teeth, scientists added that people changed livelihood strategies from growing large grains like wheat and barley to small grains like millet and rice when rainfall decreased, leading to the decline of this civilization [16].

The Indus civilization had many ethnic groups. Prominently in the north were the Naga people, one of India's oldest peoples, worshipping the Naga serpent deity. This belief influenced other religions. Additionally, there were the Dasyus people, considered to have provided the first version of reality before the Vedas. They believed in polytheism, worshipped nature, organized fertility ceremonies, valued body purification, and had some basic ideas about life after death and reincarnation. These may be foundations for the Vedas and post-Vedic religions to develop later [17, page 21]. In the

south were the Dravidians with dark skin, large noses, and black eyes, considered the masters of the Indus civilization. The Dravidians followed matriarchal systems, gathered into villages and developed into communities forming tribes. They had their own beliefs worshipping earth, trees, and animistic deities, such as the monkey god Hanuman, the bull god Nandi, and the tree god Yaksha. All these deities were incorporated into Vedic scriptures later [18, page 45]. The Dravidians went through the Stone Age and Bronze Age until the beginning of the Iron Age, when they were invaded by the Aryans from the north around the 15th century BCE.

1.1.3 Aryan migration

The Aryans were nomadic people with light skin, tall stature, straight noses, originating from Central-Near East regions, speaking Indo-European languages, healthy, resilient, rough, and brave. Some hypotheses hold that the Aryans entered Indian lands from the northern region, advancing east of the Indus River and along the Ganges. No evidence of wars was found, so most scientists now believe that Aryan penetration was peaceful, with them gradually assimilating with the indigenous people over time. At that time, the Aryans were somewhat less civilized, so they were influenced by the community nature, land policies, taxation, and beliefs of the Dravidians.

After a long period, the Aryans and Dravidians assimilated with each other, learning from each other. The Aryans transitioned from nomadic herding to settled agricultural life [19]. The Aryans brought Sanskrit language along with their religious system given in the **Vedas**. From here, India began a new civilization, the Vedic civilization. The Vedas were initially of the Aryans, then absorbed many religions from the Indus Valley civilization and were completed around the 12th century BCE. People know of this civilization through legends, most notably the two epics Ramayana and Mahabharata.

The Indus civilization had different scripts although the language might be the same. That was Sanskrit (Sanskrit) of the Aryans. Thus, the same language was Sanskrit but had many scripts, including Brahmi script, the precursor of Devanagari script of India today. Standardization among peoples of the Indus was not limited to language. Studies also show that brick sizes for house construction and architecture of ancient cities in this region were also uniform [16].

1.1.4 Formation of castes

A distinguishing feature between India and other civilizations is very clear caste distinction, which not only occurred in the past but also persists to this day. In Sanskrit, *caste* is called *varna*, meaning *color*. It may be because the Aryans, with light skin, migrated south to encounter dark-skinned indigenous people, so they discriminated, looked down on them as inferior, giving rise to castes. There are four main castes. (1) The highest caste is Brahmin [Brahmins],[5] consisting of priests and monks whose duty is to care for the soul, the most essential part of humans. (2) The Kshatriya caste [Kshatriya] consists of princes and warriors whose duty is to oversee state affairs and maintain social order. (3) The Vaishya caste [Vaishya] consists of merchants, landowners, and free people whose duty is labor and production to create wealth in society. They must also pay taxes and serve as soldiers for the state. (4) The Shudra caste [Shudra] consists of laborers, farmers, and slaves whose duty is to unconditionally serve the upper castes. They are not allowed to participate in religious ceremonies except for worship exclusive to their caste. Beyond the four castes above, there is also a class of outcasts, low-caste, untouchables called Pariah [Pariah], today also called Dalit, only permitted to do the lowest work in society.

[5]Brahma is the deity who created the world. Brahman is the absolute soul pervading the entire universe. Brahmin is the caste of priests and monks. Brahmana is the Brahma sutra, appeared later, explaining the Vedas. Brahmi is the ancient Brahmin language.

A person's caste determines their most important matters. First, caste determines marriage. From ancient times to today, marriage is mainly arranged marriage. Men and women are not free to get to know each other but are arranged by families. People of a certain caste are only allowed to marry people of that caste. Marriage between people of different castes is strictly prohibited. If they intentionally marry, both are demoted in caste. In 1950, the Indian Constitution officially abolished caste discrimination. However, recognition and adherence to castes are still accepted by the majority of Indians. It is not a matter of law but a matter of ethics and spirituality. Second, caste determines work. Each caste is allowed to do work that caste is permitted. The highest, most elite caste is the priests, who care for the soul (atman) and teach people of lower castes. Third, caste determines a person's social function. People of lower castes have the responsibility to obey people of upper castes. Castes are treated differently in society in terms of law, life, and spirituality, favoring the upper castes.

There are several hypotheses about caste distinction in India. The first hypothesis is the racial hypothesis proposed by British administrator Herbert Risley in the late 19th century [20]. Although there was assimilation between Aryans, the ruling minority, and indigenous peoples to form slave states, the Aryans did not want their race to mix with the indigenous. They believed that if there was mixing with the indigenous, because these people were more numerous, after just a few generations their Aryan lineage would disappear. The first three castes were three Aryan groups, distinct from the indigenous Pariah. Between these two groups were Shudras, consisting of two types: first, indigenous people who cooperated with Aryans; second, the mixed offspring of Aryans and indigenous peoples. This hypothesis was supported by many European scholars but opposed by many, including Indian scientists, because it supports racist viewpoints.

The second hypothesis is the accidental formation hypothesis,

that castes formed due to specific historical conditions. This hypothesis was championed by one of India's outstanding figures, Bhimrao Ambedkar. Born from the lowest untouchable Dalit caste, he rose to become one of the most influential politicians. He is also considered a bodhisattva because he converted from Sikhism to Buddhism and attracted approximately 600 thousand people to follow, reviving Buddhism in India. He was also the drafter of India's constitution.

DNA studies by Indian scientists in 2013 showed that the accidental hypothesis is correct. They examined genes of 73 population groups from various places in the country, then compared them with genes of two common groups: northern Indians (similar to Aryans) and Indians in the south (similar to Dravidians). The result was that mixing of genes of these two groups appeared from approximately 2.2 thousand years BCE until the beginning of the Common Era, then decreased sharply [21]. The caste system of India is considered to have emerged from 1500–500 BCE, coinciding with the Vedic period, long before the cessation of bloodline mixing, up to 600 years. This shows that bloodline could not be the reason for forming the caste system. The cessation of bloodline mixing that occurred may have been because it was encoded into the scriptures. Thus, caste formation prevented mixing between groups, not that castes formed because of fear of mixing. Gene studies are the most accurate studies, compelling evidence, conducted on a very large dataset, confirming the accidental formation hypothesis of India's caste system.

The caste system is a very large barrier to social development. The more civilized and modern society becomes, the more caste distinction should decrease. However, studies in 2021, that is, after nearly 75 years of independence, show that caste distinction has not decreased but shows a tendency to increase [22]. There are two affected groups: people of the lowest Dalit caste, numbering up to 232 million people, and minority Adivasi people, numbering approximately 120 million out of a total of 1.4 billion people. Thus, more

than a quarter (25.1 %) of India's current population lacks development conditions in health, education, employment,... Although state policies ban discrimination, discrimination still occurs everywhere, both urban and rural.

The Vedic tradition began to form in India with forms of deity worship. Archaeological evidence shows a developed civilization known through legends, most notably the two epics Ramayana and Mahabharata. This period is also called the Vedic – Epic period. The Vedic – Epic period is the first development stage of Hinduism. At this time, Aryans began to invade Indian lands and became masters of Vedic civilization.

From the 11th – 8th centuries BCE, the Vedic – Epic period developed into Brahmanism, also called Yajur-Veda, where ritual performance was specialized for Brahmin priests and the Shudra caste was excluded from rituals, transitioning from manual ritual performance to deep research into the meaning of scriptures about human – universe, human – human, human – individual relationships. During this period, Aryans advanced into the Ganges basin and transitioned from nomadic herding to settled agricultural life, building states according to slave-owning systems, distinguishing social castes. From the 8th – 6th centuries BCE is called the classical period. Indians realized that behind natural and social phenomena there is a powerful, invisible force creating and governing the universe. The polytheism of previous periods gradually faded and was replaced by a supreme cosmic spirit, the supreme deity Brahman. This period is also called the Upanishad age, transitioning from mythological religious thinking to philosophical thinking. Philosophy takes humans as the center to solve other philosophical problems. Indian religion explores and exploits the *self*, the spiritual essence of life.

Upanishad holds that humans consist of three parts: material body; consciousness including thoughts, emotions, experiences; pure spiritual soul. The soul part in Sanskrit is *atman*, sharing the same

root with English and Latin *atmosphere*, meaning air – air in the human body, also called the small self, the self. Importantly, each individual's atman is identical with *brahman*, the supreme, transcendent reality of the universe, also called the *great self*, the cosmic soul. At the deepest level, each individual's soul merges with one cosmic soul. Like a great banyan tree, lush but growing from a seed in a banyan fruit that the naked eye cannot see, that is brahman within all entities. Atman, the soul, the self of humans is invisible, difficult to grasp. Any object also comes from the essence of invisible brahman, including humans. In Sanskrit *Tat tvam asi*, meaning *Thou art That* or *You are one*, is the most famous saying of Indian philosophy to indicate the omnipresence of brahman, within each human individual.

The self atman and the great self brahman seem to be two but become one, with no difference. An individual perceives the world through the senses, thereby thinking there is something called "I" and something called "outside me," i.e., external objects, which is not yet correct. One must look deep inside to see a latent, invisible entity within each individual. The body attached to consciousness are impermanent entities; after death, the body and consciousness also die, but the self atman is permanent. Atman can be in one person, can also be in another person. Even atman is not in the human body but can be in animals, ghosts, deities,...

What determines atman being in a specific form is karma. Individual actions leave karma (karma) for each atman. This life receiving this body is due to karma from previous lives bringing it. Actions in this life then leave karma for the body of the next life. Thus the cycle of birth and death repeats endlessly. However, if humans know that atman is brahman, they can escape the cycle of reincarnation samsara because then the self atman can approach brahman to transcend the body and consciousness, from which karmic retribution can no longer affect atman. Upanishad holds that merely

recognizing that atman and brahman are one is not enough; people also need to contemplate and directly perceive brahman. When a person directly perceives the supreme reality, it is called attaining *ananda*, a state of boundless joy, which no reason or language can describe.

Early Vedic period worshipped polytheistic deities like Indra (thunder), Agni (fire god), Varuna (sky),... deities with many aspects equivalent to ancient Greek deities [23, page 27]. By the Upanishad period, Brahman is the supreme non-personal, invisible, infinite reality, the essence of the entire universe from which all things, including deities, are born. Old deities are not erased but lose their supreme position. They are no longer the most powerful but are seen as manifestations or different aspects of Brahman. During the early centuries SDL, the Trimurti was clearly formed: Brahma – Creator God; Vishnu – Preserver God; Shiva (predecessor was the deity Rudra in the Vedas) – Destroyer God.[6]

Later, Buddhism does not deny the existence of deities but denies the existence of a Creator. The universe operates according to natural laws like Dependent Origination. Deities (celestial beings) have very long lifespans and have more powers than humans, but they are not immortal. They still must die and be reborn in reincarnation, not yet liberated. The Buddha, an enlightened human, is considered the teacher of both heaven and humans. The wisdom and enlightenment of a Buddha surpass all deities.

1.1.5 Anti-Vedic religions

In the 6th century BCE, India saw some thoughts opposing the monopoly of the Brahmin caste and the caste system, opposing complex rituals. These religions are called Anti-Vedic religions (anti-

[6]The notion of predecessors had appeared earlier, but systematic formation and consolidation are usually associated with the Gupta period around the 4th – 6th centuries BCE.

Vedic), also called Sramana (ascetic) religions, and they developed distinct philosophies and practices. The six Anti-Vedic religions are Buddhism, Jainism, Ajivika, Carvaka (Lokayata), Ajnana, and Ajita Kesakambali. The common characteristic of these movements, as their name suggests, is opposing Vedic traditions. The first characteristic is opposing the strict distinction of humans by caste and lineage of Brahmanism. Second is opposing the practice of complex rituals and the monopoly of the Brahmin class in ritual performance. Instead, they focused on simple individual or community forms of practice. Jainism, Ajivika, and some schools of Buddhism practiced asceticism. Carvaka and Ajita Kesakambali followed materialist schools, denying the existence of the soul, afterlife, and supernatural forces.

The religions Ajivika, Carvaka, Ajnana, and Ajita Kesakambali no longer exist today as independent, organized religions. Their philosophies and thoughts only remain in historical documents and no longer have significant influence in the modern world, so this book does not focus on them.

Here, we mention **Jainism** (Jainism), initiated by the teacher Mahavira in the 6th century BCE, holding that to escape reincarnation, humans must restrain themselves from sensual desires and material things. This is the religion with the highest asceticism. Although born almost simultaneously with Buddhism, Jainism followers hold that the history of the religion extends to infinity because it is an eternal religion, beginningless and endless. The universe consists of many eons, each eon lasting millions of years. In the current eon, teacher Mahavira is the 24th founder. In Sanskrit, the founder is called *jina*, meaning *victor* or *tirthankara*, meaning *one who opens the path to escape rebirth.* Jainism disciples do not need to worship deities, only need to follow the example of founders to free the soul from bondage with the physical. Each person must take responsibility for their own actions, so they need to follow 5 principles: non-violence, truthfulness, non-possession, non-adultery, and

celibacy. The most important principle is non-violence (ahimsa), not harming any living being, no matter how small. Other principles are mainly to abandon desires and material things to focus on learning, reciting scriptures, and teaching.

The thought of not worshipping deities, humans must take responsibility for their own actions, following founders in abandoning desires and material things to achieve liberation are thoughts very similar to Buddhism. The only important difference is the method of practice through asceticism. The Buddha once practiced asceticism for a period and realized it was not an effective method, so Buddhism uses the middle way, neither following asceticism nor following materialist hedonism.

1.2 Brief history of Buddhism

1.2.1 The birth of the Buddha

The Buddha was named *Siddhartha*, meaning *One who achieves all wishes*, of the Gautama clan. Later he was also called *Shakyamuni*, in Sanskrit meaning *Sage of the Shakya clan*. Vietnamese transliteration becomes *Thích ca mâu ni*, so Buddhist clergy often take his surname Thích as their title. He was born in Lumbini garden, northern India, now in Nepal, around the 6th century BCE. The time of his birth has much controversy. Some documents comparing events with the time of Asoka's ascension calculate that the Buddha entered parinirvana around 400 BCE [24, page 42] The World Buddhist Congress unified that he was born in 624 and died in 544 BCE, living 80 years. His father was Suddhodana, his mother was Maya, king and queen of the Shakya kingdom, with capital at Kapilavastu.[7] The noun that later people called him after enlight-

[7]Legends about the Buddha in traditional books and academic documents have some differences. As mentioned above, the Buddha's lifetime according to academic documents is hundreds of years later [23, page 32]. Some Vietnamese

enment is *Buddha*, meaning *Enlightened One*, transliterated from Sanskrit to Vietnamese as *Bụt* or transliterated from Chinese to Vietnamese as *Phật*.[8] During his lifetime, the Buddha called himself *Tathagata*, meaning *One who has come from the realm of suchness*, Sino-Vietnamese transliteration called *Như lai* to avoid him calling himself "I" or "me" when teaching.

Legend says that when Prince Siddhartha was born, he had golden skin, 32 excellent marks, 80 extraordinary beauties. Although the Buddha did not reveal as in other religions, there were also special manifestations before and after the Buddha's birth. Both Hindu and Buddhist traditions acknowledge reincarnation, so the birth of Prince Siddhartha must have been very special. Before that, Queen Maya dreamed of a white six-tusked elephant, trunk holding a large white lotus flower, flying down from heaven and entering her right side. Sixty-four Brahmin prophets came to interpret the dream, saying she would give birth to a great emperor or a renowned sage. The Buddha's birth is quite similar to the birth of Mahavira – the leader of Jainism. Both men lived in the same period, belonged to Anti-Vedic religions (sramana), creating prestige by attracting followers in Northeast India [27].

Right after Queen Maya gave birth to the prince, a famous monk of that time, Asita, came to read the prince's physiognomy. The monk said the prince would become a Wheel-Turning Sage King ruling all of India or become an Enlightened Sage. King Suddhodana wanted Prince Siddhartha to become a Wheel-Turning Sage King, not wanting the prince to become a monk, so he built for him a life

scholars also have similar observations [25]. His family background is also like that. The Sakya tribe lived in a small republic in northern India ruled by a council of elders. The Buddha's father may have been only a warrior [23, page 32] or one of the elders of this small state [24, page 43]. The democratic nature of the small state may have influenced the democratic nature of Buddhism when compared with Hinduism.

[8] Venerable Thích Nhất Hạnh uses Bụt instead of Phật [26]. This noun is less common, so in this book I use the noun Phật, transliterated from Chinese.

of luxury and wealth, not letting the prince come into contact with life's suffering. At age 18, the prince married Princess Yasodhara of the Koli Kingdom, and at age 29, the two had a son.

However, luxury, wealth, and sensual pleasures could not keep the prince. Due to unavoidable causes and conditions, after four times going out the four city gates and seeing an old person (east gate), a sick person (south gate), a dead person (west gate), and a sage (north gate), the prince resolved to leave home and become a monk at age 29. In the early period, the prince learned meditation from some monks and practiced asceticism with Kondanna's group of five [Kiều trần như] and achieved some results but could not yet be liberated.

Siddhartha decided to abandon the ascetic practice to return to normal eating. He went to the Bodhi [bồ đề] tree at Bodh Gaya [Bồ đề đạo tràng] to meditate. After 49 days of continuous practice, he attained enlightenment of the Buddha Way. That was the dawn of the 8th day of the 12th lunar month in 589 BCE, when the Buddha was 35 years old.

1.2.2 Buddhism forms

After enlightenment, the Buddha met again with Kondanna's group of five to transmit his teachings at Deer Park. In this first teaching, he proposed the **middle way**, which rejects two different extremes of Hinduism at that time: the ascetic school and the materialist hedonist school. He taught the Four Noble Truths, Noble Eightfold Path, no-self, impermanence, dependent origination, cause and effect, and many other teachings. Then, in the second teaching, all five attained the fruit of arhat[9] [A la hán or La hán] (liberated being). The five people of Kondanna's group are the first disciples of

[9] Arhat or Arhant literally means *worthy one*. In Southern Buddhism, Arhat is one who has attained the fruit of Nirvana (nirvana) and escaped reincarnation. In Northern Buddhism, arhat is one who attains Nirvana, escapes reincarnation but has not yet attained Buddhahood.

Buddhism. From here, the world has the **Three Jewels** consisting of Buddha, Dharma, and Sangha.

A person following Buddhism, i.e., a *Buddhist* is one who *takes refuge in the Three Jewels*, generates the Bodhi mind, i.e., generates compassion and altruism to have a noble mind. Taking refuge in the Three Jewels is not only participating in a formal refuge ceremony with a monk as a guide to become a Buddhist, but more importantly, that person contemplates and reflects[10] on the Three Jewels even when there is no guide, absolutely trusting Buddha, Dharma, and Sangha.

The first of the Three Jewels is *Buddha*, the founder, the one who guarantees the truth and trustworthiness of the teachings. The noun Buddha is not to refer to a proper name but means an *Enlightened One*, one who thoroughly understands all problems of reality. Speaking of Buddha means speaking of two people: one human embodiment, flesh and blood, Siddhartha, and one supernatural being, Perfectly Enlightened, capable of illuminating all laws of the universe, Buddha. Buddhist followers seek every way to reduce the secular nature of Buddha. Later, *Buddha* is not only a noun to speak about the Buddha, nor only the Enlightened One, but also used to refer to *Buddhahood* based on the level of awakening, of enlightenment.

The second of the Three Jewels is *Dharma*. Depending on each aspect, Dharma can be understood in different meanings. Dharma can be considered the law governing all things, the teachings of the Buddha. For Buddhist disciples, in life Dharma is also righteousness, virtue that disciples must follow. Dharma can be considered the Buddhist path, teachings, scriptures, truth in Buddhism. Absolute truth is permanent, eternal, existing throughout the world and universe. Whether Buddha exists in the world or not, whether sentient beings know and believe or not, that truth still exists and

[10]Reflection is looking, observing very deeply at a thing and phenomenon to be able to understand it thoroughly.

governs all things. That truth cannot be described in words but needs self-practice to attain. What the Buddha said is only to help disciples know how to attain truth.

The third of the Three Jewels is *Sangha*, the Sangha or monastic community. The Sangha consists of disciples who take refuge in the Three Jewels and keep the precepts,[11] those who leave home keep more than 5 precepts, i.e., monks and nuns. Noble monks are the most excellent symbols of the Sangha. In the remaining 45 years until the Buddha entered parinirvana, he traveled many places, to many regions of India to preach the Dharma. He lived by going for alms, eating whatever was offered, not staying in one fixed place. Later, followers of Hinduism said that Siddhartha was the ninth incarnation of the god Vishnu, although Buddhism denies many powers of the Vedic scriptures, such as denying the permanent, unchanging self atman, denying the supreme position of Brahma, and holding that deities are also sentient beings in the three realms, not omnipotent, not immortal.

1.2.3 Southern Buddhism

When the Buddha was still alive, he taught many disciples according to methods depending on circumstances and each person's level, using different methods and tools to teach. Even the language used for teaching is only a means to clarify for listeners. Therefore, the content of teachings and methods of transmission already had diversity and flexibility from the beginning. The method of teaching was oral lectures, not yet recorded in writing. After the Buddha entered parinirvana, his disciples compiled his teachings into the Dharma, but the compilation was also not in writing. Monks divided into groups, each group responsible for memorizing different parts of the

[11] Depending on the school, those who leave home must follow up to hundreds of precepts. Laypeople must follow the five precepts: do not kill, do not steal, do not commit adultery, do not lie, do not use intoxicants.

teachings to recite by memory. Although they frequently met to memorize or modify content, differences still arose. In the first few hundred years, teachings were not written down but only transmitted through recitation [28].

As the sangha gradually grew, divisions in the sangha could be the result of following different vinaya codes rather than different interpretations of teachings. Different understandings of teachings are usually not too important for causing sangha divisions. In many cases, monks of different schools joyfully practice together following the same vinaya code [23, page 78]. Not to mention that in some periods, some individuals, due to desire to seize leadership, also caused Buddhism to divide. Those causes made Buddhism's later development very diverse. Compared with other monotheistic religions in the West, although there are divisions, the number of sects as well as teachings are relatively stable over a long time. Meanwhile, Buddhism is a religion divided into very many different schools. Some schools exist to this day, but some schools no longer exist. Even within one school, there are many groups with different teachings and scriptures, making the number of scriptures very large.

After the Buddha entered parinirvana, Buddhism went through many ups and downs and changes and has existed to this day for more than 2500 years. The history of Buddhism can be divided into 4 periods, each period about 500 years (except the most recent period is 1000 years). According to Edward Conze [29], there are four important periods in the history of Buddhism:

1. First period: Lasting until the 1st century BCE, Buddhism developed within the territory of today's India, focusing on individual liberation to attain the arhat fruit, abandoning desires, escaping reincarnation. Philosophy focused much on psychological issues. That period had many schools, but only one school exists to this day – Southern Buddhism (Theravada), also called

Southern School.[12] Currently, the countries Sri Lanka, Myanmar, Thailand, Laos, Cambodia, and southern Vietnam follow Southern Buddhism.

2. Second period: Northern Buddhism, also called Great Vehicle (Mahayana means Large Vehicle) or Northern School, formed from around the 1st century BCE until the 5th century. This period Buddhism developed to East Asian countries, focusing on liberation for many people, transitioning from the arhat fruit to the Bodhisattva fruit saving sentient beings, philosophy influenced by local areas. Today, the countries and regions Tibet, China, Korea, Japan, and northern Vietnam follow Northern Buddhism.
3. Third period: is the development period of Tantric School and Zen School lasting until the 11th century. Philosophy discussed many problems of the universe with the ideal being Siddhas (Siddha) in Tantric School, those who achieve complete harmony with the universe, with no limits whatsoever.
4. Fourth period: Buddhism developed to many countries worldwide.

Each development period of Buddhism is linked with the birth of new scriptures. Most scriptures were born after the Buddha entered parinirvana but are still considered the Buddha's words. This is completely different from many other religions where scriptures gradually stabilize.

When the Buddha was still alive, he mainly taught in northeastern India. After the Buddha entered parinirvana, Buddhism was

[12]Lesser Vehicle (Hinayana) – a Sanskrit term meaning *Small Vehicle*. This term was used by thinkers of Mahayana (Great Vehicle) to call many Buddhist schools that only focused on individual liberation (becoming Arhat). Today, *Lesser Vehicle* is often seen as disrespectful or pejorative because it implies that the path of these schools is inferior. Most Hinayana schools have disappeared over time. Southern Buddhism Theravada is a Pali term meaning *Path of the Elders* or Elders' School, one of the schools in the Hinayana group that still exists today.

only one of very many schools in India at that time and developed in the west. After about 100 years, it expanded to some places further south. Only after King Asoka [King Asoka][13] took refuge in the Buddha Dharma and continuously propagated the true path did Buddhism emerge to become a religion followed by many people.

The sangha could not maintain unity forever and soon divided into very many schools. The Buddhist tradition of India usually speaks of 18 schools (possibly more), but to this day none exist except one school later called Southern Buddhism (Theravada). After the Buddha entered parinirvana about 140 years (340 BCE), Buddhism began to split into the *Elders' School* (Sthaviravada), also called the Elders' School with conservative characteristics and the *Great Assembly School* (Mahasanghika) with democratic characteristics [29].

The Elders' School tried to follow exactly as the Sangha of the initial period, still viewing the Buddha Dharma as a means to create a few arhats living isolated in monasteries. They considered themselves orthodox because monks need to strictly follow the regulations of the precepts. The Elders' School acknowledged that the Buddha had transcendent qualities but still had weaknesses of a secular human, such as physical pain when sick or irritation when things were unsatisfactory. The teachings were also simply "avoid all evil, do good, and keep the mind pure" by practicing *precepts*, *meditation*, *wisdom*. Precepts are keeping the precepts that monks and laypeople must follow. The *Five Precepts* for laypeople include avoiding killing, stealing, lying, adultery, and using intoxicants. For monks,

[13]Asoka was the third king of the Maurya dynasty, ruling India from 273 – 232 BCE, having the merit of unifying a territory approximately the Indian subcontinent through violent conquests. In 263 BCE, Asoka invaded the Kalinga state (now Odisha state of India) in a war killing 100,000 people and expelling 150,000. After that tragedy, he repented and voluntarily took refuge in the Three Jewels to become a fervent Buddhist follower. Thanks to Asoka, Buddhism developed from a religion of a few monks to a multi-national religion. Asoka's role with Buddhism is similar to the role of Roman Emperor Constantinus with Christianity.

the precepts are even stricter, such as must live celibate, avoid using luxuries, not eat after noon,... Meditation is meditation, keeping the mind peaceful to see the true nature of all things. Wisdom is wisdom to thoroughly understand the Four Noble Truths, Noble Eightfold Path, dependent origination, cause and effect, karmic obstacles.

The Elders' School held that the world has three characteristics: impermanence (anitya), suffering (duhkha), and no-self (anatma), called the *three marks of existence*. All sentient beings are constituted by two factors: name (nama) – the non-material component, and form (rupa) – the material component. The four non-material characteristics are feeling, perception, mental formations, and consciousness.

Some monks wanted to oppose the influence of arhats, holding that arhats should not be respected like celestial beings (celestial beings) and wanted to relax the precepts, so they separated from the Elders' School to establish the Great Assembly School. This school existed until the end of the first period. Decisions of this school were decided by majority, not by a small group of monks, so it became a bridge for many people to follow Buddhism.

The Great Assembly School held that the Buddha is a transcendent being with no defects or impurities whatsoever. Siddhartha in history is a miraculous manifestation of the Buddha to teach sentient beings and save humanity. At this time, the Buddha became the ideal of faith. After entering parinirvana, he did not disappear; thanks to *compassion*, he transforms into many forms to help sentient beings. The Great Assembly School also held that not only a few monks can understand the Buddha's philosophy, but sentient beings can also directly receive his help through previous manifestations in the form of Bodhisattvas. Regarding philosophy, the Great Assembly School held that human thought is fundamentally pure. Indeterminate things only arise suddenly and cannot affect that purity. Some people in this school not only doubted knowledge brought by the senses but also doubted knowledge obtained from language

and concepts. Everything in the world like reincarnation, Nirvana are illusions, not real. Only what transcends everything, quiescent in all dharmas, i.e., *emptiness* is true. The Great Assembly School is the basis for later developing into Northern Buddhism in the second period.

The second division is the separation of Pudgalavada, also called Pudgala or Vatsiputriya School, from the Elders' School around 280 BCE. The Person School held that *self* must still be considered alongside *no-self*. The self is a reality, a factor connecting different lifetimes to finally still attain Buddhahood. The Person School is a reaction of knowledgeable people against the uncertainty of rigid theories about dharmas. It is precisely the predecessor of Northern philosophy. The self is close to the emptiness of Madhyamaka philosophy later.

The third division is dividing the Elders' School into the *All Exists School* (Sarvastivada) and the *Differentiation School* in 244 BCE. The All Exists School held that things have an essence, this essence always exists, so reality includes past, present, and future. This theory studies the law of cause and effect and reasons that if there is an action in the past that can lead to good or bad consequences many years later, how will that happen? If we hold that only the present exists and the past disappears immediately after the action occurs, i.e., at the present moment, the past does not exist. Then something that does not exist can create an effect at the present moment is illogical. To overcome that illogic, the All Exists School held that reality includes past, present, and future all existing together.

Besides the three school divisions mentioned above, there are many divisions due to different understandings of other problems like Nirvana, death,... The schools of the first period occurred mainly within the territory of India and had not yet reached other countries. An important mark for Buddhism's development was King Asoka, a fervent disciple who sent missions to propagate Buddhism to many

places in the world like Greece, Sri Lanka,... in 250 BCE. However, the missions' journeys did not bring many results. Only the mission to Sri Lanka was successful when that country made Buddhism the state religion. For a long time, Buddhism in Sri Lanka maintained close relations with India. In the 3rd century BCE, the Elders' School was introduced to this country, and in the 1st century BCE, the canon of the Theravada school was compiled and recorded for the first time in Pali language of Sri Lanka so the Buddha Dharma could exist long-term. Asoka combined the divine authority of Buddhism with the legal authority of a king, thereby making Buddhism a religion developing strongly in many countries worldwide.

Later, Northern Buddhism became very popular, but Southern Buddhism still exists and develops to this day. The concept of Bodhisattvas, which originally appeared in Northern Buddhism, was absorbed by Southern Buddhism and became prominent regarding the Buddha's previous lives. The origin was parables and myths from India's rich folk story treasury, later modified to fit stories about Bodhisattvas. Small details about compassion previously were emphasized to become one of the virtues associated with Bodhisattvas. Southern Buddhism also acknowledged that the age had deteriorated, no longer the age of arhats escaping rebirth, leading to secondary purposes like rebirth into realms of celestial beings, of Maitreya (the future Buddha currently in the celestial realm) also being valued. Southern Buddhism developed to its peak around 400, then existed in India until around 800 but almost had no new additions.

1.2.4 Great Vehicle Buddhism

From around the beginning of the Common Era, a new school named **Great Vehicle Buddhism** (Mahayana) appeared in northwest and south India, places influenced by external thoughts like from

Greece, Rome, Mediterranean, Iran. Great Vehicle Buddhism developed through two stages. First from the beginning of the Common Era to 150, where Great Vehicle development was not yet systematic. The second stage from 150 to 500, where Great Vehicle was systematized to become two separate schools: Madhyamaka School (Madhyamaka) and Yogacara School (Yogacara). There are two causes for Great Vehicle to develop. First are teachings and Sangha: increasingly fewer people attained the arhat fruit because disciples lost motivation in the old teachings; laypeople wanted more influence and equality compared with monks [23, page 119]. Second are external thoughts penetrating India, making teachings need to change accordingly.

Great Vehicle wanted to reform teachings to maintain vitality for the religion. Although introducing many new thoughts, Great Vehicle still presented them as the Buddha's words. If Southern Buddhism mainly spread and influenced within India and South Asia, Great Vehicle Buddhism is the transformation of Buddhism so it could be accepted by East Asian and Northeast Asian civilizations and cultures. Therefore, later Great Vehicle was called Northern Buddhism. Changes of Northern Buddhism include the following characteristics:

1. Transition from the arhat ideal to the **Bodhisattva** ideal. If arhat focuses on liberating one individual, then Bodhisattva (Bodhisattva)[14] focuses on saving sentient beings, saving many people. Bodhisattvas also want to be completely enlightened like Buddha; they have enough wisdom to do that, but they voluntarily remain in the three realms to help others.
2. Adding a liberation method is **compassion**, placed equal with wisdom. Because Bodhisattvas have compassion, they did not remain in Nirvana but reached out to help sentient beings through sufferings.

[14] *Bodhi* means *enlightenment*, *sattva* means *sentient being*.

3. Sacred objects expanded to humans, i.e., *saints*. Previously, deities almost had no origin from humans; Northern Buddhism focused on deities with origins from ordinary humans. In Southern Buddhism, the role of deities and celestial beings was very limited; in Northern Buddhism, the number of celestial beings increasingly grew, many of whom are saints.
4. Saints have a completely new virtue: *skillful means* (skillful means), placed equal with, even above, wisdom. Skillful means is the ability to develop the highest spiritual capacity in each person through actions adjusted to suit that person's needs, abilities, and awareness. This virtue is flexibility to maximize individual capacity to promote sentient beings' liberation.
5. Teachings were systematized and supplemented much about ontology to explain reality with concepts of *emptiness*, *suchness*.

Northern Buddhism holds that there are 3 levels of liberation. Arhats and Pratyekabuddhas have attained the first 2 levels, which are *self-benefit*, self-liberation and indifference to worldly problems, called Lesser Vehicle – small vehicle (Hinayana). Buddhas, including Bodhisattvas, attain level 3, which is *benefiting others*, attaining liberation for oneself but still pursuing liberation for others, called Great Vehicle – Large Vehicle, also called Great Vehicle Buddhism. In Bodhisattvas, there is consistency between compassion and wisdom. Compassion to sacrifice oneself to serve others. Wisdom to recognize the non-existence of self and to penetrate the true reality of all things. There are six practices to attain compassion and wisdom, called the six perfections (paramita means the method to cross to the other shore), also called the six perfections. These six methods are giving, precepts, patience, effort, meditation, and wisdom. Compared with Southern Buddhism, those who want to practice to become Bodhisattvas need patience and generosity. Patience is not only enduring suffering and not being angry but also a virtue of wisdom. Bodhisattvas go through countless lifetimes to practice

perfections, never discouraged. The highest wisdom is precisely recognizing emptiness, which is the only reality.

The theory of emptiness (emptiness) and suchness (suchness) of Northern Buddhism was developed from the theory of the Great Assembly School.

Northern Buddhism holds that all dharmas are empty. That is, dharmas themselves are empty, and within dharmas is also empty, so all dharmas are the same, all are unreal. Emptiness can be called suchness if people can see the truth correctly. Suchness refers to the true, real, and unchanging essence of all things and phenomena. Reality is empty. If so, then the relative is the same as the absolute, the perceived is the same as what transcends perception, reincarnation is the same as Nirvana, all are the same, all are empty. To understand the true nature of reality, one must transcend both subject and object, both affirmation and negation. The above reasoning of Northern Buddhism is very difficult to understand and grasp, but it can be said that the deepest place of the teachings is precisely silence.

Madhyamaka School, a Buddhist school based on Madhyamaka philosophy, is one of the two main schools of Northern Buddhism, established around 150 by Nagarjuna in southern India. Nagarjuna (Nagarjuna), later called Bodhisattva Nagarjuna, is one of the greatest minds of India and the second most important figure after the Buddha. The most important philosophy of Madhyamaka is precisely skepticism. Nagarjuna held that all statements are the same in that nothing is certain, including the absolute. The Buddha chose the middle way, neither asceticism nor hedonism. The middle way of the Madhyamaka School is a dialectical method or an attitude of awareness that does not accept reality and does not accept non-reality, only accepts relativity.

Nagarjuna's doctrine is **emptiness** (sunyata). All dharmas have no self-nature; all things in the world do not arise and do not cease, non-origination. According to the two truths, at the absolute level,

dharmas have no self-nature, while at the conventional level, dharmas still arise dependent on conditions; therefore, emptiness is not synonymous with nihilism. There is no difference between reincarnation and Nirvana. Nirvana is precisely absolute emptiness. This school distinguishes two types of truth: conventional truth and ultimate truth. Conventional truth (samvrti-satya) is relative truth according to experience; it is an illusion of the senses brought by ignorance and delusion.

The law of cause and effect is an example of relative truth, conventional truth. Ultimate truth (paramartha-satya) is absolute truth, emptiness pervading all things, the empty essence of the world. These two truths do not contradict each other; in essence, relative truth is the manifestation of absolute truth.

Later generations all evaluate the Madhyamaka School as creating a turning point for Buddhism; it is the central, core theoretical system of most later Buddhist schools, while also establishing a solid foundation for Buddhist thought. The Madhyamaka School changed Buddhist thought, transforming from experience and dogma to dialectical critique. Before Madhyamaka philosophy, Buddhism followed precepts, had reverence for the Buddha as a revered human but not yet having any cosmic stature. Even his existence after entering parinirvana was still a question. After the Madhyamaka School, Buddhism developed into a true religion with a philosophical system based on dialectics.

Yogacara School appeared from Nagarjuna's time but was not perfected until the 4th century, thanks to Vasubandhu and Asanga. This school focused on yoga practice as a method to attain liberation. Monks must pass through ten stages of the Bodhisattva fruit to be able to attain Buddhahood (bodhi). The philosophy of Yoga is believing that all things in the world are illusions; only the absolute is *consciousness*. Thus, humans can attain liberation only through their own consciousness, without needing external objects because consciousness can transcend both subject and object. The

Yogacara School differs from the Madhyamaka School in that the Yogacara School holds that reality is pure consciousness, while the Madhyamaka School holds that reality is emptiness.

An important milestone for Northern Buddhism's development was King Kanishka, who was of Central Asian origin, ruling from 78 – 103, considered the Dharma-protecting King of Northern Buddhism. He organized the Buddhist council at Jalandhar Kashmir so that Northern Buddhism became the largest school. Also in his era appeared great philosophers like Nagarjuna, Asvaghosha (Mã Minh), and Vasumitra (Thế Hữu). Northern Buddhism was brought by monks to Central Asia and China. This was the golden age of Buddhism in India.

Later, Buddhism continued to develop together with Hinduism, coexisting and influencing each other. Some Buddhist schools like Tantric School took many elements from Hinduism, and conversely, Hinduism was also strongly influenced by Buddhism. Especially the Gupta dynasty, ruling in western India and Bangladesh from 320 – 550, a prosperous period where both religions were respected and developed with many new Buddhist centers built.

From around 500 to 1000 was the development of Tantric School and Zen School. **Tantric School** (Tantra) is the final development of Indian Buddhism to then leave development to other countries. From its birth, Buddhism abandoned important elements of Brahmanism and became one of the schools competing with Hindu religions. At its peak, Buddhism could stand alone with the original primitive teaching system. However, in difficult times, Buddhist schools like Tantric School appealed to supernatural, mystical powers to help them. Tantric School supplemented mystical elements into Buddhism with the purpose of achieving liberation more easily. Especially deities, spirits, demons,... were introduced into stories expressing the thoughts of the common class. Buddhism previously needed to practice many lifetimes to be able to attain Buddhahood; Tantra did not expect a distant future, aiming at achieving libera-

tion in the present, in this very body, thanks to new means that are quick and easy. The ideal of arhat was replaced by an ideal of a Siddha (Siddha), similar to a Bodhisattva, with supernatural powers of transformation.

Previously, scriptures and treatises were written, and Buddhists followed scriptures; in Tantra, types of scriptures developed very quickly and very many. That is because types of scriptures do not apply to all but are *secret*. Each selected person will have a teacher teaching in an appropriate way. Compilers of Tantra scriptures do not connect the content of scriptures with the Buddha but hold that there was a legendary Buddha who taught them the Buddha Dharma from long ago. Actually, the foundation of Tantra texts is taken from Yogacara School scriptures, but Tantra compilers often forget to mention those who taught them. Tantric School often uses *mantras*, also called secret words, or true words *mantra* (meaning *true words*), these phrases are introduced into scriptures. Later, people also supplemented mysterious drawings *mandala*, hand gestures *mudra* to enhance the effectiveness of mantras. Especially, Tantric School is famous for Thangka paintings in monasteries, stupas, and royal palaces (Figure 1.1).

A branch of Tantric School is *Vajrayana* (Vajrayana) accepting many hypotheses about cosmic and spiritual powers and combining them with human actions to achieve liberation. All activities of body, speech, and mind are means to advance toward enlightenment. Thus, sublime thoughts about liberation are identified with the most ordinary actions of the body; the most transcendent becomes the most mundane. Enlightenment is the result of combining wisdom and skillful means, presented as the union of male and female in the ecstasy of love, showing that enlightenment and ecstasy are indescribable (Figure 1.1). Tantric School, Vajrayana presents the path of body – speech – mind (three secrets) to unify means – wisdom.

Figure 1.1: (Left) Thangka is a type of painting hung in Vajrayana monasteries. (Right) The union of male and female in Vajrayana. Both images taken by the author in a monastery in Nepal.

1.2.5 Buddhist scriptures

After the Buddha entered parinirvana, the teachings were compiled by his disciples. The first council (Rajagaha, right after the Buddha entered parinirvana) was presided over by Mahakassapa (Ca Diếp), with Ananda reciting sutras and Upali reciting vinaya. Initially, there were only two collections: Sutra Pitaka and Vinaya Pitaka. The Abhidhamma Pitaka appeared later. This council was based on recitation mechanisms, standardization, but had not yet been recorded.

Three Baskets (Tipitaka) consist of 3 main parts: Vinaya Pitaka, Sutra Pitaka, and Abhidhamma Pitaka. Later, Theravada recorded that the Three Baskets were written in Pali on palm leaves in the 1st century BCE as the standard canon of Southern Buddhism. Other Buddhist traditions have parallel sutra – vinaya – abhidhamma (Chinese canon, Sanskrit), not identical with the Pali Three Baskets.

Vinaya Pitaka (Vinaya-pitaka) records precepts and rules of life, can be considered the constitution of the Sangha, appearing from

the first council but going through many years of development, and the current form could only appear at least 100 years after the Buddha's parinirvana [23, page 95].

Sutra Pitaka (Sutta-pitaka) of Southern Buddhism consists of 5 Nikayas:

1. Digha-nikaya (Long Collection, similar to Dirgha-agama in Sanskrit and Chinese canon): Long sutras, philosophy and ethics, like the Great Nirvana Sutra.[15]
2. Majjhima-nikaya (Middle Collection, Madhyama-agama): Medium-length sutras, practical, like the Satipatthana Sutra (Four Foundations of Mindfulness).
3. Samyutta-nikaya (Connected Collection, Samyukta-agama): Short sutras grouped by theme: aggregates, sense bases, dependent origination,...
4. Anguttara-nikaya (Numerical Collection, Ekottara-agama): Sutras arranged by number of teachings (1-11 dharmas).
5. Khuddaka-nikaya (Minor Collection): Collection of many small sutras that cannot be included in the above collections.

Abhidhamma Pitaka, also called Abhidhamma or Superior Dharma is the third part of the Three Baskets, considered the pinnacle of analytical thought and systematization in the Theravada tradition, while also being the basis for forming later philosophical schools in both Southern and Northern Buddhism. If the Sutra Pitaka presents the teachings using metaphorical language and practice, the Abhidhamma Pitaka expresses using conceptual language and analysis. The Theravada tradition holds that the Buddha himself taught the Abhidhamma Pitaka after enlightenment, then transmitted it to Sariputta (Xá-lợi-phất) to systematize. However, modern historical and linguistic research holds that the Abhidhamma Pitaka was gradually compiled from the 3rd to 2nd centuries BCE, when the Sangha developed strongly and needed a unified theory. Some parts

[15]Later, we agree when citing: Digha-nikaya is Pali Three Baskets, Dirgha-agama is Sanskrit Three Baskets, Chang A-han is Chinese Three Baskets.

of the Northern Abhidhamma Pitaka developed independently. The content of the Abhidhamma Pitaka relates to problems about dharmas, mind, conditions, and karma, concepts about liberation.

Mahayana sutras (Northern Buddhism) appeared around the 1st century BCE – 1st century SDL, developed strongly in the 2nd to 5th centuries, with the doctrine of expanding the Bodhisattva ideal and the concept of universal Buddhahood. Scriptures were written in Sanskrit (possibly a mixed language close to Sanskrit) and translated into Chinese, Tibetan, and Central Asian languages. The main Mahayana Sutras include [23, page 147]:

1. Prajnaparamita Sutra (Perfect Wisdom), abbreviated as Prajnaparamita Sutra, discusses the new wisdom of Northern Buddhism, is the foundation of the thought of Emptiness and non-self-nature of dharmas.
2. Lotus Sutra (Saddharma pundarika Sutra) advocates one vehicle (the only path to Buddhahood), elevates faith and compassion, skillful means.
3. Pure Land Sutra (Amitayurdhyana Sutra) explains the origin of the Pure Land realm of Amitabha Buddha, also called the Buddha of Infinite Light. The method of faith – vow – practice (reciting the name), avoiding saying "just reciting" can attain the realm of Ultimate Bliss (Western Paradise).
4. Avatamsaka Sutra (Flower Garland Sutra) presents dependent origination of the dharma realm, is the basis for the Chinese Huayan School, later transmitted to Japan and Korea.
5. Vimalakirti Sutra (Vimalakirti-nirdesa Sutra) presents non-dual thought and lay wisdom, emphasizing that enlightenment is not only for monks.
6. Lankavatara Sutra presents Yogacara theory (Yogacara), analyzes mind and alaya-consciousness.

The original scriptures of the Madhyamaka School are precisely the Prajnaparamita Sutras, used by Nagarjuna as the basis to write the Madhyamaka Treatise (Mulamadhyamakakarika, also called the

Middle Treatise). The Yogacara School is based on the Lankavatara Sutra, the sutra of consciousness. The sutras listed above are only the main axis of Northern Buddhism; in reality, it is much larger.

1.2.6 Buddhism in Tibet and China

China received Buddhism via the Silk Road very early, around the 1st century BCE. But throughout the Han dynasty, although some monks came here to teach, some scriptures were translated into Chinese, Buddhism had no place in society because of the dominance of Confucianism at that time and Buddhism contradicted the tradition of ancestor worship of the indigenous people. Even the upper class considered Buddhism a barbarian religion. Until the end of the Han dynasty, society was chaotic, the elite were disappointed with Confucianism, so they sought other religions. Mouzi (Mou-tseu) compared Buddhism with Confucianism and Taoism and affirmed the philosophical superiority of Buddhism, thereby creating trust with Chinese intellectuals at that time.

By the Wei dynasty, Buddhism developed strongly and was made the state religion. Foreign monks were protected by the court; Buddhist works were built and flourished. From then on, Chinese people not only expected Buddhist monks to come to their country but also actively sent people to major Buddhist centers in India to collect Buddhist sutras and organize translation into Chinese. Many translations have great literary value and became classic literary works of China. Buddhism influenced Chinese people's lives very deeply. Buddhist thoughts about rebirth, reincarnation, cause and effect, karmic retribution penetrated very deeply into worldviews, life views, and aesthetics. In turn, China also developed its own Buddhist schools and itself became a major Buddhist center of the world, from which spreading the Buddha Dharma to other countries like Korea, Japan, and Vietnam.

For example, after developing in India, Tantric School was trans-

mitted to Tibet, China, and Japan. The two branches of Tantric School, Mantrayana and Vajrayana, were transmitted to China around the 7th century and flourished in the 8th century. Around the 8th century, the Vajrayana branch was transmitted to Tibet and merged with Northern Buddhism already present there to create a special school only in Tibet: *Tibetan Buddhism*, also called *Lamaism* (Lamaism). Tibetan Buddhism has a spiritual leader, the Dalai Lama [Đạt lai Lạt ma]. In Mongolian, *Dalai* means *ocean* and *Lama* means *teacher* (SAN guru). Lamas are considered reincarnations of Bodhisattva Avalokitesvara (Avalokitesvara) to help sentient beings. The first Lama lived in the 14th century, and to this day is the 14th Lama currently living in exile in India.

Chinese Zen School is a Buddhist school that can be considered a school of Northern Buddhism, but many Zen followers consider themselves not belonging to Southern or Northern Buddhism. Zen School recognizes Venerable Mahakasyapa (Ca Diếp) as the first patriarch and transmitted to many subsequent patriarchs in India.[16] The event of the Buddha transmitting the robe and bowl[17] to Venerable Mahakasyapa has never been recorded in Southern or Northern Buddhism, so this event may not be true.

Patriarch Bodhidharma (Bodhidharma) came to Luoyang, China, at the beginning of the 6th century. Bodhidharma is precisely the connection between Chinese Zen School and the tradition in India, and Chinese Zen School greatly values that tradition, although later teachings were changed very differently from the original teachings. According to Zen School, the Buddha transmitted the teachings to Mahakasyapa and then to other patriarchs through *transmitting mind to mind, not using words*. Zen School had strong development steps and divided into 2 stages: the Tang dynasty period and

[16] Among the patriarchs of Zen School, some are very famous like Ananda, Nagarjuna (Long Thọ), Vasubandhu, and Bodhidharma.

[17] Transmitting the robe and bowl means transmitting the authority to manage the school from one patriarch to another patriarch.

the Song dynasty period. Later, some Chinese monks brought Zen thought to Vietnam. Previously, in the 3rd century SDL, a Vietnamese Zen branch had been transmitted to China by Vietnamese Zen master Khương Tăng Hội.

1.2.7 Buddhism in Vietnam

Buddhism entered Vietnam from the 1st century with the legend of Chử Đồng Tử learning the way from an Indian monk. Around the 2nd century, Buddhism continued to develop with a center at Luy Lâu (now in Bắc Ninh) with legends about Thạch Quang Phật, Man Nương Phật Mẫu, and the system of beliefs worshipping the Four Dharmas. Regarding names, **Bụt** is an early Vietnameseization of Buddha, popular in folk literature like *Tấm Cám*; later, under Chinese influence, **Phật** became the standard way of calling in scriptures. During the Lý – Trần dynasties, Buddhism was strongly sponsored by the state and held a central role in spiritual life. By the Lê dynasty, Confucianism prevailed; Buddhism narrowed its influence in court politics but continued to develop in village communities.

In 580, Vinitaruci (Vinitaruci) came to Vietnam, founding the Zen branch at Dâu Temple, Bắc Ninh. This school transmitted for 19 generations and had great influence on Lý dynasty kings. Some research records that Vietnamese Zen masters later practiced Tantric School; however, Vinitaruci's transmission line is Zen. In history, some Zen masters like Vạn Hạnh, Từ Đạo Hạnh, Nguyễn Minh Không, Dương Không Lộ, Nguyễn Giác Hải used Tantric School alongside Zen practice, so the boundary between Zen – Tantra in folk practice sometimes overlaps.

In 820, a Zen master belonging to Huệ Năng's school, Vô Ngôn Thông from China, came to Vietnam and practiced at Kiến Sơ Temple (now in Hà Nội). This master's school lasted 17 generations with some important Zen masters like Nguyễn Minh Không, Nguyễn Giác Hải. By 1069, a Zen master, Thảo Đường, was captured by King

Lý Thánh Tông when conquering Champa. Due to special abilities, the Zen master was released and appointed as National Teacher, abbot of Khai Quốc Temple (now Trấn Quốc Temple, Hà Nội). He was the Zen master who founded the Thảo Đường Zen School and transmitted for 5 generations.

By the 13th century, King Trần Nhân Tông unified the three Zen schools: Vinitaruci, Vô Ngôn Thông, and Thảo Đường to establish **Trúc Lâm Yên Tử**, practicing at Yên Tử (now in Quảng Ninh), the first Zen school founded by Vietnamese people. This Zen school developed prosperously for the first 3 generations, including Điều Ngự Giác Hoàng (Trần Nhân Tông), Pháp Loa, and Huyền Quang, then gradually lost its position but still transmitted for 15 generations after Huyền Quang. By the Lê Trung Hưng period, Venerable Chân Nguyên Tuệ Đăng and Ngô Thì Nhậm had the merit of collecting, preserving, and publishing important works related to this Zen school.

Later, Zen School in Vietnam also developed based on importing Zen branches from China, such as Caodong School introduced by Zen Master Thông Giác Thủy Nguyệt in the 17th century, Linji School introduced by Zen Master Viên Văn Chuyết Chuyết during Lê Trung Hưng. About 150 years ago, Zen School gradually lost influence and was replaced or influenced by Pure Land School and Tantric School.

A school that was developed by Chinese people and became the most popular school in China, Japan, and Vietnam today is **Pure Land School** China, founded by first patriarch Huệ Viễn (334 – 416).[18] The purpose of this school is to be reborn in the Western Paradise, the Pure Land realm. Followers of this school emphasize faith – vow – practice: faith, making vows, practicing recit-

[18]Some documents hold that Pure Land School was born in Kashmir and Central Asia. Philosopher Nagarjuna, who is revered by tradition in China as the founding patriarch of Pure Land School, mentioned Amitabha Buddha [23, Chapter 12].

ing the name "Namo Amitabha Buddha," in which the other-power of Amitabha Buddha and the self-power of practitioners mutually assist. Besides Amitabha Buddha, there are two important Bodhisattvas: Avalokitesvara (Avalokitesvara) and Mahasthamaprapta (Mahas-thamaprapta).

Bodhisattva Avalokitesvara (some read as Quan Thế Âm) is the Buddha who contemplates the sounds of the world, the Bodhisattva manifesting the compassion of all Buddhas, most widely worshipped in Northern and Southern Buddhism, not only Pure Land School. In India, he is male, but when entering China and Vietnam, he is female. The image of Avalokitesvara is also diverse, with about 33 forms. Sometimes he is shown as a Buddha mother – one hand holding a willow branch and one hand holding a vase of nectar; sometimes as an image of a thousand-armed, thousand-eyed Buddha.

Bodhisattva Mahasthamaprapta is the Bodhisattva manifesting the light of wisdom of Northern Buddhism. Together with Bodhisattva Avalokitesvara manifesting compassion, Bodhisattva Mahasthamaprapta is the assistant of Amitabha Buddha in the Western Paradise. Pure Land School may have entered Vietnam very early, around the 4th or 5th century, but by the Lý, Trần dynasties, it developed extremely prosperously. We can see the phenomenon of worshipping Amitabha Buddha at Phật Tích Temple. But some people also hold that the phenomenon of seeking external other-power is common in Vietnam, not necessarily originating from Chinese Pure Land School. Pure Land School is the most popular school in Vietnam today. In most temples of northern Vietnam, in the second row, Amitabha Buddha stands in the middle, Bodhisattva Avalokitesvara stands on the left, and Bodhisattva Mahasthamaprapta stands on the right of Amitabha Buddha.

Today, Buddhism is the fourth largest religion in the world with about 488 million followers, behind Christianity, Islam, and Hinduism. Statistics in recent years, among major religions in the world,

the number of people following Buddhism is increasingly decreasing while other religions are increasing. However, Buddhism is increasingly influential in the West as a philosophical religion. Leading world scientists all research and trust Buddhism.

1.2.8 New values of Buddhism

In the context of Hinduism in the 6th century BCE, Buddhism brought the following new values:

1. Four Noble Truths: suffering is placed as the center of Buddhism to deeply understand the cause, cessation of suffering, and the path to cessation of suffering.
2. Middle Way principle: the right path is the middle way, not leaning toward the extremes of hedonism or asceticism. The middle way is manifested in the Noble Eightfold Path, returning to wisdom, precepts, and meditation.
3. Dependent Origination principle: the mutual dependence of all things and phenomena. This leads to the three marks of existence (Northern Buddhism), the three characteristics of existence: impermanence, no-self, and Nirvana. No-self is the central concept, completely opposite to the self-view of Hinduism. Buddhism criticizes permanent self-nature (svabhava) but does not deny convention, cause and effect – the most important difference compared with other religions and philosophical doctrines.
4. Criticism of castes: in the Sangha, everyone can leave home and practice regardless of lineage; human value is established by precepts – meditation – wisdom, not by ancestry.
5. Elimination of deity worship: the existence of objective cause-and-effect laws, so worshipping Buddhas is honoring and reminding of the Buddha Dharma, not superstition and seeking help.

These are the new thoughts that Buddhism brought, which will be presented in detail in this book. Eliminating castes and deity worship in Buddhism is similar to Anti-Vedic religions, but only Bud-

dhism and Jainism succeeded and exist to this day. The elimination of castes is a revolution against the caste system in Hinduism. Not only are humans equal in practicing the Buddha Dharma, but even all living beings in the world can have the capacity to practice like humans. This is revolutionary equality that only exists in Buddhism, as a religion thought up by humans but not placing humans at the center like many other religions.

Previously, anyone wanting to pray to deities needed to go through the Brahmin priest class. Buddhism holds that everyone has the right and ability to practice to attain Nirvana without needing to go through any intermediary class. That is the great value that Buddhism brought to Indian society at that time. Buddhism's equality is not limited to humans; other living beings can also practice to be able to attain liberation. Humans have advantages in awareness compared with other living beings, but that is not only a human privilege. A question arises: do living beings have consciousness and wisdom to be able to practice to attain liberation? This book cannot answer directly but can provide readers with clues about the wisdom and awareness of all species to gradually approach the answer.

Another value of Buddhism is eliminating deity worship. Hinduism puts reverence for the way, worshipping deities as one of three ways to practice the right path; Buddhism holds that this is not necessary because all things in the world occur due to causes and conditions, due to our actions in the past and present. Worship in Buddhism is honoring and reminding Buddhists about teachings, not worship to seek help.

1.3 Principles of Buddhism

This book presents three basic principles of Buddhism according to the author's thinking, with the purpose of clarifying the emptiness of all things. They are dependent origination, suffering, and the

middle way. The Dependent Origination principle – also called the principle of interconnectedness – says that all things lie in causal relationships, are the foundation of all things, thereby leading to no-self. Due to not understanding no-self, attachment arises, thinking there is a self-nature, independent, unchanging self, leading to the second principle: suffering. To be liberated from suffering, one must practice the middle way, practice without going to extremes, even transcending extremes.

Understanding Buddhist principles as above may not be the same as understanding principles recorded in traditional texts and documents about Buddhism. Many documents all agree that suffering, Noble Eightfold Path, liberation, Nirvana are basic principles of Buddhism [30]. However, presenting principles varies depending on the purpose of documents. For example, to apply Buddhism to oppose war and bring peace to places with armed conflicts, some scholars hold that non-violence, middle way (through Noble Eightfold Path), equality are important principles [31]. Some others hold that ethics, wisdom, and practice are basic principles of the three main schools: Lesser Vehicle, Great Vehicle, and Vajrayana when researching in Thailand [32]. The presentation of principles in this book is to prove emptiness in material, biological, and consciousness worlds, not aiming to refute ideas of any understanding or thought of any school.

1.3.1 Dependent origination

The Vedic tradition, especially the Self Theory of Upanishad, holds that there is a permanent, unchanging entity called the soul, small self, atman, independent of the external world. Atman always exists, reincarnating from this lifetime to another in the cycle of reincarnation. Atman only attaches to the body and consciousness temporarily, not depending on that body. Atman is identical with brahman, the fundamental, supreme reality that is the foundation of all things.

Table 1.1: Differences between two Indian philosophical traditions: Self Theory and No-Self Theory.

Self Theory (Upanisad)	**No-Self Theory** (Buddhism)
There is a permanent, unchanging entity called the soul or self, independent of the external world.	No permanent, independent soul exists; all things always transform, cannot be separated from the external.
The real existent, also called true form, is permanent.	The real existent is changing, a false thought with no reality.
Much concern about the universe.	Not concerned about the universe but criticizes experience itself.
There is an absolute self called true self, i.e., Brahman.	Self is the pinnacle of ignorance from which suffering arises. Denying self is the key of Buddhism, the teaching for liberation.

The famous saying "Tat tvam asi" (That is also you) expresses the concept of unity between individual and universe. The most important point of Buddhism is rejecting a permanent, unchanging atman, i.e., rejecting the main viewpoint in Upanishad of Hinduism (Table 1.1). Hinduism seeks eternal reality; Buddhism seeks reality without self-nature and dependent on conditions. The view of no-self liberates humans from the illusion of self, thereby breaking the cycle of suffering.

No-Self Theory of Buddhism holds that all things in the world cannot naturally appear; they need causes (hetu) and appropriate conditions (pratitya). Nothing exists without **causes and conditions**; when causes and conditions end, things no longer exist. Nothing exists independently or permanently because the existence of any thing must have causes and conditions from other things, and in turn, that thing is a cause and condition for other things. Such mutual relationships between all things are called **dependent origination**, also called the law of *interdependence* (dependent origination, dependent arising, SAN pratitya samutpada).

The Buddha taught in the Majjhima-nikaya (MN 3.63):

> "Because this exists, that exists; because this arises, that arises. If this does not exist, that also does not exist; if this ceases, that also ceases."

Dependent origination of early Buddhism focused on explaining the origin of suffering and rebirth, applying to sentient beings without needing to assume a permanent self. Dependent origination is presented in many forms, notably the twelve links of dependent origination. Theravada Buddhism developed dependent origination into a detailed model consisting of 24 types of conditions (paccaya), mind arising and ceasing moment by moment, actions – karma – results forming interdependent chains. However, there were already signs exceeding the scope of sentient beings, used as a principle for all conditioned phenomena (conditioned phenomena), not just the mind-body process of an individual, but this meaning was not emphasized in early Buddhism.

Buddhism holds that all things and phenomena lie in causal relationships with each other. The principle of cause and effect is universal everywhere, at all times, repeating. The principle of cause and effect has two components: laws and conditions of occurrence. In Buddhism, laws are called causes (hetu), conditions of occurrence are called conditions (pratitya). Causes and conditions is a commonly used term of this religion. In science, laws are precisely natural laws; conditions of occurrence are boundary conditions. A scientific problem needs these two components; lacking one of the two, the problem cannot be solved. If there are the same cause and the same boundary conditions, Buddhism and science give the same judgments, with repetition, therefore verifiable.

According to Buddhism, for humans, the cause of much suffering comes from the three poisons: greed (greed), hatred (anger), delusion (ignorance). To be able to practice the Noble Eightfold Path, one needs to clearly understand the **twelve links of dependent origination** leading to suffering. Starting from (1) ignorance,

i.e., stupidity, in the mind appear the three poisons greed, hatred, delusion, from which create (2) volitional formations, actions creating karmic retribution. Ignorance and volitional formations are two causes of the past, from which create present suffering results: consciousness, name and form, six sense bases, contact, feeling. (3) Consciousness is awareness obtained because volitional formations are the cause; consciousness is the condition for appearance. (4) Name and form: name is mental, form is physical. When there is name and form, (5) six sense bases appear. Six sense bases include six sense organs and six sense objects. Six sense organs are eyes, nose, ears, tongue, body, mind, which will condition with six sense objects: forms, sounds, smells, tastes, touches, dharmas, from which (7) feeling arises. Feeling is sensation, including types of suffering, pleasure, and neutral. Next are 3 present causes. (8) Craving is desire, including desire for five sense objects (related to 5 senses), desire for existence (craving for status, realm of existence,...), desire for non-existence (wanting to eliminate suffering through extremes,...) From craving arises (9) grasping, desire to then want to possess (10) becoming. The three present causes create two present results: (11) birth and (12) aging and death.

The twelve links of dependent origination are dependent origination applied to the mind-body of sentient beings – the cause of suffering. Actually, the elements of the 12 links connect with each other continuously, forming a circle, a cycle without separation and interruption, without a beginning and end point. It is not only from ignorance leading to aging and death as the end, but aging and death again form ignorance in a non-stop cycle. The cycle of the 12 links has no beginning and end point, meaning all elements can be the beginning and end. Any element in the 12 elements leads to the cycle of suffering. Liberation in Buddhism is not only liberation from one element but liberation from all cycles of the twelve links creating suffering.

* * *

Nagarjuna of the Madhyamaka School expanded dependent origination, not only applying to the mind-body process of sentient beings but to all dharmas in the universe. He developed it thoroughly, leading to all dharmas having no self-nature, i.e., "emptiness." The Madhyamaka School liberated dependent origination from ethical cause and effect to become an ontological principle: no things exist independently, and instead, all dharmas exist through relationships.

The Yogacara School added an epistemological dimension to dependent origination: focusing on the stream of consciousness and the concept of storehouse consciousness (alaya-vijnana: alaya-consciousness) containing seeds. Thus, dependent origination is not only relationships between external dharmas but the interdependent structure between internal states of consciousness [Peter Harvey]. Huayan (Huayan) put forward the thought that all dharmas not only dependently originate but also interpenetrate and inter-enter; each dharma "contains" all other dharmas. Dependent origination is Indra's net, the holistic interconnection between all phenomena.

Continuing the tradition of the Madhyamaka School, a Tibetan sage living in the 14th century, Tsongkhapa, brought dependent origination to this systematic level. Tsongkhapa respected Nagarjuna but held that dependent origination was misunderstood when only viewed as cause and effect. Therefore, he stratified dependent origination operating on three levels [33, 34]: (1) causal dependence by causes and conditions; (2) dialectical, mutual dependence of parts on the whole and vice versa (whole–part dependence); (3) dependence on conceptual imputation, i.e., all phenomena exist because consciousness assigns labels and concepts to reality. Causal dependence is the most familiar: things and phenomena A in the current state come from causes and conditions B before, and in turn, B is the cause of things and phenomena C afterward. Cause and effect emphasize the constant change of things and phenomena – *imper-*

manence is dependent origination in time. Whole-part dependence shows that a phenomenon only exists as a whole when it depends on constituents. The whole has no independent self-nature apart from components – *compositionality is dependent origination in space.* Dependence on conceptual imputation explains that a thing and phenomenon exist depending on our consciousness labeling them by naming them for convenience of use – *conceptual imputation is dependent origination in mind.* When understanding dependent origination at these three levels, we see all phenomena are **without self-nature** (nihsvabhava).

According to Northern Buddhism, dependent origination leads to three important characteristics called **three marks of existence** (three marks of existence), not existing independently of each other but, on the contrary, closely related, supplementing each other:[19]

1. *Impermanence* (impermanence): nothing is permanent, eternal. All things always transform, change continuously, not depending on our wishes.
2. *No-Self* (no-self): there cannot be a self, permanent, independent, self-existent because all things that exist have causes and conditions. Because there is dependent origination, all things are interconnected, dependent on each other, together transforming continuously.
3. *Nirvana*: When clearly seeing impermanence and no-self, the mind does not grasp, attains a state of absolute tranquility. Nirvana is not a separate realm or a permanent entity in the ontological sense but is an unconditioned dharma, transcending birth-death due to dependent origination.

In the sutra *Simile of the Elephant's Footprint* (Mahahatthipadopama) of Majjhima Nikaya (MN 28), the Buddha said that "whoever sees [the principle of] dependent origination sees the Dharma; whoever

[19]Southern Buddhism holds that the Three Marks consist of impermanence, suffering, and no-self (see page 31).

sees the Dharma sees [the principle of] dependent origination." Northern Buddhism, especially the Madhyamaka School, has a different explanation of dependent origination. Dependent origination is mutual interdependence between dharmas. Dharmas depend on each other to exist; they cannot exist independently; they have no self-nature – that is precisely **emptiness** (emptiness, SAN sunyata). Nagarjuna held that emptiness is manifested in that all things and phenomena are non-permanent, non-intrinsic, non-independent.

Emptiness in Buddhism is similar to the principle of universal interconnection in Marx's philosophy, one of the two principles of this doctrine. The principle views things and phenomena as lying in binding relationships, mutually interacting. These relationships have three properties: objectivity, universality, and diversity. In addition, Buddhism and Marx's philosophy also have in common viewing things and phenomena in dialectical relationships with each other. We will examine the manifestation of emptiness in biological systems at three levels: molecular, cellular, and individual in Chapter 4.

1.3.2 Suffering

The Four Noble Truths, also called the four supreme truths (four noble truths), are acknowledged by all Buddhist schools as the most fundamental truth of Buddhism. In the Setting in Motion the Wheel of Dharma Sutra, the first teaching after enlightenment, the Buddha spoke about the **Four Noble Truths**, consisting of the truth of suffering, the truth of the origin of suffering, the truth of the cessation of suffering, and the truth of the path. The truth of suffering says that all conditioned phenomena carry the nature of suffering because they arise and cease, are impermanent, no-self, cannot bring lasting happiness. The truth of the origin of suffering speaks of the cause of suffering, which is ignorance, human stupidity not recognizing that all pleasures and sufferings are illusions, not recognizing

that all things have emptiness. The truth of the cessation of suffering says that suffering can be ended through liberation to attain the state of Nirvana. The truth of the path presents eight methods to end suffering, which is precisely the Noble Eightfold Path.

The first noble truth is *the truth of suffering* (Truth of suffering, SAN dukkha), holding that life is an ocean of suffering. Suffering is an inherent part of existence, from birth to old age, sickness, and death, all are suffering. Human consciousness about self and world comes from the *five aggregates* consisting of form, feeling, perception, mental formations, and consciousness. Because the five aggregates are impermanent, transforming continuously, the body with five aggregates is the first suffering when born. Death is the dissolution of the body, the dissolution of the five aggregates, so that is also suffering. Besides physical suffering like old age and sickness, which are easy to recognize, there is also mental suffering. Not achieving desires leads to suffering, which is also easy to understand, but after achieving desires, thinking there will be happiness forever, but after a short time, it will disappear. Humans again have other desires replacing previous desires; if not achieved, again suffering. And thus suffering repeats throughout human life.

Buddhism points out three types of suffering: physical suffering, mental suffering, and suffering from ignorance [35]. Physical suffering from pain is the most easily recognizable type, manifested in birth, old age, sickness, and death. This is typical of **physical suffering**. The Buddha recognized four types of physical suffering to abandon the luxurious life of a prince to leave home and seek the way. The second suffering is **mental suffering**, suffering at a more subtle level. Mental suffering relates to pleasure, sensations brought by the brain's reward system. However, this sensation quickly becomes numb to external stimuli. A stimulus like achieving fame, power, even emotions creates immediate pleasure, but humans will quickly feel bored, even become disgusted with newly achieved things. The third suffering is **suffering from ignorance**; this is

the most subtle suffering, most difficult to grasp, and also the most difficult to overcome; this is the suffering of life. The origin of suffering stems from ignorance, stupidity not recognizing that the entire world is impermanent, not understanding that everything is created by causes and conditions and also ends due to causes and conditions. What has a beginning must have an end, so desires to possess and seize will necessarily lead to suffering.

The second noble truth is *the truth of the origin of suffering*(origin [of suffering]), holding that desires and cravings are precisely the cause of suffering. Craving (tanha) is human stubbornness wanting to have what they like materially and mentally. If humans get what they like, all problems in life will dissolve. Desires for rebirth, satisfying desires, everything being permanent are causes of suffering. Not only craving but conservatism in life is also a cause leading to suffering. This is a very special point in Buddhism because other religions require strict adherence to dogmas; the Buddha rejected that and held that mere belief is not enough but must also understand, verify, and practice to be able to end suffering.

The third noble truth is *the truth of the cessation of suffering*(cessation [of suffering]), holding that when knowing the cause of suffering, suffering can be eliminated by eliminating its cause, eliminating craving and desire. When eliminating all craving, eliminating all causes of suffering, humans can attain a state of full tranquility called **Nirvana** (nirvana literally means *blowing out* the candle of desire). Nirvana is permanent, unchanging, transcending the law of cause and effect that all things must follow dependent origination, thereby escaping reincarnation. Nirvana is absolute truth, not in any other place but precisely in this world. It does not lie outside the experience of the senses.

The fourth noble truth is *the truth of the path*(path [of the cessation of suffering]), holding that there are 8 steps to reach the cessation of suffering, called the **Noble Eightfold Path** (noble eightfold path). The Noble Eightfold Path is divided into three main aspects

Table 1.2: The Noble Eightfold Path is 8 steps to be able to end suffering.

Wisdom	1	Right View	Having correct views and understanding
	2	Right Intention	Eliminating delusion, increasing wisdom
Precepts	3	Right Speech	Not speaking, not hearing evil things
	4	Right Action	No actions of greed, hatred, delusion; practicing correctly according to the five precepts
	5	Right Livelihood	Earning a living by honest profession
Meditation	6	Right Effort	Effort to practice for liberation
	7	Right Mindfulness	Awareness of self and surroundings in the present moment
	8	Right Concentration	Maintaining right mindfulness

of life: wisdom (SAN prajna, ENG wisdom), precepts (SAN sila, moral virtue), and meditation (SAN samadhi, meditation). Wisdom is understanding and determination, including the first two right paths: right view and right intention. To keep precepts, people need to practice the next three right paths: right speech, right action, and right livelihood. Meditation is mind training, including the last three right paths: right effort, right mindfulness, and right concentration (Table 1.2).

1.3.3 Middle way

One of the most important principles of Buddhism is the **middle way** (middle pathway, SAN madhyama-pratipad). The Buddha's own life is the life of a human living according to the middle way. Living according to the middle way is living the path between two extremes, not too luxurious and hedonistic but also not too austere and self-mortifying. There are many legends about this period of the Buddha. According to them, in the first 30 years of life, because he belonged to the upper class, Siddhartha enjoyed all worldly pleasures; there was no material thing in the world that he wanted but

did not get. He had a wife, children, and a materially complete life with many servants. That was a hedonistic life in the true sense. However, the material pleasures of life could not keep the prince, and due to special causes and conditions, he left home. After leaving home, the Buddha once practiced asceticism for a long time but still did not see the light of liberation. The Buddha rejected the view that all things have no meaning held by ascetic monks and the hedonist school.

Southern Buddhism holds that the middle way means avoiding two extremes of sensual pleasure and asceticism, practicing according to the Noble Eightfold Path to escape reincarnation. Northern Buddhism expands the middle way to avoid fixed reality (eternalism) and nihilism (annihilationism). Existence and non-existence are only relative. The middle way is a consequence of the dependent origination principle, the principle of mutual interdependence of all things and phenomena because reality is not fixed and also not nihilistic. Reality is dependence, and that dependence changes continuously. Dependent origination leading to emptiness is correct awareness of the dependent and non-fixed nature of phenomena. The middle way combines with the Bodhisattva ideal to guide humans toward compassion, wisdom, and altruism as ways of living and practice.

The middle way is similar to the Tao in Taoism in the sense of following nature. Many Northern teachings hold that the middle way transcends even nature, transcends both grasping existence and grasping non-existence, seeing the existence of all things but not accepting it as it originally is. To avoid falling into metaphysical and unverifiable definitions, this book accepts the middle way in material, life, human, and consciousness worlds as an optimization of contradictory aspects. For example, in physics, the principle of least action, minimizing the energy of a system, is precisely the middle way. In evolution theory, natural selection operates according to the principle of the existence of the fittest. Fitness here is mini-

mal resources (matter, time, space) that organisms must expend to achieve the same result.

Not limited to the middle way of living, Buddhism also expands to middle way thinking. The middle way is not simply choosing to go in the middle but needs correct awareness of the interrelationship between all things to be able to have correct action. Northern teachings, especially the Madhyamaka School, expand the middle way principle not only as a way of living but become a profound philosophical system with a spirit of tolerance and non-attachment.

* * *

In the Madhyamaka Treatise, Nagarjuna wrote: "The Buddha's teachings are completely based on the two truths" [36, 24.8]. The two truths here are two types of truth: relative truth, also called conventional truth or conventional truth (relative truth, SAN. samvrti-satya), and absolute truth, also called ultimate truth or ultimate truth (ultimate truth, SAN. paramartha sataya). The Buddha himself did not give any specific reference to the two truths and the difference between them. Only Abhidhamma (Abhidhamma) mentioned and spoke of the difference between these two types of truth based on researching the Buddha's thoughts. Later, many Buddhist schools like Sarvastivadin (Sarvastivadin), Sautrantika (Sautrantika), Madhyamaka (Madhyamaka), and Yogacara (Yogacara) acknowledged the existence of the two truths and held that without understanding them, one cannot attain liberation [37].

According to Vasubandhu (Vasubandhu) of the Sarvastivadin school, relative truth manifests relative entities in three aspects: (1) it is reduced physically and logically when that entity is destroyed, divided physically, and disappears from the mind through logical analysis; (2) its identity does not exist independently but comes from components that make it up; (3) it is the result of mental structure, not existing in essence. For example, a pot is a relative entity because we can use physical methods to reduce the pot into

iron pieces when cutting it. Iron pieces no longer function for cooking, so the pot no longer exists. The pot's identity comes from the combination of iron pieces, and the existence of that identity is not essence but human consciousness about a utensil with a cooking function.

In modern science, all entities are reduced to other smaller components. Atoms are formed from nuclei and electrons. Electrons are fundamental particles, but nuclei are not; they are formed from protons and neutrons. In turn, protons and neutrons are formed from quarks. Fundamental particles like quarks and electrons cannot be reduced further, but they are always being born and disappearing. Viewed from afar, they appear as unchanging particles with some properties like mass, charge, spin,... According to Sarvastivadin, entities, atoms, nuclei are relative concepts, conventions because they can be reduced physically and logically into smaller particles. Their identity is obtained by the collection of fundamental particles determined by causes and conditions. Human consciousness assigns them identities, but actually they have no inherent attributes. Fundamental particles are absolute concepts because their nature is not borrowed from other entities. However, many modern scientific theories like quantum field theory and string theory do not hold that electrons, quarks, and neutrinos are fundamental particles. Therefore, we do not examine the two truths from the perspective of Sarvastivadin.

1.3.4 Emptiness

We have mentioned emptiness in previous sections but not fully. This section presents emptiness and the principle of reflecting emptiness into four worlds: matter, life, humans, and consciousness in subsequent chapters of this book. The Buddha was the first to mention the concept of no-self, an element closely related to emptiness. He held that there is no fixed, independent, self-existent self be-

cause all things and phenomena arise due to causes and conditions. Human consciousness about all things is obtained because of the five aggregates. Because the five aggregates have no independent, self-existent, unchanging nature, awareness of self and all things is essentially a temporary combination brought by the five aggregates.

Some researchers hold that the seed of emptiness existed from the Buddha's time, although at that time emptiness mainly focused on no-self.[20] In the Great Emptiness Sutra (Mahasunnata sutta), Majjhima-nikaya, the Buddha mentioned achieving *internal emptiness* and *external emptiness*. The Buddha explained internal emptiness as follows: "While I attend to internal emptiness, the mind delights, has faith, abides, [is] directed toward internal emptiness." As for external emptiness, the Buddha said: "While that person attends to the immovable, the mind delights, has faith, abides, directed toward the immovable." Thus, in essence, emptiness in Southern Buddhism refers to purity, concentration, having no distracting thoughts in the mind. Abiding in emptiness is eliminating greed, hatred, delusion; it is the emptiness of the six sense bases and the senses.

Buddhist thinkers developed the Buddha's no-self into emptiness. The person with the greatest merit for the development of Northern Buddhism and emptiness is Nagarjuna (Nagarjuna) with the Madhyamaka School. Nagarjuna used dialectics to analyze errors about reality. First, all things have causes and conditions, i.e., things exist thanks to causes and conditions. When causes and conditions exist, things exist; when causes and conditions end, things also end. Causes and conditions are many and always interact, changing continuously, leading to all things, including self, cannot exist indepen-

[20]In the 5th century, the Buddhist center Nalanda (Nalanda mahavihara) was formed and became a large and world-famous research center. The Tang dynasty monk Xuanzang went to India to research Buddhism and stayed at Nalanda for 2 years out of a total of 5 years. Around 1200, the invasion of the Islamic country named Turko-Afghan burned most documents in Sanskrit and massacred many monks and nuns of the center. Therefore, researching documents in Pali can bring much important information about early Buddhism.

dently and unchangingly. All things that appear are results of causes from before; in turn, they are causes of other things. That process is continuous and inseparable, creating a chain of cause and effect. For humans, Nagarjuna held that consciousness also has no self-nature; it comes from constituent elements, the five aggregates. These elements do not exist independently but only exist in connection with other elements.

Applying emptiness to sentient beings, we see the body is a combination of the *five aggregates* (skandha) consisting of form, feeling, perception, mental formations, and consciousness. The form aggregate is a concept referring to material elements of things and phenomena, including the human body. Buddhism holds that matter is formed from the four great elements: earth, water, fire, and wind. Material forms creating sense organs like eyes, ears, nose, tongue, body, and their objects like images, sounds, smells, tastes, touches all belong to the form aggregate. The feeling aggregate is a concept referring to sensations that humans obtain through the sense organs. They can be pleasant, unpleasant, or neutral sensations. The perception aggregate is a concept referring to recognition, discrimination, and assigning meaning to things and phenomena brought by the sense organs. The mental formations aggregate is a concept referring to states, mental activities, will, and spiritual tendencies. The consciousness aggregate is awareness, i.e., the mind's ability to know and recognize objects through the sense organs. It is awareness of the presence of things and phenomena. The five aggregates do not exist independently but always interact and depend on each other, creating human experience. No component in the five aggregates can exist independently and self-existently. This leads to no-self, a concept the Buddha presented in his teachings.

The analytical method of the Madhyamaka School includes contradictory and interacting aspects to prove that all concepts about reality have no fixed nature. Nagarjuna often used these contradictions to point out that all concepts cannot be clearly determined

and therefore have no self-nature.

> **The absence of a self-nature, independent, unchanging nature of all things and phenomena is called emptiness**.'

This is a complex and profound concept in Buddhist philosophy. From the Buddha's no-self to Nagarjuna's emptiness, Buddhism achieved very significant development. After the Madhyamaka School, Buddhism was propagated throughout the world until today. Emptiness and dependent origination are closely related because their presence coexists, one is the basis for the other to exist. The Madhyamaka School equates dependent origination with emptiness and holds that emptiness is absolute truth. The opening verse in *Contemplation of Dependent Origination* expresses the entire content of the Madhyamaka Treatise [36, 1.1]:

> "Neither arising nor ceasing,
> Neither permanent nor annihilated,
> Neither one nor different,
> Neither coming nor going."[21]

The four lines in the verse all indicate errors: arising and ceasing refer to appearance and disappearance (contemplation of the existence of dharmas), permanent and annihilated refer to things existing as permanent and brief (contemplation of time of dharmas), one and different refer to unity or difference (contemplation of space of dharmas), coming and going refer to movement or stillness (contemplation of the operation of dharmas) of things and phenomena are all errors. This is precisely the thought of *eight negations* (bát bất) in Buddhist teachings [38, page 62].

In the Yogacara School (Yogacara), Asanga (Vô Trước) and Vasubandhu (Thế Thân) examined emptiness based on the concept

[21]Chinese-Vietnamese transliteration: "Bất sinh diệc bất diệt, Bất thường diệc bất đoạn, Bất nhất diệc bất dị, Bất lai diệc bất xuất."

of consciousness-only in the 4th–5th centuries. All things and phenomena are results of consciousness, i.e., mind. Besides consciousness, there is nothing real. This is typical of idealism in philosophy. This school holds that humans have eight consciousnesses: eye, ear, nose, tongue, body, mental consciousness, manas consciousness, alaya consciousness (storehouse consciousness). Storehouse consciousness is the foundation of all phenomena. Storehouse consciousness contains seeds (bija), which are karmic seeds, waiting for sufficient causes and conditions to manifest as thoughts. Those thoughts combine with ignorance and self-grasping to create the self. Emptiness in the Yogacara School is understood as the negation of self-nature and all phenomena are manifestations of consciousness. The negation of self-nature is similar to the Madhyamaka School's view, but emptiness being the sole manifestation of consciousness differs. The Madhyamaka School holds that the Yogacara School overemphasized the importance of consciousness. In this book, we use the concept of emptiness from the Madhyamaka School to contemplate the worlds of matter and life.

Besides the scriptures of the Madhyamaka School, Northern Buddhist schools present many types of emptiness when applied to different objects in the *Prajnaparamita Sutra* (Prajnaparamita Sutra), *Lankavatara Sutra* (Lankavatara Sutra), *Avatamsaka Sutra* (Avatamsaka Sutra) are fundamental and important documents to understand the concept of emptiness [39] (see some Northern Buddhism sutras on page 42). The concept of emptiness in Buddhism is very rich and diverse, including many different types to clarify different aspects of reality. These types of emptiness are very important in helping to clearly recognize the nature of phenomena and existence, thereby achieving enlightenment and liberation. For example, person-emptiness is the emptiness of the individual, i.e., the negation of a fixed and independent self or ego. Dharma-emptiness is the emptiness of dharmas (phenomena), i.e., the negation of any self-nature in all phenomena. Emptiness refers to all phenomena having

no self-nature. Great emptiness is the emptiness of the entire universe. Ultimate emptiness is the emptiness of correct recognition of the emptiness of all phenomena. Self-nature emptiness is the emptiness of self-nature essence. Object emptiness is the emptiness of objects of cognition having no fixed and self-existent nature.

Differences in the number and names of types of emptiness in Buddhist documents can be explained through the scope of teaching, traditions and schools, as well as the context and history of those documents. This book uses emptiness as a property that is not independent, has no self-nature, and is always changing. From there, we contemplate different worlds from the microcosm of fundamental particles to the macrocosm of the universe, from the world of matter without life to the world of living beings with high consciousness.

Emptiness shows that all things have no self-nature. But no self-nature is very difficult to measure and determine. For convenience in comparing with science, the author approaches emptiness through two observable manifestations: compositionality and impermanence. These are two pieces of evidence inferring emptiness – **the core thesis of this book**. These two manifestations will be presented through the physical, biological, human body, and consciousness worlds.

To prove emptiness appears in any thing, we will prove that thing has the following two properties:

1. **Compositionality** is emptiness in space. Any thing is created by constituents and is a constituent of larger things. In turn, constituents are created from even smaller constituents. Properties that a thing and phenomenon have come from results brought by interactions between constituents. Properties of things arise from interactions between constituents, so they are not independent, have no self-nature of their own.
2. **Impermanence** is emptiness in time. Interactions between things,

between constituents of things always change continuously over time, leading to properties of things being impermanent. Properties of things when we observe come from approximations of senses and consciousness about things in a certain space and time.

Page 54 speaks about dependent origination manifesting at three levels: impermanence, compositionality, and conceptual imputation. Because dependent origination is emptiness, emptiness is dependent origination, emptiness also manifests at three levels. Here, we only examine the first two levels: impermanence and compositionality because they align with scientific viewpoints. Conceptual imputation – assigning concepts to a thing and phenomenon has a subjective nature, so it is not examined in this book.

We may wonder: if we only examine constituents creating things, for example the human body, while ignoring the interaction process between this person and others, is it sufficient? The answer is no. We do not ignore it because if we stand from a more macroscopic viewpoint, a thing like a family, a nation, then humans are constituents. Interactions between humans create characteristics of families and nations. At this time, the object of examination is families and nations, not a specific human. Therefore, examining the compositionality of a thing is sufficiently general for all things and phenomena.

Another question: if we continue dividing things into constituents, can we reach a smallest limit in space? The answer is there may be a limit. Quantum mechanics shows there is a smallest limit in space, which is Planck length with size 1.6×10^{-35} m. To imagine this length, we visualize magnifying a proton to the size of Earth, then Planck length is smaller than atomic size by 1000 times.[22] Currently, human experiments are still far from touching that tiny limit. The reverse process is the process of enlarging things, is there a limit?

[22]The ratio between Earth size 10^7 m and proton 10^{-15} m is 10^{22}. The ratio between proton size and Planck size is 10^{20}. Atomic size is 10^{-10} m.

There may be. That is precisely the size of the observable universe, estimated at about 93 billion light-years (the distance light travels in one year). That is the region of the universe we can observe. There may be other regions of the universe so far from us that light from them never reaches Earth. To this day, these are only theoretical calculations with no way to verify.

Impermanence is continuous change over time. So is there a smallest time interval that cannot be divided? That is precisely Planck time 5.4×10^{-44} s. This time is extremely small compared to the time of the most accurate clock in the world that scientists at the University of Colorado, Boulder created from strontium atoms in June 2024, which is 10^{-19} s [40]. It will probably be a long time before humans can approach Planck time. The reverse process also has a limit. The longest time is precisely the time since the universe formed, about 13.8 billion years ago. We do not know what was before the Big Bang, so the time from then to now is the longest time humans can access.

Human experience comes from senses with not very high resolution: about 10 µm in space and a few ms in time. Smallest space and time intervals like Planck time only appear in mathematical equations, theoretical models that the brain can think of. Scientific experiments can allow humans to access higher precision but are still far from reaching the above limits. That is why examining compositionality and impermanence is still considered appropriate at the beginning of 2026, when this book is completed. Certainly in the future, the development of science and technology will allow us to access reality with higher precision. Then we will continue calculating. For now, let us be satisfied with our rational and scientific capabilities.

Emptiness manifested by compositionality and impermanence leads to no-self. The relationship between the principle of dependent origination and emptiness is so close that Nagarjuna said **emptiness is dependent origination**. Contemplating emptiness of Bud-

dhism into the worlds of matter, life, humans, and consciousness means examining those worlds using the two factors presented above. If we can prove the two properties of compositionality and impermanence, we have proven emptiness lies behind all things in the universe.

1.3.5 Emptiness of emptiness

Some opinions hold that the logic of the Madhyamaka School is contradictory: if everything is empty, why are there conventional truth and ultimate truth. The Madhyamaka School explains that the two truths are merely two different ways of viewing reality. Conventional truth is the image of reality through experience from the senses, while ultimate truth views reality at a deeper level, recognizing the emptiness of all phenomena. Emptiness does not contradict the two truths; rather, it is the foundation for understanding and reconciling these two types of truth. Understanding emptiness helps us not cling to the conventional truth of conventional truth, but also not fall into nihilism when recognizing ultimate truth. Emptiness, conventional truth, and ultimate truth do not contradict but support each other in explaining the nature of reality.

Another question: is there emptiness of emptiness (sunyata sunyata) or not? In other words, if all things are impermanent and no-self, then emptiness itself – a property describing the nature of dharmas – does it change? This is a self-referential paradox commonly encountered in logic and mathematics. Self-reference is a phenomenon in which a proposition, system, or logical structure refers to itself. When this reference leads to internal contradiction or logical circularity, we have a self-referential paradox.

The most obvious paradox is the liar's paradox. When you state the proposition: *I am lying*, that is a proposition that cannot be determined as true or false because the content of the proposition always contradicts the act of stating that proposition. If this propo-

sition is true in content, then you are telling the truth about the fact that you are lying. Conversely, if that proposition is not true in content, then you are lying about the fact that you are lying (i.e., you are telling the truth). In either case, the content of the proposition and the act of stating the proposition contradict each other.

Self-referential paradoxes are encountered very often in science and philosophy: Russell's paradox in set theory, Gödel's paradox about the completeness of mathematics, Turing's halting problem paradox, the barber's paradox,... [41] These paradoxes are similar to the paradox of emptiness of emptiness.

There are two ways to resolve paradoxes. The first is to push the paradox outside the language being used to state the paradox. For example, Bertrand Russell established levels for sets and propositions. A set cannot contain itself because it belongs to a higher level. He prevented the paradox by stratifying language, a different language at a higher level, to explain the proposition *I am lying*. Alfred Tarski proposed separating object language and meta-language. A proposition can only speak about other propositions, not speak about itself. This prevents direct self-referential loops. Any self-referential statement needs a language above it to explain its content. Both Russell and Tarski push the explanation of statements outside the language they use.

The second way proposed by Nagarjuna is completely different from the dualistic thinking of Western scientists. Logic assumes there is a knowing subject completely separate from the object of knowledge. *Emptiness* with content as a proposition is precisely the object. But *emptiness* as an action is the acting subject. So if *emptiness-subject* acts on *emptiness-object* whose content is negation, it will lead to paradox. Russell and Tarski eliminate *emptiness-subject*, i.e., it cannot act on itself, which will eliminate contradiction.

Nagarjuna does not start from dividing subject–object; on the

contrary, he holds that division is an illusion arising from ignorance. Nagarjuna negates emptiness being both subject and object of knowledge. The Madhyamaka School does not establish a foundational reality, nor does it negate reality in the sense of nihilism. When saying *all dharmas are empty*, Nagarjuna only criticizes how we understand existence, not proposing a doctrine about what exists [36, 15.10].

> "Saying *exists* is precisely eternalism,
> Saying *does not exist* is precisely annihilationism.
> Therefore, whoever has wisdom,
> Does not abide in either existence or non-existence."

Madhyamaka also does not ask how we know reality because the very concept of reality is a product of dualistic discrimination. Nagarjuna does not distinguish knowledge and reality; he holds that they are interdependent, have no self-nature, and together dissolve in the view of dependent origination. Therefore, he does not build a theory of knowledge but breaks down all assumptions hidden in the concept of knowledge.

If negating both the ontology of objects and the method of knowledge of subjects, how does the Madhyamaka School explain emptiness of emptiness? This is precisely the crux of the issue, the point of difference between Nagarjuna's explanation compared to others. Nagarjuna stands at a meta-philosophical viewpoint, i.e., transcending both ontological and epistemological categories. Emptiness is a dialectical method of breaking attachments: emptiness is not a type of existence, nor is it a type of knowledge, but is a method of contemplation, helping to recognize all concepts. Concepts of *subject*, *object*, and *method of knowledge* all have no self-nature. Therefore, emptiness does not belong to a specific field of philosophy in the Western understanding but is a negating movement in the entire structure of thinking. Nagarjuna's middle way view is not between the two poles of existence or non-existence but transcends

that entire field of opposites. Those two categories only have relative meaning in conventional language (conventional truth). Nagarjuna does not explain reality or knowledge but liberates us from clinging that there is something to explain.

Nagarjuna does not reduce emptiness to ontology, nor does he identify emptiness with knowledge, but views emptiness as a method of contemplation that is meta-ontological and meta-epistemological – a middle way. Emptiness is not what is known, nor is it the knower, but is the dissolution of both in the insight of dependent origination. In classical logic systems, every proposition must be true or false. But the Madhyamaka School holds that this is only true in conventional truth. When examined at ultimate truth, reality transcends the category of true–false. Here, the middle way does not mean between two poles but transcends the entire field of opposites. Therefore, when he negates both existence and non-existence, that is not refusing to answer but showing that question is structurally wrong.

In subsequent chapters, we will examine emptiness according to the Madhyamaka School in the worlds of matter, biology, body, and consciousness.

1.4 Hinduism

1.4.1 The Vedas

For a caste-distinct society to be stable long-term, the society's structure needs to be religiousized so that each individual in society voluntarily complies. These regulations were incorporated into **Hinduism** (Hinduism). Hinduism is not a term referring to a single religion as commonly understood but is a term used to refer to a group of traditions and indigenous beliefs in India that have similarities to each other but also many differences. Some traditions have existed almost intact for thousands of years until today. It can be

said that Hinduism is a way of life and has undergone many changes over time. Some changes transformed it into a different religion like Buddhism and Jainism (Jainism); some changes were just enough to transform it internally. Hinduism also transformed into atheistic and monotheistic branches. In 1995, the Indian Supreme Court wrote as follows:

> "Hinduism has no single prophet, worships no single deity, has no single dogma, believes in no single philosophy, and follows no single set of rituals. Indeed, it does not seem to be a religion or faith in the narrow traditional sense. It can be considered a way of life and nothing more."

In 2016, after many opinions requesting change to the above viewpoint, the Indian Supreme Court voted to maintain the above statement and emphasized "Hinduism is not a religion but a way of life and a state of mind" (Hindutva was not a religion, but a way of life and a state of mind) [42].

The concept **Brahmanism** (Brahmanism) was introduced by Europeans in the 16th century to refer to Hinduism during the Vedic period from approximately 1500–500 BCE, when the Vedas were the center of religion focused on rituals and Vedic deities and the ritual performance of Brahmin priests. Later, Hinduism developed focusing on specific deities such as the Trimurti: Brahma, Vishnu, and Shiva. In the 19th century, Western scholars coined the term Brahmanism to refer to the period when Brahmin priests had important roles. Some scholars describe this period as a distinct religious tradition. However, Indians generally consider Brahmanism a period of Hinduism [43, 44]. Buddhism emerged at the end of the Brahmanism period.

Although it is a very diverse system of beliefs and religions, Hinduism still has the following common characteristics:

1. The *cycle of rebirth* (SAN samsara) of the *soul* (SAN atman) in the cycle of birth and death and the desire to be *liberated* (SAN moksha [Moksha]) from the cycle of rebirth. Hinduism holds that the soul is permanent and unchanging, used to distinguish a self from the external world. This is the greatest difference between Hinduism and Buddhism when Buddhism holds that the soul cannot be separated from the external world; it always transforms and changes.
2. To achieve liberation, one must follow *dharma* (SAN dharma), short for *sanatana dharma* (SAN sanatana dharma). Dharma can be the law of nature or the virtue that humans must follow. There are three ways to practice dharma: the path of wisdom is correct understanding and knowledge; the path of action is correct actions and behavior; the path of devotion is reverence for deities. Buddhism abandoned complex rituals in revering deities and applied the law of cause and effect to achieve liberation.
3. A consequence of the cycle of rebirth is the theory of *reincarnation* and the theory of *cause and effect*. Each individual's soul, atman, can be carried in different bodies. Each person is born in their current body as a consequence of actions they performed in previous lives. Actions accumulated are called *karma* (karma). There is good karma and bad karma. If one performs much good karma, the next life will have a better social position; if one performs much bad karma, the next life will have a lower position.
4. Hinduism has very many deities, but three are called the Trimurti (Trimurti), the most important: Brahma (creator god), Vishnu (preserver god), and Shiva (destroyer god), called *Trimurti* (trimurti). The Trimurti points to a *supreme deity* higher than all. Hindu sects also worship very many smaller deities, possibly local or personal deities with different responsibilities and powers. Deities also frequently transform into different forms, making the number of Hindu deities reach hundreds of millions. Each deity is an aspect of reality. Behind the diversity of reality are unchanging

principles. Behind unchanging principles is a single, permanent reality—that is dharma.

5. Hinduism is based on the foundation of the **Vedas**, the oldest scripture. Later scriptures such as Brahmana (Brahmanas) and Upanishad (Upanishads) also only interpret the Vedas. Some other books supplement additional information about mythology, history, and beliefs, such as the two epics Mahabharata and Ramayana.

The Vedas (veda) are the core of Hinduism, considered by Indians to be revealed truth[23] for humanity. In Sanskrit, *veda* means *knowledge*, transliterated into Vietnamese as Veda or Veda. The Vedas comprise 4 collections: the oldest collection is Rigveda (Rig Veda), considered by many scientists to have emerged around the 15th–12th centuries BCE, although some studies suggest an earlier date in the 25th century BCE. In Sanskrit, *rig* means *prayer*. Rigveda consists of 10 books with 1028 hymns praising knowledge and the deities.

The other collections of the Vedas are Yajurveda containing steps for ritual practice, Samaveda containing chants and sacrifices, Atharvaveda explaining the meaning of the three collections above. Many studies of deity names show that the Vedas have many similarities with ancient religions in Iran and relate to religions in Greece and Rome. The Vedas have a natural polytheistic character regarding the worldview and human outlook of Indians.

To explain the existence of castes, Rigveda explains that the 4 castes were all born from the primordial being called *Purusha*. The deity divided itself into 4 castes. The deity's mouth is the Brahmin caste, the deity's two shoulders are the Kshatriya caste, the deity's two thighs are the Vaishya caste, and the deity's two feet are the

[23] In Sanskrit, *sruti* literally means *that which is heard* to refer to the Vedas, Brahmanas, Aranyakas (Aranyakas), and Upanishads, written by priests and monks who listened to revelation. *Smriti* literally means *that which is remembered* to refer to books like the two epics Mahabharata and Ramayana. Although not as noble as sruti, they are also very important.

Shudra caste. Later legal codes of ancient Indian society recognized and protected this caste division system. A person born into a caste will remain in that caste for life. A person in the Shudra caste, no matter how good or how hard they try, will always be Shudra. Conversely, a person in the Brahmin caste can do many bad things and still be Brahmin. Brahmins are considered people of the deity, so even if they commit grave sins, they are only exiled elsewhere but still retain their property. Upper castes are not allowed to approach or associate with lower castes to avoid contamination.

Chapter 2

Matter

Summary. Emptiness in matter manifests in: (1) At different scales, matter has constituents. Objects are formed from molecules, molecules are in turn formed by atoms. Atoms consist of electrons, protons, and neutrons. (2) Protons and neutrons are in turn created by valence quarks. Valence quarks are not fixed but continuously exchange, appear and disappear, leaving only residual quarks. (3) Electrons and quarks are fundamental particles in the standard model, but they arise from excitations of corresponding fields, namely the electron field and quark field. String theory hypothesizes that fundamental particles are formed by extremely small strings living in space with up to 11 dimensions. Vibrations of strings create particles. Particle and wave are two coexisting aspects of entities that make up matter. They are completely inseparable.

As presented in Chapter 1, the Madhyamaka school equates dependent origination and emptiness: dependent origination is emptiness, emptiness is dependent origination. To prove emptiness, things must exhibit two characteristics: (1) **compositional nature** – things are made from smaller components; (2) **impermanence** – relationships between components continuously change without ceasing (see page 67 in Chapter 1). This chapter will demonstrate the emptiness of matter by pointing out these two characteristics at different levels of matter, from atoms to fundamental particles.

2.1 Atoms

2.1.1 Atomism

When asked to state the most important sentence in all of humanity's scientific knowledge, the answer of world-renowned physicist Richard Feynman[1] was "Everything is made of atoms" [45, Chapter *Atoms in Motion*]. Today almost no one questions that statement, but in the late 19th and early 20th centuries, atomic theory was not widely accepted.

The idea that matter is made from fundamental elements to create the material and spiritual world appeared in some schools of Hinduism, very close to modern theories about atoms that we have presented above. The philosopher Maharishi Kanad, in the book *Vaisesika Sutra*, proposed the concept of atomism. Matter is formed from particles called **anu, paramanu**. Anu is an extremely small particle, invisible to the eye; paramanu is the ultimate unit, indivisible, immutable, eternal, proposed by the Vaisesika school, later adopted by Nyaya. Paramanu combine to form dvyanuka (two particles), tryanuka (three particles), from which the material world is formed. There are four types of paramanu: earth, water, fire, wind. Thus, this is an ontological materialist system – believing that the world is constituted from immutable particles [46].

In Buddhist scriptures, the Abhidhamma speaks of materiality, of matter, and holds that matter is never independent but exists in aggregates called rupa-kalapa (a cluster of material elements rupa) [47, page 64]. A kalapa can contain from eight to thirteen rupa. These elements are considered as atoms to form the human body and mind. Moreover, the arrangement of atoms also creates energy, past karma, nutrition, and thought. These elements do not exist

[1] Richard Feynman (1918 – 1988), American physicist with many contributions to quantum electrodynamics, superfluid properties of liquid helium, path integral method and Feynman diagrams. He, together with Julian Schwinger and Shinichiro Tomonaga, received the Nobel Prize in Physics in 1965.

independently but are in relationships and continuously transform. Properties of elements come from properties of atoms. These properties are intrinsic, the unchanging essence of atoms. Atoms combine from two or three, then form the matter of the world [48].

Rupa-kalapa are not independent, immutable entities. They are formed by dependent origination and continuously transform without ceasing. This aligns with impermanence that we will see in modern scientific discoveries about atomic structure. The view of rupa-kalapa has suggested a different perspective: matter is never independent but exists in aggregates in relationships and continuous transformation.

* * *

The concept of atom comes from Greek philosophers naming *atomon* meaning *indivisible*. The 5th century BCE was an important milestone in world history. In an ancient Greek city-state called Miletus, now in Yadin province of Turkey, a great revolution in thinking appeared, signaling a first stage for scientific thought [49]. That was the *rational revolution* where religious belief in the great powers of gods in myths was replaced by rational thinking and observations. Rational thinking was honed through logic and debate to find truth. Debate without distinction of old and young, great and small, teacher and student could discover worlds behind things and phenomena of the external world.

Around 450 BCE, Greek philosopher Leucippus boarded a ship migrating from Miletus to flee political crisis and arrived at Abdera in present-day Greece. There he met a student named Democritus. These two put forward the idea of atoms. Leucippus's writings no longer exist today, but we know his thoughts through the works of his student Democritus. Atomism of Leucippus and Democritus holds that the universe was created in infinite, unlimited space and time, no up, no down. Matter existing in that space and time is

made by countless atoms in ceaseless motion. They have no properties except shape. All properties we can sense do not appear at the scale of atoms but come from our sensations of them. This is the difference between Greek atomic hypothesis compared to the Indian hypothesis. Greece held that atoms have no inherent properties while India held that these are properties attached to atoms. Atoms move ceaselessly. Their combination and collision create everything including us. Thus, we are a *random* combination of atoms.

Greek atomic theory left two very important consequences. First, if atomic motion is *random*, then humans are also a random product. Perhaps this is one of the most breakthrough and different consequences compared to teleological theories of Plato and Aristotle. Remember, more than 2 thousand years ago, most of humanity believed in gods and supernatural forces. Human presence was understood as the desire of gods. The idea that human presence is random due to chaotic motion of atoms was truly a difficult idea to accept. Second, atoms moving in space create matter. Then space has nothing, space is empty, is *vacuum*, completely opposite to the four elements theory where there is no vacuum at all. This was very difficult to understand for others: they feared vacuum [50]. Zero was created by Indians from the 5th century BCE and was introduced to Europe through Alexander's conquest in the 4th century BCE. No wonder atomic theory was criticized by Plato and Aristotle and fell into oblivion for more than 2 thousand years.

The first to revive atomic theory was Antoine Lavoisier[2] but unlike the abstract atoms of Greek philosophers, he held that atoms are the smallest components of matter that cannot be divided by chemical methods. He did not use the term atom in his famous work *Elements of Chemistry* (Traité Élémentaire de Chimie) published in

[2]Antoine de Lavoisier (1743 – 1794), French chemist who discovered the law of conservation of mass, theory of oxidation of substances, considered one of the greatest chemists in history. He was executed in the French Revolution for suspicion of involvement in aristocratic activities.

1789 but used the term "corps simples" which was later understood as *element*. John Dalton saw that in a chemical reaction, the masses of substances participating in the reaction always have a certain ratio, not arbitrary. That led him to conclude that each element contains an integer multiple of a fundamental mass unit. Dalton continued Lavoisier's idea about atoms by holding that each element is formed from one type of atom with its own chemical properties. A characteristic of atoms is their mass, compared with the mass of hydrogen atoms. Up to here, scientists only understood atoms through their chemical properties. Atomic structure was gradually discovered through physical experiments.

2.1.2 Discovery of atomic structure

In 1897, J. J. Thomson at the Cavendish Laboratory[3] of Cambridge University discovered that cathode rays are not electromagnetic waves but are created by particles with mass because their direction of motion is bent in a magnetic field. The mass of this particle is only 1/1836 the mass of the lightest atom, hydrogen. He named them corpuscles, later called electrons. Thomson built the plum pudding model of atomic structure. In it, electrons are distributed scattered in the atom. However, this model was later refuted by one of his students also at Cavendish Laboratory, Rutherford. From 1908 to 1913, New Zealand physicist Ernest Rutherford and two students Hans Geiger and Ernest Marsden conducted experiments using alpha particles emitted from radioactive Radon to bombard a thin gold foil, discovering the atomic nucleus with positive charge, very large mass, very small size inside the atom, thereby building the

[3]Henry Cavendish (1731 – 1810) was a British scientist famous for measurements determining the gravitational constant to thereby weigh the Earth. He was also the discoverer of hydrogen gas and air composition. The world-famous Cavendish Laboratory was named after him and his relative William Cavendish (who once held the position of chancellor of Cambridge University). Many renowned scientists have worked here: James Clerk Maxwell, J.J. Thomson, Ernest Rutherford, James Chadwick, James Watson and Francis Crick,...

Rutherford model: nucleus with positive charge at the center of the atom, electrons with negative charge distributed around the nucleus. From the gold foil experiment mentioned above, Rutherford calculated that the size of the atom is 10^5 times larger than the size of the nucleus.

Niels Bohr once came to Cavendish Laboratory in 1911, when Rutherford discovered the atomic nucleus, but Bohr did not leave an impression on Thomson, so in 1912 Bohr returned to Copenhagen to get married and built the planetary atomic model based on improving the Rutherford model, also called the Bohr model (Figure 2.4). From studies by Rutherford, Bohr, and Henry Moseley, Rutherford held that the atomic nucleus must be an integer multiple of a smallest unit with positive charge, the proton, and a smallest unit without charge, the neutron. In 1932, James Chadwick discovered the neutron as predicted by Rutherford.

Discoveries about atomic structure show the emptiness of components that ancient philosophers thought were smallest and indivisible. Atoms are made of electrons and nucleus; in turn, the nucleus consists of protons and neutrons. However, division does not stop there. Later we will see that protons and neutrons are not fundamental particles; they are made of quarks. Electrons, quarks, neutrinos are fundamental particles, not made of other particles. But that is not the end; fundamental particles are not simply indivisible, without compositional nature. Strangely, at the most subtle level, they are not independent, immutable, and self-existent. They are born and disappear in an endlessly repeating dance in space and time.

2.1.3 Quantum mechanics

In the 1920s and 1930s, scientists built atomic models based on quantum mechanics. The quantum concept originated from Max Planck's research on the hypothesis of energy quantization to ex-

plain blackbody radiation in 1900. Like a glass of water, viewed from afar, water seems like a continuous medium, meaning we can divide arbitrarily. A glass of water can be divided into two glasses, two glasses into four glasses,... By dividing like this, we can have infinitely many small parts of the original glass of water. But modern science shows the limit of the division process is to a water molecule. Smaller than that, the molecule is no longer water but becomes other chemicals like hydrogen and oxygen. We say the water molecule is the smallest particle of matter that has the chemical and physical properties of water. Similarly, previously people thought energy could be divided arbitrarily, but Planck thought differently; he held that energy is limited by a smallest quantity called **quantum** of energy given by the formula $E = h\nu$, where E is the energy of the quantum, h is Planck's constant, ν is the frequency of energy. Energy is quantized, discretized. That discrete quantity has a value proportional to its frequency.

But we had to wait until 1925, from studies on the photoelectric effect by Albert Einstein (1905) showing light has particle properties, the hypothesis of Louis de Broglie (1923) about electrons having wave properties, Werner Heisenberg and Erwin Schrödinger built quantum mechanics, then the modern atomic model was formed and used to this day. Quantum mechanics is based on axioms completely different from classical mechanics created by Newton from the 17th century. There are many axioms and principles, but here we study those axioms and principles that leave philosophical meanings. Other technical issues are not fully presented.

First axiom: the state of a physical system is characterized by a **wave function** containing information about coordinate x and time t, usually denoted $\psi(x, t)$. From the wave function, we can calculate the probability of finding the particle at a position at a given time. Classical mechanics allows calculating the state of a particle with arbitrary precision, while quantum mechanics only allows calculating probability. That is the most important difference. Uncertainty

arises from the axioms of the theory itself, not from lack of information related to boundary conditions. Cause and effect according to classical mechanics is deterministic by cause, here laws are determined. Probability comes from lack of information of conditions, i.e., initial conditions. Once cause is determined and conditions are complete, the event must occur.

But cause and effect according to quantum mechanics has uncertainty right from the cause, in the law itself, i.e., the wave function, so even when cause and effect are complete, we can only know the probability that things and phenomena can occur.

Second axiom: the wave function of a physical system is determined from Schrödinger's equation, an equation equivalent to Newton's second law. This is a linear differential equation, first order in time and second order in space. Because it is a linear differential equation, if the equation has some particular solutions, their linear combination is also a solution. For example, we consider an electron in a hydrogen atom: the state corresponding to minimum energy is ψ_1, the state corresponding to excited energy is ψ_2, then the linear combination of these two states $\psi = a\psi_1 + b\psi_2$ is also a solution (a, b are constants indicating the probability of each state). This property of Schrödinger's equation leads to a very special principle, the **superposition principle** of states: microscopic particles can be in both states simultaneously until we measure them.

The superposition principle of states has caused controversy since quantum mechanics was born. It is part of a debate about the completeness of quantum mechanics, known as the **Bohr – Einstein debate** named after the two leaders of the two schools. Issues related to this debate: superposition, completeness, quantum entanglement, locality, probability of quantum mechanics. Einstein held that quantum mechanics is incomplete; particles are all localized at definite positions; probability comes from lack of information about boundary conditions. Bohr, on the contrary, accepted all the strange properties of quantum mechanics because he held that humans are

macroscopic beings, so microscopic experiences do not necessarily match our perception. In the example of an electron in ground or excited state mentioned above, Bohr's explanation is that the electron is not determined in state ψ_1 or ψ_2 until we measure it. At the moment of measurement, the electron decides its state. When not measured, the electron is not in any state, or can be in both states simultaneously. Contrary to the difficult-to-understand explanation above, Einstein held that the electron is in one of two states before measurement. Measurement only reveals information about the state the electron already had before. The superposition of electrons is a strange property, unique to microscopic particles that Bohr and his followers believed.

The superposition principle of states has a profound meaning about emptiness. In classical mechanics, a particle must be in a definite state at each moment. But in quantum mechanics, before measurement, the particle is not in any state, or we can say it is in all states simultaneously. This reflects the emptiness of physical properties: there is no fixed, self-existent state. States only appear when there is interaction with the observer, i.e., when there is dependent origination. This is precisely the manifestation of emptiness in the microscopic world: there is no fixed essence; everything depends on observation conditions, on cause and effect.

Quantum mechanics accepts the **uncertainty principle** proposed by Werner Heisenberg in 1927. This principle holds that the uncertainty in position Δx and uncertainty in momentum Δp are linked together with a certain precision limit, Planck's constant. The mathematical formula for this principle is $\Delta x \cdot \Delta p = h/(4\pi)$, where $h = 6.626 \times 10^{-34}\,\mathrm{J\,s}$ is Planck's constant. The tiny value of Planck's constant makes the uncertainty principle only apply to microscopic particles like electrons or atoms. Quantum effects of macroscopic particles like balls or humans are very small, unobservable with the most modern scientific equipment. If uncertainty in position is large, then uncertainty in momentum will be small.

Momentum equals mass times velocity $p = mv$, so uncertainty in momentum is proportional to uncertainty in velocity. Besides position and momentum as a pair of quantities following the uncertainty principle, in quantum mechanics there are also some other pairs of quantities, typically energy – time $\Delta E \cdot \Delta t = h/(4\pi)$. If we want to measure position accurately (i.e., uncertainty in position Δx is very small) of an electron, then the error in velocity will be large (uncertainty in momentum Δp is large) and vice versa. The uncertainty principle limits the precision with which we can observe the microscopic world.

People once thought vacuum is nothing, but that idea is wrong. Vacuum is a seething sea of particles appearing and disappearing very quickly. Vacuum also has its own energy, called zero-point energy. At a very short time Δt, uncertainty in energy is very large. Large uncertainty causes energy to appear from vacuum. At any sufficiently short time, energy can be large enough to create particle-antiparticle pairs [51]. The appearance of particle-antiparticle pairs appears and recombine to annihilate each other so quickly that their existence almost leaves no significant causal effect on the surroundings, so we call them virtual particles and virtual antiparticles. The appearance of virtual particles is a consequence of the uncertainty principle of quantum mechanics.

The existence of vacuum energy is a crucial key showing the manifestation of emptiness of matter. Compositional nature – one of two manifestations of emptiness we discussed in the previous chapter (page 67) may be limited at fundamental particles. By definition, fundamental particles like electrons, quarks, neutrinos are indivisible. They are not made of other smaller entities. If so, compositional nature may not apply. Here, zero-point energy plays a role to save compositional nature. Because this energy is nonzero, the uncertainty principle allows vacuum to temporarily borrow to appear virtual particles and antiparticles. Virtual particles will interact with fundamental particles, making them not fixed, indepen-

dent, but in a continuous creation-annihilation process, resulting in what we observe. That is fundamental particles – a consequence of approximate observation from afar. We will examine this process in detail in later sections.

Pauli exclusion principle is rarely mentioned in popular physics and science books. However, this is a very important principle for building the entire material world we see. The exclusion principle holds that the state of each electron in an atom is different; no state coincides with any other. For example, the lowest energy state of a hydrogen atom allows two electrons to be present, denoted $1\,\mathrm{s}^2$. These two electrons have a quantity called spin opposite to each other, so they can coexist. If we add another electron to the atom, that electron must occupy a different state and cannot be together with one of the two previous atoms. The exclusion principle makes each electron only able to be in one state, so it determines the arrangement of electrons in atoms and creates the order of elements in the **periodic table**. From the periodic table, we have the material world as seen.

The exclusion principle is opposite to the **indistinguishability principle**, also called the principle of identical particles. If the exclusion principle carries information about self to distinguish this particle from that particle, then the indistinguishability principle erases that self. If we do not consider electron spin, we cannot distinguish two electrons in the $1\,\mathrm{s}^2$ state of a hydrogen atom. Then, they are completely identical and lose their individual characteristics. To identify particles, we need an additional measurement. Because the difference lies in spin, the measurement must relate to this quantity, which is interaction with a magnetic field. When interacting with a magnetic field, the spin of two particles gives different results, from which particle characteristics are revealed.

We have seen human understanding of atoms change greatly over time. From initial hypotheses about indivisible particles by Kanad and Leucippus, scientists of the early 20th century discovered their

complex structure consisting of neutrons, protons, and electrons. The compositional nature of things (page 67) is manifested in atoms, structures almost smallest of matter. Moreover, the principle of indistinguishability of particles makes particles lose their self-existent nature. If two electrons are in a quantum system, they lose their individual identity. This is a manifestation of no-self in the material world at the most fundamental, most subtle level. Their self is only maintained in the state they can exist. That is a relativity where electrons relate to the surrounding environment, here the nucleus. Electrons are distinguished from each other by their state, i.e., by the potential energy that the nucleus brings to the electron, not by self-existent values that the electron has.

Quantum mechanics presented above is classical quantum mechanics because it has not integrated with special relativity when particles move at speeds close to the speed of light. Paul Dirac introduced special relativity into quantum mechanics to create quantum field theory (quantum field theory – QFT). It is called quantum field theory because this theory holds that objects we think are particles come from oscillations of fields. Electron particles appear due to oscillations of the electron field. Other fundamental particles are the same. Quantum field theory does not distinguish particles from fields; they are one and can transform into each other.

Versions of quantum field theory are: (1) quantum electrodynamics (QED) describing electromagnetic interaction between charged particles (like electrons, positrons, protons) through photon exchange; (2) quantum chromodynamics (QCD) describing interaction between quark and gluon particles; (3) electroweak theory (ET) governing beta radioactive decay and many other nuclear processes (see Figure 2.5). Among them, quantum electrodynamics is considered the most accurate theory humans have ever conceived. A calculation of electron magnetic moment in 2023 showed experimental and theoretical values differ by one part per trillion [52].

2.1.4 Problems of quantum mechanics

Quantum mechanics has explained and predicted many experimental results like the periodic table, chemical properties of atoms, molecules, fundamental particles in the standard model of particle physics. However, this theory has contradictions with intuition from our experience. One of the problems is the question of what *measurement* is. In classical mechanics, measurement does not affect physical events. If we roll a ball from a table to the floor and measure the landing position of the ball, then looking at the ball by shining light on it does not affect the ball's trajectory. We can observe and measure the ball with arbitrary precision and unlimited number of measurements.

The situation will be completely different with microscopic particles. If we have an electron source and suppose we can emit only one electron at a time. This is completely possible with today's experimental equipment. Then according to quantum mechanics, the electron's state is described by wave function $\psi(x, t)$, a sphere spreading in space with radius R equal to electron velocity times time. Suppose at time t_0 we detect the electron at position x_0 on a fluorescent screen, then quantum mechanics explains that at exactly the measurement time t_0, the electron's wave function collapses, or reduces to a single point. This process is called **wave function collapse**. The large sphere with radius R immediately collapses into a point. What is special is that collapse is instantaneous, immediate, without needing a time interval for information to travel. What about the electron itself? Before measurement, the electron does not exist. It only exists when interacting with the fluorescent screen. Only through that interaction does the electron manifest its presence. Einstein did not agree with that explanation and asked "Does the moon exist when no one looks at it?" [53]

An imaginary experiment named *Schrödinger's cat* (Figure 2.1) raises the problem of state superposition. In a closed box, there is

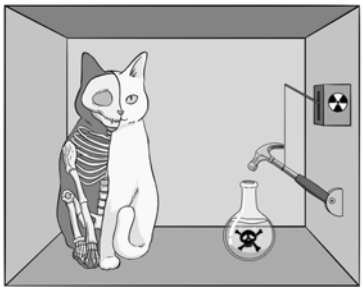

Figure 2.1: Diagram of the imaginary experiment *Schrödinger's cat* (proposed 1935). Decay of a radioactive substance nucleus in the quantum world causes a hammer to fall onto a bottle containing poison. The broken bottle releases poison that can kill the cat. Superposition of decayed and not decayed states makes the cat also in a superposition state between alive and dead.

a radioactive source. Radioactive nuclei can decay randomly, and this decay follows quantum laws. When a nucleus decays, it signals a receiver called a Geiger counter. The counter clicks each time it detects a decay particle. From there, the system can activate so a hammer breaks a bottle to release poison. When poison is released, the cat dies. The wave function describing the nucleus has two states: decayed and not decayed, two states of the microscopic world. These two states correspond to two macroscopic states of the cat: dead and alive. Before the observer opens the box, quantum mechanics holds that the cat is in a superposition state between alive and dead. That is, a combined state both alive and dead. Only when the observer opens the box does the cat's state become determined as alive or dead.

The cat experiment mentioned above has acknowledged that the observer or measurement system is separate from the measured object. Bohr held that the measurement system is classical; only microscopic systems are quantum. This is the weakest point of Bohr's explanation, also called the Copenhagen interpretation: why is the measuring device not counted in the physical system? If the entire observer and box are placed in another larger box, that could also

be considered a quantum system. That imaginary experiment was named *Wigner's friend* proposed by Eugene Wigner in 1961 [54].

Wigner's friend experiment is an extension of Schrödinger's cat experiment. In the cat experiment, when no one observes the box, the cat is in a superposition state between alive and dead. A friend of Wigner named Jack observes the box in a closed room. Outside the room is Wigner – another observer (Figure 2.2). There are two levels of observation corresponding to two physical systems: physical system 1 consists of the box containing the cat, and physical system 2 consists of physical system 1 and Jack. Jack is the observer of physical system 1, and Wigner is the observer of physical system 2. The paradox here is that Jack's and Wigner's observation results are different. If Jack has observed physical system 1 by opening the box, he will know for certain the cat is alive (or dead), while Wigner has not observed physical system 2, i.e., has not opened the door to ask Jack about the result, so he firmly believes the cat is in a superposition state between alive and dead. Wigner cannot separate the box containing the cat from his friend Jack. For Jack, the cat is alive; for Wigner, the cat is half alive and half dead. So who is right? Two observers use the same measurement method, conduct on the same experiment, but bring two different results just because they stand inside or outside the room.

The paradox above comes from our classical intuition, we macroscopic humans. For both Wigner and Jack, the physical system is the same. Standing inside or outside the room, even placing the room and Wigner inside a building, the physical system remains the same. But quantum mechanics of microscopic objects is different; the physical system can change without any contradiction with our intuition. In an experiment with 6 photons, Wigner's friend paradox with a scenario with 4 observers (not 2 as in the original version) has been confirmed. If we assume assumptions about locality (photons can only be at one position at one time), then this result implies that quantum theory should be interpreted in an

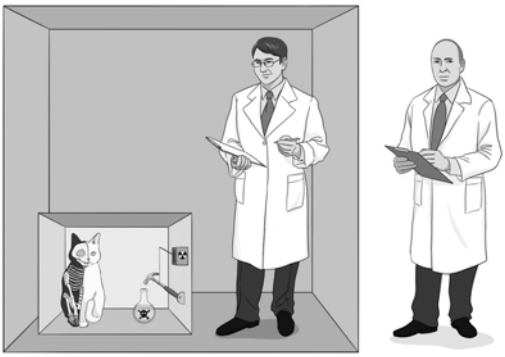

Figure 2.2: Diagram of the imaginary experiment *Wigner's friend*. Physical system 1 has Schrödinger's cat. Physical system 2 consists of physical system 1 and a friend of Wigner named Jack. Wigner outside observes physical system 2.

observer-dependent way [55].

Wigner's friend paradox relates to the concept of no-self in Buddhism in how can we define a self? Here it is the physical system! Why is the box containing the cat an object of study and not the room consisting of the box and Jack? With such reasoning, the next question appears: Why is Wigner not counted in the physical system? Before quantum mechanics, those questions seemed silly because separating the researcher from the object does not affect measurement results. But here it is different; separating the observer from the observed object is a fundamental problem because it will determine the results the observer can obtain. Determining the boundary of the physical system in the microscopic world is not simple. This is similar to determining the boundary of self in Buddhism. Due to the holistic connection of dependent origination, everything is in intricate organic relationships, very difficult to separate. That is no-self, a manifestation of emptiness in human psychology.

* * *

According to calculations from classical mechanics, the time an

electron moves is only about 10^{-11} s before losing energy and falling into the nucleus, so atoms cannot be stable [56, page 72]. Therefore, to explain the existence of atoms, we must abandon one of two properties: locality and reality. **Locality** (localized): at a moment, the electron must be at some position in space. Reality: the electron does not exist until there is observation. Most scientists accept non-locality to save reality. This is a property contradictory to human intuition when having to admit that electrons can be nowhere as well as can be at many different positions in space at one time. Non-locality is a consequence of quantum entanglement, a fundamental issue of the Bohr – Einstein debate [57] we mentioned above. Bohr's explanation seems to have won when John Bell put forward theoretical calculations [58] and physicists conducted experiments proving the existence of non-locality of microscopic particles. Starting from 1927 at the Solvay Conference, this debate reached its peak in 1935 with the EPR paper by Einstein–Podolsky–Rosen and led to many interesting physical consequences like quantum entanglement, quantum computing, quantum teleportation, quantum cryptography,... In 2022, the Nobel Prize in Physics was awarded to experimental scientists proving that Bohr seems right, Einstein seems wrong [59]. It seems reality at the microscopic scale only appears properties when observed. We must accept the counter-intuitive properties of microscopic particles.

Although there are many opinions that quantum mechanics is incomplete, to this day no theory describes the microscopic world better than this theory, so we are forced to acknowledge that quantum mechanics, a product of high-level reason, can describe reality better than sensation and experience brought by sensation. The foundations of the theory manifest emptiness of Buddhism in different aspects. Electrons and other microscopic particles have wave-particle duality, meaning they manifest both wave and particle properties depending on how they are observed. Electrons have no fixed essence; they are particles or waves depending on how they are ob-

served. The wave function describing electron information only tells the probability that the electron can be in definite space and time. Instead of specific values, the wave function reflects the emptiness of properties like position and momentum cannot be simultaneously determined accurately. Wave function collapse shows that the quantum system's state is not determined before measurement. The specific state that the quantum system collapses to depends on the measurement process: there is no fixed and self-existent state, but depends on interaction in the measurement process. The superposition principle shows that the quantum system's state has no fixed essence before measurement. Before measurement, the electron can simultaneously be in all possible states. Only at the exact moment the electron interacts with the measuring device does the electron determine its specific state.

The uncertainty principle only allows determining position and momentum with some precision, not arbitrarily. Position depends on velocity and vice versa; they cannot be independent of each other. The uncertainty principle leads to **quantum vacuum**: or zero-point energy, it is not empty but full of virtual particles appearing and disappearing in very short time intervals. Virtual particles have no fixed existence but always transform, appear and disappear continuously by borrowing energy from vacuum. Viewed from afar, we see vacuum as quiet, but if we zoom in space and shrink time, it seethes with particles borrowing energy to appear then disappear.

The indistinguishability principle shows that the essence of microscopic particles is not fixed and has no separate identity. Particles only exist and are determined by the quantum state they occupy, not as independent entities. The Pauli exclusion principle shows that the state of each particle is constrained by the presence of other particles. An electron in the ground state in a hydrogen atom can only be in that state if it is still empty. If the ground state is occupied, the electron must be in other states. This simple principle determines the chemical properties of elements in the Periodic

Figure 2.3: Among the four fundamental forces of nature – gravitational, electromagnetic, strong nuclear, weak nuclear – humans only perceive the electromagnetic force.

Table, thereby creating the matter we see. The indistinguishability principle and the exclusion principle show that no particle exists independently and fixedly. Particles have self because of the process of interaction with the surrounding world; without interaction, there are no properties, no self. The essence of microscopic particles is no-self; self-nature only appears at a moment, obtained by cause and effect that the particle has. That is emptiness that quantum mechanics brings to matter.

2.2 Modern model of atoms

2.2.1 Fundamental forces

Our understanding of atoms is very different from more than 2 thousand years ago. Although there have been many advances, it seems the more we study, the harder it is to grasp atomic components. Protons and neutrons inside the nucleus are held by a force called **strong nuclear force**. Protons are stable particles with lifetime

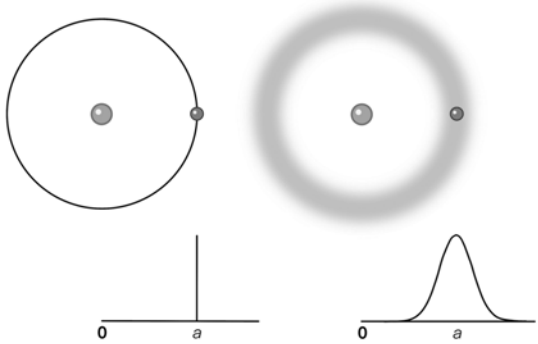

Figure 2.4: (Left) Bohr's atomic model: electrons move on definite orbits at distance a from the nucleus. (Right) Quantum mechanical model, probability of electron presence is highest at distance a from the nucleus.

measured by a parameter, mean lifetime, of about 10^{34} years (this is a lower limit; proton decay has not been detected) [60], much more than the age of the universe of about 13.8×10^9 years. Neutrons are not as stable as protons with mean lifetime of about 900 seconds [61], then decay into a proton, an electron, and an antineutrino. The force controlling this process is **weak nuclear force**.

The two nuclear forces only have short range within the size of the nucleus. Beyond that range, this force decreases very quickly, so only long-range forces remain: electromagnetic and gravitational. Two short-range forces and two long-range forces are the four fundamental forces of physics (Figure 2.3). Electrons move around the nucleus due to **electromagnetic force** at speeds of millions m/s to form a probability **cloud** (Figure 2.4).

The last force among the four fundamental forces is **gravitational force**, the force controlling the macroscopic world of planets, stars, galaxies, and the universe. Gravitational force has the smallest value among forces but plays a decisive role in the operation of the macroscopic world. According to relativity theory, gravitational force appears from deformation of spacetime around objects with mass. Modern physical theories have unified the two nuclear forces and electromagnetic force into a single type of force but have not yet

unified gravitational force. Leading scientists are trying to build the Theory of Everything presented on page 111. In this book, we only discuss electromagnetic force, the force directly bringing sensations, operating our physical body and consciousness.

2.2.2 Observer precision

The nature of reality depends on observer precision. If from a distance of billions of km, Earth looks like a point in the photograph *Pale Blue Dot* (pale blue dot), then we will not consider its internal structure.[4] From that distance, we usually care about Earth's position in its motion around the Sun and ignore other information. If we increase resolution by approaching closer, Earth has much more information about a planet with complex structures including biosphere, oceans, land, Earth's crust movement, climate,... Continuing to increase resolution, we will see mountains, rivers and lakes, and living organisms on them. Closer still, organisms are formed from cells, the building blocks of life. In turn, all cells are formed from atoms. Under an electron microscope, atoms appear as particles arranged in a certain order in solids. The size we say is of atoms is precisely the space where electrons can move, the uncertainty in electron position with magnitude 10^{-10} m we just mentioned above.

The Pale Blue Dot photograph was taken by Voyager 1 spacecraft in early February 1990 when this artificial object was 6.1 billion km from Earth, so far that it took five and a half hours to transmit the photo with 640,000 pixels to Earth at light speed [62, page 30].[5] The photo shows our beloved Earth is just a small dot with

[4]Pale Blue Dot (pale blue dot) is a famous photograph of Earth taken by Voyager 1 spacecraft on February 14, 1990, at a distance of 6 billion km. The photo shows Earth is just a faint blue dot a few pixels in size in vast space, inspiring the book of the same name by astronomer Carl Sagan [62].

[5]On November 20, 2025, Voyager 1 was 25.4 billion km from Earth, equivalent to 169.8 astronomical units (distance from Earth to Sun), light travel time

the size of a few faint blue pixels. All human history, everything humans know, lies on that small lonely dot in the darkness of a vast universe. If something happens to that small dot, no one, no civilization nearby can help.

It can be seen that depending on the observer's perspective, there will be corresponding precision. Different precision requires scientific theories to change accordingly. **Rational perception can contradict sensory perception.** But humans can accept and correct those mistaken perceptions because they still lie within human imagination. Perception about Earth's position in the solar system, of the Sun in the Milky Way, though very different from initial perception, we can still understand because it fits imagination brought by experience. However, not all scientific theories are based on such assumptions. Modern science presents theories very difficult to imagine.

For example, relativity theory holds that time will be stretched and size will be contracted when objects move at high speeds close to the speed of light in vacuum. Modern experiments with high precision all confirm relativity theory and cannot refute results of this theory. Although not similar to everyday experience, results of relativity theory do not contradict themselves and do not contradict our experience. Time dilation is only because sensory ability is not precise enough to observe. For example, a car moving at about 72 km/h has time stretched more than a person standing still on the road by a time interval of 2.2×10^{-15} seconds, i.e., more than 2 femtoseconds.[6] In that time interval, light only travels a distance equal to the width of a tuberculosis bacterium (Mycobacterium Tuberculosis) in humans. With the ability of our senses only recognizing

is 23 hours 32 minutes. Reference: https://voyager.jpl.nasa.gov/mission/status/

[6]Time of car is t moving at velocity v, time on road is t_0 at rest, then the relationship between them is $t = t_0/\sqrt{1 - v^2/c^2}$. Applying Taylor expansion at low velocity $v \ll c$ then $t \approx t_0[1 - v^2/(2c^2)]$. Substituting velocity value $v = 72\,\mathrm{km/h} = 20\,\mathrm{m/s}$, speed of light $c = 3 \times 10^8\,\mathrm{m/s}$ will obtain the difference $t - t_0 \approx 2.2 \times 10^{-15}\,\mathrm{s}$.

differences of order 10^{-3} seconds (ms), that short time interval is imperceptible.

But do we accept theories that not only differ but also contradict our perception ability and experience? Electrons moving around the nucleus on curved orbits will emit energy and fall into the nucleus. The uncertainty principle can be used to explain why electrons do not fall into the nucleus. When electrons are moving in atomic space with size 10^{-10} m, uncertainty in position is precisely the size of the atom, i.e., $\Delta x \approx 10^{-10}$ m. If electrons fall (we do not discuss how electrons *fall*) into the nucleus, i.e., electrons must be within nucleus size, then uncertainty in position will be limited by space with magnitude $\Delta x \approx 10^{-15}$ m and cannot escape, a region 10^5 times smaller than before. If uncertainty in position decreases, uncertainty in momentum and thus velocity (because $p = mv$) increases 10^5 times. Because electron kinetic energy increases proportionally with velocity squared ($K = mv^2/2$), energy increases very large. Here we have not mentioned electron relativity when moving near light speed. This contradicts an important principle of physics, the *principle of least action*, which holds that in all physical processes, the action of a physical system is minimal. Because inside the nucleus electrons have large energy, outside the nucleus electrons have small energy. The uncertainty principle and the principle of least action force electrons to be outside the nucleus. Thus, when precision is high, different from humans' innate observation ability, rational perception can be very different from sensory perception. That may suggest that sensory perception is an approximation of objective reality.

2.2.3 Standard model of particle physics

The standard model of particle physics based on quantum field theory (theory combining quantum mechanics with special relativity) is the best model describing fundamental particles that create the material world. This model has 17 fundamental particles and does not

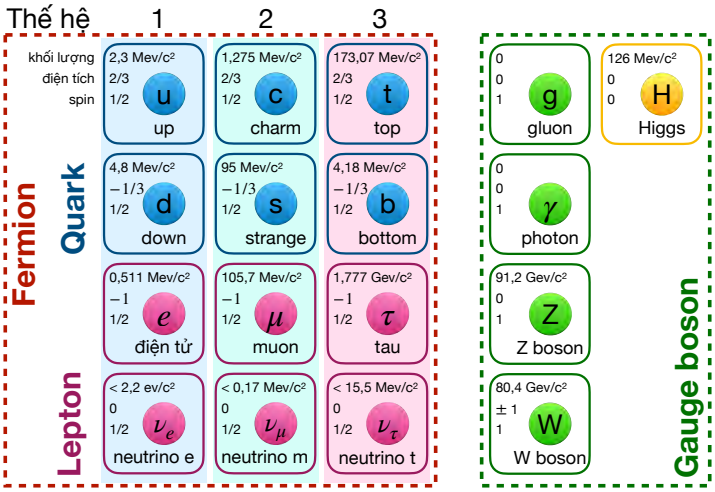

Figure 2.5: Standard model of fundamental particle physics. (Left) 12 particles directly creating matter (fermions) divided into 3 generations 1, 2, and 3. Each generation has 2 quarks and 2 leptons. (Right) 5 interaction-carrying particles (bosons) creating strong nuclear force (g), electromagnetic (γ), weak nuclear (Z, W) and mass (H).

count antiparticles. Among them, 12 particles create ordinary matter divided into 3 generations. Electrons are fundamental particles belonging to the lepton group (Figure 2.5), meaning electrons are not made of other particles. If we magnify an atom to a room with size about 10 m, then nucleus size is only like a pinhead (100 µm). If so, with human eye perception ability, the nucleus is only like a dust particle at atomic scale. The nucleus is made of protons and neutrons – they are not fundamental particles but are created by a sea of fundamental particles like quarks, antiquarks, gluons. Proton structure is nearly like a planet with a core having pressure greater than any form of matter ever known [63].

Protons are not simple particles; their structure is so complex that people call it the **proton sea**. The standard model holds that they are formed from quarks (Figure 2.5). In 1964, Murray Gell-Mann and George Zweig independently proposed what is called the quark model: protons, neutrons consist of three quarks (Gell-Mann was the one who named quarks). Mesons are formed from one quark

and one antiquark. Quark particles have three generations. We care about generation 1 consisting of two types: up quark (u) and down quark (d). With up and down to indicate the direction of quark particle spin. Spin is an intrinsic property of fundamental particles, having similarity (but not identical) to the self-rotation of those particles.

Earth rotates around the north–south axis one turn then returns to the initial position, we call Earth's spin 1. A King of Spades in playing cards only needs to rotate half a turn – 180° then it returns to itself; the King of Spades has spin 2. Similarly, electrons have spin $\frac{1}{2}$. Spin is an intrinsic quantum property of particles, not rotational motion in the classical sense. To illustrate, people often say that a particle with spin $\frac{1}{2}$ needs to rotate 720° (2 turns) to return to the initial state, but this is only a mathematical analogy, not describing actual motion. Quarks have spin $\frac{1}{2}$ like electrons. However, note that the names up and down of quarks are not related to spin direction but to an intrinsic property isospin related to SU(2) symmetry.

Up quarks have charge $+\frac{2}{3}$, down quarks have charge $-\frac{1}{3}$. Protons are made of two u quarks and one d quark, denoted uud, so proton charge is $\frac{2}{3}+\frac{2}{3}-\frac{1}{3}=1$. Neutrons are made of udd, so neutron charge is $\frac{2}{3}-\frac{1}{3}-\frac{1}{3}=0$. However, modern studies show that those 3 quarks of protons are also only an approximation.

To study proton structure, people bombard protons with electrons accelerated to high energy to see what bounces out. Bombardment results puzzled people. The more carefully studied with high resolution, the more products from the bombardment process. Two colliding particles produce countless particles. Increasing electron kinetic energy, the number of particles produced increases and seems unlimited [64]. Thirty years ago, a version of quantum field theory for strong interaction, quantum chromodynamics, predicted the proton sea model showing uud are three valence quarks, the average number of a seething swirling sea consisting of countless quark and antiquark particles continuously appearing and annihi-

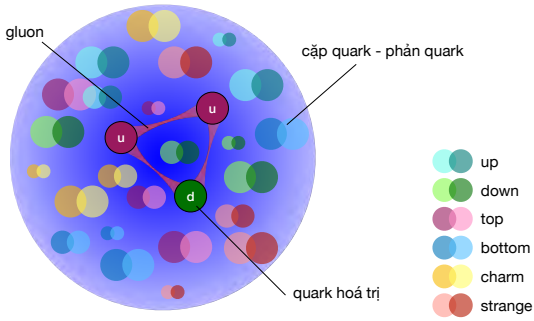

Figure 2.6: Proton sea: proton structure is not simple with countless particles and antiparticles appearing and disappearing continuously, leaving only 3 valence quarks. Quarks are linked together by gluons.

lating each other in a short time interval (Figure 2.6). The number of quarks equals antiquarks, so they annihilate each other, leaving only 3 valence quarks creating proton properties.

The proton sea model seems not yet correct because observations in 1991 showed that the number of particles shot out include particles heavier than electrons, protons, deuterons (nuclei of hydrogen atoms consisting of one proton and one neutron). Moreover, the number of down antiquarks is more than the number of up antiquarks. To explain this phenomenon, people had to update the proton sea model by adding a shift between protons and neutrons through exchange of a pion (made of one quark and one antiquark). Thus, besides protons being a sea of quark particles, protons themselves transform into neutrons and vice versa. This conversion process appears a pion particle, and this particle has down antiquarks, so experiments observe more down antiquarks than up antiquarks. However, problems continue to arise. Specifically, three valence quarks only contribute 30% of proton spin. The remaining 70% still unknown where it comes from. That awaits new calculations and experiments.

People once ignored proton gravitational interaction, but recently studies of interaction between electrons and protons have revealed that proton gravitational force can bend the structure of

spacetime around the particle. When bombarding protons with electrons, electrons exchange a photon with a quark and are deflected. However, there is a probability of one in a billion that a quark absorbs that photon and emits another photon. Studying the difference between the two incoming and outgoing photons will reveal information about pressure and force inside the photon. From that information, people can calculate interaction between photons and gravitons. These studies are being conducted, and we expect more time before results.

Studies of proton structure also discovered another strange point. Physical quantities related to protons are charge, spin, and mass. Because protons are not fundamental particles, the above quantities appear through complex interaction between proton constituents. Charge and spin are determined by quarks and studied in detail through experiments on electron-proton scattering [65]. But proton mass mainly comes from gluons, particles representing the glue binding quarks together. Quarks have mass from interaction with a field called the Higgs field (Figure 2.5). In 1964, Peter Higgs proposed a field present throughout the universe, and interaction between matter particles like quarks, electrons with this field will create mass for those particles. In 2013, the Nobel Prize in Physics was awarded to scientists who put forward this theoretical model. However, quark masses in protons only create about 1% of proton mass. Most mass comes from energy between quarks through gluons.

Quarks are like marbles tightly bound together by spring strings, gluons. Because springs are very strong, people have never observed independent quark particles. Gluon binding energy is so large that if converted to mass, that mass accounts for 99% of proton mass. Scientists at MIT have collaborated with artists to try to illustrate the proton sea. At large time scales, protons look like spherical particles. Shorter time scales make three valence quarks appear. Shrinking time scales will see a sea of particles appearing and disappearing continuously [66]. Information about protons depends on the preci-

sion with which we observe them.

With the standard model, quarks, electrons, and neutrinos (extremely small particles that almost do not interact with other matter particles) are fundamental particles. Three quarks creating protons are not fixed; they are the remainder of countless quarks and anti-quarks appearing and disappearing continuously. The contribution of 3 quarks only creates 30% of nucleus spin, showing that remaining quarks do not simply create protons; the internal process has many complex issues scientists need to explore. If **compositional nature** of protons manifests through 3 valence quarks (as defined in Chapter 1, page 67), then **impermanence** manifests by those 3 quarks not being independent, immutable, but only the remainder of the creation-annihilation process of virtual particles inside protons. Satisfying compositional nature and impermanence, protons have emptiness.

Viewed from afar, we see a stable, immutable, continuous, self-existent proton particle, but when approaching closer, magnifying size in space, shortening observation in time, it turns out proton particles are not simple at all. They are results of interaction of countless components creating them, quarks, and that interaction continuously changes over time. Compositional nature, impermanence, two characteristics creating emptiness manifest very clearly in protons. Similarly, the 3 quarks of neutrons are also results of continuous transformation of fundamental particles.

Fixed self-nature that we assign to protons with mass, charge, and spin is only a convenient approximation when studying in classical quantum mechanics. Approaching closer, we know protons are made of three quarks *uud* is also an approximation. Continuing closer, protons are a sea of creation-annihilation of particles where three quarks are only remnants of continuous transformation. No quark is fixed, independent, so protons are also no-self.

In classical physics, when two electrons approach each other, they repel. The repulsion of this electron to that electron through

the electric field that electrons create. Modern physics then holds that these two electrons exchange a photon making them pushed apart. This is like two people (representing two electrons) on two small boats on a lake throwing (exchanging) balls (virtual photons) to each other. Throwing balls makes the two people gradually move away from each other. Because electrons have charge, they will create a disturbance in the electromagnetic field (called photon field) around them. Therefore, instead of imagining electrons appearing as ripples rising in the electron field, we should visualize electrons continuously converting (combining) between two fields: electron and photon.

Thus, electrons are ripples of two fields: electron and photon. Those ripples continuously alternate; sometimes they are ripples in the electron field, other times ripples of the photon field. Physical statement: a real electron can transform into a virtual photon and a virtual electron, then return to a real electron. Even alternating like that, electrons still have mass and definite energy, momentum values, so if we observe them from afar and with sufficiently long time, they look like particles. Modern physics describes fundamental electron particles as a continuous transformation process from an electron field to a photon field. In each field, electrons only reside for an extremely short time interval, then immediately shift to another field.

* * *

I want to pause here to reflect on emptiness in the material world. If protons and neutrons are non-fundamental particles, obviously having compositional nature and impermanence, then fundamental particles like electrons, quarks, neutrinos – particles not created by any particles – do they have compositional nature and impermanence?

The complexity of protons shows emptiness at the deepest level. When we say proton, we are referring to an entity that only ex-

ists as an approximation at a certain observation level. At more microscopic levels, protons are no longer fixed particles but a dynamic process, a sea of continuous creation-annihilation. This aligns with Buddhism's view of emptiness: no entity exists independently, immutably. Even what we consider particles are only temporary appearances in a continuous transformation process. Protons have emptiness because they have no fixed self-nature – their properties only appear in relationships with the environment and observation conditions.

If fundamental particles are objects of quantum mechanics, then fields are objects of quantum field theory. In quantum mechanics, the most fundamental entity is a particle; in quantum field theory, the most fundamental entity is a field. Fields and particles do not exist independently of each other; they continuously transform into each other.

Quantum field theory holds that electrons appear as excitations, ripples rising from a corresponding field, the electron field. Not only electrons, other fundamental matter particles like quarks, neutrinos all have corresponding fields: quark field, neutrino field. When studying an electron, what information do we need to know? Position, velocity, momentum, energy,... are dynamic variables (their values can change over time) – external properties. Electrons also have what we usually call intrinsic properties: mass, charge, and spin. We usually think intrinsic properties are immutable attributes attached to electrons.

Modern physics then explains that intrinsic properties of electrons are not so simple. Like the proton sea, viewed from afar, it looks like a particle, but when approaching closer, it is a seething sea of interacting particles, creation-annihilation in an endless cycle. Electron mass is not simply an attribute of electrons but comes from interaction of electrons with a field filling empty space called the Higgs field. Fundamental particles interacting more with this field have large mass; interacting less have small mass (photons

have no interaction so have no mass). Electron mass is a consequence of the interaction process between electrons and the Higgs field (called Yukawa coupling); mass is not a self-existent attribute. However, once formed, under normal conditions, electron mass does not change, so it operates like an intrinsic property.

Similarly, charge and spin also come from properties of space and time. Charge appears through a local symmetry called gauge. Charge is not self-existent but because electrons carry a specific gauge representation. Once again, charge does not change in all interactions, so we still call it intrinsic. Spin is an inevitable consequence of geometric structure – symmetry of Minkowski spacetime, Lorentz symmetry in Einstein's special relativity. Spontaneous breaking of Higgs symmetry, gauge symmetry, and Lorentz symmetry are structures of empty space, structures of transformation possibility, and geometric structure of spacetime. Thanks to these structures, electron properties can be generated.

If electron properties are all determined by symmetry and interaction, what does the electron still have that is truly intrinsic nature? In quantum field theory, a strong view is that intrinsic properties are only how we describe parameters that do not change within the current symmetry scope. In essence, they are manifestations of deeper structures of space and time. This leads to intrinsic and extrinsic properties not being absolutely separate. What is considered intrinsic is actually a result of universal relationships between fields and symmetry. Electron nature does not exist as an isolated entity but is a result of symmetry relationships.

Intrinsic properties of other fundamental particles are also like electrons; they are not immutable attributes but results of relational structures. Modern physics denies self-nature of fundamental particles at the deepest level. From there, composite particles like protons, neutrons, and atoms, thereby creating all matter we see daily, all have no self-nature, cannot be independent, and are not immutable.

Standing from afar, at low energy levels, at large time intervals, we see electrons as particles with properties we assign to them like mass, charge, spin. Although these properties are blurred by the indistinguishability principle, we still try to reason that they have those attributes. The Pauli exclusion principle brings self-nature of fundamental particles like electrons through unique quantum states. Each electron can exist in a different state; no state is like any other. Individuality, or we can call electron self-nature, is obtained through that unique state. But that state is obtained through the relationship between electrons and the nucleus. This relationship has created the periodic table of chemical elements we use today. Without them, science would certainly be different.

At smaller observation distances, higher energy, shorter time, we see electrons as oscillating waves. Waves propagate in space; they are non-localized energy, so assigning self-nature to waves is more difficult than assigning to particles. That observation results depend on observation size of things is a strange result. In physics, the **renormalization group** explains how physical parameters (like mass and interaction constants) of a system change according to the energy scale or distance we observe. At different levels, entities manifest different properties. They do not have a definite value. The definiteness of an entity completely depends on how we observe.

Think about this strangeness; it is completely different from how we think about science as an objective, independent method (separating object from subject). Physics, the most fundamental science, at its deepest theory, does not allow separation between subject and research object. What we observe are quantities of a particle in experiments. Because there are many interactions between fundamental particles and virtual particles and antiparticles in vacuum, when observed from afar, obtained values are effective, a way of averaging that observation process. If we believe quantum mechanics is correct and complete (still so to this day), then fundamental particles are completely not independent, have no self-nature, and

are not immutable. **Compositional nature** of electrons manifests through two factors: electron field and photon. **Impermanence** manifests through continuous alternation between these two fields (see definition in Chapter 1, page 67). Thanks to these two properties, emptiness manifests in fundamental particles like electrons, quarks, neutrinos.

2.2.4 Theory of everything

Nature has four fundamental forces: strong nuclear, weak nuclear, electromagnetic, and gravitational. The standard model has unified the first three forces, but gravitational force still stands apart from other forces in modern physical models. Unifying fundamental forces is the most important problem in physics today. The standard model has not yet met this requirement. A candidate for a theory of everything is **string theory**. This is a theory attracting much attention from leading physicists. This theory holds that ultimate reality is strings with very small size, on the order of Planck length 1.62×10^{-35} m, much smaller than proton size 0.8×10^{-15} m. Strings vibrate with different modes to create particles like electrons, quarks, neutrinos. Not only vibrating, strings have complex structure with many dimensions. Like a thread, viewed from afar is a 1-dimensional string, but if we use a magnifying glass to enlarge, we will see other dimensions of that thread. Strings in string theory are the same. They have more complex structure than their appearance at first glance because besides 4 macroscopic dimensions (3 spatial and 1 temporal) that we usually see, they also have other dimensions up to 11 dimensions. With everyday experience, humans cannot visualize a 4-dimensional world, let alone a world with up to 11 dimensions. But that is the world of string theory. The weakness of this theory is that it can explain many physical phenomena but has not yet made any predictions that can be verified by experiment. Predictions of string theory are all at sizes too small compared to

atomic and nuclear sizes, so with current experimental capabilities, it will likely be a very long time before humans can know whether this theory is correct or wrong.

String theory was extended into brane universe theory. Each vibrating brane creates a universe. Branes do not interact with each other, making universes exist independently of each other. Some fundamental forces only exist within one brane, making the existence of this brane not affect other branes. But gravitational force is not limited to one brane; this force can *leak* partially to other branes. This explains the very small value of gravitational force compared to other forces. Moreover, it can explain dark matter. If we compare the electric force between two electrons close together, the electric force between them is 10^{42} times larger than gravitational force. No wonder just brushing a plastic pen on hair can attract a piece of paper on the table. The tiny electric force between the plastic pen and that piece of paper has overcome Earth's gravitational pull on the paper. Brane universe theory explains that gravitational force in our universe leaks to another universe, so it becomes so small [67].

2.2.5 Dark matter and dark energy

Observations show that ordinary matter (creating stars and planets) only accounts for 5% of the total energy of the universe (matter and energy are equivalent through Einstein's formula $E = mc^2$, so when we say energy, we also mean mass). The remaining energy comes from dark matter and dark energy.

From 1933, Fritz Zwicky discovered that the rotation velocity of stars at the edges of many galaxies, including the Milky Way, is too fast compared to gravitational force created by matter calculated from stars. The faster the velocity, the stronger gravitational force must be to keep stars on galaxy orbits. If velocity is too fast, gravitational force is not strong enough, then stars will be flung out

of galaxies. According to calculations, matter from luminous stars only accounts for a small part to create gravitational force, so people hypothesize that a type of **dark matter** exists in galaxies keeping stars rotating so fast. It is called dark matter because besides gravitational force, it does not interact with ordinary matter through electromagnetic force. Currently there are some hypotheses about the origin of dark matter, such as the hypothesis that dark matter is made of weakly interacting massive particles (WIMP) or particles heavier and slower than neutrinos called neutralinos.

Multiverse theory holds that each universe, including the universe we are living in, exists on a universe brane. Among the four fundamental forces, only gravitational force can transmit between branes. Gravitational force of our universe can leak to other universes, making gravitational force the weakest among fundamental forces. Conversely, gravitational force from neighboring universes can act on our universe, and the result is dark matter.

Dark energy accounts for most of the universe's energy with 68 % (Figure 2.7). Opposite to ordinary matter and dark matter, dark energy makes galaxies move away from each other with accelerating speed, making the universe expand faster. To explain dark energy, scientists put forward some theories like the existence of Einstein's cosmological constant characterizing negative pressure of vacuum; gravitational force changing depending on universe scales; or the existence of a dynamic field (quintessence) varying with space and time. The above theories all aim to explain the accelerating expansion of the universe over time but have not yet been confirmed by experiment.

So what is ultimate reality? Fundamental particles in the standard model? Strings in string theory? Branes in brane universe theory? Or other realities? The famous 20th-century philosopher Karl Popper held that human sensory and rational perception (manifested through scientific theories) are only conjectures that our brain builds to become consciousness of objective reality. Compared to

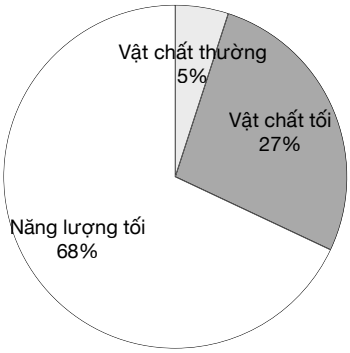

Figure 2.7: Proportion of ordinary matter, dark matter, and dark energy in the universe.

Buddhist doctrines, objective reality is ultimate truth; our consciousness of reality is relative truth. Then relative truth is only images, models the brain creates for convenience of existence and development. Similar to Popper's view, relative truth is conjectures. And if they are conjectures, they are impermanent, have no long-term correct value. Any conjecture is only correct in some context, then other conjectures will replace the initial conjecture. All conjectures can be refuted. We will study more about this issue in the Consciousness Chapter.

Depending on the precision the observer desires, conjectures are used appropriately. Newton's theory, though known to be less accurate than relativity theory, is still used in calculations of space flights. Simulating the operation of a ship on water, a version of classical mechanics is continuum mechanics, still used. That is because their calculations are good enough for practical use. In this aspect, we can say scientific theories have pragmatic nature equivalent to the idea of skillful means in Buddhism. Like the point mass that Earth appears as in *pale blue dot* when viewed from Voyager is an approximation. The point mass that atoms appear as when we look at electron microscope photos, that is also an approximation. The point mass that nuclei appear as when magnifying atoms

to room size, that is also an approximation. Each time we observe things and phenomena, we will examine with a certain precision. With each precision, the brain puts forward a conjecture – an approximate model of reality so that it fits. Thus, conjectures, i.e., relative truth, are approximate, frequently changing to serve the observer's purpose.

The above shows two properties of atoms and subatomic particles, particles creating matter: **compositional nature** and **impermanence** (as defined in Chapter 1, page 67). Compositional nature manifests by any particle being formed from smaller constituents. Smallest are fundamental particles, but these particles are not fixed, always appearing and disappearing continuously. String theory also holds that fundamental particles are due to vibrations of extremely small strings. To this day, people have not found a way to test their existence. Impermanence manifests through continuous change of fundamental particles. The proton sea is an example showing that though protons are made of three quarks, these three quarks are not fixed; they are three valence quarks. The appearance and disappearance of quarks, antiquarks in the nucleus leave three quarks. Research on matter structure from cosmic scale to subatomic all shows emptiness of matter.

2.2.6 Atoms and chemical properties

Up to here, we can summarize what was presented above into a modern model of atoms consisting of nucleus and electrons. Nucleus size is very small compared to atom size but contains most of the atom's mass. Electrical neutrality of atoms comes from balance between positive charge of nucleus and negative charge of electrons. The nucleus is formed from protons and neutrons. Although protons carry positive charge, they are held in the small space of the nucleus by strong nuclear force (equivalent to gluons in the standard model). Protons and neutrons are formed from quarks, but nucleus

mass obtained does not only come from quarks but mainly from gluons binding quarks together. Quarks and electrons (together with neutrinos) are fundamental particles in the standard model creating atoms to form matter around us. Those fundamental particles do not exist independently and immutably. When observed with very short time, they frequently appear and disappear without ceasing. Only when observed with low precision do they manifest as particles with characteristic properties like mass, charge, and spin.

An atom has many electrons moving around the nucleus. Electron motion is described by a probability cloud with different states called orbitals. Each orbital contains a certain number of electrons. For example, s orbital can contain at most 2 electrons, p orbital can contain at most 6 electrons, d orbital can contain at most 10 electrons,... The arrangement of electrons in orbitals creates periods of elements in the Periodic Table. Period 1 has two elements corresponding to two possible states s^2. Periods 2 and 3 have 8 elements including two elements s^2 and 6 elements p^6. Periods 4 and 5 have 18 elements because besides 8 elements like previous periods, we need to add 10 elements d^{10}. Similarly, periods 6 and 7 have an additional 14 elements f^{14}, so they have up to 32 elements. Orbitals have a certain shape: s spherical, p dumbbell-shaped,... Orbital shape affects chemical properties of atoms. The size of the outermost orbital is precisely the size of atoms. It is not that having more electrons means larger size; it can even be the opposite. In a period, i.e., a row in the Periodic Table, atom size decreases when the number of electrons increases. In a group, i.e., a column, size increases with the number of electrons. All properties of the Periodic Table can be explained by quantum mechanics [56, page 491].

The Periodic Table of 118 known elements arranged neatly on one page is considered the most important page in scientific history. That page, with all symbols, numbers, shapes, colors,... comes from very vague and non-sensory axioms of quantum mechanics. What is special is that those axioms lead to many results very consis-

tent with principles of dependent origination, no-self, and emptiness. Denying fixed essence in quantum mechanics is very consistent with no-self, no self-nature of Buddhism. In quantum mechanics, before measurement, particles do not have a pre-determined state (for example: no fixed position or momentum). They only exist as a wave function describing a set of possibilities. Specific states only appear when there is observation. The indistinguishability principle blurs the boundary of a particle understood in ordinary experience. The superposition principle makes particles able to be in many different states simultaneously.

Based on the Periodic Table, we determine chemical properties of elements (elements are substances formed from one type of atom). We can divide into 3 types of chemical bonds: metallic, ionic, and covalent. Characteristics of these bonds are exchange of electrons between atoms when they participate in bonds. Additionally, between atoms and molecules there are four types of physical interactions, also called molecular attraction forces due to charge imbalance of atoms and molecules. Arranged from strong to weak: (1) ion – dipole (salt dissolves in water because salt ions are attracted by water molecule dipoles), (2) dipole – dipole (water molecules in liquid and solid states attract each other), (3) dipole – induced dipole (polar water molecules attract nonpolar oxygen molecules, making oxygen dissolve in water), and (4) instantaneous dipole – induced dipole (oxygen molecules are instantaneously polarized, attracting each other when liquefied). Interactions (3) and (4) are called **van der Waals forces**; interaction (4) is also called London dispersion force.

Characteristics of physical interactions are no electron exchange, and they are always attractive forces. This attraction is also governed by spatial symmetry and molecule size. Molecules with low symmetry and large size will have strong interaction and vice versa. For example, asymmetry of water molecules makes dipole – dipole interaction strong, so water is a liquid under normal conditions

while carbon dioxide molecules are gas. Carbon dioxide molecules are larger than oxygen molecules, so carbon dioxide can dissolve in water more. That is why people use carbon dioxide to dissolve into carbonated drinks.

The second type of interaction is dipole – dipole occurring between H and one of three elements with high electronegativity: N, O, and F, called **hydrogen bond**. Actually it is a physical interaction, but because this interaction is strong, it is called a bond. Hydrogen bonds play an important role in biology, especially between atoms in DNA. Hydrogen bonds between ATGC bases are strong enough for DNA to be stable but also weak enough to separate the double helix in gene transcription and replication processes. Van der Waals and London interactions, though weak, are important to create high-order structures of macromolecules like proteins or between phospholipid molecules creating cell membranes. All biological processes of life mainly rely on covalent bonds and the 4 interactions mentioned above.

2.3 Time

> "Spring passes, a hundred flowers fall,
> Spring comes, a hundred flowers bloom.
> Before eyes, things pass endlessly,
> On head, old age arrives.
> Do not say spring ends, flowers fall completely,
> Last night in the front yard, one branch of plum."[7]

Like many other peoples, Hinduism holds that time relates to motion and transformation through the presence of **Trimurti** of Hinduism: creator Brahma, preserver Vishnu, and destroyer Shiva.

[7]Poem by Zen master Man Giac (Ly Nguyen Giac) of the Tran Dynasty: "Xuân khứ bách hoa lạc, Xuân đáo bách hoa khai. Sự trục nhãn tiền quá, Lão tòng đầu thượng lai. Mạc vị xuân tàn hoa lạc tận, Đình tiền tạc dạ nhất chi mai."

These deities manifest the cycle of birth, existence, and disappearance of all things in the universe. Buddhism uses two concepts to refer to time: ksana [instant] to refer to short time and kalpa (aeon) to refer to long time. Ksana refers to a very short time interval, but original Buddhism does not specify how long. Later scriptures try to specify specific time intervals. For example, according to *Dizang Fashu*, in one thought (one thought, thinking) there are 90 ksana. Kalpa is also an estimated number, equal to 16.8 million years, called a small kalpa. Additionally, original Buddhism also has medium kalpa equal to 20 small kalpas, great kalpa equal to 4 medium kalpas.

Buddhism holds that time is beginningless, endless, denying ultimate time. The cause-and-effect principle forces all things to move continuously, so there will be no starting point without a cause. This is a controversial point because modern physics holds that the universe, including time and space, began from the Big Bang 13.8 billion years ago. However, many reputable scientists including Roger Penrose hold that the Big Bang is only one among infinite explosions that have and will occur. Thus, for us, the Big Bang is the beginning of the universe, but for the universe, that is only one event among countless events that have and will occur [68]. Of course, Penrose's hypotheses are only ideas with no experimental evidence yet.

Buddhist perception of time is very special, acknowledging time is real, i.e., exists objectively, but sensation of time differs. Human senses bring illusions about the world, i.e., **relative truth** (as presented in Chapter 1). There, time is divided into three periods: past, present, and future. For saints who have been enlightened to **ultimate truth**, there is no distinction of time within time; they transcend the bondage of time, realizing the emptiness of time. This view is very close to understanding time in the **block universe** of relativity theory. There, time is like space creating a spacetime continuum without past – future anisotropy. Anisotropy comes from lack of information, lack of understanding about the world (see *Com-*

putational Irreducibility by S. Wolfram on page 136). If we know information at all levels from microscopic to macroscopic, time no longer has anisotropy. Here ignorance is understood scientifically as absence of information about a system's state. Due to finiteness of computational ability and time needed to calculate information, time appears as we sense: anisotropic and irreversible.

Suffering is a pervasive concept, the essence of existence in life. The goal of humans is cessation of suffering, liberation from all suffering. When saying so, we have acknowledged an anisotropy when comparing suffering with joy, two opposite concepts, creating the **anisotropic concept pair** joy – suffering. Modern psychology proposes six basic emotional states of humans according to the definition by psychologist Paul Ekman put forward in 1998 [69, 70]. These are emotions: fear, joy, disgust, anger, sadness. Not only humans, animals also have similar emotions through expressions of individuals when full or satisfied with some desire [71]. Happiness and suffering in Buddhism are probably broader in meaning or combinations of some sensations mentioned above. Only one of six basic sensations relates to joy, that is happiness. Most sensations relate to suffering.

Buddhism views joy – suffering as the essence of existence. That is truth, what is correct in the correct – wrong anisotropic pair. Not only that, anisotropy also appears in good – evil (ethics), beautiful – ugly (aesthetics) to form the concept of truth – goodness – beauty of Buddhism. So where does that anisotropy come from? How does it appear? This section presents anisotropy; it appears in nature in non-biological and biological systems related to order – disorder manifested through a fundamental concept of physics: entropy. Not only that, this anisotropy is considered related to time anisotropy and how organisms experience the flow of time in the universe. Not only anisotropic, time has irreversibility, only going in one direction toward the future and cannot return to the past. This sounds crazy, but let us study why scientists hold that entropy brings anisotropic and irreversible properties of time, of life, of truth – goodness –

beauty.

2.3.1 Arrow of time

Time has direction; people call it the arrow of time. Time is a universal concept. Anyone has consciousness of time. Whether that consciousness is the same or not, ordinary perception of time can be summarized through some characteristics.

First, time does not manifest for us to perceive directly but can only be indirect through transformations of things and phenomena. Vision plays an important role in time perception because this sense shows us ceaseless motion of things, and that motion is in space. Eyes can see distance between two things; that is space between them. But no human sense directly measures time. Human consciousness of time is indirect. Hearing helps form the concept of time through listening to intervals between heartbeats, the rhythm of footsteps. Emotional states also help perceive time differently: when in danger, time feels slow; when joyful, it flows fast. Senses bring humans consciousness of time in the brain. Time is like other concepts, products of the brain, but the brain senses time through transformations of things and phenomena coming from senses. No sense directly senses time.

Second, time flows ceaselessly, and that flow is irreversible, flowing in one direction, from past, to present, and toward future, cannot return, even for a moment. The description of time flowing in one direction is for approximate visualization; that is not an accurate description because time does not manifest.

Third, the flow of time seems uniform everywhere, although modern physics shows time can be stretched, locally bent by matter and velocity. Human intuition shows space and time are background frames on which things and phenomena occur, but modern theories show time cannot be separated from space and matter.

Due to the above properties, scientists call it the **arrow of time**,

or psychological arrow of time. For a long time, time has been an object of study by philosophers and scientists. Even in fairy tales, holy books, poetry, music, people try to understand, explain what time is.

Time is the measure of motion and development of things and phenomena. The first person to have statements related to time was Greek philosopher Heraclitus. He held that all things always change ceaselessly through the famous saying *One never bathes twice in the same river*. It can be understood that time does not flow, but all things flow in time. In other words, all things always transform, and that transformation occurs in time. Aristotle also had the same view and defined time as the measure of motion of all things. All things always move, and that motion follows time. Even so, what time is remains a mystery. He held that time is the most indeterminate among indeterminate things.

Later philosophers like Leibniz defined space as the order of entities coexisting; time is the order of entities succeeding each other. Thus Leibniz linked time with space. Hegel held that humans are in time, and time does not exist outside humans. Therefore, humans are time, and time is humans. Hegel's idealism manifests everywhere, even in explanation of time. Engels held that basic forms of all existence are space and time. Through that, Marxist – Leninist philosophy links time with space; both are linked with matter. This view is precisely modern physics' view of spacetime and matter. Those three entities coexist and cannot be separated. Eastern philosophers do not specifically state definitions of time, but Hinduism, Buddhism, Taoism, Confucianism all share the view that all things transform, and transformation relates to time. Previous philosophers did not state precise definitions of time but linked time with motion.

In Vietnamese fairy tale treasure, there is a story about change in time flow. That is Tu Thuc, a young man lost in the fairy world. Only living about 3 years there corresponds to three hundred years

on earth. That is probably the only imagination of ancient Vietnamese people about being able to go to a place where time flows slower than normal life. According to relativity theory, all entities move at light speed in spacetime. If moving slowly in space, they will move slowly in time and vice versa. Therefore, the slowness of Tu Thuc's time corresponds to a reference frame moving fast in space like on a spaceship. Could ancient people have encountered an extraterrestrial civilization?

There are many **definitions of time**, but in summary, they all refer to a measurement between events. This definition fits the definition of space as a concept referring to measurement between things.

> "Time reveals distance between events; space reveals distance between things."

The arrow of time is linked with an important principle in the universe, the cause-and-effect principle. The cause-and-effect principle tells the influence of this thing, this event, which we call cause, on another thing, another event, which we call effect. All cause-and-effect processes occur in time, following the arrow of time. Understanding the arrow of time will lead to understanding the cause-and-effect principle, an important principle in nature. Buddhism emphasizes the cause-and-effect principle, and this principle occurs following the arrow of time. Therefore, studying time is important for Buddhism.

The arrow of time is linked with the cause-and-effect principle, an important principle in Buddhism. Buddhism's view of time is very special: enlightened beings do not distinguish time within time; they transcend the bondage of time. This suggests that the irreversibility of time we sense may only be an illusion, a relative truth, coming from lack of information about the microscopic world. When we study entropy and time, we will see that irreversibility may come from limitations in our perception, not an inherent nature of

time. This is precisely the emptiness of time: there is no absolute, fixed time. Time only appears in relationships with entropy, with information, and with the observer.

2.3.2 Time assigned to entropy

Time is assigned to entropy; time anisotropy is studied through entropy anisotropy. Why do we see time as irreversible? Why is the flow of time irreversible? Scientists use scientific theories to explain time. But how do they achieve that when time is subjective perception, completely non-existent to perceive through senses, while scientific theories expressed through mathematical equations are objective knowledge?

This is a very difficult problem, so despite much scientific progress over hundreds of years, studies on time have not had many significant results. Scientists' approach is to find a correspondence between properties of time and some equations in science and examine whether those equations can give verifiable results. To do this, physicists search in nature for equations describing an irreversible process and assign its irreversibility to irreversibility of time. Carlo Rovelli put forward three equations that can be considered [72]:

1. Second law of thermodynamics: entropy of a closed system always does not decrease $\Delta S \geq 0$. This is the only equation in physics showing continuous irreversibility and is also a very important equation governing the universe so we see it as it is and not otherwise.
2. Non-commutative relation in quantum mechanics leads to **Connes flow** proposed by Alain Connes. This flow relates to thermodynamic flow manifested in the second law and is assigned to the flow of time.
3. Wheeler – DeWitt equation does not contain time in loop quantum gravity theory, one of theories unifying quantum mechanics with relativity theory. This theory differs from other theories in

not placing things and events in space and time but holding that space and time co-arise from spin networks.

Among the three equations above, most scientists rely on Arthur Eddington's proposal when holding that time is linked with increase of entropy, so the second law of thermodynamics can be used to describe time. The irreversibility of equation $\Delta S \geq 0$ creates the **arrow of time**.

Entropy is a physical quantity proposed by Rudolf Clausius in 1865 when studying thermodynamics. In the context of steam engines emerging, requiring scientists to find ways to optimize the process of converting heat into work efficiently. People knew heat relates to temperature, but what is that relationship? That is precisely the definition of entropy that Clausius put forward. He held that when giving a very small amount of heat $\mathrm{d}Q$ to a physical system at temperature T, entropy of the system changes by a very small amount $\mathrm{d}S = \mathrm{d}Q/T$. Entropy is a physical quantity relating heat quantity and temperature. From there, scientists developed the **second law** of thermodynamics, which Sadi Carnot had previously put forward in initial form: heat always loses a part when producing work. To this day, we state the second law of thermodynamics in the form that entropy of a closed system always does not decrease $\Delta S \geq 0$. Other completely equivalent versions of this law: cannot convert all heat into work; cannot manufacture perpetual motion machines of the second kind; heat cannot transfer from low temperature to high temperature.

Clausius's definition does not tell the microscopic nature of entropy and of course has no relation to disorder. In 1870, Ludwig Boltzmann brought meaning to entropy when he held that entropy is a physical quantity measuring the number of microscopic configurations (i.e., the number of possibilities of arranging gas molecules in a physical system; we will see a specific example below) W of the system $S = k \log W$ (k is Boltzmann's constant). **Entropy is a physical quantity characterizing the number of possible mi-**

croscopic states of a physical system. Boltzmann's definition relates macroscopic entropy of Clausius with microscopic states, i.e., states of atoms in the physical system, and the relation coefficient is precisely Boltzmann's constant. The number of microscopic configurations is very large but is reduced through the log function, a function inverse to the exponential function.

***

Boltzmann was a brilliant but unfortunate theoretical physicist. He was one of the first to accept atomic theory, molecular theory, and connect properties of atoms with macroscopic properties of physical systems. His research work is very important for classical physics but is a transition to modern physics: (1) generalization of velocity distribution of systems consisting of gas atoms by James Maxwell, which we now call Maxwell – Boltzmann distribution, (2) derivation of formula for Stefan's law on blackbody radiation, which we now call Stefan – Boltzmann law, (3) building equation for entropy around 1872 – 1875, (4) explaining the second law of thermodynamics based on atomic theory by adjusting mechanical laws into statistical laws applied to motion of atoms, (5) founding quantum mechanics by accepting discrete energy levels in physical systems. However, when alive, his view on applying atomic theory was rejected by many influential people at that time like Ernst Mach and continuously attacked, making him severely depressed. In 1906, Boltzmann hanged himself during a vacation with his wife and daughter. Only a few weeks after his death, his research was confirmed by experiment. To commemorate Boltzmann's contribution, the equation $S = k \log W$ is engraved on his tombstone in Vienna, Austria.

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Because entropy is proportional to the number of microscopic states of a physical system, and this number increases over time according to the second law, entropy increases over time. Example of

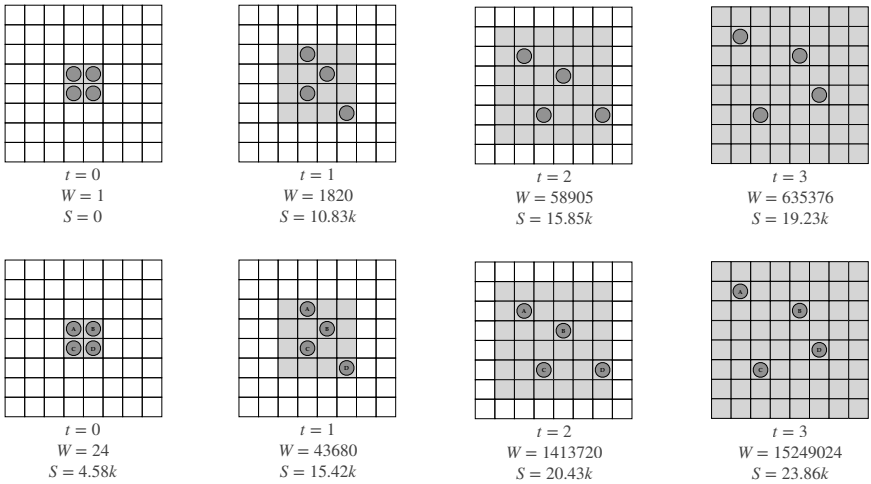

Figure 2.8: 2D diagram of ink drops diffusing in water over time increasing from left to right. (Top) not distinguishing ink molecules. (Bottom) distinguishing ink molecules. Depending on information needed, entropy can differ.

ink drops diffusing in water over time in Figure 2.8. At time $t = 0$, four ink molecules are at positions close together, each molecule at one position. At this time, there is only one way to arrange molecules; we call that microscopic configuration $W = 1$, substituting into the entropy formula $S = k \log 1 = 0$. This is the state with lowest entropy. As time increases, ink molecules can diffuse around, so the number of possible states that ink molecules can move to increases. For example, at time $t = 1$, there are 16 possible states for 4 ink molecules to reside; at this time, the number of microscopic states $W = \frac{16!}{4!(16-4)!} = 1820$, therefore, $S = k \log(1820) = 10.83k$. Similarly, we can calculate entropy of the system after any time interval t as indicated in Figure 2.8.

Note, entropy is the number of possible microscopic states, not yet related to disorder. Disorder only arises when we consider meaning (or information) about some specific configuration that we call ordered configuration. Suppose we call the state corresponding to $t = 0$, four ink molecules close together forming a square, *ordered configuration*; other configurations are called disordered configura-

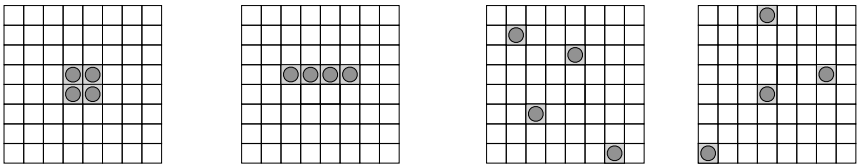

Figure 2.9: Configuration that we determine as order is arbitrary. Configurations not like above are all considered disordered.

tions. We expect at other times $t = 1$, the system still in this ordered configuration. Boltzmann said that the probability of having that ordered configuration is only 1/1820. As time increases, this probability becomes smaller because possible configurations increase over time. Therefore, expecting the system to be in ordered configuration at time $t > 0$ is very unlikely. The probability that the system is in configurations different from ordered configuration, which we call disordered configurations, is very high. That is precisely the content of the second law, saying that entropy of a closed system always does not decrease. Another statement: **disorder of the system always does not decrease**. Important note: when stating about disorder, we need information related to order. That is, we must determine what is order.

What is ordered configuration? We can define a different ordered configuration. It can be ink molecules arranged in a straight line horizontally in the middle, or can be any configurations we want as in Figure 2.9. Therefore, our choosing some configuration that we call ordered configuration has subjectivity of the observer. That is assigning information to a physical system. Each time we define ordered configuration, configurations not like that are disordered configurations. In most cases, disordered configurations are much more than ordered configurations. Therefore, the possibility of observing disorder always increases over time.

Does ordered configuration necessarily have to be subjective? In nature, some configurations preferentially develop life like structure of DNA, proteins, amino acids,... or necessary for development of

consciousness like structure of human body, nervous system, brain,... This ordered configuration is a configuration that can create a chain of definite causal powers. All those structures existed before humans raised the issue of entropy and asked questions about order in physics. Only today, with development of science and technology, humans understand and analyze that order, so we can say order of systems does not necessarily have to be subjective order. Those are objective orders necessary for specific purposes of living organisms. But whether order is objective, serving some purpose, or subjective order assigned by scientists to physical systems, that ordered configuration is always much smaller than disordered configurations. Therefore, in practice, we always observe disorder of physical systems increasing over time.

Defining what is order has subjectivity of the observer. This reflects the emptiness of the concept of order: there is no absolute, objective order. Order only has meaning in relationships with a specific purpose or a specific observer. The concept of order has no independent self-nature because a configuration considered ordered for this person can be disordered for another person, depending on their purpose and perspective. This shows emptiness not only applies to matter but also to abstract concepts like order and entropy.

Author Carlo Rovelli gives a very interesting analysis in his book *The Order of Time* when holding that the first law of thermodynamics, the law of conservation of energy, is actually not as important as the second law [72]. The law of conservation of energy is important and taught from secondary school. However, if energy is always conserved, why in daily life do we always have to add energy for ourselves, cars, motorcycles,... and all other systems? The second law is truly important. It tells us no process is perfect according to our wishes; energy is always dissipated, lost to places we do not want. That is why daily activities of the entire world, humanity, are only to find energy sources supplementing that loss. In the end, all basic activities of human life, economic, political, social activities, are to

satisfy needs for food, energy for each individual in a country. And that energy is energy compensating for loss occurring at all times everywhere in all things around us. That is an important law of nature that needs to be taught carefully from secondary school, not waiting until university.

Another noteworthy point is defining microscopic states through dividing space to calculate entropy. Figure 2.8 shows a microscopic state through a cell area. A cell area has at most 8 nearest cells. If in time from t to $t + 1$, ink molecules can move from some cell to the nearest cell, then it only has 8 possible cells allowed. This is a very rough approximation of microscopic state of molecules because our dividing space into cells like that is also arbitrary. If we divide space into cells with size much larger than molecule size, then the number of possible microscopic states will differ when we divide space into cells with size equal to molecule size. If dividing smaller, the number of microscopic states will increase very much. Because determining microscopic states is arbitrary, the number of these states is infinite because we have acknowledged classical physical systems are continuous.

This problem is solved by quantum mechanics. Quantization leads to finiteness of the number of microscopic states. A chemist at Hebrew University Jerusalem, A. Ben-Naim, is the one who applied the uncertainty principle and the indistinguishability principle of particles in quantum mechanics to calculate entropy value [73]. From the definition of information entropy by Claude Shannon put forward from 1948, he calculated the value of thermodynamic entropy. Dividing space into cells cannot be arbitrarily small but is limited by the uncertainty principle. With uncertainty of momentum $p = mv$ will lead to uncertainty in position. Uncertainty in position limits the minimum value that atoms, molecules can have. Thus, there is a minimum cell area for atoms and molecules to reside. The number of areas proportional to the number of possible microscopic states is finite. Even so, the number of microscopic states is

still very large and almost impossible to calculate for macroscopic systems.

In practice, scientists do not calculate absolute values of entropy but calculate changes in entropy. Like when working with potential energy on Earth. A ball on a table has a certain potential energy relative to the floor; that potential energy is completely determined easily. Actually, that is the difference in potential energy of the ball when it is at two different positions. Physicists do not need to calculate absolute potential energy relative to Earth but only need to determine the difference between potential energy on the table and potential energy on the ground. Similarly, entropy is also the difference between state at some time compared to state at another time. Entropy value in the past is always lower than in the present.

2.3.3 Problems when assigning time to entropy

The increase of entropy makes the physical system must have low entropy at some time in the past. So why must entropy in the past be low? This is a problem many scientists put forward controversial hypotheses. Some hold that there is a universe entropy with value smaller in the past and reaching minimum when the universe formed. This is called the **past hypothesis** by American philosopher David Albert put forward in 2003 [74]. The past hypothesis has been mentioned again by many physicists in explanations about time. At the moment of universe formation at the big bang, entropy is lowest. After that, together with universe expansion, entropy gradually increases over time.

Where can entropy increase reach? Can the universe reach a state of maximum entropy? That is what scientists do not yet know, but previously Lord Kelvin proposed **heat death** describing the most chaotic state of the universe. Currently no one can calculate universe entropy. If there is such an entropy, determining its value is impossible. Therefore, the hypothesis that universe entropy must

be small at the big bang moment is still controversial. Thus, from the big bang with low entropy, the universe progresses over time and entropy gradually increases. That process creates the flow of entropy, an anisotropic and irreversible process. That is precisely the origin of time, the connection with entropy. Entropy flow leads to time flow.

After determining the state we call ordered state, entropy is a physical quantity proportional to disorder level, and that disorder increases over time. The increase of disorder has nothing mysterious. That is precisely the number of ordered microscopic states is much smaller than the number of disordered microscopic states, so the probability that the system can be in disordered state is higher than ordered state. The equation $\Delta S \geq 0$ is an equation similar to time in that it is irreversible. Therefore, scientists assign time change with entropy change.

Below are some notes when studying **entropy** that many people often confuse. Entropy is not a measure of disorder; it is a measure of the number of possible microscopic states. Only when someone observes the physical system, comparing it with a special state we call order, does the concept of disorder appear. Entropy always increases over time in a closed system. Entropy is like other thermodynamic functions, like other physical quantities having definiteness at a moment, so when saying it increases, it is according to some variable. Time is a variable for entropy to increase. But if entropy increases over time, can it be equated with time? Entropy does not have a definite value. What is determined from experiment is entropy variation. Assigning that variation to variation of microscopic states is truly a difficult problem. Suppose there is a relation between entropy and time, then we need to clarify some problems below. How are entropy and time related? Do both increase at the same rate? We see time flows relatively similarly for each person, but disorder increases here, decreases there. Time is universal everywhere, but entropy does not seem universal like that. Two physical

concepts moving together but one fast, one slow, how can they be assigned to each other? Entropy is only determined for a thermodynamic system, a thermodynamic process. Therefore, if only limited to increasing entropy of some subsystem will be very unreasonable because there are countless ways to choose such a subsystem.

If we choose a subsystem, for example a refrigerator, and place a clock inside it, we do not see time decrease. We can reason that entropy of the refrigerator decreases, but entropy of the room containing the refrigerator will increase. But if we cool the entire room, time also does not decrease, so we need to consider a larger system. A subsystem does not work; a large system, if expanding the thermodynamic system to the entire universe, then perhaps the increase of universe entropy will be assigned to the increase of universe time, in which, there is time of a subsystem we are living in. Thus entropy must be of the universe. The universe we mention is precisely the entire universe including visible and invisible parts. The visible universe part has size about 93 billion light-years (diameter). Some studies show that if the entire invisible universe equals Earth, then the observable universe part is smaller than a proton 100 times (see page 154). Determining universe entropy is impossible under current conditions.

Suppose we hold that the entire universe is a system to calculate entropy, then how will entropy increase in some part of the universe contribute to entropy increase of the universe? The change of universe entropy increase will certainly relate to increase of a part in it. Because the universe is very large, how will entropy increase in some part of the universe be informed to the entire universe if we suppose information transmits at light speed? Entropy increase relates to disorder increase. And disorder is determined from order, a special state for some purpose or for some observer. An observer's consciousness will determine which state is the special state called order. Other states are called disordered. If so, entropy increase is determined by subjective consciousness, not objective reality. If it

is some purpose not related to observer consciousness, then no one will raise whether that state is special or not special. If order is subjective, then a blind person not seeing chaos, do they sense time flow? Certainly yes. Therefore, we can say assigning time flow with entropy increase still has problems.

Boltzmann is the one who assigned entropy with disorder. He explained the second law: entropy increase is precisely the number of microscopic states increasing over time. The biggest objection of other scientists contemporary with him was that Boltzmann had discarded determinism in Newtonian mechanics to put forward a probability of thermodynamic systems. We say Boltzmann gave birth to statistical mechanics is at this point. Entropy increase is precisely because the probability that microscopic particles are in any state is much larger than some special state. The world was shocked when a scientist held that the previously deterministic universe was now replaced by a universe full of probability. Although probability here is probability due to lack of knowledge of information about boundary conditions, physical laws are completely deterministic.

* * *

The arrow of time can also be explained by **non-commutative geometry** theory (non-commutative geometry – NCG). This is a branch of mathematics aiming to solve geometry using non-commutative algebra. The pioneer in this field is French mathematician Alain Connes (won Fields Medal in 1982). We all know ordinary calculations have commutative property $ab = ba$; people define commutator $[a, b] = ab - ba$. Commutator equals zero for terms with commutative property and nonzero when they do not commute. In quantum mechanics, scientists must work with Hilbert space, where there are physical quantities that may not commute with each other. For example, coordinate x and momentum p of a particle $[x, p] = xp - px = i\hbar$. Order of terms is important for calculation; changing order will change results.

Non-commutative geometry works with non-commutative space. We can visualize non-commutative space consisting of two continuous space layers like two sides of a sheet of paper, and between those two layers is a discontinuous non-commutative space interval. The set of non-commutative variables creates a mathematical structure called non-commutative von Neumann algebra. In this structure appears an ordered flow called Connes flow. Connes flow closely relates to thermodynamic time flow.

Alain Connes held that behind complex calculations always hide physical realities, and physical realities are precisely mathematical gems. To study them, we need extremely powerful microscopes, that is non-commutative geometry. Non-commutative geometry is the basis to study fundamental physics like generation of Higgs particles is a result of discontinuous structure of manifolds, of non-commutative geometry of manifolds, not a particle brought from outside like the standard model of fundamental particle physics. Divergence of physical quantities in renormalization theory is also explained by non-commutative geometry, precisely the cosmic Galois group, one of fundamental problems of mathematics. Moreover, non-commutative geometry also has the ability to build the theory of everything, unifying quantum mechanics with relativity theory [75]. Non-commutative geometry also leads to Connes flow; this flow relates to thermodynamic time. Time determined by microscopic states and time determined by quantum non-commutativity are two aspects of the same phenomenon [72, page 134]. Thus, irreversibility of the second law can be the basis to study irreversibility of time.

Loop quantum gravity theory (loop quantum gravity – LQG) with basis is Wheeler – DeWitt equation does not appear time as an independent variable. This theory combines quantum mechanics with relativity theory, holding that ultimate reality is spin networks, also called spin foam with size on the order of Planck length. That is the smallest size of space; distances smaller than it have no meaning.

Thus, LQG holds that space is quantized by Planck length 1.62×10^{-35} m; those are atoms of space. Corresponding to it is the atom of time, Planck time 5.39×10^{-44} s. Nodes of spin foam are finite loops from which space, time, matter are formed. Time is the succession of physical events following each other like the block universe of relativity theory.

LQG equations are modern versions of Wheeler – DeWitt equations completely not containing time. Variables describing fields forming matter, light, electrons, other components of atoms, and gravitational field – all at the same level. Quantization of gravitational field is precisely quantization of space. Matter is not embedded in space like other theories, but these equations themselves generate space through gravity. In the macroscopic world, with approximation of human perception, we see time flows and is irreversible. According to LQC, time with properties that humans sense is only an approximation when viewed from afar without seeing quantization of quantum foam and quantum uncertainty of physical quantities. Time according to LQG is vague, lack of information about the microscopic world when standing in the macroscopic world [72].

2.3.4 Irreversibility is computational irreducibility

In 2023, Stephen Wolfram,[8] wrote a book about the second law proving that he understood this law after 50 years of thinking (when writing the book he was 62 years old, so he thought about the second law from age 12) [76]. Wolfram uses **cellular automata** (cellular automaton), a concept proposed by John von Neumann from the 1930s, to simulate natural processes. Cells interact with neighboring

[8]Stephen Wolfram (1959 – present), British – American physicist and computer scientist. He researches theoretical physics and has important research work from age 15, then switched to computer science research and is famous for creating Mathematica software. He spends most of his recent time researching cellular automata.

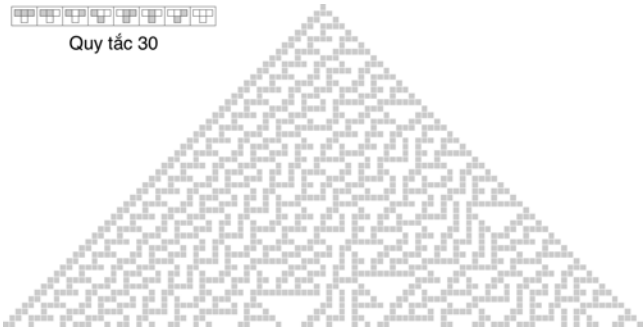

Figure 2.10: Cellular automaton according to rule 30. Living state (1) and dead state (0) are colored gray and white. From a simple initial rule, after only a finite number of development steps, system order becomes so complex that calculation cannot reverse the past (source: https://www.rule30prize.org).

cells according to different rules. A rule named number 30 can create systems so complex that initial conditions cannot be calculated from any state after that.

Wolfram put forward a one-dimensional array consisting of infinite cells (represented by a cell). A cell can be in one of two states: dead (0) or alive (1) (Figure 2.10). At any moment, the state of a cell in the future (below) is determined by the state of two neighboring cells at present. For example, rule number 30 hypothesizes that if at present two neighboring cells are alive, then at the next moment the cell being considered will die due to overpopulation. Conversely, when two neighboring cells are dead, then at the next moment that cell also dies because too empty. A cell can only live when there is a sufficient amount of surrounding cells alive [77]. Many simulations create results with order, easy to predict, but rule 30 gives very random results, and if time is sufficiently large, we cannot go back to determine the initial state. Therefore, he put forward explanations for the second law coming from limited ability of the observer (observer theory) about perception through senses and computational ability (computational irreducibility) of the brain.

Observer theory of Wolfram is as follows: human observation ability is limited by senses only allowing us to perceive and calculate

the world at human-sized space and time. Here, Wolfram views the human brain as a prediction machine, but not everything can be predicted; it can only predict within the framework of space and time suitable for human ability and experience coming from senses. We cannot perceive and calculate objects like atoms, molecules at smaller space, faster motion, so we can only perceive and calculate macroscopic objects in macroscopic time. That is computational boundedness about human observation ability.

We only have information about things and phenomena at the present moment to predict what happens in the past and future. That prediction depends on two factors: *distance* from present and *precision* desired. The farther the distance and the higher the precision, the more difficult prediction becomes, and vice versa. Finiteness of computational ability makes human prediction ability in the future finite. Classical physics shows that finiteness comes from too much information of boundary conditions, while quantum physics tells that finiteness comes from the laws themselves. Not only is the future vague, but even information of the past is not easy. The farther into the past, the more difficult determining information becomes. Wolfram's simulations show that though rule number 30 is very simple, after only a sufficiently distant time interval, we also cannot reverse to find that rule. Wolfram calls it computational irreducibility.

Three important current theories: relativity theory, quantum mechanics, statistical mechanics are derived from mutual interaction between computational irreducibility and limited ability of the observer leading to perception of continuous space and physical laws as we have today. Limits of perception and calculation lead to disorder. So, disorder is due to human ability, and that ability comes from evolutionary conditions to become us. Thus, if there is a conscious organism that can live in a tiny world, can see atoms, can sense time at very small moments, then they may not see the second law.

Computational irreducibility is the limitation of computational ability of the brain and current computers not allowing us to trace initial conditions. This limitation makes humans feel any physical system becomes more chaotic over time. Thus, there is another hypothesis to explain the second law of thermodynamics. That is computational irreducibility, the complexity of natural processes so large that initial conditions cannot be traced due to limitations of computational ability of the brain and computers.

If we glance, the second law and the arrow of time seem related, but that assignment is very vague and many fundamental questions appear. However, many scientists, including many famous scientists, believe that time increase relates to entropy increase of the universe like M. Gell-Mann, S. Hawking, L. Susskind,... We temporarily set aside problems of this assignment to see how thinking about time continues. If entropy of the universe increases, then in the past the universe must have low entropy, and it will be lowest at the moment of universe birth – the big bang. Hawking once hypothesized that we are in the expansion phase of the universe, so entropy increases. If in the contraction phase, entropy will decrease. However, later Hawking changed his hypothesis and held that whether in expansion or contraction phase, entropy increases.

2.3.5 Entropy and life

One of the most influential books about life was written by a quantum physicist in 1944, Erwin Schrödinger. The book *What is Life?* summarizes his presentations at Trinity College, Dublin [78], speaking about life from the perspective of physics and chemistry. The book presents many issues, but one important idea is the ordered state of a living body. Living bodies are thermodynamic systems exchanging with the external environment. In living systems, order level is very important, so entropy of living systems decreases. Schrödinger calls it negative entropy (negentropy). Life needs en-

ergy because entropy decrease is not a spontaneous process. The energy source for most organisms on Earth is precisely the Sun. Note, spontaneous process we want to mention here is a process according to thermodynamic statistics with high probability. Entropy decrease in a system can occur in a spontaneous process but with low probability.

Spontaneously, disorder of all systems increases over time, but in some component systems, disorder can decrease. Those are living systems. Entropy increase in large systems is always larger than entropy decrease of component systems. Because entropy decrease process is not spontaneous, for living systems to exist, two conditions are needed: (1) cannot be isolated but must be in material exchange with the environment; (2) need an energy source to be able to create order because order needs energy. On Earth, except for a few organisms using geothermal energy in the Earth's interior, most living organisms rely on energy sources coming from the Sun. Plants and some organisms can absorb light energy through photosynthesis. Animals digest plants to convert energy that plants absorb into energy for them to operate.

Not only animals, most daily activities of humans come from solar energy. Cars use gasoline, power plants run on coal, which are fossil fuels created by trees from the end of the *Carboniferous Period* more than 300 million years ago [79]. Electricity in households and factories can come from hydroelectric plants like Hoa Binh or Son La using water from rivers and streams, but water comes from evaporation from lakes and oceans thanks to solar energy. Of course, solar electricity comes directly from the Sun. Wind electricity comes from air convection in the atmosphere. That convection is obtained because the Sun shines unevenly on Earth. Except for geothermal energy and nuclear energy, energy sources supplying human activities all come from fusion reactions in the Sun's inner core.

Each time soaking in hot water at Kim Boi, let us think about the rare heat source coming from geothermal energy, not from the

Sun. That geothermal source may have nurtured life at initial times about 4 billion years ago. Additionally, humans also develop nuclear energy serving commercial purposes. France is the country consuming the most nuclear electricity in the world. Thanks to energy sources from the Sun, geothermal energy, life on Earth can develop as we see today. Without them, there would be no living systems and of course no high-intelligence organisms asking questions about where life comes from.

Life is a system that can maintain order, reduce local entropy by increasing entropy of the surrounding environment. This shows the emptiness of life: there is no clear boundary between living and non-living, between order and disorder. Life does not exist independently but completely depends on the environment. It has no fixed self-nature; properties of life only appear in relationships with the environment and with other systems.

2.3.6 Spacetime is relative

Einstein's relativity theory shows spacetime is a unified entity. Spacetime can be stretched, bent, even fragmented discretely. **All things move in spacetime at the same speed**, that is the speed of light. This is truly a strange result. Daily life all shows people walking on the road move slower than people in moving cars, so how can we say they have the same speed, and it is the speed of light? Here we do not only limit motion speed in 3-dimensional space but motion speed in 4-dimensional spacetime. If an object has high speed in space, then speed in time will decrease and vice versa.

The story of twin brothers: the younger brother standing still on Earth is stillness in space, not stillness in time. Even standing still in a house or office, the younger brother is moving in time from present to future. We are aging is an unavoidable fact. This aging is faster in people moving slowly. The older brother on a spaceship moving at speed close to light speed, i.e., moving fast in space, then

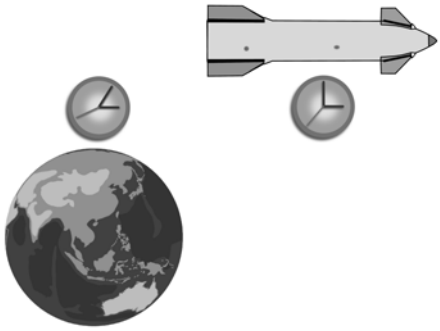

Figure 2.11: Time on Earth flows faster than time on a moving spaceship.

the older brother's speed in time will decrease. That is precisely time dilation in the moving reference frame (Figure 2.11).

In daily life, we can easily see differences in motion speed between objects, but why do we not see differences in time? A person sitting in a car sees themselves moving faster than a person standing on the road, but why do those two people not see their time flowing differently? Car motion is very small compared to light speed, so we cannot sense that difference. Not only motion deforms spacetime; gravitational force can also have similar effects. Time at the foot of a mountain flows slower than time at the mountain top because gravitational force at the foot of the mountain is stronger than gravitational force at the mountain top. But like motion of a person sitting in a car, with ordinary perception ability of senses, we cannot recognize this difference, but precise scientific equipment can prove relativity theory.

In 2010, scientists at NIST conducted an experiment measuring time difference according to speed and altitude simultaneously: speed difference is 10 m/s, and altitude difference is 1 m. Atomic clocks can measure that tiny difference based on Al^+ ions with working frequency up to petahertz (PHz) 10^{15} Hz and error only a few mHz (equivalent to error 10^{-18}). With that precision, people measured time dilation due to motion and due to Earth's gravitational force as 10^{-16} s [80]. Human senses only recognize time differences

of about a few milliseconds, so this value is too small to recognize.

2.4 Information

2.4.1 Entropy is hidden information of matter

In the previous section, we learned about the concept of thermodynamic entropy that Clausius put forward. After that, Boltzmann explained the statistical meaning of this concept. In 1938, Claude Shannon proposed information theory in which measuring the amount of information needed through a quantity called information entropy [81]. Physicists also put forward other definitions like von Neumann's quantum entanglement entropy, Bekenstein and Hawking's black hole entropy.

Stephen Hawking agreed that time asymmetry is considered linked with asymmetry property of entropy in the second law of thermodynamics [82]. Asymmetry of entropy – disorder is an **arrow** of time.

According to Leonard Susskind,[9] entropy is also understood as hidden information and proposed a law even more fundamental than the 4 known thermodynamic laws. He named it law -1, standing before even law 0, more fundamental than law 0. This law states that information of a system is always conserved. For physicists, information of an object like an atom or molecule is precisely coordinates and velocity of particles. A system with how many particles has that much information about coordinates and velocity. We can say for an ordinary physical system, the amount of information about that system is very large. For example, data on a computer hard drive is information we use in daily life. If deleting information on it, that information does not disappear but is converted into heat

[9]Leonard Susskind, American physicist considered one of the founders of string theory. He has many contributions to quantum field theory, quantum statistics, and quantum cosmology.

energy when the hard drive head acts an electromagnetic signal to make that information disappear. That heat energy is transmitted to the external environment and increases entropy. For us, information on the hard drive disappears; for physicists, information does not disappear but is converted to other entities.

The most important discovery is precisely the relationship between entropy and information. Entropy is the amount of undetermined information of a system. If entropy is low, information about the system is much, and vice versa. When we look at a classroom and see students sitting neatly in order matching student names, then information needed to know where a certain student sits in the class will be little. Conversely, when students sit randomly, we need to know more information. In 1961, physicist Rolf Landauer proposed a principle later named after him, Landauer limit: theoretically, the minimum limit E to change state of one bit of information is $E \geq kT \log 2$. This is a very small value derived from the second law of thermodynamics and recently verified by experiment [83]. Based on those relationships, many scientists put forward the idea that information is the most important entity. More important than both matter and consciousness. It is the foundation for the world to appear; that is precisely *panpsychism* theory in philosophy.

2.4.2 Information is conserved

When studying **black holes** in the universe, initially people held that black holes would not emit anything because gravitational force of the singularity at the black hole center is very large, so large that light cannot escape. If we place a gas mass near a black hole, that gas mass will fall into the black hole. Viewed from outside, we see disorder, i.e., entropy decreases. We can reason that entropy outside the black hole decreases, but entropy inside the black hole will increase, and entropy of the entire system increases. However, black holes do not let light escape, so how can we know entropy of the

black hole? What properties do black holes have that observers outside can know entropy of the black hole increases? Ordinary thinking is that when matter falls into a black hole, the volume of the black hole will increase because mass of matter is proportional to volume. Therefore, entropy of the black hole if it exists will be proportional to the volume of the black hole, more precisely proportional to the volume of the event horizon of the black hole.

In 1972, a graduate student at Princeton University, Jacob Bekenstein, put forward the hypothesis that the area of the event horizon is precisely the measure of black hole entropy. When matter falls into a black hole, the area of the event horizon increases. Entropy of the gas mass outside the black hole may decrease, but total entropy of the gas mass and black hole increases thanks to the increase of event horizon area. The noteworthy point here is that entropy is not proportional to volume but proportional to the area of the event horizon, opposite to ordinary thinking. When a black hole receives one more bit of information, the area of the event horizon increases a bit. One bit of information occupies an area nearly equal to Planck area, $10^{-70}\,\mathrm{m}^2$, a number so small it is unimaginable.

Up to here, another problem arises: if black holes have entropy, they will have temperature. Black holes have temperature, then like all other objects with temperature, they will emit energy. But by definition, black holes are objects that do not emit anything, including light. Bekenstein's proposal solves the entropy problem but creates the problem of black hole emission. This contradicts research by Bekenstein himself and his teacher John Wheeler[10] who held that "black holes have no hair." The term *no hair* (no-hair theorem) was put forward by Wheeler to say that black holes only have classical

[10]John Archibald Wheeler (1911 – 2008), American physicist, famous for research on nuclear fission reactions when he participated in the Manhattan Project. He is the one who put forward the term black hole (black hole), quantum foam (quantum foam), and wormhole (wormhole). Besides Bekenstein, he is the teacher of many talented physicists like Kip Thorne, Hugh Everett, Richard Feynman, Charles Misner,...

properties like mass, charge, and angular momentum (for rotating black holes), not depending on the nature of objects collapsing into black holes. If we have two black holes placed to compare with each other, all information about previous history to form black holes is absent. There is only information about mass, charge, and angular momentum. Therefore, a large amount of information of matter before collapsing into black holes will disappear.

So black holes will emit if they have entropy, and because they emit, they will evaporate. In the universe, all objects will evaporate; it is only a matter of how long. At least black holes will also evaporate even when they do not emit energy. Here we speak about forms of energy like photons or particles with mass. Relativity theory shows that an object with mass will emit gravitational waves. But gravitational waves carry small energy, so this evaporation process will be very long. Black holes certainly emit gravitational waves, but here black holes do not only emit gravitational waves; because they have entropy, they will emit forms of energy like particles with mass or photons.

In 1974, Hawking proposed the theory that black holes are not too black because they emit energy, which people later call **Hawking radiation**. Hawking radiation is the combination of quantum mechanics and gravitational force at the boundary region of the event horizon. This is the first theoretical calculation combining the two most important theories of modern physics: quantum mechanics and relativity theory [82].

According to the uncertainty principle of quantum mechanics, vacuum is not an empty space but always seethes with a sea of virtual particle-antiparticle pairs appearing and disappearing in a very short time. These virtual particles can violate the energy equation in short time intervals thanks to the uncertainty principle, but total energy of the particle-antiparticle pair still equals zero. Under normal conditions, a virtual particle-antiparticle pair appearing from vacuum will not exist long because they meet each other and anni-

hilate each other, returning energy borrowed from vacuum thanks to the uncertainty principle.

But in the space region near the event horizon, the gradient of spacetime (due to strong gravitational force) can separate a virtual particle-antiparticle pair before they can annihilate each other. When a virtual particle-antiparticle pair appears near the event horizon, a situation can occur: one particle falls into the black hole, while the other (its companion) escapes outside. The particle falling into the black hole can have negative energy in the black hole's reference frame, making the black hole lose mass. The particle escaping outside has positive energy and is considered by distant observers as emission from the black hole. For an observer outside, emission of these particles seems to come from the black hole surface. Over time, usually very long, black holes will gradually lose mass and finally can evaporate completely. However, Hawking radiation has not yet been directly observed in practice due to very weak intensity, especially for black holes with large mass. Thus, black holes emit energy and therefore have a certain temperature. According to the formula, black hole temperature is inversely proportional to its mass; the smaller the black hole, the larger the temperature.

A black hole with mass equal to solar mass will have temperature about 3×10^{-6} K and needs a time interval of 2.1×10^{67} years to evaporate completely. Very low temperature, near absolute zero, is almost impossible to determine because cosmic background temperature is about 2.7 K. Evaporation time is also much much longer than the age of the universe determined about 13.8 billion years. Suppose a mini black hole with mass about 100 kg has very high temperature and evaporates in a time equal to a few thousandths of a second. Black hole evaporation so fast can be equivalent to the explosion of a bomb much larger than an atomic bomb.

From here, Hawking calculated entropy of black holes more accurately than the formula Bekenstein put forward before. Today people call this formula the Bekenstein – Hawking formula. Accord-

ing to this formula, black hole entropy is inversely proportional to *Planck area*, a quantity with value equal to the square of Planck length. Thanks to this formula, we recognize the physical meaning of Planck length. That is the length to record minimum information of a fundamental particle. Black holes have entropy, so they will emit and will evaporate. If black holes evaporate, then matter previously sucked into black holes together with all their information will disappear forever. Thus, law -1 of thermodynamics is violated. In 1997, Hawking bet with Kip Thorne and John Preskill about information disappearing in black holes. Hawking and Thorne bet that information will disappear forever and law -1 of thermodynamics is violated, while Preskill bet that information is still conserved. Later research results showed Preskill won; information is always conserved; law -1 is correct.

2.4.3 Holographic principle

Holograms are a technique for photographing and creating 3-dimensional (3 dimensions: 3D) images of an object invented by Dennis Gabor in 1947. Holograms are images obtained when letting a laser beam pass through a splitter to get two coherent laser beams. Together with laser development, holographic technique is increasingly popular and widely used. To obtain holograms, object and reference beams must be coherent light[11], so only when laser technique developed did holograms receive much research.

If we look at any painting, for example *Em Thuy* by painter Tran Van Can, we see the painting only carries 2D information from the painter's viewing angle. Other information is completely absent. We cannot go behind the painting to see what the girl's hair is tied with. The painting does not carry all information of the

[11]Coherent light is light beams with the same direction, same frequency, and constant phase difference. If in phase, they reinforce each other; if out of phase, they cancel each other. Laser is light with highest coherence.

Figure 2.12: (Left) Painting *Em Thuy* by Tran Van Can painted in 1943 (reprinted with permission from the author's family). (Right) Painting *The Weeping Woman* by Picasso painted in 1937 (source: Wiki Commons).

3D entity. This is the limit of ordinary image recording methods like painting or photography. Some painting schools like Cubism[12] (cubism) try to express much 3D information on a 2D surface by bending 3D shapes to spread on a 2D plane. That expression has artistic nature, expressing subjective views, not objective information recording (Figure 2.12). For example, Picasso's painting *The Weeping Woman* tries to express many viewing angles of 3D entities on a 2D surface, which is the principle of the Cubist school: describing the right side but still seeing the eye and some information of the left side of the woman.

When looking at film obtained by ordinary photography methods, even if it can be negative, we easily notice similarity between film and object. Holographic film is not like that; holographic film is very different from the real object; it is like interference images of many wave sources when throwing a pile of stones onto water

[12]The Cubist school is one of the important painting schools of the Enlightenment period created by Pablo Picasso and Georges Braque. The painting principle of this school is not observing 3D objects from a fixed viewing angle but from many different viewing angles and expressing results of viewing objects on a 2D painting surface.

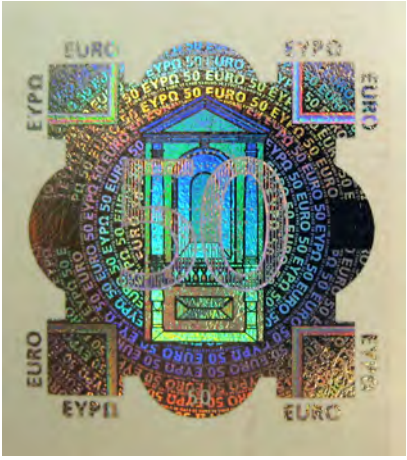

Figure 2.13: Hologram image on 50 euro coin. Source: Wiki Commons.

surface (Figure 2.13). To reconstruct the image, people shine a laser beam onto the holographic film to obtain a 3D image floating in air. Because it is a 3D image, we can walk around it and look at it from various angles like a real object. We can even stick our hand through that 3D image easily because there is really nothing there except light.

In 1994, Gerard 't Hooft proposed the **holographic principle** and proved that information when black holes evaporate is not lost [84].[13] Not long after, Leonard Susskind developed Gerard 't Hooft's idea about the holographic principle to put forward the mathematical basis of the holographic principle based on string theory [85]. The holographic principle of black holes roughly holds that 3D information of objects when passing through the event horizon of black holes will be encoded on the 2D area of the event horizon. Extending to multi-dimensional space, the holographic principle says that a physical system with $(n+1)$D can be correspondingly described by its boundary space with lower dimension nD. The area of one

[13]Gerard 't Hooft (1946 – present), Dutch physicist famous for research on quantum structure of weak interactions, for which he won the Nobel Prize in Physics in 1999 together with his advisor Martinus Geleltelt. Hooft is the first to propose the holographic principle for black holes.

bit of information on boundary space still according to Bekenstein equals Planck area.

When black holes evaporate due to emission of particles, information from that event horizon area will be released outside, not disappear together with the black hole. Because entropy, a physical quantity characterizing information falling into black holes, is proportional to the area of the event horizon, when black holes evaporate, event horizon area gradually decreases, information returns to the universe outside black holes, and the second law of thermodynamics is still obeyed. In 2004, Hawking and Thorne accepted defeat before Preskill, ending the famous bet between leading world scientists.

The holographic principle shows something strange: 3D information of objects can be encoded on a 2D surface. This suggests that 3-dimensional space we sense may not be fundamental reality but only a representation of a deeper reality in lower-dimensional space. In Buddhism, there is a similar view: the phenomenal world we sense (relative truth) is only a manifestation of a deeper reality (ultimate truth). The holographic principle may suggest that the 2D world can be more fundamental reality.

The above principle of Gerard 't Hooft and Leonard Susskind is a special case of a more general principle called anti-de Sitter/conformal field theory correspondence, abbreviated as **AdS/CFT duality** (anti-de Sitter/conformal field theory correspondence) proposed by Juan Maldacena[14] in 1997 [86], so it is also called Maldacena duality or gauge/gravity duality. AdS/CFT duality is a hypothetical relationship between two physical theories. One side is spaces named anti-de Sitter used in quantum gravity theory built on calculations from string theory or M theory. The other side is confor-

[14]Juan Maldacena (1968 – present), Argentine physicist, currently working at the Institute for Advanced Study, Princeton University, famous for proposing the AdS/CFT correspondence principle about the holographic principle. In 2020, his paper on AdS/CFT had 20,000 citations. Some people consider him the brightest physicist today.

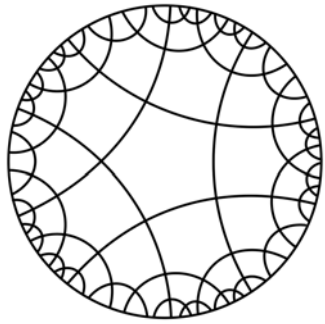

Figure 2.14: AdS/CFT duality. 2D information inside the figure is encoded on its 1D boundary around it. Redrawn according to https://ncatlab.org.

mal field theories, which are quantum field theories. AdS/CFT duality is extremely effective in high-energy physics, particle physics, and holographic cosmology. It may be the bridge between the two most important physical theories: gravity theory and quantum mechanics.

AdS/CFT duality affirms that there is a mathematical correspondence between 3D space inside black holes and a 2D surface of the event horizon (Figure 2.14). Inside black holes is anti-de Sitter space with negative curvature having gravitational force projected onto the boundary surface of black holes, which is the surface of the 2D event horizon not having gravitational force but only strong interaction of quantum fields. AdS is space with negative curvature, containing gravitational force, and has 1 more dimension than CFT. AdS needs negative curvature because only space with negative curvature has a corresponding boundary surface from which the holographic principle appears. AdS contains weak gravitational interaction, while CFT contains strong interactions, allowing scientists to study more easily. If CFT interactions are too complex, physicists can switch to studying particles in AdS or studying the event horizon surface allows knowing information in space inside black holes. Because there is duality between inside and surface, all information inside can be studied through the event horizon surface.

Suppose we have 2 space travelers A and B approaching a black hole. A is on the ship, and B will pass through the event horizon to explore inside the black hole. We assume B voluntarily does this because he knows there is no chance of survival and returning to the ship when passing through the event horizon. His body will be torn apart because the difference between gravitational force at the part near the black hole and far from the black hole is too large. This difference of the Moon and Sun on Earth causes the tidal phenomenon. The Sun has mass larger than the Moon nearly 30 million times, but distance to Earth is larger 400 times. Because *tidal force* is proportional to the cube of distance, tidal force of the Sun is only half that of the Moon. For supermassive black holes, because size is very large, A can safely pass through the event horizon of supermassive black holes. A observes B approaching the black hole by shining light on B; this is feasible when B is far from the event horizon. Suppose B still lives and passes through the event horizon; he will not feel anything different. Space is still 3D; sensation of space and time still does not change. When B passes through the event horizon, A will no longer be able to observe B because light cannot escape the gravitational force of the black hole. For A, B completely disappears; all information about B if any is only the increase of event horizon area. Because there is AdS/CFT duality, if somehow A studies this information, he can know information about B.

The strange point here is that all information about B that A wants to know is expressed on a 2D surface while B really still lives in 3D space. Like holographic film, information of objects in 3D space is encoded on 2D film, and information about these two worlds with different dimensions has 1–1 correspondence. So are we living in 2D or 3D space? To answer this question, we need to see whether the universe we are living in is a holographic universe? Research on the holographic principle of black holes can be extended to the entire universe. Black holes have boundaries and correspondence

of inside space onto black hole boundaries. Does the universe have boundaries?

2.4.4 Holographic principle applied to the universe

To this day, scientists all agree that the universe has no boundaries. The universe we observe has size about 93 billion light-years [87, page 294], equivalent to length 8.8×10^{26} m, while universe size according to estimates is 10^{51} m. If the entire universe shrinks to Earth, then the observable universe part is smaller than proton size 100 times.[15] We can see the observable universe part is much smaller than the entire universe. An obvious question appears: if the age of the universe is about 13.8 billion years, why is the observable universe part 93 billion years? That is because the universe expands with increasingly accelerating speed; the farther galaxies are, the faster they expand away from each other, so if counting both accelerating expansion speed, the size we can observe is larger than the age of the universe.

To this day, scientists also agree that the universe has no boundaries but is finite, although the degree of finiteness of the universe compared to us is truly very large. Suppose somehow we build a spaceship that can travel much faster than light speed to explore the entire universe, then if we keep going, going, at some moment we return to the starting point. Like Neo in the movie Matrix stuck between the real world and the simulated world, where only the train driver (trainman) can help him escape that deadlock. At that time, Neo tries to run after the train, but each time he runs, he returns to the starting point. The universe is the same; keep going, then return to the initial point.

Why is the observable universe much smaller than the calculated

[15] The ratio between universe size 10^{51} m and visible universe size 10^{27} m is 10^{24}. The ratio between Earth size 10^{7} m and proton size 10^{-15} m is 10^{22}.

universe? That is due to inflationary expansion of the universe right at the moment of universe birth. The *cosmic inflation* model proposed by Alexei Starobinsky, Alan Guth, and Andrei Linde[16] in the early 1980s. This model was put forward to explain the uniformity of the universe at large scales. According to this model, at the moment right after the big bang occurred, the universe expanded at very, very fast speed, much faster than light speed. In the time interval from 10^{-35} to 10^{-32} seconds, after each 10^{-34} seconds, universe size triples. In that short time interval, it is enough for universe size to increase 10^{50} times and its volume to increase 10^{150} times. After that, the universe expands at slower speed. That terrifying exponential expansion period was called the inflation period by Alan Guth, like monetary inflation in some countries with economic problems.

Because expansion speed is much faster than light speed, many universe regions are isolated from each other. This isolation not only isolates light but also isolates gravity. That means we cannot know the existence of the universe part next to us because that universe is too far that light from that universe can never reach our universe. Gravitational force also cannot affect because gravitational waves also transmit in vacuum at light speed.

In 2006, Shinsei Ryu and Tadashi Takayanagi applied AdS/CFT duality to find the relationship between **entanglement entropy** calculated from CFT with the area of the event horizon in AdS space [88]. They saw that quantum entanglement in CFT creates distance in AdS space. Two space regions closer together in AdS are created by components more entangled in CFT. Recent studies suggest that the above relationship can be applied to our universe. What keeps space in our universe from moving away from each other into separate space regions is their quantum entanglement. Quan-

[16]Three scientists Alexei Starobinsky (Russian), Alan Guth (American), and Andrei Linde (American of Russian origin) won the Kavli Prize in 2014 for pioneering in cosmic inflation theory. Western science often considers Alan Guth the father of the cosmic inflation idea. Actually, the roles of the two are equal in this theory.

tum entanglement in lower-dimensional CFT space creates space structure in our universe. Space is not fundamental but comes from quantum entanglement. But quantum entanglement does not come from this world but from another world with lower dimensions. That is how space is generated. What about time? How is time generated from string theory? Currently, physicists are still researching this issue and have no answer for it.

In 2015, Engelhardt found the relationship between entanglement entropy and geometric form of AdS spacetime: AdS spacetime **co-arises** from CFT quantum entanglement. This not only happens in black holes but also in our universe [89]. On the CFT side, we can consider quantum subsystems, which are smaller quantum systems describing fields or particles. When subsystems are not entangled with each other, there is no spacetime in AdS. Without spacetime, relativity theory is not suitable to apply. But when quantum subsystems are more entangled, entanglement entropy will correspond to appearance of spacetime pieces in AdS. Spacetime pieces are physically separated from each other, so if wanting to go from piece A to piece B, we must leave A to reach B. These discrete pieces obey relativity theory. Now if entanglement continues to increase, spacetime pieces will connect with each other to create continuous spacetime. Spacetime is the geometric representation of entanglement of quantum systems. Without quantum entanglement, there is no spacetime. Some scientists hold that space and time co-arise from those fundamental realities [90].

For example, water as a liquid has properties: can flow, form drops, form waves, freeze or evaporate. Those properties come from co-arising of many water molecules. Individual water molecules H_2O cannot create the above properties. The liquid property of water is completely different from properties of individual molecules. Similarly, macroscopic space does not necessarily have to be formed from space atoms with the same properties. Atoms of space are not the smallest elements of space. More precisely, atoms of space are con-

stituents of space. Geometric properties of space are new properties, collective in nature, and an approximation of systems consisting of many space atoms [91].

Thus spacetime is not fundamental reality; it is a consequence of deeper things: quantum entanglement. If spacetime co-arises from the degree and nature of quantum entanglement from a lower-dimensional space, then that means that lower-dimensional world is more real than the 3D world. Thus we can see, quantum mechanics, relativity theory, string theory bring space, time, matter a close relationship with each other. No entity can exist independently separately. They are as we see daily due to cause and effect, due to combination of components with each other. That is precisely emptiness of space and time.

The view that spacetime co-arises from quantum entanglement has a profound meaning about emptiness. If spacetime is not fundamental reality but only appearance from quantum entanglement, then this means spacetime has no independent self-nature. This aligns with Buddhism's view of dependent origination: everything appears due to conditions, due to causes. Spacetime is not an absolute entity but only appearance from specific conditions. Spacetime we sense is only relative truth, while fundamental reality is quantum interactions in lower-dimensional space.

2.4.5 Rational and sensory perception

Scientists use mathematics, a product of reason, as an effective tool to solve their problems. Although some scientists like Stephen Wolfram try to abandon mathematics to build a new science not based on mathematics but based on rules for *cellular automata* (cellular automaton) [92], most scientific work today is based on mathematical axioms and results. Before solving problems, they build models, then apply mathematical tools to get results. But mathematics is not an omnipotent tool; it has limits that cannot be overcome. The

one who proved incompleteness of mathematics is mathematician Kurt Gödel put forward in 1931, when he was 25 years old.

Incompleteness theorem not only applies to mathematics but also to logic and sciences based on logic. Gödel proved in a complex and rigorous thesis; we do not go into details but only present a summary of consequences of the incompleteness theorem. The theorem speaks of limits of all systems manifested in 2 theses: (1) there is no absolutely perfect mathematical theory both complete and consistent, i.e., there always exist propositions that cannot be proven, cannot be negated; (2) cannot prove an axiom system using only that axiom system, i.e., we need another axiom system to prove correctness of the axiom system being considered. Gödel himself commented on his theorem as follows: "in every arithmetic system built rigorously from an axiom, there exists at least one proposition that cannot be proven, also cannot be negated" [93].

The incompleteness theorem is very important in science and mathematics. Einstein and Gödel both worked at the Institute for Advanced Study of Princeton University, and though much older than Gödel, Einstein considered Gödel equal to himself. He even said he came to the office only to have the privilege of walking home with Gödel. Gödel was the first to receive the Albert Einstein Award for outstanding theoretical physicists in 1951. To the outside world, the incompleteness theorem as well as Gödel's name are not famous, perhaps because the incompleteness theorem lies at deep levels of mathematics and logic that mathematical tools used normally do not yet need. That work shows that wanting to explain everything by reason is impossible and that relying only on reason cannot be complete. This is also an issue where Buddhism can accompany science, can use personal experiences of meditation processes to help science understand truth.

The incompleteness theorem speaks of limits of logic and reason. In his famous book, Japanese mathematician Fujiwara Masahiko held that problems of the world, especially of humanity, cannot be

solved only by reason but also need emotion and form [94]. Reason and logic have brought success in history, and they become universal values worldwide. But are reason and logic enough to solve problems of humanity? What defects do reason and logic have? Can emotion and form supplement those defects?

Reason itself has limits because the dependent origination principle leads to mutual dependence of too many factors in modern society, so reason cannot know all influencing factors. Using reason when understanding is not complete can bring unexpected results and cannot be reversed. Neoliberalism that we discuss on page 282 is an arrogance of reason and logic, so it needs to be balanced again by emotion and form. Emotion is subjective, sensory, often looked down upon when compared with reason, but it is an important supplement for reason. Japanese emotion is emotion with nature, with ordinary things happening around like gardens, trees, even drinking tea, appreciating flowers, reading books. Japanese people also value emotion with the transience of human life and quickly fading beauty. That emotion is described by a word: *mononoaware*.

Vietnamese people are the same. Using reason but always accompanied by emotion: "One hundred reasons do not equal a bit of emotion." Only a little emotion is enough to understand each other, to sympathize with each other, will help very much in solving complex problems of life. Zen master Huyen Quang, the 3rd patriarch of Truc Lam Yen Tu Zen school, describes sensing beauty even in insect sounds as another example of emotion about beauty in the most ordinary things in the poem *Mountain Rain* (translation by Nguyen Lang) as follows:[17]

> "Autumn wind at evening blows western veranda,
> Mountain house desolate leaning on green leaves,

[17] Sino-Vietnamese transcription: Son vu. "Thu phong ngo da phat thien nha, Son vu tieu nhien cham luc la, Di lu thanh thien tam nhat phien, Cung thanh tuc tuc vi thuy da."

Devotion from those early times,
Crickets chirping sadly, for whom?"

Huyen Quang's ideas are the middle way lifestyle, combining reason and emotion, content and form. Clearly recognizing the dependent origination principle about mutual dependence of things to see emptiness everywhere to be able to behave appropriately, humbly with nature.

Saying so is not to deny the role of mathematics, logic with science, but we need a thorough view of limits of mathematics in science. Most scientific fields need mathematics, and mathematics is still an effective tool to help scientists think and solve difficult problems. The incompleteness theorem also does not negate achievements of mathematics; it shows that a scientific theory questioned to the end will have a proposition that we must accept cannot be proven true or false. Like René Descartes doubting everything but still had to leave one thing: we are doubting. To this day, mathematics is still an effective tool to solve scientific problems. Is reason, through scientific theories and mathematics, truly the way to approach truth? And reason comes from consciousness, a product of the brain. Emptiness of consciousness will be presented in the *Consciousness* Chapter, showing emptiness manifests in all worlds from matter to spirit.

* * *

This chapter shows that matter (at the most fundamental level), space and time have two properties: compositional nature and impermanence. Compositional nature of fundamental particles manifests through continuous conversion between fields and particles; impermanence can be clearly seen through ceaseless interaction. Particles with some properties attached to particles are an approximation when standing far, low energy, long time. Under other observation conditions, particles are no longer independent, immutable,

self-existent, completely different from particles according to ordinary views. Compositional nature shows time is assigned to entropy and information. And entropy and information are linked with matter and motion of matter in space. Moreover, space and time also stretch, bend depending on motion and matter lying in them. Precision and position of the observer also determine properties of space and time. Impermanence manifests through change of matter and observer lying in space and time. Because of ceaseless change of objects inside, space and time themselves transform accordingly. Therefore, emptiness appears not only in matter but also in space and time themselves.

Chapter 3

Life

Summary. Emptiness manifests in living organisms: (1) Life is self-sustaining chemical systems that change according to Darwin's theory of evolution. Gene segments are self-replicating entities, evolutionary subjects; they rely on living machines, individuals, to exist and develop. (2) Evolution theory proves that all organisms are related to each other through cooperation and competition from levels like molecules, genes, proteins to individual and species levels. (3) The origin of life shows important relationships about the birth of elements in stellar cores, asymmetry of fundamental forces of physics, movement of Earth's core layers all can affect life as we see.

3.1 Definition and origin of life

3.1.1 Definition of life

Most people can recognize something as living or non-living, but to give a precise definition of what life is is not easy. Scientists need to define life because that is necessary and affects their work. For example, if the definition of life relates to carbon, then searching for extraterrestrial life will exclude planets not containing this element. Not only biologists but scientists in other fields and philosophers all try to state a definition of life most suitable for their work. Life we

mention in this book is life on Earth, related to human existence.

The most common way to define life is listing elements necessary for life. For example, (1) life is a highly organized ordered system maintaining low entropy, (2) formed from organic molecules based on carbon chemistry, (3) pre-programmed in structure and function in genetic material, (4) can interact and adapt to external environment, (5) has ability to reproduce and evolve [95]. With such a definition, robots or organic robots (for example xenobots) [96] are not considered living objects because they do not have the ability to self-evolve but must rely on human intervention. Viruses are not considered organisms because they do not exchange matter with the environment. Similarly, mules (offspring of horses and donkeys) cannot reproduce subsequent generations. Edward Trifonov analyzed language of 123 definitions to find basic similarities between them: life is (self-)reproduction and evolution (variations) [97].

For the purpose of this book, we use the definition of life by NASA following Carl Sagan's suggestion [98]:

> **Life is a self-sustaining chemical system capable of Darwinian evolution** ([life is a] self-sustaining chemical system capable of Darwinian evolution).

This is a concise, memorable definition good enough for us to discuss related issues. This definition includes three main characteristics:

1. Can maintain itself through metabolism with the environment
2. Has ability to reproduce to create subsequent generations.
3. Can change and evolve over time through natural selection mechanism (Darwinian evolution).

To perform the above processes, living systems must have ability to store information and transmit information to each other and to future generations. Information relates to entropy, so studying life cannot be separated from entropy and information. Another definition removes the evolution aspect but adds the thermodynamic

aspect when considering living systems as systems far from equilibrium (far from equilibrium system), where information flow exchanged between organisms and environment is the object of study. This chemical system has self-catalytic ability from which genetic ability through genes is created [99].

Some people hold that the above definition still lacks some important properties [95], but we still use it in this book. NASA uses this definition to study life on Earth and in the universe. For example, in the 1990s, some scientists held that meteorite Allan Hills 84001, a fragment of Mars, falling in Antarctica contains fossils of ancient microorganisms of Mars. The basis of the proposal is structures with repetitive nature on the surface of this meteorite. However, many scientists refuted that because they held that those bases are vague as they contradict existing definitions of life. Some philosophers like Carol Cleland hold that trying to put forward a definition of life can be counterproductive to understanding what it is because if we want to search for life outside Earth based on life on Earth, that would be unreasonable [100].

Such properties will never be exhaustively listed because human understanding of life will expand in the future. We only need some minimum properties that can still determine life; that is good enough to use it for issues this book will discuss. For example, NASA's definition says nothing about negative entropy (will be discussed below) or DNA genetic information system. That does not make the definition need to change because from the contents of the definition, entropy and DNA can naturally follow.

Buddhism does not have a definition of life in the biological sense above, but in Abhidhamma there are related concepts like *bhava* (bhava), a concept describing the nature of existence, arising, and continuation of sentient beings [101]. Another concept is *vitality* jivitindriya (life faculty, vitality): jivita means life, life force; indriya is maintenance and ability. Vitality consists of two parts: material

and mental, tightly interwoven to create one of seven mental factors. An important characteristic that Buddhist scriptures mention about life or vitality is the unity of body and mind. Due to continuous interaction, life is not a state but a process. Bhava in Theravada Buddhism is *becoming*, the process of mind and body interacting, due to intention and conditions.

NASA's definition states that self-sustaining chemical systems can determine the self-nature of a system with a definite boundary in space at a definite moment in time. The boundary is not closed but has structure so the system can exchange matter with the environment. Not only that, this system is not immutable because it has the ability to change to adapt according to Darwin's theory of evolution, consistent with two concepts related to life and vitality in Buddhism. NASA's definition and many other definitions by biologists do not include psychological or consciousness elements in the definition of life due to fragmentation of current science. We will mention consciousness in the final chapter of this book. Therefore, in this book, I use NASA's definition.

After understanding the definition of life, a natural question arises: where does life come from? This is one of the greatest questions of science. Is life created by a supernatural force, or can it arise naturally from inorganic matter? Answering this question is not only important for understanding our origin but also clarifies emptiness: life has no fixed self-nature but is only a stage in the evolution process of matter.

3.1.2 Life arises from inorganic matter

Can life arise from the abiotic world? And if possible, how does that process occur? When someone says life forms randomly from the abiotic world, immediately that person is viewed with suspicion because most of us feel there is a deep chasm separating the abiotic world of earth, rocks, oceans from the smallest living organism like

worms, crickets. Human understanding of life is still finite; there are still too many things we do not know, but if we only look at the development of life science in 165 years, from when Darwin wrote *The Origin of Species* to now, humans have traveled a very long path, and we can trust that we can understand the nature and formation of life.

Some scientists hold that life comes from outside Earth, also called **panspermia theory**. This hypothesis is also only temporary because we push the answer to somewhere not Earth – our home. What is that outside Earth? Perhaps another planet where life already exists. Did that life already exist there, or did life also begin from the abiotic world? Due to the universality of the abiotic world compared to the biological world, we must necessarily hold that if life has an origin, it must be from the abiotic world. Even humans have conducted a very large project building Arecibo Observatory in Puerto Rico to hunt for signals of extraterrestrial life from 1963. This is a giant device weighing 900 tons, with radius 305 m, assembled from nearly 39,000 aluminum panels with size 1×2 m^2. After 57 years of operation, on December 1, 2020, this observatory had an accident and collapsed [102]. Throughout its operating life, Arecibo Observatory had many contributions to astronomy but could not detect any sign that seems to be of intelligent organisms outside Earth. This does not affirm that extraterrestrial life does not exist but can only say that if it exists, it is very rare, or if life is common, then intelligent life that can actively communicate with people on Earth is even rarer.

The rarity of life in the universe is often linked with *Fermi paradox*. It is called a paradox because the vastness, immensity of the universe seems to support high-intelligence life capable of informing other civilizations through radio signals, with lack of results in searching for those signals. With such a vast universe, receiving signals from aliens should have appeared long ago. After manufacturing atomic bombs, scientists of Los Alamos Laboratory led by En-

Figure 3.1: Biological evolution on Earth. History of about 4.5 billion years compressed into one 24-hour day, each hour corresponding to 190 million years. Life begins at 4 AM; humans appear in the final seconds of a day.

rico Fermi together with Edward Teller and Herbert York discussed unidentified flying objects, UFO. One of their hypotheses was that UFOs could come from extraterrestrial civilizations. Fermi, famous for quick estimation ability, held that with data about the number of galaxies in the universe and the number of planets that could have life, Earth must have been visited long ago and many times [103]. However, to this day we have not received any signs coming from extraterrestrial civilizations.

Different from panspermia theory, **abiogenesis theory** searches for the origin of life from inorganic matter on Earth. Earth was formed about 4.5 billion years ago. In the first few hundred million years, Earth's surface temperature was too hot; water could only be in vapor form. After that, Earth's surface gradually cooled to be an environment where life could arise. Some later experiments simulating conditions of Earth in that primitive period showed the possibility of existence of hydrogen cyanide (HCN), water (H_2O), amino acids, and nucleotide bases. Fossils of the most ancient life are about 3.5 billion years ago [104], showing existence of prokaryotic

single-celled organisms. This suggests life could have been formed earlier than that time, about 4 billion years ago (Figure 3.1). 2 – 1.8 billion years ago, Earth appeared eukaryotic organisms with nuclear membranes to separate nucleus from other intracellular organelles [105]. Complexity of eukaryotic organisms increased, making organisms use resources more efficiently.

Single-celled organisms always evolve and diversify, but macroscopic morphological changes had nothing special until about 650 million years ago when the first multicellular organisms formed. About 400 million years ago (Devonian period), some animals moved from underwater to land [106]; 230 million years ago (Triassic period) dinosaurs appeared; 65 million years ago dinosaurs went extinct. Genus Homo appeared on Earth 2 million years ago, and humans Homo sapiens only appeared about more than 300 thousand years ago.

To this day, scientists have found, classified, and named about 1.8 million different species. People predict the number of species could be several times that number waiting to be discovered and explored. Studying species is not only to satisfy human curiosity but helps us understand, preserve, and protect *biodiversity* of ecosystems where humans are one species. The number of humans increased very quickly in a short time, seriously affecting ecosystems on Earth. In the past 50 years, human activities have lost 20% of humus layer on Earth's surface, 30% of forest area, and made about 90% of large ocean fish species disappear [107, Part 1, page 11].

There may be many different forms of life somewhere in the universe, but on Earth, all forms of life only rely on one cycle: existence, development, and reproduction. This life cycle has been present on Earth for a long time, about 3.5 billion years. All activities of organisms follow laws of physics and chemistry. Though seemingly complex like eating, working, thinking, we all follow chemical reactions. Of course, when speaking of chemistry, we need to speak of elements. Let us see what elements the most basic compounds of

life are formed from.

3.1.3 Origin of elements

Elements can reveal clues about the life formation process. We know that the periodic table of chemical elements put forward by Mendeleev in 1869 only had about 60 elements. To this day, scientists have found 118 elements, but only 94 elements exist in nature. The rest are artificial elements, unstable, only appearing for a very short time in laboratories. Although the number of elements is so many, the number of most common elements in organisms on Earth is very few; only 6 elements must be present in all living organisms. That is CHNOPS.

The first element is O, accounting for about 65% of human body mass, mostly existing in water form as a solvent for the body to operate. Oxygen is also the most abundant element in Earth's crust and accounts for about 20% of air composition. Next is C, the most important element of the organic world because of C's ability to form bonds with other elements like H, O, N, and with itself. C can form chains of atoms for organisms to use like proteins, genetic information. C accounts for about 20% in organism bodies. Total of 6 elements necessary for life account for 98% of human body mass [108].

H is the simplest element consisting of only one proton at the center and one electron around it. This is also the most common and oldest element in the universe. H can form covalent bonds with many elements in chemical reactions like photosynthesis and metabolism. Especially H reacts with O to create water, the most important compound of life. H accounts for about 10% of organism mass.

N is the most abundant element in the atmosphere, accounting for about 80%, but the triple bond of nitrogen molecules N≡N is very strong, so special techniques are needed to fix nitrogen in air. N creates amino acids, proteins, and is an indispensable component

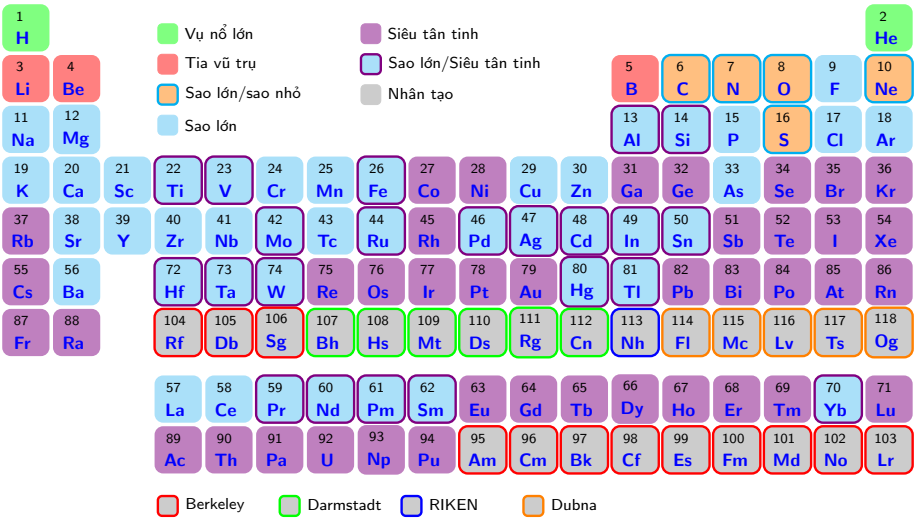

Figure 3.2: Origin of elements on Earth: 94 elements existing in nature come from the big bang and stars. Elements from 95 onward are artificial elements [56].

in DNA. P is used to create phospholipids of cell membranes. Cell membranes are formed from phospholipids strong enough for cells not to break but also flexible enough for cells to exchange matter with the external environment. P is also present in ATP, the material for cells to operate. Through ATP, cells can use energy stored in organic compounds. P is present in phosphate molecules together with sugar groups to create the backbone of two DNA strands. Thus, 5 elements forming DNA are O, C, H, N, and P. S is present in two important amino acids: cysteine and methionine, helping create proteins to stabilize structure and self-healing ability in body parts.

Figure 3.2 shows the periodic table and origin of elements. Elements common in organisms are all elements tending to form covalent bonds, located on the right side of the periodic table. Except H is an element though placed in group 1 but is a non-metal element. H at the first position is the most common element in the universe. Elements C, N, O, and S all form in large or small stars. Only P forms in large stars. H, C, N, O are light elements present

in living organisms because they are very common in the universe and on Earth. However, the presence of P and S, elements much less common, is an issue that needs to be raised and may suggest the origin of life. Especially P, compounds of P are usually very difficult to dissolve in water. And water is the solvent for most organisms to operate metabolism. Moreover, P is present in the most basic component of life, DNA, and ATP material for cells to operate. The presence of P in the most important places of life can tell us that life could have originated where there is much P.

Life comes from dust of dead stars. That is precisely what science has discovered in about the past 100 years. The first person to hold that the origin of elements comes from stars was Arthur Eddington put forward in the 1920s [109]. Except H and He formed after the big bang, subsequent elements up to Fe in the periodic table come from furnaces in stellar cores. Because nuclei carry positive charge, they will repel each other according to Coulomb's electrostatic law. Fusion energy of stars like the Sun helps nuclei have sufficient kinetic energy to overcome electrostatic repulsion between them. When close together to a distance about 10^{-15} m, strong nuclear force will cause fusion. This nuclear fusion reaction creates a large and stable amount of energy sufficient to help the Sun exist for several more billion years. The fusion process from light elements like H and He does not continue forever but only stops at Fe because binding energy of this element is largest. Therefore, iron is a quite common element in the universe.

Elements heavier than iron are rare because they are only created in a very short moment before the star dies and explodes in what is called a supernova explosion. In the short time of the explosion, energy is so large that nuclei of elements heavier than iron continue to fuse to create heavy and unstable elements. Elements standing behind Fe in the Periodic Table, i.e., elements with atomic number larger than 26 like gold, platinum, natural radioactive substances, only form in that short moment, so they are very rare.

After supernova explosions, elements are shot into space to form next-generation stars like the Sun and Earth, the basis for life to be born [109]. Recent studies on neutron star collisions tell that many heavy elements can also be formed from these collisions [110].

Life prioritizes using available things, so light elements usually appear in organisms. However, Chapter 4 we will see a microorganism named cyanobacteria knows how to use compounds between Fe and Mo (standing 42nd) to catalyze fixing nitrogen in air. This is one of the very special cases of organisms when using elements heavier than iron in biological activities. We clearly see the relationship between life including us with stars in distant universe, with explosions of supernovae. It seems that vast space creates separation that cannot affect each other. Yet they determine human existence in every moment. Compositional nature of life manifests here. Life is created by atoms of elements. Atoms are born in furnaces of stars. Some organisms also use heavy elements only born in supernova explosions.

Water is a compound found very much in the solar system. When Earth first formed, not only because temperature was too high but there was no water in liquid form; Earth at that time also had very little water. After the Theia collision[1] about 4.5 billion years ago, Earth gradually cooled. Water vapor condensed into liquid water. To have a large amount of water covering about 71% of Earth's surface area, a hypothesis accepted by many people is that water came from meteorites, asteroids, and comets[2] in ice water form about 3.8 billion years ago. However, there is also a hypothesis that water was present

[1]Hypothesis about planet Theia with size of Mars colliding with Earth to create the Moon [111].

[2]Meteorites, asteroids, and comets are objects existing since the formation of the solar system. These objects did not participate in planet formation, so they wander in space between planets. Meteorites and asteroids lie near the Sun, so main components are earth and rocks. Comets lie far from the Sun, so besides earth and rocks, they also have other components like water ice, frozen gases.

on Earth right from formation in the form of hydrated minerals. H has three isotopes: ^{1}H (denoted H, protium), ^{2}H (denoted D, deuterium), and ^{3}H (denoted T, tritium). Protium is the main isotope of hydrogen; people study the D/H ratio to determine the origin of hydrogen. The D/H ratio near the Sun is smaller than the D/H ratio far from the Sun; therefore, comparing heavy water isotope D_2O with H_2O can tell the origin of water on Earth. In 2014, Rosetta probe carrying Philae lander[3] landed on comet 67P/Churyumov-Gerasimenko to take soil and rock samples and analyze hydrogen isotopes, giving results that most water on Earth comes from meteorites and asteroids. Only about 10% of water comes from comets far from the Sun; the rest comes from meteorites near the Sun. The condensation process for water in liquid form on Earth occurred about 3.8 billion years ago [112].

Compositional nature shows that life of organisms is formed from elements. Elements are obtained by fusion reactions in stellar cores. Supernova explosions shot these elements into the universe to create our Earth. Due to special conditions, only Earth has life formed. Emptiness of Buddhism is again manifested in the most unexpected places.

3.1.4 Life: entropy and information

One of the founders of quantum mechanics, Schrödinger, wrote a famous book "What is Life" that has very great influence on scientists studying the origin of life [78]. Two important contents were presented: negative entropy (later French physicist Leon Brillouin shortened to negentropy [113]) and genetic information. Genetic information will be presented in later sections. This section focuses on the entropy concept related to biological systems.

[3]Rosetta probe was launched by the European Space Agency in 2004 carrying Philae landing module. After 10 years, the probe approached comet 67P/Churyumov-Gerasimenko to study the origin of the solar system.

We know that entropy is a physical quantity related to disorder of a system; conversely, negative entropy will relate to order. High order of living systems manifests by the order of atoms forming DNA, of proteins, and of biological and chemical processes for a cell to operate so complexly. Any disturbance or disorder can lead to serious mistakes that can be fatal. For example, only changing one letter in DNA of gene HBB, where –GAG– is changed to –GTG–, leads to sickle cell disease [114]. Sickle cells do not carry oxygen, easily deform, causing blood vessel blockage, chronic pain leading to early death. Only one letter wrong in DNA can cost life. That shows how strictly the body must follow order.

Page 121 presents anisotropy and irreversibility of time and assigns it to anisotropy and irreversibility of entropy. Spontaneous processes all tend toward entropy increase and locally will progress to equilibrium state. Biological systems have definite structures necessary for existence. That is precisely order of microscopic configurations because components forming biological systems must have. Any disorder affects system operation; some effects are large enough to affect existence. Thus life is not a spontaneous process. If so, it is a process requiring energy from outside, so life cannot be a closed system. To maintain low entropy inside, energy must be supplied to the system. Entropy decrease inside biological systems does not compensate for entropy increase outside, so the second law of thermodynamics is always satisfied.

When living systems need order, then anisotropy appears. Especially with living systems with consciousness, especially high-level consciousness like in humans. Anisotropy is not only sensory, vague, but anisotropy is elevated to scientific perception. High-order states that life exists and develops based on make appear states not like it, called disordered states. That is the birth of order – disorder anisotropy, from which is the basis for other anisotropic pairs, first of which is precisely the second law of thermodynamics. We can say life is entropy decrease, opposite to ordinary processes which

are entropy increase processes. Living systems need energy through metabolism with the external environment.

The second law of thermodynamics says that spontaneous processes of a closed system have entropy increasing over time. So does entropy decrease in living systems violate the second law? The second law is considered one of the most fundamental and solid laws of modern physics. At the macroscopic level of systems we can observe with the naked eye, there has never been any experimental evidence showing the second law is violated. A closed system is a system not exchanging energy or matter with the external environment. But living systems (cells, organism bodies, ecosystems) are all open systems. They absorb ordered energy (low entropy) from the environment (sunlight, food, chemical gradients,...) and release energy in heat form or substances with high entropy through metabolism. Thanks to that, living systems maintain a high order level inside, i.e., internal entropy decreases. Energy exchanged to the environment increases entropy outside more than the decrease inside. Therefore, the second law is still conserved at the global level.

The second aspect that Schrödinger mentioned is information: **life conserves and propagates information through generations**. This is similar to John Wheeler's hypothesis about information "it from bit" [115] when he held that physical systems have information as their nature. In organisms, information is encoded by gene segments in DNA *transcribed* into messenger RNA (messenger RNA – mRNA) in the cell nucleus, then transmitted outside the nucleus to synthesize proteins through the *translation* process at ribosomes. The one-way information process DNA $\rightarrow$ RNA $\rightarrow$ protein found by Francis Crick in 1957 is called the *central dogma* of molecular biology [116]. The reverse process does not occur, i.e., information from proteins cannot return to DNA. Later, people saw some special cases where some enzymes allow information from RNA to return to DNA.

Information stored in DNA, neatly contained in the nucleus, con-

tains up to 3.1 billion ATGC characters, encoding about 22 thousand genes. That is a huge and important book, so it cannot be used arbitrarily. When cells need a gene segment, it will be brought outside the nucleus for other organelles to perform. This is similar to copying a few pages from the book to give to people outside to execute commands written on those pages. DNA of a gene is transcribed into shorter mRNA segments and brought outside the nucleus to a specialized complex called ribosome. There, every three nucleotides, called codon, encode one amino acid. To translate, another type of RNA called *transfer RNA* (transfer RNA – tRNA) is used to assemble proteins. Each assembly, three nucleotides of a codon read from mRNA will produce one amino acid in the protein. For example, codon CGU produces amino acid Arg (arginine), CUU produces Leu (leucine),... Amino acids form proteins to create cell framework, transport oxygen and nutrients, protect the body, provide energy for the body,... All activities of life at cellular scale and thereby bringing macroscopic properties are all directed from information encoded in genes. Thus genes have two main tasks: replicating themselves and creating proteins. The replication process only occurs when cells divide, while the protein creation process in cells is a regular task.

Through examining entropy and information in life, we have seen that life is a complex system, always exchanging energy and information with the external environment. But how can complex molecules like DNA and RNA form from simple compounds in Earth's primitive environment? We will examine the "primordial soup" hypothesis – a special chemical environment where the first chemical reactions creating life could occur.

3.1.5 Primordial soup

The first chemical reactions creating life on Earth. To be able to study the origin of life, people must examine the composition of

Earth's surface from when life originated. According to many studies, about 3.5 billion years ago, Earth's surface already had oceans; the atmosphere did not have O but had hydrogen cyanide (HCN), carbon dioxide (CO_2), methane (CH_4), ammonia (NH_3), together with volcanic activity, lightning creating conditions for chemicals to combine with each other to make what is called **primordial soup** (prebiotic soup). From this primordial soup, life formed.

Scientists have simulated Earth's conditions in that period to study how compounds necessary for life formation occurred. The famous experiment by Stanley Miller and Harold Urey conducted in 1953 simulated some conditions similar to ancient Earth, telling some important components of primordial soup. They had electric sparks shine into a solution containing some initial chemicals as mentioned above to obtain some organic compounds with more complex structures like glycine, alanine, aspartic acid, and aminobutyric acid [117]. Later, scientists continued studying the soup from the Miller–Urey experiment and found more than 20 types of amino acids, more than the initially published number.

But how do amino acids link together to become larger molecules like peptides? Perhaps besides chemicals, microscopic surfaces of earth and rocks can play an important role. In 1949, John Bernal hypothesized that clay surfaces have roles as catalysts to create peptides, even DNA. Clay, for example kaolinite [kaolin], has very important characteristics: because it has layered structure, surface area to absorb large molecules is large, can be templates to synthesize macromolecules, protect molecules from sunlight (especially ultraviolet rays) [118]. Later computer simulations also confirmed Bernal's above proposal.

Under the effect of an energy source that could come from geothermal energy or sunlight, after a long time, a molecule with self-replication ability appears randomly, a *self-replicator* we mentioned above [119, page 51]. The probability to randomly form a molecule with ability to create copies of itself from initial primitive molecules

is certainly very small. However, if we wait long enough, we can observe some events. Time from when Earth formed until the first life appeared is about 1 to 1.5 billion years (Figure 3.1), longer than human life, even longer than the time that intelligent humans *homo sapiens* have been present on Earth (about 200 thousand years).

First chemical of life: RNA world. Today we think all organisms are very complex, even the simplest organisms like bacteria, but life could have begun from simpler things. Russian biochemist Alexander Oparin held that life on Earth originated from simple organic compounds. Under the effect of energy sources from lightning, volcanoes, sunlight,... those organic compounds synthesized into molecules necessary for life. Many scientists hold that initial self-replicators were only molecules, and that was RNA molecules. The theory that initial self-replicators were RNA is called **RNA world theory** put forward in the 1960s [120]. After that, RNA became a template to synthesize DNA. The model of that theory is called the monomer model, where RNA formed first, DNA formed later. Conversely, the chimeric model holds that initial self-replicators did not only consist of RNA but were mixtures of DNA and RNA monomers together creating life [121]. Recently, in 2021, scientists in Germany used transfer RNA (tRNA) as a building block to create life. They showed that tRNAs can self-assemble with each other to create more complex systems, and in turn, these systems have replication ability [122].

One billion years is a very large time interval for some random chemical reaction to accidentally form an RNA or RNA – DNA molecule. This molecule differs from other chemicals in only one characteristic: they have the ability to create copies of themselves. This is precisely the key point of the problem. It seems a bit forced to say that randomly an abiotic chaotic world of primordial soup appears a self-replicator. Self-replication ability requires that molecule to have much more order than other molecules.

A cup is too difficult, but a smaller object like a molecule, the

probability to create an ordered molecule will be larger, so the possibility of observing that can be shorter. Returning to forming a molecule that can self-replicate itself requires the order level of that molecule to be higher than other molecules. So how can a chaotic world form some molecules with a certain order level to thereby have self-replication ability? According to the second law of thermodynamics, entropy – disorder of the entire system always increases is an averaging process over time. If examining some subsystem (here a molecule with self-replication ability) for sufficiently long time, then entropy of that subsystem can decrease (although entropy of the entire system always increases). Entropy decrease of a subsystem does not violate the second law. Forming a molecule, or a structure with high order to be able to self-replicate them is possible in sufficiently long time. An energy source from outside will help the local entropy decrease process easier. Like a refrigerator needs an external power source, primordial soup supplied energy from the Sun or from geothermal sources in Earth's interior will create favorable conditions to form molecules with self-replication ability.

For a molecule with self-replication ability to form is already rare; how can we have countless self-replicators we see today? Creating a self-replicator is only difficult initially, but afterward it is much easier. Suppose at some moment a first self-replicator appears, then from one initial self-replicator will appear 2, 4, 6,... versions of it. If this happens, it will be very fast; self-replicators will self-replicate to be present everywhere in primordial soup. Perhaps initially self-replicators were short molecules created by inorganic compounds like clay or possibly from organic compounds [123]. But to this day, DNA's dominance shows that if self-replicators formed from inorganic compounds existed, they must have given way to organic compounds. Initially self-replicators had short molecular lengths, but if between their ends chemical reactions can occur with each other, then self-replicators can be very long molecules like DNA molecules we see today.

3.1.6 Biological self-replicators

The purpose of life is an issue not only limited to the scientific domain but also related to philosophy and religion, a topic human generations have searched for a long time. Especially religions always hold that the purpose of life lies in supernatural forces. We only stop at discussing the purpose of life within the limits of natural science. At the molecular scale, life has a very special purpose. That is the process of transmitting information (of DNA, RNA) from one material entity to another material entity, also called a **self-replicator**, more precisely a *biological self-replicator* to distinguish from some other self-replicators we will discuss below. Under this view, viruses are also considered to have life, to be organisms. Everyone can easily notice the concept of self-replicator is quite familiar from the proposal of Oxford University biology professor Richard Dawkins.

Biological self-replicators can also be extended in the future to computer software – machines that can perceive or characteristics of a culture. There are many similarities between genes of life and genes of culture that Dawkins calls **memes** [119, page 284] – *cultural self-replicators*. Artificial intelligence, computer programs with the purpose of spreading information they want, are also a type of self-replicator. Different from organisms, where transmitted information is encoded in DNA, information in computer programs is encoded by command lines; we call them *digital self-replicators*.

These self-replicators all have the common characteristic of being self-centered; they act unconsciously for themselves, completely without plans, without long-term vision. Note that *self-centered* is a concept we introduce to personify self-replicators; self-replicators do not have consciousness in general and completely do not have consciousness about the self-centeredness we are mentioning. Similar to sociality of viruses mentioned in the section above, defective viruses seem to have no plans and long-term vision, but natural se-

lection seems to have provided viruses with perception to be able to deceive and exploit normal viruses. Viewed from afar, viruses are self-centered. Similar reasoning can be applied to self-replicators. Self-replicators need tools to achieve replication goals. For biological self-replicators, those are **survival machines**, i.e., biological bodies from bacteria, plants to humans are survival machines, tools for genes to use for replication purposes. NASA's definition of life mentioned above is precisely the definition of a living system, Dawkins's definition of survival machines. For cultural self-replicators, replication machines are cultural products we often see: musical pieces, songs, poems to transmit information about love, happiness,... For digital self-replicators, replication machines are computer software, computer viruses to transmit information encoded by command lines.

* * *

In some aspect, the complex structure of bacteria is a tool, a **survival machine** to help self-replicators replicate. Surely you recognize familiar concepts: self-replicators, survival machines, memes,... somewhere. That's right, those are contents in the very famous book *The Selfish Gene* by Richard Dawkins. Although there are many points of disagreement, I acknowledge that this book is influenced by Dawkins's thought at important foundations. Especially evolutionary subjects and human bodies are survival machines. Self-replicators – gene segments in each cell of ours are masters residing in body survival machines. According to NASA's definition of life: life is self-sustaining chemical systems with ability to follow Darwin's theory of evolution. With that definition, self-replicators are considered living systems.

That explains why life has almost no significant changes from 3.5 billion (fossils of single-celled organisms in Pilbara, Australia). At that time, resources were still abundant; competition was not fierce, so evolution occurred slowly. Until the Cambrian explosion

539 million years ago, diversity of organisms occurred increasingly fast through appearance and disappearance of very many forms of organisms with increasingly complex levels [124]. Cooperation and competition in evolution helped form organisms we see today.

Most studies show that the more complex self-replicators are, the longer their ability to create copies. In 2021, a group of scientists at Groningen University also researched self-replicators and they introduced another factor into the self-replication process: "death." And surprisingly, when introducing death, complex self-replicators have replication and survival advantages [125]. 55 years ago, American biologist Sol Spiegelman held that complex systems will meet disadvantages in the replication process. So how can complex systems like life evolve? Team leader Sijbren Otto researched self-replicators to create fiber strands formed from monomers. This is like building strands created from rings connected with each other. Otto used different rings, some small rings, some large rings to create strands. They proved that if rings are small, strands created from small rings grow faster and longer than strands created from large rings. This is easy to understand because large rings are more complex than small rings, so strands created from large rings grow slower than small rings.

Scientists introduced a chemical that can break rings and were very surprised to see that introducing the chemical makes strands created from large rings more stable. This is a discovery that could be very important because a complex system that does not transform will be more disadvantageous than a simple system, but a complex system that can transform can have advantages in self-replication over simple systems. Transformation is not arbitrary. Transforming much does not create complex systems, but not transforming, complex systems have no advantages compared to simple systems. Therefore, complex systems can evolve stably with appropriate transformation. This is the most recent research on experiments proving that from simple compounds, complex systems can

form and evolve. Balancing between short rings developing fast and long rings developing stably could be the *middle way* not going to any extreme in Buddhism.

In nature, can experiments like the above be performed? Through biotechnology, scientists can cut genes of a cell and study their growth process (equivalent to making rings smaller). When reducing half the number of genes, cells lose stability and flexibility, but if only keeping one gene, can **minimal cells** exist and develop? Scientists studied *Mycoplasma mycoides* cells for 300 days and 2 thousand generations (equivalent to 40 thousand years of humans), one group lost 50% (only 493 genes remaining) and one group lost very much, only minimum gene number (80 genes), and saw that cells with minimum genes evolve 39% faster than the other cells [126]. Research shows that even with very few genes, cells can still exist and develop through gene mutations to adapt to the environment, for example presence of antibiotics.

Up to here, most readers are still not satisfied about the origin of life and the process of life developing from inorganic compounds. Let me remind that there is no human theory that clarifies this process. All hypotheses presented above are only hypotheses, though based on models and experimental evidence, still not strictly confirmed. This is a difficult problem because life could have formed over a long time of millions of years, far exceeding the time of experiments. But that does not discourage scientists. Many theories continuously emerge related to the problem of the origin of life. Assembly theory is a new theory emerging in 2023 but has potential to clarify the problem due to a very different approach [127]. This theory still relies on laws of physics and evolution, but objects that these laws act on change. Instead of considering objects, here could be chemicals, bacteria, individuals,... as points, they are considered entities with possible formation history. Each entity is determined by boundaries and has evidence of natural selection. Scientists must determine two parameters: number of constituent objects and as-

sembly index. Constituent objects are components that cannot be further reduced to form entities, while assembly index is the minimum number of steps to form that entity. For example, to create an organic molecule, the number of objects is the number of C and H atoms together with bonds between them. Assembly index is the minimum number of steps to link constituents together. Assembly theory allows simulating and calculating formation of living entities by computer and comparing with experiments. Though newly developed, this theory has the ability to make predictions that can be verified by experiment.

According to Worldometer Organization, the Covid-19 pandemic raged with nearly 704 million cases and about 7 million people from 229 countries and territories died [128]. SARS-CoV-2 virus is precisely the self-replicator that used survival machines, the victims, to spread throughout the world. This is the world's largest pandemic ever and is the clearest example showing that complex survival machines with superior intelligence like humans must struggle painfully, laboriously with simple self-replicators like SARS-CoV-2 virus. The process above seems vague; to this day, is there any evidence of the existence of these primitive self-replicators? After nearly 4 billion years of evolution, traces of those self-replicators no longer exist; they have changed, developed into extremely complex systems: ourselves. Our genes are precisely self-replicators, and our bodies are survival machines for those self-replicators.

3.1.7 Left-handed life

Such hypotheses about the origin of life seem too simple, so many scientists refute this hypothesis. One of the bases to refute is that among about 100 amino acids humans know today, only 20 types are present in life. All amino acids share one property: they are **left-handed** due to Louis Pasteur, the first to discover in 1860 (Figure 3.3). We all know Pasteur as a microbiologist, but few know

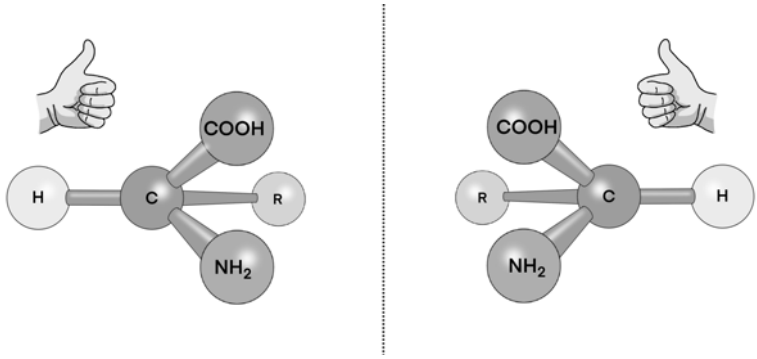

Figure 3.3: Amino acid molecules can have left-handed or right-handed structure. They are mirror images (chiral) of each other. R is an organic functional group.

he also had a very important discovery: finding biological organic compounds with left-handed property. Specifically, **amino acids are left-handed**. He wrote that "this is one of the relationships between life on Earth and the universe" [129].

In 1815, French physicist Jean-Baptist Biot discovered that when letting normal unpolarized light pass through some organic solutions, that light becomes polarized, but he did not understand the cause of that polarization. Later, German chemist Eilhard Mitscherlich saw that tartaric acid extracted from grapes rotates polarized light, while paratartaric acid produced from industry does not.[4] At that time, people thought these two substances had the same chemical composition. Pasteur did not believe two identical chemical compounds could have different polarization results, so he spent time researching these two compounds. He let these two compounds freeze into crystals and carefully examined their shapes under a microscope, and he discovered that although chemical formulas are the same, their structures show symmetry is not the same. Crys-

[4]Light is electromagnetic waves in the visible region, alternating oscillation of electric and magnetic fields. The oscillation direction of the electric field is called polarization direction. Normal light has polarization direction arbitrary in space. However, when passing through some crystals or certain solutions, only light with certain polarization direction can pass through. That is polarized light.

tals of tartaric acid have left-handed symmetry, while crystals of paratartaric acid have both left-handed and right-handed types. Left-handed structure is precisely the mirror image of right-handed structure. Only looking at an ordinary microscope and still seeing crystal symmetry has scientific basis. For example, table salt crystals NaCl have cubic structure, so when we look carefully, we will see salt particles have this structure. Ice is the same; because ice crystals have hexagonal shape, when freezing, we can see snowflakes also have hexagonal shape.

Amino acids formed in chemical reactions have equal probability for left-handed and right-handed. But in living organisms, from viruses, bacteria, fungi, plants, animals, and humans, these amino acids are all left-handed. Left-handed amino acids form proteins, while DNA has right-handed double helix structure. Miller's experiment created both left-handed and right-handed amino acids; then why is life created from left-handed molecules? Why is the double helix not left-handed? The answer is surprisingly from research by high-energy astrophysicists.

Many scientists hold that DNA being right-handed is a coincidence of nature. Perhaps right-handed DNA configuration was created first, so it has numerical advantages over left-handed DNA. The evolution process increases differences, and to this day only one type remains. We had to wait more than a century for scientists to have deeper insight into this problem. In 2020, the research group of Noémie Globus of New York University put forward an explanation for DNA's right-handedness [130].

They hypothesized that from the very beginning, life could create both left-handed and right-handed double helices in equal numbers. If life like that could evolve, then to this day we could observe both types. However, cosmic rays made this balance disappear. Every second, cosmic rays from deep space continuously shine down on Earth, passing through our bodies. From before until now, almost no one has researched their effects on life because cosmic ray energy

is too large compared to chemical energy of life. Cosmic rays continuously bombard Earth's atmosphere and create pion particles. Pion particles, also called pi mesons, are particles created by one quark and one antiquark. These particles are unstable and quickly decay due to weak nuclear force controlling; this is the only fundamental force with mirror asymmetry property.[5] Pion particles bombarding the atmosphere create showers of muon particles (particles like electrons but with larger mass) and electrons. Because pions have mirror asymmetry, they create muon particles with identical *magnetic polarization* going down to Earth's surface. Here, muon particles will ionize (shoot out an electron) right-handed double helices more than left-handed double helices, making right-handed double helices have more mutation ability, therefore evolving faster. Although the difference between these two structures is very small, millions, even billions of muon particles can create more than one electron in right-handed strands.

The next problem to research is whether ionization preference for right-handed DNA creates more mutations still needs further research. This is quite time-consuming because mutations are not frequent occurrences, but people can test on microorganisms, which can give faster results. Another hypothesis was put forward when analyzing chemical composition of amino acids taken from meteorite samples; NASA astrobiologist Jason Dworkin showed that left-handed amino acids are more than right-handed by about 20%, and excess of left-handed amino acids could be transmitted to Earth to thereby create right-handed DNA [131].

Another hypothesis holds that left-right imbalance is caused by Earth's magnetic field. We know that Earth's magnetic field has direction from north pole to south pole. Although the direction of this magnetic field can change after every 300 thousand years [132], that time is long enough to create preference for left-handed molecules.

[5] CPT symmetry (charge, parity, time: charge, mirror, time) is conserved in weak nuclear force, but component symmetries are not.

There is an RNA precursor molecule named ribo-aminooxazoline (RAO) that can develop RNA nucleotides in a single direction, either left or right. On Earth, there is always presence of magnetite compound Fe_3O_4 inside containing super-small magnets due to iron atom spin. Under the effect of Earth's magnetic field, these super-small magnets orient according to the magnetic field direction. Therefore, they have a certain preferred direction. When RAO develops into RNA, they interact with super-small magnets of magnetite, so they also orient according to the preferred direction. The probability to create initial difference is 60%, but later reaches 100% in later stages [133].

Very recently, February 2024, American scientists proposed a completely different mechanism creating left-handedness [134]. Based on a set of chemicals containing sulfur that could exist in Earth's initial period, capable of linking individual amino acids to form dipeptides, the basis to create proteins. When carefully studying a chain of reactions, they saw that the ratio of left-handed amino acids is larger than right-handed is 60:40. The reason why there is initial difference is currently unclear, but that difference leads to subsequent chain reactions being amplified to create left-handed dipeptides. This research differs in not relying on asymmetry-causing agents from outside like cosmic rays or Earth's magnetic field but relying on asymmetry in some initial chemical reactions, then subsequent processes are amplified.

* * *

Whether the origin of left-handedness of amino acids or right-handedness of DNA comes from asymmetry of weak nuclear force or from anisotropy of Earth's magnetic field, it gives us a lesson: everything can very likely be related to each other in unexpected ways. The dependent origination principle, or emptiness manifests in all processes of biology. From the origin of elements appearing in furnaces of stars to arrangement of atoms to form molecules with

definite ordered structures also come from universal relationships between all things in nature. Once again, emptiness of Buddhism appears in understanding the origin of life.

Buddhist scriptures say nothing about the origin of life, but if life relates to reincarnation, there is no first starting point. In the Samyutta Nikaya, Nidana-samyutta, Chapter 10 Samsara, the Buddha said: "The first point of reincarnation cannot be traced. This life has rolled from beginningless time; we cannot say 'from here is the beginning."' So does Buddhism's view contradict the origin of life in modern science? If we only stop here, we can say the two views are fundamentally contradictory. However, before final conclusions, emptiness of Buddhism suggests to us impermanence over time of organisms – evolution theory.

Next we study Darwin's theory of evolution – one of the greatest discoveries in scientific history. Evolution theory not only explains how organisms change over time but also clarifies emptiness of life: life has no fixed self-nature but always changes and evolves, depending on environmental conditions and natural selection processes.

3.2 Evolution theory

3.2.1 Natural selection

Perhaps the greatest discovery in history is **evolution theory**, telling us the deep relationship between organisms existing in nature. This discovery completely contradicts human thinking, especially previous philosophers and theologians when holding that humans have a very special position in the universe. In Darwin's time, people already knew how to collect and classify animal and plant species based on characteristics of each species. Lists of species increasingly expanded over time because biologists found many new species. Especially many organisms were found in fossil form – a type of trace of organisms stored in rocks. People began asking whether a

species can self-change depending on conditions of the environment where that species lives?

Charles Darwin[6] was a British naturalist, abandoning medical studies at Edinburgh University to study biology at Cambridge University. He spent 5 young years following the ship *HMS Beagle* traveling around the world from 1831 – 1836. On that journey, he collected and recorded characteristics of many animal and plant species, especially birds from the Galapagos Islands. He noticed that physical characteristics of birds were slightly different when going from one island to another. Later experts confirmed those birds belonged to different species, making Darwin put forward the relationship between geographical characteristics of islands affecting changes in characteristics of birds. He continued researching for more than twenty years, until 1859, publishing *The Origin of Species* expressing the doctrine of evolution of organisms based on **natural selection**.[7]

Darwin's theory explains some characteristics about physics or behavior of an individual belonging to a population changing over time. Through the process of genetic variation, some individuals have characteristics different from other individuals. These characteristics are transmitted to offspring generations; offspring generations again transmit to grandchild generations. If continuing like that, after some time there will be very many individuals with changed characteristics. But this phenomenon is not common in na-

[6]Charles Robert Darwin (1809 – 1882) was a British naturalist famous for evolution theory based on natural selection. His doctrine holds that high-intelligence animals like humans also evolved from lower animals. Evolution theory is one of the most influential and also most controversial scientific theories to this day.

[7]The idea of evolution theory began from Jean-Baptiste Lamarck (1809) when he held that organisms are not immutable but gradually transform from simple to complex forms. In 1858, Darwin and Alfred Wallace together published evolution theory with natural selection as the basis. However, Darwin is considered the main author of the theory because of his book in 1859 with full evidence he had collected over decades.

ture because not all characteristics can be inherited; only individuals with characteristics best adapted to the environment have opportunities. Individuals with unsuitable characteristics have difficulty having opportunities to survive to adulthood and cannot transmit those characteristics to new generations. Selection of natural environment only allows individuals with most suitable characteristics to survive; those characteristics can then be multiplied over time. Darwin called this process **natural selection** (natural selection). The term *survival of the fittest* was proposed by Herbert Spencer after reading Darwin's work and was accepted by Darwin for use in later editions.

Darwin held that organisms as **individuals** are what are evolving; he placed individuals as the center to examine changes over time in a certain environment. Evolution originates from competition between individuals. Competition is due to finiteness of resources of the surrounding environment; it is struggle between desire to have many resources to serve needs of individuals and limitation of resources in a certain environment. The idea of survival competition originates from a pastor named Thomas Malthus. In the book *An Essay on the Principle of Population* (1797), he warned that most policies designed to help the poor will fail due to unceasing pressure of population growth. Darwin realized that animals and plants also suffer similar population pressure, so proposed evolution theory occurring over time by natural selection. Specifically, characteristics of individuals that adapt to the environment will survive.

Biologically, evolution is change in heredity, also called variation, in organism populations over many successive generations. Most variations do not bring competitive advantages to organisms, but some variations make those organisms most suitable to the new environment to exist. The fittest does not mean the largest, biggest. Usually large goes with strength, but large also goes with consuming resources to nourish the body and also goes with easier prey ability. Excellent characteristics inherited through many generations grad-

ually become common for the entire population, called adaptation to the surrounding environment. The fittest also does not mean the most complex. Very many organisms with simple structures have existed for billions of years until today without much change. But there are also the fittest becoming extremely complex like us humans.

Artificial selection is a process similar to natural selection, where individuals with physical and behavioral characteristics most suitable for human needs will survive and be inherited to other generations. Artificial selection has faster effects, easier to observe than natural selection. Humans made the agricultural revolution transitioning from hunting, gathering to domesticating animal and plant species to serve their needs about 10,000 years ago. Goats and wheat are species cultivated earliest in today's southeastern Turkey and western Iran [135]. About 90 % of energy humans obtain comes from crops and livestock domesticated from about 9500 – 3500 years BCE. For us humans, thousands of years seem long, but for artificial selection, such a long time is needed. Over thousands of years, animal and plant species have evolved according to human wishes. Goats became gentle, obedient following control of herders; wheat produces many grains after each harvest. Almost no notable animal or plant species has been domesticated in the past 2000 years.

3.2.2 Evidence of evolution

Evidence of evolution can be seen everywhere: fossils, similarity of body structure, geographical distribution, embryo development, similarity of DNA sequences of species. With ordinary perception, we cannot notice organisms, including humans themselves, have and are changing because the time frame of one lifetime, or a few human lifetimes, is too small compared to the time frame in life history. Since discovering that organisms evolve, we have found evidence of evolution everywhere [136]. Scattered pieces of evidence, scattered

in alleys and corners, are connected to tell one of the greatest stories in human history: humans are related to all organisms that have and are living on Earth. Not only distant relatives, but the influence of those relatives on activities of the human body in every moment is very strong and exists until today.

Fossils are traces of ancient organisms existing in earth, rocks, amber, ice. They provide information about characteristics of each species, geographical distribution, transitions between species. Through that information, we can build evolutionary relationships between species in the past and until today. One of Darwin's worries was the absence of intermediate forms between some groups of organisms in time order. For example, fish $\rightarrow$ amphibians $\rightarrow$ reptiles $\rightarrow$ mammals $\rightarrow$ primates. Fossil results gradually fill those gaps, showing consistent changes over time and finding no reversals. For example, fossils of *Tiktaalik roseae*, living 375 million years ago, is an amphibian species with fins and gills like fish but with solid bones to be able to crawl onto land. Fossil evidence shows Tiktaalik is a transitional fossil from fish to amphibians [137] and later became ancestors of species existing to this day like crocodiles, horses, bats, and humans.

Comparative anatomy studies similarity of body structure between species starting long ago. Although living conditions are very different, skeletons (and other organs) of humans, mice, whales, and bats are similar in structure [138]. Biogeography contributes to evidence of common ancestry. With about 250 thousand plant species, 100 thousand fungal species, and 1 million animal species already described and named (this process is still ongoing). Each species occupies an appropriate ecological environment and position. From the position of current species and comparing with fossil evidence in the past will tell geographical distribution of species.

One of the enthusiastic supporters of evolution theory is German biologist Ernst Haeckel. He observed similarity in embryo development stages between different species, supporting the hypothe-

sis of common ancestry. However, his recapitulation theory holding that embryo development fully recapitulates evolutionary history has been refuted. Even so, similarity in early embryo stages (for example, early stages of embryos in fish, reptiles, birds, mammals all have gill slits and tails [139]) is still important evidence of common ancestry, though not complete recapitulation of evolutionary history.

In about the past 20 years, with development of biotechnology and gene decoding of organisms including humans, data show similarity of DNA in different organisms is undeniable evidence of relationships between species. For example, the human genome has up to 8% genes, corresponding to 100 thousand DNA fragments of viruses that have invaded throughout the entire evolution process [140]. These genes do not lie meaninglessly in the human genome but can activate the immune system to protect the body from attacks of other viruses. The human immune system consists of two systems: innate immunity and adaptive immunity. Innate immune system exists from birth. Adaptive immune system is obtained through the body's learning process when confronting external disease-causing agents. A study by scientists at Utah University showed that if removing viral genes, the body's immune system resists much weaker than when not removing these genes due to inefficient transcription [141].

At the molecular scale, the most basic activities of all living organisms all rely on using amino acids to create proteins, breaking sugar molecules to release energy, transporting nutrients across cell membranes. Those activities all share one principle; all are directed from DNA. When studying DNA sequences, people will see differences and the formation process creating differences. After analyzing DNA, people rebuild evolutionary relationships and verify with fossils, embryo development to reach conclusions about evolution between species. For example, two types of proteins present in most multicellular organisms are hemoglobin and myoglobin. Myo-

globin receives oxygen from hemoglobin and stores it in tissues until needed. Structure and amino acid sequences of the two proteins are determined and show ordered relationships between mammals, birds, reptiles, amphibians, fish, worms, and mollusks. Research by Nebraska, Lincoln University based on ancient proteins shows that hemoglobin of vertebrates evolved from an ancient monomer to become a tetramer as today [142].

Scientific evidence shows that organisms on Earth all share a common first ancestor (Last Universal Common Ancestor – LUCA) [107, Chapter 3]. Evolution theory shows that human presence is the result of a long process of millions of years related to very many different forms of organisms from viruses, bacteria, lower animals, higher animals. Even if humans have different intelligence, that difference comes from evolution of other organisms not only in the past but also existing until today. That affirms one thing: we are not born separate and independent; we are nurtured by Mother Nature to achieve differences thanks to symbiosis with all other organisms. If so, humans need to view and treat other organisms appropriately, not cruelly and arbitrarily as currently. That is precisely what Buddhism teaches us.

Darwin put forward evolution theory based on observing external traits of organisms. But internal causes, internal factors for the evolution process to occur, he did not know. There is still an important missing piece to complete the evolution picture: genetics.

3.3 Genetics

3.3.1 Classical genetics

Speaking of genetics (genetics) is speaking of Gregor Mendel[8], but ideas about heredity did not begin from Mendel but originated from Greek philosophers Hippocrates[9] considering the body as a whole, not a combination of separate parts. Therefore, reproductive material must be collected from all parts of the body to transmit to subsequent generations. This heredity theory existed for 23 centuries without significant changes until the 19th century. In 1886, Darwin put forward *pangenesis theory* (Pangenesis) holding that all animals and plants secrete tiny cells distributed throughout the body, then they fall into reproductive organs, so traits are transmitted to subsequent generations. It is unfortunate that Darwin did not know or did not properly evaluate Mendel's research because when Darwin put forward this theory, Mendel had published genetics theory more than 20 years earlier, and some scientists had cited Mendel's work.

In 1865, Mendel published his research on heredity of pea plant traits, and he was the first to put forward the concept that later Danish botanist Wilhelm Johannsen called gene. He crossed pea plants with clearly contrasting traits, then counted hybrids in different groups. From there, he built crossing formulas based on strict mathematical foundations to find heredity laws. For example, he tested crossing between two types of peas, one tall and one short. The first generation F1 is always tall pea plants; he called dominant traits. Short plants do not completely disappear; they appear in F2 plants, the next generation of F1. For every 3 tall plants, there is one short plant, called recessive traits. Tall-short traits of pea plants

[8]Gregor Johann Mendel (1822 – 1884), Austrian priest and scientist considered the father of modern genetics based on his research on hereditary characteristics of peas in 1865.

[9]Hippocrates of Kos (460 – 370 BCE), greatest physician of ancient Greece, considered the father of world medicine. He is the author of *The Hippocratic Oath* that all doctors today follow.

do not affect other traits, for example fruit color or seed smoothness. Pea traits seem to operate independently of each other, but dominant, recessive traits and 3:1 ratio are the same for all types of traits.

Mendel presented his discoveries in 1865, but listeners were disappointed when seeing that research as mathematical work. Mendel printed and sent his research to famous scientists at that time, but no one was interested. Mendel's research results came from traits independent of each other, so he could put forward laws that can be described mathematically. If they are independent, they must be controlled by some cell element, which today we call genes – the basis of heredity. By the early 20th century, scientists fully recognized the importance of genetics, and genetics became a research field not only limited to biology but also expanded to other fields.

Up to here, we again see a relationship between mathematics and genetics. The neglect of scientists contemporary with Mendel was because they thought there could be no necessary relationship between these two fields. Mathematics belongs to pure rational thinking, yet can describe or make humans understand genetics, while at that time people thought it belonged to mysterious supernatural. Through patience in experiments and observation ability, Mendel bridged the gap between the above two fields. Certainly genetics needs to be studied deeper, but not starting from Mendel but from other scientists independent of him.

Swiss biologist Friedrich Miescher was the first to isolate nucleic acid in 1869. German physician Albrecht Kossel showed that nuclein includes a part with protein and a part without protein. The part without protein is precisely what Miescher researched, precisely nucleic acid, including 3 parts: (1) a sugar group with 5 carbons, a phosphate group, and a nitrogen-containing base. Kossel and collaborators found 4 types of nitrogen-containing bases forming DNA:

cytosine (C), guanine (G), adenine (A), or thymine (T).[10]

A student of Kossel, Phoebus Levene, is credited with many firsts in research on this issue. For example, he was the first to discover the order of three main components of a single nucleotide (phosphate-sugar-base); the first to discover the carbohydrate component of RNA (ribose); the first to discover the carbohydrate component of DNA (deoxyribose); and the first to precisely determine how RNA and DNA molecules are combined together. Especially the ribose sugar group can lose one oxygen at an important position to become deoxyribose. He added the letter D in the word DNA and held that DNA is genetic material of cells [143].

By the late 19th century, people knew 10 different types of amino acids to form proteins. So what contains genetic information, protein or DNA? Because DNA only differs by 4 bases while proteins are formed by dozens of amino acids, most scientists held that protein is genetic material of cells because the number of amino acids is much larger than the number of bases. Diversity of characteristics of species must certainly have many instructions. Like an alphabet to write instruction texts, an alphabet with many letters of amino acids will write more diverse instruction commands than letters of bases. With such thinking, research by Miescher, Kossel, and Levene gradually received less attention, but people returned to Mendel's almost forgotten discovery.

3.3.2 Modern genetics

The birth of modern genetics began from the 20th century. In 1900, simultaneously 3 European scientists revived Mendel's genetics research and quickly made this research become a science

[10]In South American indigenous language, bird droppings are guano. This compound is named guanine because initially it was extracted from bird droppings. Adenine in Greek is pancreas because this compound was extracted by Kossel from pig pancreas bought from slaughterhouses. Thymine from thymus glands. Cytosine has root from cytoplasm, meaning cell.

called genetics. In 1901, through microscope observation, people discovered that chromosomes are genetic material in the nucleus. In 1905, Nettie Stevens discovered sex chromosomes X or Y, chromosomes determining sons or daughters. Thomas Morgan[11] put forward genetics theory linked with chromosome heredity. He experimented with fruit flies and quickly discovered dominant and recessive traits together with 3:1 ratio like Mendel experimented with pea plants. He and collaborators held that genes are arranged in vertical rows on chromosomes; positions of genes on chromosomes are called locus. They discovered linked heredity supplementing Mendelian heredity. Because chromosomes form pairs, each locus consists of two genes located on two chromosomes called allele pairs. The combination of Mendelian genetics and Morgan's chromosome genetics creates *modern synthesis* theory – classical genetics still used by biologists to this day. However, the question of whether DNA or protein is genetic material was still unresolved. People realized that DNA extraction methods of Miescher and Kossel using acid could break DNA molecules. Perhaps these molecules are very large to be able to contain all genetic information.

Erwin Chargaff was the first to prove that DNA molecules are very long molecules. Depending on each species, the ratio of bases in DNA differs, and he found the rule named after him, *Chargaff's rule*, holding that A always pairs with T, G always pairs with C in 1:1 ratio. This ratio is the same in all organisms.[12] Nucleotide composition of DNA differs between species. In other words, similar nucleotides do not repeat in the same order, as Levene proposed, so

[11]Thomas Hunt Morgan (1866 – 1945), American geneticist famous for discovering the role of chromosomes in heredity of organisms, about linear arrangement of genes on chromosomes, and playing a role in perfecting chromosome heredity theory. He won the Nobel Prize in Medicine in 1933.

[12]Erwin Chargaff (1905 – 2002), American biochemist credited with discovering the rule named after him. This rule contributed importantly to discovering the double helix structure of DNA.

DNA could be genetic material of cells. In the early 1950s, Alfred Hershey and Martha Chase researched how bacteriophages (phages, also called bacteriophages) attack cells. These viruses approach cell membranes and inject genetic material into cells to replicate. They designed a clever experiment using radioactive isotopes of phosphorus and sulfur to find genetic material. Because phosphorus is in DNA while sulfur is in protein, if genetic material of bacteriophages after injecting into cells contains phosphorus, then material injected into cells is DNA. Conversely, genetic material injected will be protein. Checking radioactivity is very simple, using a Geiger counter; it will make sounds if there is radiation. Results are very clear: genetic material that bacteriophages inject into bacteria is DNA, not protein.

3.3.3 The race to find DNA structure

Hershey and Chase's experiment opened many important questions and urged scientists to research DNA in 1951. The first important thing is what DNA structure is like. In 1952 – 1953, there were three research groups racing to find DNA structure: L. Pauling at Caltech, R. Franklin at King's College London, and Watson and Crick at Cambridge University.

At King's College London, Maurice Wilkins was the first to propose that DNA structure could be found by X-ray diffraction, but he did not have knowledge of this technique and proposed finding someone with knowledge to assist. Rosalind Franklin[13] with very good knowledge of crystallography succeeded in photographing DNA diffraction. *Photo 51* is the most important photo helping

[13]Rosalind Franklin (1920 – 1958), British Jewish biophysicist famous for X-ray diffraction results of DNA, from which Watson and Crick built the double helix model. She died early from cancer, possibly due to working too much with X-rays. If Franklin were still alive, she could have won the Nobel Prize in Biomedicine in 1962 because of important contributions in finding the double helix structure [144].

scientists discover DNA structure (Figure 3.4). This is the most important document because previously all experiments to photograph DNA diffraction had failed due to poor DNA extraction, and Franklin was an expert in X-ray diffraction so knew how to adjust equipment to have sharper beams. However, due to conflicts between Wilkins and Franklin, they did not cooperate in research.

Linus Pauling was already a renowned scientist pursuing many scientific projects but also researched DNA structure. He could not see Franklin's X-ray diffraction results because the U.S. government refused to grant him a visa to go to England due to suspicion that he was a Communist Party member because of activities against nuclear weapons development, though he had never been a party member [145]. Pauling had to rely on results from other colleagues at Caltech. But those results, due to poor sample preparation, were often hydrated, so diffraction images were greatly distorted. That could be one reason why Pauling could not propose an accurate model.

Two young scientists at Cambridge University were James Watson, 25 years old, and Francis Crick, 30 years old,[14] people without much experience, also began researching DNA structure. At that time, people already knew DNA had sugar molecules and phosphate groups linked together to form long chains. However, how many chains DNA had, where bases were located on chains, and how they interacted was still unclear. In early 1953, after a period of struggling but unable to solve the problem, Watson thought he needed help and went to London to see Franklin's results. When he arrived, he did not meet Franklin but met Wilkins. Wilkins took the X-ray diffraction results of DNA, Photo 51 (Figure 3.4), showed them to Watson without Franklin's consent, and said this was a

[14] James Watson (1928 – present), American biologist, and Francis Crick (1916 – 2004), British biologist famous for proposing the double helix DNA structure. They won the Nobel Prize in Biology in 1962 together with Maurice Wilkins, a New Zealand physicist.

double helix. That was the critical moment for Watson to realize the model. He returned to Cambridge to work with Crick, and a few months later published the DNA structure to the world in the journal Nature [146], in the same issue with two other proposals from Pauling and Franklin. After the results were published, Pauling[15], though then the world's leading chemist, also admitted he was wrong. The results of the two young scientists were certainly correct, and from then on he used his influence to popularize the double helix structure.

3.3.4 Genes are the subject of evolution

Darwin held that the subject of evolution is an individual. The survival of the fittest is the survival of individuals with physical characteristics and behaviors best adapted to the surrounding environment. If we consider individuals as the subject of evolution, we cannot explain many problems about behaviors of species in nature where many individuals must sacrifice their lives to serve other purposes. Especially the biological nature of selfishness and altruism of each individual. Selfishness brings immediate benefits to individuals, but altruism, for example altruism of parents for children, siblings, relatives,... how would it be explained from a biological perspective. After Darwin, many famous biologists held that the subject of evolution could be *species* or a group of individuals rather than an individual. EO. Wilson,[16] who was called the New

[15]Linus Pauling won the Nobel Prize in Chemistry in 1954 for proposing the electron hybridization theory to explain properties of carbon atoms, won the Nobel Peace Prize for relentless activities against nuclear weapons proliferation and war in 1962. He is the only person to win complete Nobel Prizes (not shared with others) twice.

[16]Edward O. Wilson, commonly called EO. Wilson (1929 – 2021) was a famous American biologist and naturalist. He is considered the most famous biologist of the 21st century, the New Darwin, the father of the concepts *social biology*, *biodiversity*. He specialized in ant research so was also called with the nickname "the ant man".

Darwin, enthusiastically supported group selection theory and held that kin selection is not needed [147].

Richard Dawkins[17] opposed the idea of group selection and held that the subject of evolution is genes lying deep inside each cell of organisms. George Williams[18] defined a gene as a DNA segment on chromosomes that can exist and replicate through many generations over a sufficiently long time, then is considered a unit of natural selection [149], a living system, a self-replicator, the subject of evolution. These genes have selfish properties in the sense that they are completely for themselves, for the extension of genes so they can spread most widely, longest, and become *immortal*. Therefore Dawkins' theory is called the theory of *selfish genes* or *immortal genes*. The shorter the gene length, the higher its selfishness and vice versa. So do genes have consciousness about selfishness or altruism? No, genes do not have consciousness about those things, genes have no plans, they have no future. Everything we see is human perspective on selfishness and altruism; genes simply find ways to replicate themselves as much and as long as possible. Even more surprising is that activities that seem purposeful have causes from random phenomena, following known physical and chemical laws.

In Darwin's time, he did not yet know about genes and DNA because science at that time had not discovered this knowledge, but we understand how evolution occurs at the molecular scale. The molecule we speak of here is DNA (Deoxyribonucleic acid),

[17]Richard Dawkins (1941 – present) is a famous British evolutionary biologist with the book *The Selfish Gene* [119] who held that the subject of evolution is genes, not individuals, groups of individuals, or species. Dawkins and the ant man EO. Wilson are two leading opponents on gene theory representing two continents. Wilson is a biologist with reputation and many scientific works in prestigious journals. Dawkins is a philosophy of biology scholar so cannot publish like Wilson, so Wilson called him a "journalist" [148]. However, Dawkins' influence on society and philosophy is very great to this day.

[18]George C. Williams (1926 – 2010) was an American evolutionary biologist who, together with W. D. Hamilton, John Smith, Richard Dawkins, held that the subject of evolution is genes, refuting the idea that the subject of evolution is groups of individuals.

a large molecule with the shape of a double helix formed from two strands rotating around an imaginary axis. Each strand is a polymer formed from nucleotide monomers. Each nucleotide is formed from one of four nitrogen-containing nucleobases: adenine (A), thymine (T), guanine (G), cytosine (C) linked with deoxyribose sugar and a phosphate group.

All organisms on Earth, from viruses, bacteria, fish, reptiles, plants, mammals, and humans all have DNA formed from the same nucleotides. The difference between species, between individuals is the different arrangement order of nucleotides. It is the difference in nucleotide sequence order that creates differences between this species and that species, between this individual and that individual. In single-celled organisms like bacteria, DNA lies in the nucleus of bacteria. In multicellular organisms like humans, DNA does not lie in any specific organ of the body but lies in all cells. Adult humans have about 30 trillion (30×10^{12}) cells, most of which (except female egg cells and male sperm cells) contain a complete copy of that individual's DNA [150].

For simple organisms, DNA also consists of millions of BP, while for complex organisms there are billions of BP like that. Humans have 3.1 billion BP multiplied by the length of one BP of 0.34 nm, so human DNA length is about 1 m [151]. DNA molecules are small but very long, compactly packaged in the cell nucleus with size only a few µm [152].[19] If we connect all DNA of the human body, we have a string with length so great that it can connect from Earth to the Sun and back 300 times [153]. DNA is folded in an ordered way according to many different structural levels forming chromosomes. When needed, DNA is unwound to create copies and proteins. Humans have 46 chromosomes divided into 23 pairs; the number of chromosomes varies depending on species. If we magnify DNA molecules

[19]Some documents record human DNA parameters as 6.2 billion bases and length 2 m, double the above numbers because they calculate the quantity and length of both DNA strands [152].

by a hair with diameter 34 µm, that hair is 100 km long. Therefore, folding DNA is a complex process to store compactly and unwound when used. DNA is so long because it contains all instruction information to create the body and guide the body to operate throughout life.

DNA molecules have two important tasks: (1) they create copies of themselves each time cells replicate, and (2) create protein molecules controlling cell activities. Each person begins forming from a single cell from mother and father. From there, over time cells are doubled into 2, 4, 6,... Our body develops as it is now as a result of billions of divisions, with each division DNA is copied accurately, very few errors. Except reproductive cells where the meiosis process makes these cells have only half the number of chromosomes, other cells all have the same DNA copy stored in 23 pairs of identical chromosomes. The process of directing information from DNA to develop the entire body is a complex process that humans have not yet fully understood. Probably still need much more time to fully understand this process, but we know that one of the first instructions from DNA to the body is the protein creation process.

The process of transferring information from a gene to a protein is called gene expression. The order of ATGC bases carrying information of a gene recorded on DNA is *transcribed* (transcription) into messenger RNA (mRNA) inside the nucleus. If DNA is a double helix, RNA is a single strand, shorter but carrying complete information of the gene. RNA bases are also not completely like DNA. Thymine is replaced by Uracil, making the four bases of RNA AUGC. The mRNA segment escapes the nucleus to enter the cytoplasm to synthesize proteins. The *translation* process is the process where transfer RNA (tRNA) reads information already transcribed on mRNA at ribosomes to create amino acids. These amino acids self-arrange to form proteins. Protein structure is completely determined by the order of DNA bases through the **central dogma**

(the central dogma) of molecular biology proposed by Francis Crick in 1957 [154]: DNA – RNA – protein. The central dogma holds that information can only transmit one way, from gene to protein, with no reverse process.

Why does DNA not transmit information directly to proteins but must go through RNA as intermediary? That is because DNA is very important; it needs protection from changes during information transmission. During transcription and translation, RNA can be modified and destroyed, but DNA is always safe. A gene can be transcribed into RNA many times and create many proteins, so from one gene many proteins can be created in a short time.

Creating proteins is not yet the entire process of body formation and development, but it is the first process to do that. Proteins not only create structure for the body but also control chemical processes occurring inside cells by stimulating or inhibiting at specific times and locations. The process of transmitting information from DNA to proteins is a one-way process and only one way, with no reverse direction. There are only commands from DNA to proteins; proteins do not change DNA. The body forms to protect genes, protect them from being modified. Characteristics of an individual like strength, intelligence are not inherited to the next generation in the form of inheritance.

* * *

3.3.5 DNA structure and human gene map

DNA structure is a large polymer molecule formed from two strands twisted together, so it is called a double helix. Each backbone in the helix is a deoxyribose sugar and a phosphate group. Each rung is formed from a pair of nitrogen-containing base molecules (base pair – BP): cytosine (C), guanine (G), adenine (A), or thymine (T). These bases link with each other through hydrogen bonds according

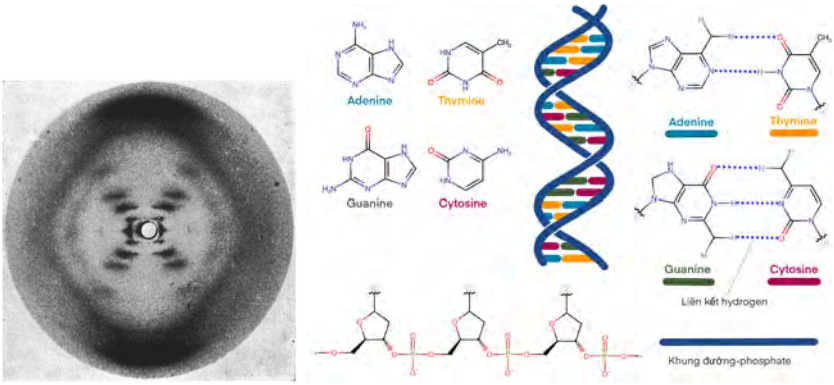

Figure 3.4: Left: *Photo 51* X-ray diffraction photograph taken by student Raymond Gosling under guidance of R. Franklin in 1952 to find DNA structure (Source: wikipedia 2023). Right: diagram of double helix structure of DNA molecule.

to Chargaff's rule: A always pairs with T and G only pairs with C through hydrogen bonds (Figure 3.4).

Hydrogen bonds[20] are physical bonds, strong enough but also weak enough. Strong enough for two helices to combine together, weak enough that when needed these two strands can open like opening a zipper (Figure 3.4). The opening and closing of the double helix allows DNA to have two main functions. One is *replicating* DNA and inheriting to subsequent generations through the meiosis process known before. Two is *creating proteins*, the daily work of DNA. Every 3 rungs in DNA create a gene code, also called a codon, responsible for a single amino acid. Amino acids are the basis to create proteins like hemoglobin and insulin.

Gene differences are mainly differences in arrangement order of nucleotides (consisting of three components: phosphate group, deoxyribose sugar, and a base); genetic information is encoded by nucleotide sequences. Strong chemical bonds along the vertical direction and weaker bonds along the horizontal direction between ATGC bases allow the two strands of DNA molecules to separate or

[20]Hydrogen bonds are electrostatic attraction forces between polarized hydrogen atoms and one of the three most electronegative atoms: N, O, and F.

combine, from which they can self-replicate. Hereditary properties and variations of classical genetics can completely be explained under the light of DNA molecules, thereby giving birth to molecular genetics.

Finding the structure of DNA is an important thing, but there are still many things to know so that information encoded from DNA can control cells; the body is still unclear. Crick spent time afterward explaining the control mechanism of genes in organisms. Daily activity of DNA is creating proteins; every 3 bases – called a codon – make a gene code to produce one amino acid. The order of bases determines the type of amino acid. For example, GGA produces Glycine, CUU produces Leucine,... He and others proved that: (1) genetic information is stored in DNA sequences; (2) messenger RNA molecules (messenger RNA) carry instructions from DNA outside the nucleus to manufacture proteins in the cytoplasm; (3) molecules later called transfer RNA connect codons and amino acids; (4) ribonucleic proteins have catalytic roles to connect amino acids into proteins. Proteins are complex and large compounds that control other biochemical processes to create phenotypes and express traits externally. Thus, scientists have understood the entire process of trait formation from the most basic level of organisms, which is genes.

The book of life consists of many letters; the order of letters is commands controlling life through proteins, but not all letters have meaning, i.e., not all base sequences are translated into proteins. Nucleotide sequences translated into proteins are called a gene. Between genes are nucleotides with no function. Here arises the requirement: how can we read all genes on DNA? To do that, people need to read all letter sequences of DNA; this process is called **DNA sequencing**.

In the 1970s, Frederick Sanger[21] put forward a DNA sequencing

[21]Frederick Sanger (1918 – 2013), British chemist, famous for finding the structure of proteins and the DNA sequencing method, for which he is the only

method consisting of 3 steps: (1) heating DNA to open the zipper from double-stranded helix to single-stranded DNA because hydrogen bonds between bases are not large, so only moderate heating can achieve this; (2) breaking single-stranded DNA by bombarding this strand with ultrasound into short segments; (3) creating complementary DNA segments of short segments after bombardment from individual bases with radioactive component atoms. By reading complementary DNA segments containing radioactive atoms, Sanger could read the sequence of DNA. The Sanger method helped him decode the entire gene of a bacterium with 5400 bases and 11 genes for the first time.

In 1986, American scientists automated the Sanger method by using fluorescent molecules instead of radioactive molecules and reading by computer instead of reading by hand. ATGC characters are displayed in different colors, and this is easily done by computer. This allows scientists to sequence DNA in large quantities. That was the basis for the U.S. Department of Energy to launch the *human gene map* decoding project in 1987. The reason the U.S. Department of Energy launched the project was that at that time people wanted to understand the effects of radioactive substances in nuclear explosions and nuclear power plants on human DNA; later the project was transferred to the National Institute of Health (NIH). The Human Genome Project was initially assigned to James Watson along with about 2000 scientists worldwide to complete in 2003, exactly 50 years after finding the DNA structure. The purpose of this project was to determine the sequence of ATGC base pairs forming DNA molecules and to determine the human gene system. The human genome contains all DNA – the genetic code of human life – is the information system controlling all activities of the human body, then the human gene map is the instruction book for all those activities.

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person to win the Nobel Prize in Chemistry twice, in 1958 and 1980.

Initially Watson was assigned as head of the project, but he was forced to resign in 1992 due to conflicts with the head of NIH at that time regarding copyright of the gene map. Watson wanted a free gene map for everyone while NIH did not. His replacement was Francis Collins[22] who led the project until completion in 2003. Initially only one NIH group participated in decoding DNA sequences, mostly done by hand, with very little computer assistance.

One person in the NIH research group was Craig Venter[23] who was impatient with the slow approach, so he separated to establish his own company called Celera Genomics, a non-profit company aimed at competing in decoding human DNA. Thus there were two groups racing to decode human DNA with different methods. The NIH group treated chromosomes like a book, divided into chapters, sections, and paragraphs. After dividing, they sent segments to different laboratories to analyze together, then assembled them into a complete book. The Celera group did not do that; they divided the entire book into small segments and sequenced those small segments, then assembled them based on overlapping parts at the beginning and end of each segment.

By 1998, despite deploying the project for more than 10 years and having to increase the budget significantly, the NIH group had only decoded about 4% of total human DNA. But from 1998, with Celera's competition, NIH was forced to use the power of computers to do their work. Within a few months, these two groups increased their working capacity to decode billions of genes, equal to all the time of more than 10 years before combined. By 2003, the human DNA map was officially completed, including about 3.1 billion base

[22]Francis Collins (1950 – present), American biologist, director of the Human Genome Project from 1992 until the project ended. He is a very devout person, as shown in his famous book *The Language of God* [155].

[23]Venter once fought in Vietnam as a medic [156, 157]. After seeing hundreds of American soldiers die in vain, he was truly disappointed and intended to commit suicide. But later, he changed his mind, earned a Ph.D. in biology, and joined NIH.

pairs (BP) written with four characters corresponding to four bases with about 22 thousand genes.

The Human Genome Project only decoded the gene map averaged from a few specific people to mix together, not a complete gene set. The number of genes after averaging equals about 92% of the total number of genes. Decoding a complete gene set must be a gene set of a specific person. This was another race in other projects of Watson and Venter when these two people decoded their personal gene maps. The number of bases that need to be sequenced is doubled because it includes both the father's and mother's genes, including both individual mutations and epigenetic inheritance. Venter launched the project from 2002 and spent up to 100 million dollars for this work. Watson launched later, in 2007, but used the most advanced technology at that time and only cost about 2 million dollars. Both projects launched results a few days apart in 2007 [158]. By 2022, scientists first decoded the entire human gene completely, including difficult regions that the Human Genome Project in 2003 had not decoded [151]. Once these two gene sets were published, scientists worldwide could access them to research and could find secrets about diseases of the biological genealogy of these two people. Venter published all gene information and medical information, while Watson hid information about the APOE gene, the gene causing Alzheimer's disease for Watson's grandfather. Research results from these two projects are the sacrifice of private information of the two leaders and their families. Scientists found many disease-causing genes that these two people did not express any diseases despite their advanced age.

Although the number of bases of genes directly encoding proteins in DNA accounts for less than 2 %, genes controlling protein-encoding genes are numerous, possibly accounting for nearly 20 % of total bases in human DNA. Regulatory genes are factors that inhibit or express adjacent genes besides epigenetic inheritance. Although humans have disease-causing genes, activities like exercise

and sports can change epigenetic inheritance to suppress the activity of these disease-causing genes. The remaining number of bases up to 80 % currently have undetermined functions, although we know they come from viruses, jumping genes. We will discuss these BPs on page 260.

3.3.6 Decoding ancient human genes

Decoding ancient human genes helps understand modern human genes. The event of decoding the human genome fundamentally changed many scientific fields. First is the molecular biology field related to the genome, but other scientific fields like anthropology have greatly benefited from this discovery. The Nobel Prize in Biomedicine in 2022 was awarded to Swedish scientist Svante Pääbo for his and collaborators' discovery related to decoding ancient humans homo neanderthal and homo denisovan to determine the migration path of homo sapiens to all over the world. Previously, research on anthropology mainly relied on analyzing macroscopic sizes and shapes of fossil bone fragments, requiring many samples and being inaccurate. Pääbo's method was to decode DNA of these bone fragments. If well preserved, ancient human DNA can survive up to 100 thousand years.

Pääbo's purpose was to find gene differences between homo sapiens and other animal species. To do that, nothing is better than decoding the genome of sapiens with ancient human species closest to sapiens. Those are neanderthal and denisovan. Initial enthusiastic publications in leading journals were quickly removed because people discovered they were modern human genes. During experiments, researchers carelessly mixed fragments containing human genes into samples. Genes of fossils are often fragmented, so arranging to create order is very difficult. Fortunately, most ancient human genes are similar to modern human genes, so gene decoding succeeded.

By comparing DNA of modern humans living in different regions

of the world and comparing with DNA of homo neanderthal and homo denisovan, scientists could know information about hybridization between modern humans and those ancient species. When homo sapiens arrived in new lands, they would hybridize with other people, and DNA of those people would leave marks in modern human genes. By finding characteristic genes of ancient humans and comparing with homo sapiens, Pääbo held that neanderthals separated from modern humans about 600 – 550 thousand years ago and lived in West Asia and Europe (farthest is southern Siberia). Meanwhile sapiens mainly lived in Africa. However, these two human species had hybridized with each other from 120 thousand years ago [159]. There is evidence of sapiens migration waves out of Africa from about 100 to 120 thousand years ago but were unsuccessful [160]. About 70 thousand years ago, sapiens left Africa to reach new lands in Asia and Europe, then humans existed and developed [161]. They had hybridized with neanderthals, and that hybridization led to the extinction of this human species.

The extinction of neanderthals about 40 thousand years ago did not come from fighting and competing for territory but from intermingling between the two human communities, and whoever had a more adaptive gene set would prevail. Modern humans (not in Africa) have about 1 – 4% neanderthal DNA but completely lack neanderthal mitochondrial DNA (genes only transmitted through the maternal line), showing that hybridization only occurred between neanderthal males and sapiens females. Hybridization between sapiens males and neanderthal females did occur, but their descendants were less successful in maintaining their lineage [162]. Pääbo also researched denisovan DNA and found that this gene appears mainly in Asian, Melanesian, and Papua New Guinean people (with the highest rate, about 4–6%) and some populations in America, but not in African people [163].

Decoding ancient human genes has shown that genes are not only genetic material but also traces, allowing us to trace the origin

and migration history of humans. Epigenetics – a scientific field researching how genes are regulated and expressed without changing DNA sequences – not only explains how cells can develop into different cell types but also clarifies the emptiness of life: even how genes are expressed has no fixed self-nature but depends on environmental conditions and epigenetic factors.

3.3.7 Epigenetics

All cells, except reproductive cells, have the same gene set located in the cell nucleus. The gene set is very conservative, only giving one-way commands from genes to the body, so how can the diversity of cell types, for example humans have more than 200 cell types, and environmental influences on the body be stored in genes? That is the content of research on *epigenetics* (epigenetics), also called extra-genetics. Chromosomes consist of DNA and proteins. DNA carries genetic genes and is very long, so proteins are needed to arrange them compactly in the nucleus. A protein like a tube for DNA to wrap around is called a *histone*. In turn, after DNA is wrapped around, histones arrange into a structural unit called a nucleosome, and nucleosomes coil to form chromosomes, thereby DNA is packaged compactly in the cell nucleus. The existence of proteins in the nucleus made scientists once think that proteins were the genetic material of cells. Now we know proteins are not genetic material but are coils to wrap DNA into neat structures fitting in the cell nucleus.

Histones in the nucleus not only play the role of a coil but also have a role in regulating gene activity. Although all cells have identical DNA, thanks to the effects of external agents either directly on DNA or indirectly through histones, gene inhibition or activation occurs differently. So how does the body control closing or opening genes? The most common method is using small molecules containing methyl $-CH_3$ to do this work.

Methyl molecules create epigenetic markers that prevent transcription, making genes not transcribed into RNA. Small molecules can also act on histones so histones do not unwrap DNA and therefore cannot transcribe and translate into proteins. Thus the body uses small molecules, commonly methyl, to directly lock bases or lock histones to suppress the development of a certain protein. By this principle, the body can control from a single gene set into specialized cells like heart cells, liver cells, muscle cells,... With this cell, only some genes are turned on, some genes are turned off. With other cells, some other genes are turned on, the remaining genes are turned off. Specialized cells form tissues, tissues form organs in the body. In 2015, NIH scientists published epigenetic factors for 111 human cell types [164] so we can understand how the body controls the body through turning genes on and off.

For example, a pair of identical twins have identical genes, but as they grow up, each person gradually becomes very different in physique and personality. In 2019, a study by NASA scientists with identical twin brothers who were two astronauts Scott and Mark Kelly. Scott was sent to the International Space Station ISS for nearly a year from 2015 – 2016 while Mark still lived on Earth. Because in space moving at high speed, according to relativity theory, Scott would be younger than Mark by about 6 minutes of lifetime. That time is too short to see differences between the two people. Scientists researched changes in epigenetic factors and found that epigenetic changes of about 800 genes of Scott were much faster than Mark's. This change disappeared when he returned to Earth [165]. That proves that although twin brothers have the same innate gene system, during development, epigenetics will have differences. If the difference is large enough, people can detect this difference through epigenetics.

What happens if Scott and Mark marry and have children? Will epigenetic inheritance accumulated during body development be transmitted to the children of these two people? Usually all epi-

genetic inheritance is washed away and removed during meiosis. Children receive genetic information from the father without epigenetic inheritance. That is in theory; in practice, there are some cases where parents' experiences leave traces on children's genes, even grandchildren's. In 2006, a study on living conditions and causes of death of a small village in northern Sweden named Överkalix proved that living conditions leave genetic traces for children and grandchildren, especially transgenerational inheritance [166]. This village recorded detailed genealogy and reasons for death of each person from the early 19th century. The birth rate of this village was very high, about 75 % of households had 5 – 10 children, the rest had more than 10 children. This is a village near the Arctic, so people's lives depend on weather and natural conditions. In the past there were periods of consecutive crop failures, sometimes up to 8 years in a row, but there were years of consecutive good harvests.

Because food shortages in this village were repeated many times in history, Swedish scientists decided to research how such food shortages affect later generations. Logically, the mother's food shortage will have major consequences for later generations, but besides that, people also saw that the father's meals have very important effects on children's health. Moreover, not only the father's living conditions but also the grandfather's affect later generations. Sons who were hungry before puberty then their grandsons have much better health and live longer than normal people. Conversely, sons who ate excessively at that age then their grandsons have 4 times the risk of diabetes than normal people.

Of course, men do not get pregnant, so the influence of fathers and grandfathers on children and grandchildren is certainly through sperm. But sperm carries DNA that does not depend on men's eating, so this influence is through epigenetics. Research also showed that not all periods of eating conditions affect epigenetics but only in the period before puberty does this have strong effects. At that age, the body develops slowly, before the body forms sperm, and when

sperm forms, factors causing epigenetic changes will appear and affect sperm. Although the mechanism of influence is still unclear and needs further research, this research shows that environmental influences and living conditions can also leave inheritance for later generations [167].

Besides eating, men's living habits also leave biological consequences for descendants. For example, if a man smokes before age 11, it is highly likely that his children, especially sons, have higher obesity than men who smoke at older ages. Smoking before age 11 still affects children even when he stops smoking later. Vietnamese people previously had the habit of eating betel nut, which has entered culture. Nutritionally, this is completely not good for their children's metabolism. Eating betel nut can cause oral cancer due to damage caused by chemicals [168]. Besides smoking and betel nut, some stimulants like cocaine, cannabis, alcohol, nicotine,... also have similar effects.

Thus, besides the gene set understood as the order of base pairs in DNA, epigenetics has very great importance to humans. Humans are an animal species with very high sociality; social activities may have made human epigenetics more numerous and stronger than other animals. Therefore, the number of human genes is much lower than many other animal species. In 2021, scientists decoded genes of lungfish (lungfish), a creature living in the Devonian period about 400 million years ago, the first creature to step on land, and were very surprised about its DNA. Fossils of Australian lungfish show they are very close to relatives of four-legged animals, with the largest gene set among animals, more than 14 times that of humans [169]. In 2024, scientists decoded the genome of South American lungfish still living today and were surprised by the huge size of 91 billion BP, 30 times larger than the human genome [170]. This is the largest genome in the animal world that humans have ever known. Jumping gene factors (also called selfish genes) account for up to 90% of BP number while that number in humans is only about

40%. However, length and number of genes do not determine the complexity of organisms if we acknowledge that humans have more complex structures than lungfish.

Human complexity may come from epigenetics. The number of genes is also important, but using genes flexibly is equally important. Epigenetics allows the human body to control turning genes on and off according to specific circumstances. The diversity of epigenetics comes from conscious activities of the brain and very high social activities of humans. The human gene set has created conditions for intelligence and collective life to appear, then in turn living habits affect genes back, but through epigenetics.

Epigenetics shows causal relationships between successive generations and across generations. If genes are usually conservative and do not change, then individual efforts throughout life will not leave any genetic characteristics for descendants. Do people who try to exercise to have good health or live healthy lives leave nothing positive for future generations? Fortunately, nature is not so indifferent and cold. Through epigenetics, efforts of an individual with certain advantages can leave characteristics through inheritance. This is also a manifestation of relationships between generations; no generation can be completely independent but lies in certain causal relationships. That is a manifestation of emptiness in genetics.

3.3.8 Gene technology

Gene technology (genetic engineering) includes two main techniques: (1) transgenesis – introducing DNA from other species, creating genetically modified organisms (GMO); (2) gene editing – precisely changing DNA, like CRISPR-Cas9.

To create a **transgenic** individual, people need to identify the gene to transfer. Usually, these are genes that help crops have good characteristics like resistance to pests, resistance to viruses, not affected by herbicides or pesticides, or higher yield and quality. This

characteristic is usually found in another plant species, so this gene needs to be isolated from the plant with that characteristic. With current biological techniques like polymerase chain reaction (PCR), people can do this relatively easily. Then, the isolated gene needs to be introduced into cells of the plant we want to transfer genes using a gene gun, creating holes with electric pulses (electroporation), or with Agrobacterium bacteria or plasmids.

Gene guns use high pressure to shoot metal particles carrying genes through membranes into cells. Electrophoresis uses high voltage to create holes on membranes, and genes can pass through. Agrobacterium is a Gram-negative bacterium causing tumors in plants. This bacterium has the function of carrying desired genes into cells to transfer genes. Plasmids are circular DNA molecules containing desired genes. These are molecules abundant in bacteria so they can exchange genes with each other. People use plasmids to carry genes into cells.

From when Watson and Crick found the double helix structure of DNA in 1953 until decoding the human gene map in 2003, scientists continuously sought ways to manipulate genes to serve human purposes. But everything seemed stuck for a long time because just manipulating genes with bacteria alone was already complex enough, not to mention manipulating genes with higher organisms. The complexity is not only in sequencing DNA, identifying genes, but in how to locate desired genes and change them at the molecular scale. This difficulty was solved when the CRISPR-Cas9 technique was developed and published in 2012–2013. In 2012, Charpentier and Doudna proved CRISPR-Cas9 could be used to edit genes in vitro, and in 2013, Zhang proved it could be applied in human cells. This gene manipulation technique created a revolution in gene technology, where scientists can **edit genes** to remove gene defects causing diseases, can revive extinct organisms, and even create humans with edited genes as a Chinese scientist conducted in 2018.

In 1989, Francisco Mojica, a biologist at the University of Ali-

cante in Spain researching genes of bacteria living on the coast, discovered that after each normal gene segment, there were 5 identical base groups, each group consisting of 30 bases. Especially, those 30 bases have reversible properties: forward reading order is the same as reverse reading order, called palindrome.[24] For example, we have a group of 6 bases in order –GAATTC–, its complementary bases are –CTTAAG– (because A pairs with T, G pairs with C). Complementary bases are precisely the original base group but read in reverse. Palindrome groups are connected by a spacer group to create palindrome-spacer pairs repeated many times, called CRISPR.

After publishing his discovery about CRISPR (clustered regularly interspaced short palindromic repeats), many other scientists also found similar DNA code segments in many other bacteria. In 2003, Mojica used BLAST software to compare spacer groups in CRISPR and discovered thousands of bacteria and dozens of viruses sharing that DNA segment. He found the DNA code controlling the bacterial immune system against viruses [172]. Although the mechanism of resisting viruses was still unclear, he provided evidence that bacteria had changed genes to resist viruses.

The mechanism of bacterial virus resistance was found by scientists of American company Danisco when they selected fermenting bacteria to produce yogurt [173]. They showed that bacteria resisted viruses in 2 steps: recognizing viruses and destroying them. Bacteria recognize viruses through DNA segments of spacer groups and consider them as fingerprints to identify viruses. If bacteria survive after virus attacks, these bacteria take DNA segments of virus spacers and store them in bacterial DNA. In this way, bacteria *remember* the type of virus that attacked them. This memory is not by virus shape but by a DNA segment of the virus. Characteristic DNA segments of viruses become spacer segments in CRISPR, which are fingerprints to identify viruses.

[24]Many science history researchers hold that Yoshizumi Ishino was the first to record CRISPR, but he did not know its importance [171].

Humans use eyes and brain to recognize and remember others. Eyes collect information and store it in the brain. When meeting someone, we also use eyes to receive images and search in the brain to see if that image is familiar. A graceful figure, a lovely face,... are examples of characteristics for us to recognize familiar people. Bacteria do not have complex structures like humans, so they use available things to do the above work. Bacteria recognize viruses through a DNA segment of viruses and remember with bacterial DNA. Spacer segments in bacterial genes are precisely copies of virus DNA. If spacer segments are too long, memory will consume many resources. If DNA is too short, there is not enough information to recognize enemy viruses. Therefore, about 25 to 65 BP is appropriate, long enough and accurate enough to do this [174]. Thus, bacteria remember enemy viruses on spacers of DNA. If a type of virus invades bacteria, bacteria use an enzyme named Cas9 to compare with remembered spacers. If the virus has DNA matching the spacer segment, Cas9 will cut the virus DNA, making the virus unable to replicate in cells.

CRISPR-Cas9 was extended to other organisms besides bacteria. The turning point in CRISPR research was the participation of two female scientists who won the Nobel Prize in 2020: Emmanuelle Charpentier at Umea University, Sweden, and Jennifer Doudna at the University of California, Berkeley in 2011.[25] These two scientists realized that Cas9's ability to cut DNA segments, previously only researched for bacteria, could completely be used for other organisms because cutting operations based on fingerprints written with ATGC characters are the same for all organisms. In 2012, Charpentier and Doudna proved that this technique could

[25]Mojica has the merit of discovering CRISPR and pointing out how bacterial immunity resists viruses by changing their genes; scientists of Danisco company pointed out the method of recognizing and destroying viruses of bacteria; Zhang proved genes could be adjusted in eukaryotic cells. However, Nobel Prize regulations only award to a maximum of 3 living people, so this has caused much debate in the scientific community.

completely be applied to humans [175], although only successful in vitro. This research opened gene editing technology including human genes.

In 2013, a young American scientist of Chinese origin working at MIT and Harvard, Feng Zhang, joined the race to use CRISPR-Cas9 technique to apply in the body, and from then on, a race occurred between Doudna's group and Zhang's group in researching and patenting human gene editing [176]. This competition was truly very valuable. Previously, treating genetic diseases only stopped at treating symptoms, but with gene technology, people can cure these diseases at the root. Not only treating diseases, gene technology can also edit genes of animals, for example pigs because of the strange similarity between pig genes and human genes.[26] People can adjust pig genes so the human body does not reject pig organs when transplanted into the human body. This technique can create organ transplantation technology from animals. Gene technology also allows creating animals that have recently become extinct, for example mammoths. People can adjust mammoth genes to adapt to a normal elephant so this elephant can carry a mammoth. And above all, gene technology can give birth to genetically modified humans to eliminate congenital diseases or to create humans with superior advantages. If possible, could eugenics rise?

Previously, when humans began organ transplantation or in vitro fertilization, ethicists had condemned excessive importance or influence of these technologies on society. To date, applying biotechnology like this has become normal and common. Using gene technology to manipulate individual human genes to create a human with advantages over others, is this ethical violation? Where is the boundary that humans can and cannot do? Have genes become the

[26] After decoding the pig gene map, people saw very large similarity between pig and human genomes. This is not because these two species have close ancestors (common ancestor of humans and pigs 80 million years ago) but similarity in evolution between pigs and primates.

final boundary of ethics? Certainly these questions cannot be answered immediately but need more time to know. But one thing we know is that humans are gradually intervening at the deepest level of life. In 2016, a Chinese scientist He Jiankui launched a project to edit genes in human embryos using CRISPR-Cas9 to change the CCR5 gene to create HIV resistance. At the end of 2018, He Jiankui self-announced that two baby girls with edited genes had been born, causing a strong wave of criticism from the international scientific community about ethics and safety [177].

Gene technology has caused concern for many people. For example, if possible, parents could adjust genes to have identical traits, leading to humans born later being similar, reducing diversity of human traits. Or if humans respect and honor intelligence, in the future they could modify genes to have super-intelligent people, creating superiority in class with remaining people. Such concerns are appropriate; however, we need to consider that human aesthetics themselves are also diverse. For example, many people like blonde-haired, blue-eyed girls and modify genes so their children have such characteristics. But when a certain aesthetic value becomes popular, people seek different aesthetic values rather than being like the majority.

In 2019, a study by British and American scientists with over 240,000 people showed that there are up to 538 genes related to human intelligence [178]. That is, intelligence is not determined by one gene or a few genes but is a combination of very many genes. Among them, some genes are also responsible for human lifespan and health. Moreover, intelligence depends not only on genes but also on living conditions and environment. Adjusting genes to have an intelligent child will be extremely difficult and certainly cannot be done in the near future.

The Covid-19 pandemic raged worldwide from late 2019 until 2022 ended thanks to vaccines. Currently there are dozens of companies successfully producing vaccines to serve the community. This

is a proud achievement of gene technology applied in preventing diseases caused by viruses. So what are vaccines, and how do they work? Vaccines are biological products introduced into the human body to train the body's immune system to resist disease-causing agents, for example viruses or bacteria. Biological products must have some molecules taken from disease-causing agents. Like preventing crime, police must have parameters about suspects for prevention to be effective. Those parameters could be photos, fingerprints, DNA, some habits,... to put into the control system.

The body's immune system does not have eyes to see shapes, does not have fingerprint scanners, but can only recognize disease-causing agents through molecules of these agents. Viruses and bacteria all have molecules for identification, which are precisely antigens (antigen), characteristic proteins usually appearing on surfaces. These are also keys for viruses to attach to cells and penetrate cell membranes. Without these proteins, viruses cannot harm cells. When completely new disease-causing agents attack the body, the body has some mechanisms to immediately resist this attack. But this process needs time, usually about $2 - 3$ weeks through reactions like coughing, sneezing, fever,... to create antibodies (antibody) to neutralize the activity of disease-causing agents.

Based on that principle, people produce vaccines according to 3 different principles equivalent to ways of training the immune system. These methods all rely on proteins, so they are called protein methods. First, inactivated virus vaccines (inactivated vaccine) are viruses that have died due to heat or acid but still have outer shells, so they can create immunity. Because they are not strong, these vaccines must be injected several times to prevent hepatitis A, rabies, flu (short-term), polio (short-term),... Second, attenuated vaccines (attenuated vaccine) are viruses still alive but greatly weakened, used to treat measles, mumps, rubella, chickenpox,... Third, surface antigen vaccines are a piece of pathogen shell but still have antigens so cells can recognize them. The above methods, though different,

must have proteins on pathogen surfaces, which are antigens for the body to recognize as foreign objects. Methods 1 and 2 have intact virus shells, while method 3 has shells broken into many fragments. When recognizing antigens, the body will *remember* those antigens to produce a special protein, antibodies, to resist pathogen activity. Protein vaccines have a disadvantage: the time from research to production is usually very long, about 4 years. Recently a new method was first used to manufacture SARS-CoV-2 vaccines, called the nucleic acid method carried by mRNA or DNA.

We already know that genes have two main functions: replication and creating proteins. The replication function is only performed when cells divide, while the protein creation function is a regular function of DNA. But DNA does not directly create proteins; it commands mRNA to do this, a process called transcription. From mRNA commands, tRNA carries bases to create amino acids. Amino acids combine with each other to form proteins; this process is called translation. Thus, proteins are created thanks to mRNA commands, so if we design information from mRNA, we can command cells to produce antibodies against bacteria. Remember that antibodies are a type of protein, so producing antibodies is also like producing other proteins, which is from mRNA. That is precisely the principle of manufacturing vaccines with mRNA that Katalin Kariko, a Hungarian scientist who immigrated to America, researched from the 1990s. However, this idea met many difficulties because when injecting virus mRNA into humans, cells will recognize this as a foreign molecule, so they will cause immune reactions to remove these chemicals.

Kariko's work met many difficulties because her proposal was not noticed or funded. Genetic techniques at that time were not developed, so people did not know how to replicate mRNA and how to deceive cells not to recognize mRNA as foreign molecules. Many years later, with technological development, the above problems were solved. DNA replication techniques using polymerase chain

reaction (PCR) allow replicating mRNA into many copies. In 2000, Kariko realized cells could be deceived by slightly modifying uracil nucleotides, then mRNA could pass through the immune system and not be rejected. That is precisely how Kariko thought to bring mRNA past the immune system to enter cells.

In 2005, a colleague Drew Weissman realized Kariko's idea was feasible, so together with her researched and found funding to establish an mRNA vaccine production company. In 2017, they succeeded when producing Zika disease vaccines in a very short research time. However, the mRNA method still had no place in the world vaccine industry because people had not seen the need to switch to producing by another method. Everything only changed when the Covid-19 pandemic occurred. Because the epidemic developed quickly and widely, vaccine production requirements also had to be fast. Previous protein vaccines needed at least 4 years to perfect products, while mRNA vaccines only needed about 1 year. In November 2020, Covid-19 prevention vaccines of Pfizer/BioNTech company produced based on mRNA of SARS-CoV-2 virus were certified to serve the community. Kariko is Vice President of BioNTech Company, currently valued at tens of billions of dollars. In 2021, Kariko together with collaborators Weissman and Pieter Cullis[27] together won the main prize of VinFuture[28] first time for their contributions to manufacturing COVID-19 mRNA vaccines. In 2023, they won the Nobel Prize in Biomedicine, a deserved reward for contributions of people who courageously and persistently pursued a scientific research direction full of risks and fatigue.

Through surveying gene technology, we have seen that humans

[27]The person who contributed to manufacturing lipid nano packages wrapping mRNA to prevent these molecules from being rejected by the body.

[28]VinFuture is a global science and technology prize with the mission "to create positive changes in the daily lives of millions of people by promoting scientific research and technological innovation," awarded for the first time in 2021.

can intervene in genetic processes to serve their purposes, from treating diseases to improving quality of life. All these scientific discoveries lead to a fundamental question: what do these understandings about life and evolution mean for Buddhist philosophy? This is precisely the issue that the final part of this chapter will survey – a synthesis of scientific discoveries with Buddhist philosophical principles about emptiness and dependent origination. This shows the emptiness of life at its deepest aspect: even scientific discoveries about life have no absolute self-nature but only have meaning in relationships with philosophical and ethical principles, a relationship showing the emptiness of both science and philosophy.

3.4 Buddhism with life and evolution

This chapter gives us the principle of dependent origination and emptiness manifested from the origin of life to the evolution process of organisms. Compositionality is manifested at all levels, from atoms forming life to chemical processes occurring between amino acid molecules, proteins,... to gene segments exchanged between organisms. Characteristics of atoms come from small molecules in cell nuclei, which in turn have a strange history in the universe. The origin of light atoms from furnaces in distant stars, to heavy atoms in an extremely brief moment of dying stars, shows the co-arising of countless causes and conditions seemingly infinitely distant until their manifestation in the body of each of us. Molecules assemble into genes – macromolecules like DNA carrying genetic information and using organism bodies as living machines to help them maintain existence and replication. Impermanence is manifested by continuous changes in interactions between atoms and molecules, making entities, at any observation level, only an instantaneous, approximate snapshot of reality. At any other moment, reality will be different.

* * *

In Buddhism, the Buddha and those who attained enlightenment are **Perfectly Enlightened Ones**[29] are the pinnacle of awareness and wisdom, knowing everything. They are not only seen as correctly enlightened people but also as **Omniscient Ones** (omniscience), seeing and knowing all dharmas, clearly understanding the operating principles of the world. More importantly, they not only enlighten themselves but also have full capacity and compassion to teach and guide other sentient beings to the path of liberation.

The capacity of Perfectly Enlightened Ones is transcendent, comprehensive, and unlimited awareness capacity, but it is not supernatural power but wisdom attained by eliminating ignorance. We need to note that omniscience is not currently knowing everything but has the capacity to know everything when needed, transcending all obstacles of space and time, only knowing what they want to know. This suggests that the entire philosophy of Buddhism aims to liberate humans from suffering, so the Buddha focused on human suffering. Other issues of all human life may not necessarily be issues he cared about. Thus, the Buddha's omniscience is perfect insight into the path leading to the cessation of suffering, not knowing all facts in the universe. Moreover, with the development level of human understanding of the world more than 2 thousand years ago, we cannot expect anyone to ask questions beyond the awareness of that era for him to answer.

The Buddha used language as a means, still taught the Dharma, but those words are only fingers pointing at the moon. The purpose of language is not to describe reality but to arouse intuitive seeing of truth in listeners. Therefore, Buddhist teachings can only be seen through direct experience, cannot be heard through language. Those who only stop at learning words without practice only understand

[29]Perfect (samyak) is perfect, comprehensive; Enlightened (sambuddha) is enlightenment, awakening, perceiving things and phenomena as their nature.

shadows of truth. Therefore, when we quote words in scriptures, we must remember that they do not encompass all truth. With such understanding, we will examine the origin of life and evolutionary biology under the illuminating eye of Buddhism.

However, the Buddha warned that the Dharma he realized is profound, difficult to see, transcendent, so language cannot fully express that reality. Listeners, though intelligent and wise, can only approach a small part of that truth through language. In Samyutta Nikaya (SN 12.23),[30] the Buddha said: "I do not always know everything at once, but when I want to know something, I immediately know that." Thus he only cared about important issues: suffering and liberation. Therefore, things he did not care about would not be taught to others.

On page 190 we already asked about contradictions between the origin of life and Buddhism's view of beginningless, endless reincarnation. In Samyutta Nikaya SN 15.1, the Buddha taught: "Monks, this reincarnation has no beginning. No first point can be found about sentient beings wandering and reincarnating, covered by ignorance and bound by craving." Reincarnation includes life, so the appearance, existence, and rebirth of all sentient beings (life) have no first point and also no final point. If omniscience knows all infinite knowledge in space and time at all different scales, it can be seen that beginningless, endless does not acknowledge that life has a beginning or end.

It can be understood differently that the meaning of time that humans of that era could observe differs from today's science. Certainly human knowledge more than 2000 years ago could not know how the universe and life were formed and developed, so beginningless, endless is only a long time compared to human lifetimes, not a specific value to compare with archaeological evidence about the existence of bacteria 3.4 billion years ago [179]. Beginningless, end-

[30] SN 12.23 is the abbreviation rule for a sutta in Samyutta Nikaya, belonging to chapter 15, sutta 23 in the Pali canon.

less refers to a very long time compared to human suffering life, not necessarily compared to the age of life, the age of the universe.

Another understanding is based on the observer's scale. Like an atom, from afar we see it like particles, but when approaching, i.e., magnifying space, blurriness begins to appear because the spatial boundary between the electron cloud (the shape of the atom is determined by the outermost electron cloud) and vacuum is difficult to distinguish clearly. The atom will not be like a bare particle but have a rather strange shape with very vague boundaries. When magnifying time, observing electron motion in atoms at very fast scales, not only is the electron cloud unclear, but they also appear and disappear without a specific position. Even how the observer (we are not discussing how to observe) will affect the results of observing electrons.

Our view of atoms described above is based on quantum mechanics, the best current theory. So has any theory described how life was formed and developed? Although many scientists believe life arose from the inorganic world in the panspermia theory, or the RNA world, they do not show us a complete picture but only scattered evidence, static snapshots at some moment in the distant past. Perhaps omniscience knows all those processes and sees that life, though appearing, is not sudden as we think. Life may evolve gradually: atoms create macromolecules that can self-replicate, a boundary created from phospholipids to create an isolated region more ordered than the surrounding environment, the merging of some isolated regions to give birth to the first bacteria,... Because not limited by space and time, omniscience will know all those processes occur according to necessary causes and conditions. Only with ordinary people, limited awareness capacity, do they see life suddenly arise. Like omniscience observing atoms sees that they have no boundaries, so does life; it has no starting point. The starting point (or ending point) is due to the finiteness of the observer's

awareness capacity, not reality. With that understanding, life does not naturally have a starting point; it continuously arises and ceases due to conditions, without a first cause.

If we accept that each living organism has a self manifested through a gene segment or a living machine is a bacterium, a body, then life can begin from an inorganic world on Earth or somewhere in the distant universe. The panspermia theory hypothesizing that life exists everywhere in the universe then lands on this Earth seems consistent with Buddhism. Perhaps on some planet, with special conditions, life arose. That hypothesis is only correct when separating soul and body because if souls can easily travel between distant planets, they do not form or are carried by matter we have known, according to which, nothing moves faster than the speed of light in vacuum.

If life has no self and self-replicators only exist momentarily, living machines are only tools co-arising due to causes and conditions, then it would be truly meaningless to discuss the beginning and end of an existence if all existence is empty in its nature. According to Buddhism, creation theory and evolution as a discussion about the beginning of existence is a wrong view. Buddhism denies the self of things but instead argues about mutual causal relationships. According to causal relationships, self and world, body and mind are not divided but understood in mutual causal relationships, mutually influencing each other [180].

When Darwin's evolution theory appeared, Buddhism easily accepted scientific theories with great tolerance because from the beginning, the Buddha held that humans as well as other organisms are equal, perhaps with a slight advantage in practicing and following the Noble Eightfold Path. Buddha is in the heart of all people, in the heart of all sentient beings in the world.

The compositionality of life comes from special arrangements of specific forms of matter. That matter in turn comes from within

distant stars thanks to nuclear reactions and supernova explosions. Activating epigenetics depends not only on a specific gene but also on coordination of other genes. The combination of genes from many different sources shows that genes have no fixed self-nature. Instead, genes exist and change depending on evolution processes and complex genetic events. Of course, accompanying compositionality is impermanence manifested by continuous changes in interactions between components. These results show that life at the level of DNA molecules, of individual organisms, all have compositional and impermanent properties. Evolution theory shows that all organisms are connected to each other through a common ancestor network and depend on ecosystems; no species, no individual can exist independently, self-existent, and immutable.

That is the emptiness of life.

Chapter 4

Human

Abstract. Emptiness is manifested in the human body at three levels. (1) The human body is a symbiotic system between humans and hundreds of trillions of other microorganisms, mainly in the intestines. These microorganisms help us digest food; support the immune system to resist attacks by viruses, bacteria, and parasites; support the nervous system (also called the second brain). (2) Most human body cells have mitochondria, which are foreign bacteria that invaded billions of years ago and have now become organelles. Mitochondria are energy factories providing energy for cell activity, are timekeepers in programming cell death, and even determine the development of human embryos. (3) Human DNA only has about 20 % related to genes expressing proteins; the rest are foreign genes co-symbiotic in the human body. At all three levels of body, cell, and DNA, the human body is a symbiotic ecosystem with other living entities. They depend on each other to exist through cooperation or competition. They cannot exist independently, separately.

4.1 Human body

> "This is not mine,
> this is not me,
> this is not my self."

— Samyutta Nikaya (SN 22.59)

According to Buddhism, the perception of an unchanging and independent self is the greatest mistake of humans; that is ignorance, the cause of all suffering in life. The principle of dependent origination shows the mutual interdependence of all things and phenomena manifested in two dimensions of time and space necessarily leads to the impossibility of an unchanging and independent self existing. This chapter shows the dependence of the human body on other organisms at three levels: body, cell, and molecule. At all three levels, our attempt to create a boundary to define the self, distinguishing us from outside us, is not easy. Moreover, even if we can define the self, we realize that self cannot exist independently. Everything lies in **symbiotic** (symbiosis) relationships with each other, one depends on the other.

In 2024, Elon Musk and his SpaceX corporation are realizing dreams of conquering space, sending humans to Mars to realize a great dream from ages past: creating an interplanetary intelligent life. ChatGPT from OpenAI, launched just two years ago, has changed the lives of hundreds of millions of people due to the intelligence of this artificial intelligence. To date, humans are the only animals on Earth realizing the dream of interplanetary life and "outsourcing" their intellectual capacity outside the physical body.

Despite such advances, many things within humans remain unclear. Perhaps scientists are influenced by rational thinking, analytical thinking, separating the research object from the subject, so many problems within humans are often overlooked or not given appropriate attention. The human body and mind that we see daily are still great mysteries to explore. That is precisely the goal of Buddhism and some Eastern philosophies when considering humans themselves as objects to understand alongside understanding the world around us.

The first thing we previously misunderstood is that the body

belongs to each individual and we have supreme authority over the material and spiritual parts of that body. The origin of suffering is precisely self-attachment, i.e., stubbornly believing there is a self of one's own. The principle of dependent origination leads to no-self, the complete denial of the opinion that things and phenomena have a kind of self-existent, independent, permanent, and unchanging nature.

Let us look at ourselves in the mirror. Consciousness of the self manifests in two aspects: material and spiritual, corresponding to two views: objective and subjective. In terms of material, the human body represents the physical self. The physical self has objectivity through characteristics such as structure, appearance, and biological functions, lying in interaction with the external environment and following natural laws: birth, aging, sickness, and death. In terms of spiritual, consciousness of the I is the mental self, having subjectivity with thoughts, emotions, beliefs, and experiences. The mental self determines the continuity of an individual through the continuity and cohesion of personal experiences. Below we will see the inseparable connection of the physical self of humans with other organisms.

4.1.1 Physical human always renewed

Matter creating the human body is always renewed but consciousness always maintains an unchanging self. The perception that the body is ours and almost unchanged needs to be seriously examined. Over time, the human body develops from small to large, and we see that development. From a child, humans gradually grow up with the care of parents and family to mature, age, and die. Birth, aging, sickness, and death are the laws of life, the four manifestations of physical suffering in Buddhism. That change can be easily observed and acknowledged.

But there are changes not easily recognized. The body always

renews itself by replacing matter from outside. In terms of mass, an adult man aged 20 – 30, weighing 70 kg has about 25 % liquid which is plasma outside cells, 7 % solid which is minerals like calcium forming bones, and 68 % substances forming 30 trillion (30×10^{12}) cells. In general, large cells live longer than small cells, so matter replacement is mainly created by small cells. For example, muscle cells weighing 1 – 10 µg have a lifespan of 30 – 70 years, intestinal epithelial cells weighing 1 ng have a lifespan of 3 – 5 days, red blood cells weighing 0.1 ng have a lifespan of 120 days.[1] Blood cells and intestinal wall cells have the shortest lifespan, so these two cell groups create most of the body's replacement matter with corresponding proportions of 49 % and 41 %. The remaining mass includes 4 % skin cells, 4 % fat cells, and 1 % muscle cells.

About every year and a half, an amount of matter equal to body mass is completely replaced (not 7 years as some previous publications). In terms of quantity, fat and muscle cells contribute very little, only 0.1 %, because they are large and have long lifespans. Red blood cells account for up to 87 % of replacement quantity because they are very small and have short lifespans. About every 80 days, our body creates a new amount of cells equal to the total number of cells currently in the body. The total daily replacement matter mass of the body is 80 ± 20 g, equivalent to 330 billion cells, of which 90 % are blood cells and intestinal epithelium [181].

Despite renewal, structure, shape, and function do not change, so we almost do not see that change. Although peripheral nerve cells can regenerate when damaged, central nervous system cells almost cannot regenerate and they do not change throughout life [182]. Our physical body is always replaced by new matter, but consciousness always maintains the self for each person. Is it because the brain creates consciousness, so they cannot change?

This may be a clue about the origin of consciousness. Dualist

[1] $1\,\mu g = 10^{-6}$ g, $1\,ng = 10^{-9}$ g.

theory holds that consciousness and body are separate from each other. Consciousness only temporarily resides in the body. If so, even if the body changes, it does not affect consciousness. Changing cells in the brain without affecting consciousness – the brain has already done that. It is not random that cells must die and be renewed. The cycle of birth, aging, sickness, and death at the cellular level helps organs operate with maximum efficiency. Old, damaged cells need to be replaced and supplemented [183]. But that process is not applied to central nervous system cells. Research on memory by Nobel Prize-winning neuroscientist Eric Kandel [184] shows that human memory is stored at neural synapses. Not only are neurons important, but synapses between them are also important for memory, and thus for consciousness.

The symbiosis theory holds that consciousness appears as a consequence of interaction between many nerve cells with each other, from which consciousness appears as a phenomenon completely different from the properties of individual cells [185]. If so, the existence of a nerve cell is not too important, but the interaction of a large number of cells is important. Renewal (if any) would therefore not affect consciousness. *Central nervous system cells do not change throughout human life* suggests to us the importance of each nerve cell to consciousness. Surely the matter creating central nervous system cells has an irreplaceable role in forming consciousness, so the body does not renew these cells. In other words, **consciousness is bound to the matter forming it** (here is the matter creating cells in the brain and spinal cord) throughout human life. In science fiction stories, people think consciousness can be moved or copied from one person to another using an advanced machinery system. But things are not that simple. Moving consciousness accompanies nerve cells, not simply copying some kind of information separated from those cells. This shows that copying consciousness seems difficult. Advanced technology in the future may accurately copy cells from one person to another through technology, for ex-

ample, 3D printing with detail down to each molecule, then copying consciousness might be feasible.

The continuous change of matter in the human body is clear evidence of the emptiness of the physical self. "My body" is not a fixed, unchanging entity but a continuous flow of matter. About every 18 months, all matter in the body is completely replaced, but we still perceive the body as unchanged. No-self is clearly manifested here: there is no fixed, independent physical self. The body is only a temporary collection of constantly changing components, maintained by continuous metabolic processes with the external environment. An unchanging self is a mistake of consciousness.

4.1.2 Role of intestinal microorganisms

The above section showed that the human self always changes over time through continuous matter replacement. But the emptiness of the self is not only manifested through the time dimension but also clearly through the space dimension. The above section showed that the human self always changes over time. Below we will see that in space, the human body is a fertile world for various microorganisms to **symbiotically coexist** (symbiosis). On the body of an adult weighing 70 kg, there are about 30 trillion cells belonging to 200 different types, but there are up to 100 trillion microorganisms including bacteria, archaea, viruses, and fungi [186]. They form a microbiome (microbiome) symbiotic with the human body. Like cells in the human body, microorganisms always change. Each day about 100 grams of microorganisms are replaced [181], corresponding to nearly 50 % of symbiotic microorganism mass.

The term parasite often has a derogatory nature, meaning only trying to live off the host; that is a mistaken view. Modern research shows that most symbiotic microorganisms on the human body are bacteria. They have important roles for humans in many activities of the nervous system, cognitive processes, social relationships, de-

velopment processes, and psychological issues. The role of intestinal bacteria is to help break down food, absorb vitamins, minerals, and strengthen the immune system. Recent research also shows that intestinal bacteria affect human psychology and neurology.

It is time to count all cells and microorganisms as our body because of the important role of microorganisms for functional activities of the body. In terms of quantity, bacteria outnumber cells, but in terms of mass, because bacteria are very small, bacteria only account for about 0.2 kg, equivalent to 0.3 % of body mass [150]. Bacteria inhabit the entire body but most are on surfaces. Not the skin has the most surface area but the intestines. The surface area of the intestinal system of an adult is very large, about $250 - 400\,\mathrm{m}^2$. In a human lifetime, nearly 60 tons of food have passed through the digestive system along with countless types of microorganisms from the external environment [186, 187]. The colon is where bacteria density and quantity are highest, hundreds of times more than other organs.

The symbiosis of microorganisms with the human body is divided into several types. First is **mutualism**, referring to long-term relationships between different species, where one species brings benefits to the other. For example, the bacterium Ruminococcus bromii (*R. bromii*) in human intestines has the function of breaking starch into smaller molecules so the human body can absorb, which is a form of symbiosis with humans [188]. Without such bacteria, humans cannot digest starch. The bacterium *R. bromii* does not do this for free; it uses part of the energy in that breaking process for itself. Each organism enjoys a bit and contributes a bit from the energy pie. Second is commensalism, referring to relationships between different species where one or some species benefit but other species do not benefit but also do not suffer harm. A large number of intestinal bacteria have no or insignificant roles for human body activities [189]. Third is competition, referring to relationships where species must compete to survive. In competitive battles, many scenarios

can occur depending on the strength of participating parties, such as fair competition or one-sided competition (amensalism where one side dominates and other sides are disadvantaged) [190]. Fourth is parasitism, referring to relationships where one side benefits while the host usually suffers harm. The human body has many parasitic species from protozoa like malaria-causing viruses, amoebic bacteria causing dysentery, multicellular animals like worms.

Most of us have aversion to microorganisms because they are often related to infectious diseases. This is not a bias of ordinary people but comes from human history of researching and fighting epidemics. In 1890, Robert Koch put forward the principle named after him, attributing infectious diseases to viruses and bacteria. Indeed, diseases causing many deaths in the 20th century like typhoid, cholera, tuberculosis, and flu are all related to them. An important achievement in fighting those epidemics is finding antibiotics, first penicillin found by Alexander Fleming in 1928. Not only helping fight infectious diseases, antibiotics also help treat wounds in war. The use of antibiotics today is overused. When taking antibiotics, not only disease-causing bacteria are killed but symbiotic bacteria in the microbiome also die. This causes body imbalance, leading to other problems [191]. Later, whenever we must take antibiotics, we should think about the drug's effect on the intestinal microbiome to make the right decision.

The symbiotic relationship between humans and microorganisms is a typical example of emptiness in biology. The boundary between "our body" and the external world becomes blurred when we realize that our body not only includes human cells but also hundreds of trillions of microorganisms. These microorganisms are not foreign invaders but indispensable components of the living system. We cannot exist without them, and conversely, many microorganisms also cannot exist without us. This is precisely the manifestation of the principle of dependent origination: no entity can exist independently. Emptiness is manifested in that there is no absolute

boundary clearly distinguishing the human body from the external environment. The body is a complex ecosystem, a network of interdependent relationships between different components.

4.1.3 Microorganisms help digest food

After understanding the presence and general role of microorganisms in the human body, we need to go deeper into specific functions they perform. The first and most important function is supporting the food digestion process – a process without which, without microorganism assistance, humans would not be able to absorb most nutrients necessary for life. Why do bacteria mainly concentrate in the intestines? To answer this question, we acknowledge that all organisms need energy for their activities, so where there is energy, organisms will develop there. It is very surprising that some very common foods humans eat daily are fiber (cellulose), mainly carbohydrates abundant in grains and fruits, yet they are not direct food for humans in the sense that the digestive system can absorb immediately like sugar and milk. This food needs some enzymes like cellulase to break large molecules into small monomers so our body can absorb. Intestinal microorganisms can produce enzymes to digest fiber, mainly by fermenting food. Products of microorganism digestion are food for us absorbed by the intestines. Intestinal bacteria have consumed about 10 – 15 % of food energy; the rest goes to humans. It is not an exaggeration to say that humans must eat the waste of microorganisms.

When food enters the body, in the process of breaking food from large, high-energy molecules into small, low-energy molecules, this occurs in many stages. Bacteria take advantage of each stage to seek energy for themselves. Let us look at an example when the body digests daily rice. Rice contains much starch (75 %); when chewed in the mouth, the enzyme amylase breaks down part of starch into sugar, so if chewed well, there will be a sweet sensation. The pres-

ence of this enzyme in the mouth is due to a DNA segment of a virus inserted into human genes (see details on page 262). When food goes down to the stomach and small intestine, starch is broken down through many stages into glucose chains, maltose, and glucose monomers. At glucose monomers, the body will absorb into blood to bring to cells. Each time breaking molecules requires the assistance of enzymes, which are proteins that assist reactions breaking molecules. These enzymes are partly produced by our body but mostly by trillions of bacteria living in the intestines.

For example, when researching digestive enzymes for polysaccharides, common food for humans including starch in grains, cellulose in vegetables and fruits, and glycogen in meat, people found 97 types helping break these macromolecular chains. Through DNA research, scientists determined that at minimum 8 enzymes and at maximum 17 enzymes have DNA encoding in the human body. The rest, i.e., at least 80 enzymes, are not produced by the human body but must rely on bacteria. Through modeling, scientists discovered 177 bacterial families, most numerous being Firmicutes (104), Bacteroidetes (29), Proteobacteria (22), Actinobacteria (12) [192]. It can be said that without intestinal bacteria, we can only digest a very small part of the food we eat daily. We can also reason that without bacteria, humans could choose only directly digestible food types, but if so, choices would be very limited and human life would certainly be very different from today. Here we do not discuss expanding food choices but only want to emphasize mutual dependence between biological kingdoms. Humans with the most complex structure have evolved to coexist with bacteria for a long time, so mutual dependence is a universal law that Buddhism calls dependent origination.

The existence of intestinal bacteria is necessary for normal body development. Newborns can only breastfeed in the first stage of life partly because the microorganism system in the digestive tract is not fully developed, although about 3/4 of microorganisms come from the mother. Even in these early stages of life, the role of

bacteria is very important. We know that breast milk has many important nutrients for young children. Fatty acids are one such substance. There is a type of fatty acid that greatly affects brain development: omega-3. This fatty acid affects bacterial composition and thereby affects brain development. Thus, the health of young children's brains depends on what intestinal bacteria eat. Another compound is polyphenol, which dieters often mention, abundant in grapes, tea, chocolate,... having anti-aging effects and enhancing memory ability. Those effects are not due to us directly absorbing polyphenol but because intestinal bacteria digest polyphenol to create other compounds causing the above effects. We depend on what we eat. Or more accurately, we depend on what symbiotic bacteria eat.

Thus, microorganisms are not simply beneficiaries from food; they play an active and indispensable role in converting food into energy and nutrients. However, the role of microorganisms does not stop at digesting food. Recent research shows they also have profound effects on other important body systems, especially the immune system and nervous system – two systems determining human survival ability and quality of life.

4.1.4 Microorganisms coordinate with immune system and nervous system

Microorganisms coordinate with the immune system, affecting health. Microorganisms also support the body's immune system. First, microorganisms are the body's first line of defense. Occupying survival space and using energy from food prevents disease-causing microorganisms from attacking the body. Competition from beneficial bacteria significantly reduces the probability of invasion by other bacteria, especially in early life stages. In addition, the microbiome helps train and regulate immune responses. Some microorganisms produce chemicals that can promote or inhibit the

activity of white blood cells and macrophages, helping them protect the host more effectively [193]. For example, archaea present in symbiotic microorganisms in the human body can recognize viruses attacking the body. If the virus has DNA already recognized by archaea before, they use CRISPR-Cas9 to cut the virus DNA [194]. Thus, archaea destroy viruses attacking the human body. A valuable repayment to the host.

Humans are a social species, manifested in symbiotic microorganisms. People living physically close like family members, friends, even neighbors can clearly affect each other's microorganism composition. Not only does the mother's microbiome affect the child during pregnancy and birth, but the microbiomes of spouses tend to become similar. A study on 10 million people showed up to 50 % of microorganism strains are the same between mother and child (not limited to newborns but also older ages), 12 % and 32 % of intestinal and oral microorganism strains are the same between spouses. A kiss affects oral microorganisms, leading to the presence of microorganism strains in partners [195]. Conversely, the symbiotic microbiome can affect our choice of partners.

Intestinal microorganisms support the nervous system. Besides digestion, probably few can imagine that tiny microorganisms living in our digestive tract can affect fear responses, affect human cognition and emotions. Bacteria can control cognition and emotions, affect mood, cause mental disorders, and even information processing [196]. Westerners have the mistaken perception that emotions come from the heart. But that is completely incorrect; the heart only pumps blood and does not affect human emotions. But the digestive system is different.

In Vietnamese, there are phrases *đau lòng lắm* (very heartbroken) or *phải lòng nhau* (falling in love) to indicate a state of sadness, disappointment, or love. Why do sadness and love relate to the digestive system in the intestines? In 2019, scientists studied two groups of mice: one group after birth developed normally with

full bacteria in the intestines, one group was given antibiotics and isolated to have no intestinal bacteria. The experiment was conducted as follows: researchers rang a bell then shocked both groups of mice so they knew that after the bell was an electric shock. This was repeated so members of both groups learned fear when the bell was sounded. After some time, researchers rang the bell but did not shock anymore to see if the ability to learn to forget, in this case not fearing the bell, differed between members. Results showed that mice with full intestinal bacteria quickly learned to forget fear while mice without intestinal bacteria could not forget fear [197].

Previously, people knew that the intestines and brain are connected by the vagus nerve, so scientists tried cutting this nerve in mice, but results showed the vagus nerve is not the only connection between intestines and brain. The presence of intestinal bacteria plays an important role in brain activities. Research found four digestive compounds produced by bacteria that affect nerves are the cause of the ability to forget fear in mice. Intestinal bacteria created compounds, and these compounds help the nervous system of mice. The group without intestinal bacteria did not have these compounds [197]. Digestive compounds are not caused by one or some types of bacteria but by many types. Bacterial diversity in the intestines is extremely important for normal body activities. If this diversity is changed, the host body will suffer very adverse psychological effects similar to psychological trauma or depression. Mice without bacteria when grown were transferred to live with mice with bacteria; after some time, they also had bacteria in intestines, but the ability to forget fear was also disturbed. This shows that bacterial diversity must be developed at the right time, especially when young. This research can be completely generalized to other learning abilities and other species like humans.

One of the important roles of microorganisms is producing neurotransmitters, which are chemicals with very important roles in inhibiting or stimulating neural signals, and are chemicals control-

ling human emotions [70]. Most neurotransmitters like glutamate, GABA, serotonin, and dopamine are synthesized through intestinal microorganisms as intermediaries. The microbiome can create precursors or catalyze to create neurotransmitters. Then compounds will follow the circulatory system or the gut–brain axis to reach places in the body, especially the central nervous system [198].

The enteric nervous system (enteric nervous system – ENS), the largest part of the peripheral nervous system, is precisely the brain of the intestines, also called the second brain of humans because it includes about 600 million neurons along the length of the intestines, from the esophagus to the rectum, and can operate independently without direct control from the brain [199]. ENS produces and regulates neurotransmitters like serotonin, an important neurotransmitter involved in regulating mood, sleep, and digestion. It also produces a significant amount of dopamine and other neurotransmitters. Changes in the intestinal microbiome can affect mood and behavior. Bacteria in the intestines produce chemicals that can affect brain activity, contributing to the development of disorders like anxiety, depression, and even other neurological disorders. Thus, the human brain is not the only factor determining emotional states; the enteric nervous system, in which intestinal microorganisms play a strong role in these sensations.

Besides bacteria, there are many viruses in human intestines. A 2021 study by European scientists using gene analysis methods showed there are up to 142,809 virus species living in human intestines, and about half of them we have never known [200]. Most of these viruses have DNA cores rather than RNA cores like the SARS-CoV-2 virus. All samples were taken from healthy people, showing that not all viruses are harmful but even have important roles for intestinal activity. This result shows that human intestines are truly a world where various microorganisms, including viruses, reside and develop. Biodiversity in terms of virus species can greatly affect human health.

The symbiotic microbiome on the human body in particular and on other animals and plants in general is not only a one-way beneficial relationship but in many cases is symbiosis beneficial to both sides so that microorganisms and host body develop. This symbiosis is so closely bound that it cannot be separated. If we try to separate, even the host will not be able to survive or will seriously affect normal activities of body functions, so individuals have little chance of survival and inheritance for the next generation.

Microorganisms not only help digest food in human intestines but also in other animals. Ruminants like buffalo, cows, goats, sheep with main food being grass rely on microorganisms like fungi and anaerobic bacteria, archaea, protozoa. These microorganisms ferment and break cellulose bonds to create glucose and some nutrients absorbed by animals [201]. Other species like humans do not have microorganisms that can ferment grass, so they cannot digest this type of food. In the fermentation process, a byproduct created is methane gas CH_4 [202]. This is a colorless, odorless gas causing greenhouse effects stronger than carbon dioxide by at least a hundred times. Ruminants release gas into the environment by belching. Fortunately, the lifetime of a molecule of this gas is only about a few decades while carbon dioxide lifetime is up to hundreds of years. But a few decades is enough to heat Earth. Therefore, the process of producing 1 kg of beef emits an amount of greenhouse gas equivalent to 18.21 kg CO_2 [203], equal to a car running 100 km.[2] If you hesitate between beef phở and chicken phở, you should choose chicken phở because 1 kg of chicken only emits 4.02 kg CO_2. We can reduce greenhouse effects with a simple choice. If you want to reduce more, choose eggs (3.02 kg), fish (2.83 kg), or vegetables and fruits. The relationship between the symbiotic microbiome and the human body is not limited to the above issues. There are many other processes like synthesizing vitamins B, B12, K, thiamine,...

[2]Burning 1 L of gasoline emits 23.2 g CO_2. If a car consumes about 8 L/100 km, it is equivalent to 18.5 kg CO_2.

that need assistance from symbiotic microorganisms.

Thus, at the body level, we have seen the emptiness of the physical self: the human body is not an independent entity but a complex ecosystem, a network of symbiotic relationships between humans and hundreds of trillions of microorganisms. However, symbiosis does not stop at the body level. When we go deeper into internal structure, from body down to cells, we will discover that even at the cellular level – the basic unit of life – symbiosis continues to manifest clearly and profoundly.

Next, we will examine a type of symbiotic microorganism, not on surfaces of the human body but inside human cells. If separated, cells will die. That is mitochondria and chloroplasts.

4.2 Human cells

Normal activity of human cells and many other animal cells depends on an organism that was previously a bacterium but has now become an extremely important organelle: mitochondria. Cells are the building blocks of life; they all have 3 parts: (1) cell membrane to separate cells from the external environment; (2) cytoplasm including cytosol and organelles; (3) genetic material is DNA. Genetic material can be wrapped by a membrane to form a nucleus. Cells with such a nucleus are called eukaryotic cells to distinguish from DNA not wrapped inside a nucleus, called prokaryotic cells. Eukaryotic cells have complex structure and are present in animals and plants. Human cells are eukaryotic cells (Figure 4.1). Each cell carries complete genetic information of the body. They can divide independently, so when cells divide, genetic information is replicated. Cells are so basic that we often think they are entities characteristic of the body. If large bodies have symbiotic microorganisms to operate together, at the most basic level, cells do not need symbiosis for their activities. But that thinking is mistaken. Even within cells,

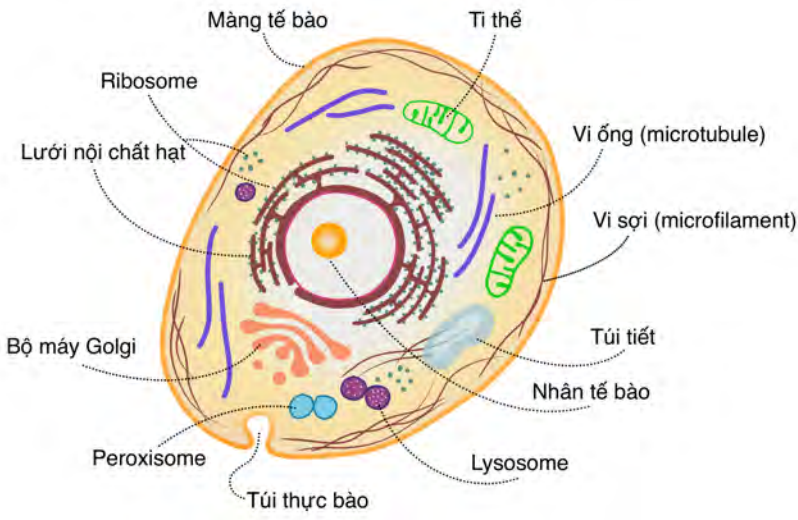

Figure 4.1: Structure of a typical human cell.

there are symbiotic components that have become indispensable elements of cells: mitochondria and chloroplasts.

4.2.1 Mitochondria – foreign bacteria

In eukaryotic cells, there is an organelle that looks like a bean floating in the cytoplasm, which we call **mitochondria** (mitochondrion, Figure 4.1) with an extremely important function: producing ATP (adenosine triphosphate) – fuel for cell activity through the *Krebs cycle*.[3] The Krebs cycle is an important part of cellular respiration in eukaryotic organisms. This cycle occurs in mitochondria and plays a main role in converting carbohydrates, proteins, and lipids into energy in the form of ATP.

Mitochondria are foreign organisms that invaded most human cells and became organelles holding many important roles in the body. Mitochondria have proteins to convert glucose into ATP through

[3]Hans Krebs (1900 – 1981), German-Jewish British biochemist, famous for discovering 2 important metabolic cycles: the urea cycle and the Krebs cycle. He won the Nobel Prize in Physiology or Medicine in 1953.

cellular respiration. This process requires oxygen, so it is called an aerobic process, but differs from combustion in that energy is released through a series of reactions (glycolysis, Krebs cycle, electron transport chain) rather than exploding heat like ordinary combustion reactions. The number of mitochondria in different types of human cells is not the same. Red blood cells do not have mitochondria, but liver cells have up to 2000 mitochondria. In general, cells that need more energy have more mitochondria.

After nutrients are assisted by intestinal microorganisms to become short molecules like glucose, they permeate through the intestinal membrane to enter blood vessels bringing them to body cells. The energy that glucose brings is precisely equal to the process of burning to create water and CO_2. However, in cells, this reaction does not occur directly, partly because combustion reactions need large activation energy, but through a series of processes to create ATP. Only when glucose is converted into ATP can cells directly use it. The glucose conversion process requires mitochondria – the energy factory of cells. Thus, when glucose enters cells, the body still needs other microorganisms to convert.

Mitochondria are not only banks providing energy currency for cells but also determine cell life and death. The above section presented the body renewing matter about every year and a half. That is, cells when born are already programmed to die. Once again, birth, aging, sickness, and death are not only manifested at the body level but also at the cellular level. Here, death is precisely mitochondria [204]. Programmed cell death (apoptosis) is a mechanism by which the body regulates biochemical processes to remove dead, infected, or unwanted cells. Removing some cells so remaining cells survive better. The body uses a protein called caspases to destroy cells. Activating caspases has two ways: external and internal. Mitochondria control the entire internal process by releasing cytochrome c to activate caspases. This mechanism was found by Kerr in 1972 [205].

Mitochondria are an excellent example of emptiness: once independent bacteria, but through billions of years of evolution, they have become an inseparable part of eukaryotic cells. The boundary between "us" and "not us" is completely blurred. Mitochondria are no longer foreign organisms but have become essential organelles of cells. This shows emptiness in space: there is no absolute boundary between living entities. All boundaries are relative and can change over time through evolutionary processes and interactions. No entity has fixed self-nature is manifested here. Mitochondria are not absolutely "bacteria" or "organelles" but are both, depending on how we view them and the context we are examining.

Some research shows mitochondria affect homeostasis through modulating bioenergetics, redox balance (through producing reactive oxygen species – ROS), absorbing and regulating calcium and epigenetic genes. Mitochondria regulate cell death pathways, which is very important to allow stem cells to preserve genome integrity and function of differentiated offspring. Research on the role of mitochondria with stem cells is still nascent and needs more experimental verification [206].

In a 2023 study, Harvard University scientists found another important function of mitochondria: controlling human embryo development [207]. Research showed that early human and mouse embryos are very similar, but then mouse embryo development increases rapidly while human embryo development increases slower by 2 – 3 times. The factor controlling fast or slow development of human embryos is mitochondria. Many mitochondria help embryos develop fast and vice versa. Mitochondria may participate in gene regulatory networks (gene regulatory network) to control embryo development time. Note that mitochondria are only inherited through the maternal line, so mitochondria here are precisely the mother's mitochondria. Mitochondria are like a clock signaling body development from the first days. Thus, mitochondria have the following important roles: (1) energy factory of cells, (2) programming cell

death, (3) determining embryo development speed.

Mitochondria are foreign organisms that invaded human cells and became organelles. In 1966, Lynn Margulis[4] was the first to propose the hypothesis that mitochondria and chloroplasts (chloroplast, also called chloroplasts) are descendants of independent bacteria. That hypothesis was too bold, so it was rejected by about 15 journals; only when one journal boldly published it did her idea gradually get verified and is now widely accepted [208]. The bacterial origin of mitochondria is manifested through many characteristics: (1) they have circular DNA like bacteria and their own protein synthesis mechanism; (2) they develop and reproduce independently from the rest of the cell. When reproducing, mitochondria genes mainly come from the mother because only maternal mitochondria genes are replicated in embryos. Mitochondria from the father also reach the egg but are marked and destroyed.

Because mitochondria are only inherited through the maternal line, scientists have used this characteristic to research human evolution. Analyzing human mitochondrial DNA, scientists found that human genes only differ by about 0.1 %, a number much lower than other primates, because we all share a common mother living 170 thousand years ago. Research by international scientists conducted in 2019, analyzing mitochondrial genes of more than 1200 indigenous people currently living in southern Africa today. They analyzed many lineages of a gene named L0 of mitochondria to go back to the past to find when those lineages began to converge. Because mitochondrial DNA is only transmitted from mother to child, this search is called finding the mitochondrial mother (mitochondrial Eve). Research results showed that the mitochondrial mother lived

[4]Lynn Margulis (1938 – 2011), also known as Lynn Sagan because she was once the wife of famous astronomer Carl Sagan, American biologist with many contributions to developing endosymbiotic theory in evolution of eukaryotic organisms. She held that mitochondria and chloroplasts are descendants of independent bacteria. She also supported Gaia theory viewing Earth as operating like a self-balancing entity.

in a region with an area of about 120,000 km^2, which today is the Kalahari Desert but was then fertile land [209].

People think that about 2 billion years ago, a small cell entered another larger cell, and from then they symbiotically coexisted. The process of one cell living inside another is called the **endosymbiotic** process. The cell inside is called the endosymbiotic cell, abbreviated as the endo cell. The cell outside is called the host cell.

Why mitochondria became organelles of cells is still unclear. Perhaps the host cell tried to eat the endo cell but could not digest it, so the endo cell got stuck inside the host cell. Or the endo cell tried to live parasitically in the host cell to avoid being eaten or to exploit nutrients. Whatever the reason, it led to cooperation between the two types of cells. Each cell contributed advantages to the other, creating mutually beneficial cooperation. Over time, this cooperation became so tight that they could not leave each other, bound tightly together as if they were one. The endo cell gradually lost independent functions and gradually depended on the host cell. The host cell delegated an important function: synthesizing ATP for the endo cell. The endo cell became an organelle with a specialized function, but they still retain characteristics of a cell: having a membrane, cytoplasm, and separate DNA.

The symbiosis of two types of cells to become a single type may have originated very long ago, and the theory about this is called *endosymbiotic theory* (endosymbiotic theory). A prokaryotic cell entered another prokaryotic cell to form a eukaryotic cell. In 2017, scientists found fossils of fungal cells in South Africa determined to be 2.4 billion years old, evidence of this merger [210].

4.2.2 Chloroplasts and nitroplasts become organelles

The endosymbiotic process of mitochondria is not unique. Another merger process for chloroplasts to become organelles of plant cells

occurred about 1.6 billion years ago. **Chloroplasts**, also called chloroplasts (chloroplast) are organelles of plant cells with photosynthetic function, places containing chlorophyll capable of absorbing energy from sunlight, converting energy to store in high-energy molecules like ATP and NADPH (nicotinamide adenine dinucleotide phosphate) while releasing oxygen and water. Chloroplasts also have some other functions like creating organic molecules, synthesizing fatty acids and amino acids, immune responses in some plants. The number of chloroplasts in a plant cell also differs, ranging from 1 – 100. Like mitochondria, chloroplasts have their own independent genome separate from the host plant.

Research from early 2024 published in leading world journals Science and Cell reported that an organelle of cells in some plants can fix nitrogen in the atmosphere and convert it into a form usable for life. Nitrogen is an important element because it is present in DNA, RNA, ATP, amino acids, proteins,... Most animals cannot use nitrogen in air because nitrogen gas is chemically inert in biological reactions. The inertness of nitrogen gas comes from the bond between two atoms in this gas molecule being a triple bond N≡N with bond energy of 949 kJ/mol, nearly double the O=O bond of 500 kJ/mol. Previously, people thought that only bacteria could absorb nitrogen in air. For example, legume plants contain symbiotic bacteria in root nodules to help plants fix nitrogen. In 1998, Jonathan Zehr at the University of California, Santa Cruz discovered a DNA segment of a cyanobacterium (cyanobacterium) that can fix nitrogen, and he named it UCYN-A. Scientists spent many years researching this bacterium as an endosymbiotic partner (endosymbiont) with organisms belonging to the alga family (alga). Two papers published in April 2024 by international scientists led by MIT showed that UCYN-A is not only a symbiotic microorganism but is truly an organelle of marine algae [211, 212].

Determining bacteria becoming organelles is not easy. We know that the central dogma of molecular biology allows creating pro-

teins from DNA in genes. If a bacterium like UCYN-A becomes an organelle of marine algae cells, it will abandon part of its DNA, and the bacterial genome will become smaller, and the bacterium will use proteins from the host (here is marine algae). Proteins that previously needed genes to synthesize will no longer exist. The bacterium will outsource these proteins from the host cell. Conversely, the host cell actively manufactures proteins and marks these proteins for the bacterium to use. Evidence in the above papers shows that UCYN-A not only lives endosymbiotically but truly becomes an organelle, called **nitroplast**, helping cells fix nitrogen. They cooperate with each other, each doing one thing helping each other survive and develop: UCYN-A fixes nitrogen for algae, algae create proteins for UCYN-A.[5] This process only began about 100 million years ago, very short compared to mitochondria and chloroplasts, with time becoming organelles of hosts estimated at nearly 2 billion years ago.

Why do animals have to symbiotically coexist with mitochondria, plants have to symbiotically coexist with chloroplasts (and also mitochondria)? This is an aspect that Darwin's original evolution theory cannot explain only through competition based on natural selection. There is another mechanism – cooperation mechanism to create advantages. The process of efficiently acquiring energy is not easy, so if any organism does this well, it will be coordinated by other organisms. It is not easy to spontaneously develop an optimal energy acquisition mechanism. Mitochondria experts have done best at synthesizing ATP; chloroplasts have done photosynthesis as well as possible. So why doesn't some organism use mitochondria and chloroplasts (if they once existed independently in the past) for the purpose of acquiring and converting energy? It can be said that organisms have specialized difficult tasks, specifically delegating energy acquisition and conversion to external experts. In return,

[5]UCYN-A has nitrogen-fixing effects, so it is called nitroplast, temporarily translated as nitroplast because it has a blue color.

organisms must provide other needs for these experts. Cooperation is not only give-and-take but so tight that they can no longer live without each other. This is the endosymbiotic process – an important process in biology.

It is not only humans who discovered that chloroplasts are energy experts. A leaf slug (leaf slug) under the sea also recognized this. This leaf slug species has a part like a syringe. They use it to pierce algae cells and suck out all contents. Instead of digesting all algae matter, the leaf slug's digestive system recognizes, separates, and retains chloroplasts. This phenomenon is chloroplast theft (kleptoplasty) and has been quite well researched for many years [213].

Leaf slugs move these chloroplasts into specialized cells along their flat back, like a leaf. When leaf slugs expose themselves to sunlight, millions of stolen chloroplasts continue to operate. They perform photosynthesis, creating sugar and fat, directly providing energy for the slug. Thanks to this mechanism, leaf slugs only need to eat algae once when young. After that, they can live up to 9 months (nearly their entire lifespan) just by running on solar energy without needing to eat anything more.

The slug provides a safe, protected environment and exposure to light for chloroplasts. Conversely, chloroplasts provide food for the slug. Chloroplasts do not exist independently; they need hundreds of proteins to repair and maintain. Most genes encoding these proteins are not in chloroplasts. Algae have taken on these genes through billions of years of evolution. But when separated from algae and in leaf slugs, there must be some biological mechanism that helped chloroplasts survive for a long time. Leaf slugs actively do the following to serve chloroplasts: (1) turn off the digestive system to spare chloroplasts instead of decomposing them; (2) the slug's immune system accepts chloroplasts as part of the body; (3) provide necessary substances to maintain chloroplast stability. The leaf slug body has specially adapted many processes so chloroplasts can survive long in its body. This is a special manifestation of the en-

dosymbiotic process changing how we think about evolution – manifestation of dependent origination and emptiness. Without chloroplasts, leaf slugs have little chance of survival; conversely, the leaf slug body has protected chloroplasts in a new environment.

* * *

Compositionality is manifested in cooperation of different organisms. Mitochondria, chloroplasts, nitroplasts were once independent microorganisms that then evolved to become organelles in other multicellular organisms. In that process, organisms depend on each other to survive, to reduce their own work and delegate some other work to partners. This cooperation is so tight that it cannot be separated. The delegated organism becomes an organelle of the host organism. Without energy experts like mitochondria and chloroplasts, animals and plants cannot survive.

Current research has only determined nitroplasts help fix nitrogen but has not discovered other functions. In the future, can we see symbiosis in other functions for nitroplasts like mitochondria and chloroplasts? This process has been and is continuously occurring. That is impermanence. Compositionality and impermanence show emptiness is manifested in organisms.

If at the cellular level, symbiosis is manifested through the merger of microorganisms into organelles, then when going deeper into the molecular level – the level of genes and DNA – we will discover a similar phenomenon but even more complex and profound. Genes of organisms are not only a single gene system but come from many gene segments created by different organisms in the evolutionary process. This means symbiosis is not only manifested at body and cell levels but also in genetic material itself – what we often think is the most core of biological identity.

4.3 Human genes

4.3.1 Symbiotic genes

Since decoding the human genome map in the early years of the 21st century, scientists have decoded genes of many animal and plant species and found that human genes have many parts similar to other organisms. For example, human genes are 98 % similar to chimpanzees, 90 % similar to musculus mice, 85 % similar to zebrafish daniorerio, 36 % similar to fruit flies, 21 % similar to roundworms, 7 % similar to *E. coli* [214]. Although the human genome has about 3.1 billion base pairs (base pair – BP), it is surprising that there are only about 22 thousand genes (accounting for 1.5 % of total DNA BP). This number is only equal to the number of genes in worms, butterflies, and many other plant species. Clearly, the activity of the human body with 30 trillion human cells is much more complex than the activity of a worm with less than 1 thousand cells, so why is the number of genes the same? Some hypotheses hold that humans also have regulatory genes, which are genes regulating the on/off of other genes, and their number is very large, accounting for nearly 20 % of total BP. Thus, the number of DNA directly or indirectly related to genes is about 20 % of total BP. If we previously thought that humans are the center of the universe, the human genome map shows that humans are also no different from other species, even the number of genes is not different from lower organisms.

A very large proportion, about 8 % of human DNA is from viruses [215]. This sounds absurd, but let us try to understand how virus DNA inserts into the human genome.

For example, a virus with an RNA core is HIV[6] causing AIDS.

[6] *Human immunodeficiency virus* (human immunodeficiency virus) is a virus that can cause *acquired immunodeficiency syndrome* (acquired immunodeficiency syndrome – AIDS) reducing the immune system's ability to function, from which bacteria and cancer develop threatening human life.

HIV activity is completely different from other RNA viruses in that they use a reverse transcriptase enzyme to convert their own RNA into corresponding DNA. Host cells treat virus DNA segments as their own genome to transcribe and translate virus genes and host cell genes to create proteins necessary to assemble new virus copies. The normal protein creation process is genetic information from DNA transcribed into RNA then translated into protein: DNA $\rightarrow$ RNA $\rightarrow$ protein, but in HIV virus, this process is reversed: virus RNA $\rightarrow$ virus and host cell DNA $\rightarrow$ virus and host cell RNA $\rightarrow$ protein. Viruses operating according to HIV's principle are called *retroviruses* (retrovirus). The important thing here is that after virus RNA is translated into DNA, this DNA is brought into the cell nucleus and integrated into host cell DNA. Thus, host cells after being attacked by viruses will change their own DNA because viruses have officially turned host cell DNA into a place to harbor their DNA. Once virus DNA integrates into host cell DNA, they never leave again.

Host cells cannot recognize the difference between virus DNA or their own DNA when DNA is integrated because the transcription process into RNA is only a process of reading ATGC characters coldly and unconsciously. That is why virus gene segments will be translated into proteins for successful virus replication. The virus replication process will drain all cell resources and cause cells to die. Virus versions will disperse and attack other healthy cells. That is how retroviruses invade the host body.

Clearly, if DNA is created first, then the transcription and translation process into protein will be faster. So why do some viruses create RNA before creating DNA? It may be because RNA is more flexible, directly creates proteins and self-replicates. That is also the reason the *RNA world theory* (RNA world) holds that the first self-replicator was RNA, not DNA. DNA in double helix form is better protected than RNA with ATGC bases protected by a backbone formed from sugar and phosphate groups. While RNA is only a sin-

gle strand with unprotected bases and directly suffers impacts from the environment. Moreover, ribose and deoxyribose molecules differ in that ribose has an $-OH$ group, i.e., has one more oxygen atom, so has stronger activation. Therefore, some types of viruses choose RNA to record genetic information even though RNA is less stable than DNA, but it can permanently stick to the host cell genome.

In the long evolutionary process, some virus genes have become useful tools for the human body, even becoming indispensable factors for human survival and development. This is precisely the manifestation of symbiosis at the gene level: no gene completely belongs to "us" or to "virus," but they exist in a network of complex interdependent relationships.

4.3.2 Symbiosis between genes

Symbiosis – hosts use genes to improve their activities. In the evolutionary process, humans were attacked by various types of viruses, and the consequence is that until today, there are very many human genes that are virus genes. Ancient virus genes do not affect much human body activities. But some virus genes still affect us. For example, the human stomach has many starch-digesting enzymes. One of the main enzymes helping this is *amylase* secreted from the pancreas. But at some point in the past, viruses changed human genes so that this enzyme appears in the oral cavity, in saliva. Thanks to that, when we eat starch, we feel a bit sweet in the mouth because the amylase enzyme in saliva has helped break down part of starch into sugar. From now on, whenever we eat rice, especially if eating rice cooked from the world's best rice ST25[7] and feel sweetness in the mouth, remember that sweet sensation is thanks to virus genes inserted into human genes and helping us have that sensation.

[7]The world's best rice in 2019 and 2023 is ST25 rice, also called Sóc Trăng fragrant rice, the result of research by labor hero Hồ Quang Cua over more than 20 years.

Not only making us eat deliciously, but perhaps what amazes us most is that the ability to carry pregnancy, nurture mature fetuses through the placenta is the result of gene integration from viruses. Among about 6.4 thousand known mammal species, there are 6 thousand species with placentas, common from terrestrial, aquatic, flying mammals,... including humans. Clearly, the prevalence of placental mammals is an advantage for mothers to keep and protect children when not yet mature in the womb. When in the mother's womb, the connection between mother and child is through the placenta, where nutrients from the mother are transmitted to the child. The ability to have a placenta is what mammals learned from retroviruses. How does that process occur?

One of the special abilities of viruses is the ability to tightly attach to host cells, by the virus membrane and cell membrane merging so the virus injects genetic material into the cell. The attachment method of embryos to the uterine wall through the placenta is the method that mammals learned from viruses. Fertilized eggs form embryos. Initially, embryos float in the uterus, then they implant. Implantation is achieved thanks to a connecting protein to form the placenta [216]. This protein is obtained thanks to a DNA segment originating from retroviruses. That is precisely the DNA segment that viruses use to attach to host cells before injecting their genetic material into the cell. Molecular biology research shows precise coincidence between mammal DNA and retrovirus DNA, almost precisely to each genetic character. This is a clear example showing that viruses inserted their gene segments into our ancestors, and later those gene segments became useful [217]. Moreover, this retrovirus DNA segment also creates proteins helping strengthen fetal immune systems by fighting some harmful bacteria.

4.3.3 Parasitism of microorganisms

Parasitism is a relationship where one species benefits while the host receives almost no significant benefits, even suffers harm in that relationship. Parasitic organisms can be viruses, bacteria, and large multicellular organisms.

Research with vole species (vole) belonging to the rodent family (rodent) shows that viruses can make voles abandon promiscuity and become ideal husbands [218]. Normally, voles are polygamous animals (polygamous) with males having habits of promiscuous relationships with many different females. Scientists infected male mice with an adenovirus type that had been gene-modified in the laboratory. After being infected with the virus, the virus inserts their DNA segments into male mouse DNA so male mice become ideal husbands only relating to a single female. The virus used a gene segment producing a protein to create a chemical called vasopressin, a reward chemical secreted during sexual relations. Virus-infected mice will be rewarded with much vasopressin when male mice relate to a single partner. The result is that the virus transformed mouse genes from polygamous to monogamous.

That is an experiment on mice in the laboratory; can this phenomenon occur with humans in nature? The human genome map shows a large number of virus genes, so viruses controlling emotions and our spiritual life through gene manipulations is completely possible. Modern human society with monogamy more prevalent than polygamy – is it due to social will or virus control? Research conducted on other animals will be conducted easily. Due to ethical reasons, experiments cannot be conducted on humans, so the answer needs more time.

Retrovirus invasion into human genes occurs in chronological order, not randomly. People see that invasion occurs in waves; there were times in the past when genes changed very much, times when genes changed very little. One invasion wave occurred right before

mammals exploded in numbers, and another wave occurred when apes appeared, ancestors of humans. Large gene changes and speciation occurring is reasonable because they go together. Different species can mate with each other, but this mating cannot produce offspring, or if offspring are produced, offspring cannot reproduce. Horses and donkeys are two different species; normally they do not attract each other, but in some cases they mate to produce mules. Mules cannot reproduce.

People took two groups of wasps (wasp) infected with popcorn bacteria and had them mate; their embryos could not develop normally. If we use antibiotics to kill bacteria, the two wasp groups can reproduce normally. Thus, popcorn bacteria had the effect of separating the two wasp groups; over sufficient time, they may evolve into two different species. Returning to human DNA changes in waves going together with mammal and ape formation, viruses may have played roles similar to popcorn bacteria with wasps. Bacteria attacked ancestors of mammals and apes so DNA of these species gradually differed to eventually become different species.

* * *

Bacteria can interfere with mammal behaviors. The bacterium *Toxoplasma gondii* (*T. gondii*) is a single-celled parasite so common that about 30 – 50 % of the world's population has been exposed to this parasite because the animal transmitting bacteria is one of humans' favorite pets: cats. For humans, when infected with *T. gondii* bacteria, they may have fever, swollen lymph nodes, aches,... however not too dangerous to life. This parasite can be transmitted from mother to child. When sick, if the body is healthy and the immune system works well, after some time the disease will heal; if not, treat with antibiotics.

T. gondii has two replication methods: asexual and sexual. When in ordinary warm-blooded animals, *T. gondii* reproduces asexually.

But when they are in cat intestines, these bacteria reproduce sexually. Whether asexual or sexual, *T. gondii* are all replicated, but they very much like sexual reproduction, so they find every way to return to cat intestines. *T. gondii* found a very special way: using mice to enter cat bellies. In 2013, a group of scientists at the University of California, Riverside showed that the parasite *T. gondii* inserted genes controlling mouse brains so mice lost fear of cats. In the external environment, this bacterium can infect many different animal species like cows, pigs, humans,... and mice. Because mice are cats' favorite food, when *T. gondii* infects mice, this parasite will create 2 proteins to control the dopamine release process, a hormone and also a neurotransmitter stimulating reward-promoting actions. When mice smell cat urine, normally the amygdala (amygdala) in mouse brains will send andrenaline signals warning of danger so mice will avoid. But in *T. gondii*-infected mice, this activity is inhibited; instead, the protein controlling dopamine secretion acts to encourage reward, making mice stimulated to seek cat urine smell. The result is that mice bring themselves to cats to be eaten, so *T. gondii* can return to cat intestines to reproduce sexually. Mice became tools for parasites to mate with each other [219].

Not only mice, primates, close relatives of humans, also became victims, tools of the parasite *T. gondii*. A 2016 study by French scientists showed that this bacterium used a similar mechanism applied to chimpanzees (chimpanzee) losing fear of their predator enemy, leopards (leopard), because leopards are cat family, so leopard intestines are the environment for *T. gondii* to reproduce sexually [220]. If chimpanzees can have emotions controlled, can humans be victims? Currently, there is no official research, but some people addicted to raising cats, raising dozens, even hundreds of cats may feel attracted by cat smell, more accurately cat urine smell. This may be a psychological issue of loving animals, but it may also be that psychology being controlled by a type of parasite. A recent study showed that *T. gondii* makes infected people become more

attractive in the eyes of the opposite sex because the parasite can spread through sexual contact. When infected people become more attractive, those people have more ability to have sexual relations with more people; therefore, the chance for this bacterium to spread will be greater [221].

In the above examples, *T. gondii* does not replicate genes, does not insert their genes into host genes, but they only steal host genes and insert their own genes to replace. But that is enough for hosts to exchange their lives. Certainly, when mice and chimpanzees seek the smell they think is extremely attractive – cat and leopard urine – they cannot think that their action is fulfilling the wish of a low-level organism: bacteria.

Similarly, some people with the disease of liking to raise many cats may also think they have free will to do what they like and cannot imagine that liking may be controlled. So suffering comes from joy, satisfaction, free will of an individual because those sensations may be unconsciously controlled. Moreover, the one giving orders to individuals is not a supreme being with infinite authority but a single-celled organism named *T. gondii*, specializing in parasitism, even without complex awareness. The awareness they have is only doing everything to survive and replicate, without any lofty ideal. If mice, chimpanzees, and we know we have such a master, should we follow that master's instructions?

* * *

If viruses and bacteria living parasitically on hosts with much more complex body structures than parasites and controlling host bodies according to their wishes, can multicellular parasites do similar things? A flatworm species named Ribeiroia ondatrae can parasitize many different reptile and bird species, but they seem to have controlled the body plan (body plan) of some frog species so that frogs infected with this worm have deformed bodies like no

legs or many legs growing at abnormal positions. Body deformation makes these poor frogs unable to move effectively and easily become prey for frog-eating animals: birds. When birds eat frogs, worms will infect birds, and worm dispersal ability will be wider. In 2002, ecologist Pieter Johnson at the University of Colorado at Boulder researched parasitic infections in amphibian species living in ponds, lakes, and swamps in some western US states, proving this [222, 223].

Ribeiroia flatworms have complex life cycles with many different hosts starting with snails. Flatworms turn this mollusk species into parasitic machines. Worms reproduce asexually inside snail bodies and make snails nightly release hundreds of worm larvae into the environment so they swim freely in water with the purpose of finding the next host with quite high accuracy: tadpoles. When approaching tadpoles, these larvae will enter tissues that will later develop into frog movement organs to deform them. The consequence is that when tadpoles transform into frogs, legs are already deformed, either no legs or many legs at abnormal positions, making frogs unable to move effectively. A previous statistic in 1999 showed that over 30 % of frogs had deformed movement functions, showing the prevalence of this parasite species. These deformed frogs easily become good prey for their predator enemies: birds. When birds eat frogs, birds become the next host of Ribeiroia flatworms. Here, the worm's purpose is revealed: sex. Only in bird intestines do worms reproduce sexually, the reproduction method they desire. When birds excrete, worm larvae return to the environment to return seeking the original host: snails. Sex is not only important to higher animals but also important to microorganisms. Flatworms have turned frogs into tools so they can reach bird intestines in the sky, where worms can reproduce sexually with each other. Poor frogs.

In the future, can humans be separated into different species due to virus invasion? Completely possible with retroviruses. HIV is currently the leading dangerous object for humans; this is a retrovirus

type, and if an HIV pandemic occurs, very many people on Earth will die. Survivors will belong to one of two groups: one group not infected with the virus and another group infected but the immune system has succeeded in fighting this virus. These two groups of people may separate because if they relate to each other, people without antibodies will die. Over time, the group of people with immune systems fighting HIV will gradually evolve but cannot avoid becoming a different species. At some point, HIV genes will insert into human genes and become an inseparable part of the human genome.

Viruses not only harm humans but can promote human development in a good direction. Borna virus (bornavirus) is an RNA virus infecting many mammal species, including humans. The time needed for this virus to insert their DNA into host DNA which is humans is short, only about 30 days [215]. Borna virus can increase DNA diversity to form and operate nerve cells and can burst human intelligence [224].

If so, in the past, there may have been viruses that transformed human ancestor genes from ordinary animal intelligence into a species with superior intelligence. When Darwin put forward evolution theory holding that humans originated from apes, he made a revolution in world perception. Now, not only did apes evolve into humans, but bacteria and virus species are also causes for the evolution process to occur to have today's humans.

Bacteria and viruses have not stopped; they continue to affect and shape that evolution in the future. Previously, many people thought that speciation or intellectual mutations came from organisms themselves or macro environments. Few people thought that symbiotic bacteria or viruses on the body could be causes of those huge changes. Compositionality and impermanence are also manifested in invisible things.

The story of symbiosis at the gene level does not stop there. There is another type of gene, not from external viruses or bacteria,

but self-replicating DNA segments right in our genome. These DNA segments, called self-replicators, occupy a large part of the human genome but seem to have no specific function. Why does the genome contain such seemingly meaningless DNA segments? The answer will clarify more about the emptiness of the genome and how life self-organizes at the molecular level.

4.3.4 Commensalism of self-replicators

In every moment, the human body continuously operates, in which cell replication to keep the body fresh takes much energy. Most DNA in our genome is unknown what it is for? Each time cells replicate, all DNA is accurately multiplied. So why does life waste resources to do such seemingly meaningless things?

Barbara McClintock[8] discovered jumping genes (mobile genetic element or transposable element), which are floating genes and randomly insert into chromosomes since the 1950s. They can jump from one chromosome to another [225]. The origin of jumping genes is still debated. They may be virus genes left in human genes from very long ago, gradually losing information, only remaining a few BP as today. They may also be jumping genes already present in human genes, then seceding from human genes to become viruses. But whatever the reason, they are currently an inseparable part of the human genome with almost no function.

For example, a jumping gene named Alu (Arthrobacter luteus) with length about 300 BP appears millions of times in human chromosomes, so they account for 300 million BP pairs, equivalent to about 10 % of the amount in our genome [226]. Alu is only one case of jumping genes; if calculating the total bases of different jumping gene types, they account for up to 50 % of the human genome [227]

[8]Barbara McClintock (1902 – 1992), American cell geneticist who won the Nobel Prize in Medicine in 1983 for discovering gene transposition (transposition), which was her research 30 years earlier.

(this number for South American lungfish is 90 % [170]). These genes are like commensal genes, not having much effect on body activities. What Alu's existence is for, we still do not know, but it completely fits Dawkins's theory – the purpose of genes (if any) is self-replication without consciousness about it. Research on genome size reveals organisms with the largest genomes: South American lungfish with 91 billion BP pairs [170] (largest in the animal kingdom), disc fern (Tmesipteris oblanceolata) with 160 billion BP pairs (largest in the plant kingdom and largest among known organisms) [228], 30 to 50 times the human genome. It seems genome size is not related to body size and complexity.

If we accept Richard Dawkins's hypothesis about selfish genes, we see that nothing is meaningless. According to NASA's definition, life is a self-sustaining chemical system capable of adapting according to evolution theory, then a DNA segment that he calls a self-replicator is life with unconscious desire to replicate itself. The diversity of life is synonymous with the diversity of DNA segments. Large amounts of seemingly meaningless BP are only meaningless to us. For self-replicators, that is existing, being replicated, or more accurately, they are living. Symbiosis is not always smooth. Some super-selfish genes appear not only with the purpose of replicating themselves but may harm hosts. Research on super-selfish genes in fruit flies shows these genes combine to create a super gene, a cluster of selfish genes on one chromosome so they can be inherited together. In sexual reproduction, genetic material DNA from father and mother combine through recombination. This is an important process of natural selection for individuals to maintain beneficial mutant genes and eliminate harmful mutant genes. Fruit flies infected with super genes stop the recombination process by killing male sperm not carrying super genes. Therefore, super genes are completely inherited to offspring. The inheritance process without recombination brings immediate benefits to super genes but will lead to species degeneration and collapse [229].

So, if we consider self-replicators as gene segments in our DNA, that is symbiosis of DNA. Like endosymbiosis of mitochondria in animal cells, chloroplasts in plant cells, bacteria in intestines. Compositionality is manifested by all depending on each other, interacting with each other, relying on each other to exist and develop. So humans themselves, the human body is endosymbiosis from macro to micro, so how can there be an independent, permanent, unchanging self? It can be clearly seen that the principle of dependent origination of Buddhism explains the most modern sciences very well. This principle leads to three marks: no-self, suffering, and impermanence. The immediate self of all things is all assembled by the five aggregates (form, feeling, perception, mental formations, consciousness). From DNA in the nucleus to cells, tissues, and organs, all symbiotically coexist inseparably, so there cannot be what is called the material *self*.

4.3.5 Human genome evolving increasingly fast

In modern life today, humans do not face dangers like avoiding other animals eating meat, competing for food, fighting to get mates, but conversely, dangers come from us eating more than necessary, food and environment being polluted, along with irregular living leading to non-communicable diseases. According to World Health Organization WHO statistics in 2019, the top three causes of human death in the world are cardiovascular (16 %), stroke (11 %), lung obstruction (6 %). The total deaths from these three diseases are 18.4 million people [230]. These diseases are all related to our diet and lifestyle. Thus, dangers for humans now are very different from our ancestors. Those dangers are no longer external but mainly related to ourselves, to how we choose to eat and live.

A famous author Harari holds that the human body evolves slower than civilization development [135]. He holds that in the past, leading causes of death were hunger, disease, and war. Civiliza-

tion developed very fast; the agricultural revolution provided enough food for humanity; progress in human rights has developed highly; the above causes are no longer worrying for humans, then humans die from eating too much. Cardiovascular diseases, stroke,... are consequences of our bodies evolving very slowly. The 21st-century body is still the body of homo sapiens living on savanna grasslands 70,000 years ago. At that time, ancient people were always in a state of hunger, so whenever conditions allowed, they would eat more than the body's actual needs to prepare for a few days, even a few weeks of food shortage, they could still survive.

The above arguments seem logical and based on an important hypothesis that the 21st-century human body is the body of homo sapiens on savanna grasslands in Africa. That is, we have almost not evolved in 70 thousand years. This is certainly incorrect. In 2018, a group of scientists led by the University of Copenhagen published research on a nomadic tribe named Bajau in Indonesia. This tribe lived in floating boats between islands in Southeast Asia more than a thousand years ago [231], so they are called Sea Nomads (Sea Nomads). Their livelihood is fishing seafood in a very manual way by using goggles, a spear, and a weight to dive into the sea. All people in this tribe can dive deep to 70 m and can hold their breath for up to 13 minutes. The unusually long breath-holding ability of Bajau people made scientists curious to find out if people of this tribe have any gene differences.

When we hold our breath and dive into cold water, blood vessels to limbs need to contract so blood can rush to important organs. The body protection mechanism under oxygen-deficient conditions (hypoxia) is activated. At this time, the spleen (spleen) needs to contract to pump red blood cells so oxygen concentration increases to 9.6 % compared to normal. The larger the spleen, the higher the breath-holding ability. They used ultrasound to measure spleen size and compare with the spleen of the Saluan tribe nearby but living mainly on land. Results showed that Bajau people's spleens are 50 %

larger than Saluan people's spleens. Scientists found a gene named PDE10A only in Bajau people. This gene is thought to control the thyroid and thyroid hormones making the spleen become larger and carry more oxygen during underwater diving. In addition, Bajau people also have a gene denoted BDKRB2 affecting human diving responses.

Assuming that Bajau people started living as sea nomads as now and have not changed habits for more than a thousand years. During that time, they did not communicate and hybridize with other ethnicities. Then, new genes appeared to adapt to local natural conditions in time only about a few hundred to a thousand years. Thus, Harari's hypothesis about the human biological body almost not evolving in 70 thousand years is unreasonable. The Bajau people case is an example of human gene changes adapting to oxygen-deficient conditions. People living on the Tibetan Plateau or very high mountains all must adapt to oxygen-deficient conditions, and their bodies have had new genes appear. Siberian people living near the Arctic in extremely cold conditions are also like that [232].

The appearance of a new gene for humans to adapt to new living conditions cannot quickly appear. How long does human gene change need? Change depends not only on time but also on population growth. The faster population grows, the more mutation ability and consequently new genes appear.

Gene changes of Bajau people or Tibetan people need thousands of years because this is a small community with not too large a population. Research on genes by scientists at Stanford University in 2016 on 3 thousand genomes of people of British origin over about 100 generations (from 2 to 3 thousand years ago) showed that time can be from a few decades to centuries is enough for new genes to form making humans adapt to digesting dairy products, taller, lighter blonde hair, bluer eyes [233]. Humans are still evolving, and that evolution seems faster than before. The human genome consists of many different components, and they always change continuously.

That is emptiness in the human genome.

Humans are evolving is a certain conclusion of scientists. That means we have been and are gradually adapting to new conditions and circumstances. Although recently evolution speed has been faster, to become a perfect species, we must wait very long, or may never be able to achieve. Changes in living and working habits in the past few decades like living concentrated in large cities, increasingly less physical activity, food with too much energy, browsing social networks without direct meetings,... have made the human body unable to change in time, leading to unprecedented incurable diseases. Intelligence and body still differ greatly. The sensation of pain that the brain brings and tissue damage are not synchronized. Finding the middle way is the path leading to harmony between body and intelligence: not too highly valuing suffering but also not despising it; not letting desires lead in a hedonistic way but also not too austere leading to excessive physical weakness.

Through examining the human body at three levels: body, cell, and gene, we have seen that emptiness is manifested consistently and profoundly. The body is a symbiotic ecosystem with hundreds of trillions of microorganisms. Cells have the most important organelles like mitochondria, which were once independent bacteria. The genome is a collection of DNA segments from many different sources in the evolutionary process. All of this leads to a fundamental question: if the body, cells, and genes all have no independent self-nature, then what is our self – what we call "I" – actually? The next section will examine this question from philosophical and neuroscience perspectives to better understand the emptiness of the self.

4.4 Self

4.4.1 Self is relationships

The self is a concept referring to *the I* as a core part of humans, focusing on self-awareness and self-recognition. This section presents self changes from the body's perspective over time and space. Over time, the human body is born, grows, gets sick, ages, and dies, following the four sufferings of birth, aging, sickness, and death. Even when mature, body matter is always continuously renewed (see page 237), but consciousness always gives us a sense of a single and unchanging self. In space, at all three levels of individual, cell, and gene, the human body is symbiosis with other organisms. Due to symbiosis, body activities depend very strongly, even determined by external organisms and genes. Our self always changes and depends on external factors, but self-awareness is always single, independent, unchanging. That is precisely suffering from not recognizing no-self in the self. So can the self be defined? Or can we answer the question *who are you?*

Western philosophy has a story named *Theseus's Ship*. According to Greek mythology, Theseus was the founding king of Athens after killing the half-human, half-bull monster Minotaur to save the children of Athens on Crete island; he returned to Delos, Greece on a ship named after him. Later generations commemorated by pilgrimaging to Delos on the ship that Theseus once sailed, which people call Theseus's ship. After many years, wooden pieces on the ship rotted, so they were replaced with new, better wooden pieces. That process was repeated many times until all old wooden pieces were completely replaced. The ship made of new wood was called the Aegeus version of Theseus's ship because it frequently traveled on the Aegean Sea. Philosopher Thomas Hobbes supplemented the story that old wooden pieces that were replaced were gathered at Piraeus (an ancient seaport near Athens) to rebuild Theseus's orig-

inal ship, which people call the Piraeus version. Unlike the Aegeus version, the Piraeus version cannot be used for travel but only has exhibition nature [234]. The problem raised here is which version is Theseus's ship. The Aegeus version was continuously used from when Theseus returned to Greece and later used for pilgrimage. Its identity was always maintained because it carries continuity of action, but materially, it no longer has any relation to Theseus. The Piraeus version is the opposite; it carries wooden pieces that Theseus once stepped on and used but does not have continuous activity history in the pilgrimage process. This problem has been raised by many philosophers for discussion but has no clear answer and is called *Theseus's Ship Paradox*.

The problem of Theseus's ship identity can be extended to human identity. In other words, what is a human's self. Besides the human body always being renewed by matter from the surrounding environment, each part on the body is completely replaced. Unfortunate people who have accidents may lack limbs or use prosthetic limbs, which certainly does not affect that person's self. Modern medical technology allows transplanting organs from donors into patients to maintain life and basic functions. According to the World Health Organization (World Health Organization – WHO), in 2022, the US had about 42,000 organ transplant cases, increasing 4.6 % compared to a year before. Kidney transplants have the highest rate, up to 60 %, followed by liver 24 %, heart 10 %. About every 10 minutes, someone registers on the organ request list [235]. Even scientists are researching organ transplants from animals; most feasible is gene-modified pigs. Pigs are chosen as organ donor animals because pig organ sizes are similar to humans, short maturation time, can be gene-modified to fit the human body [236]. Transplanting or replacing body parts does not much affect that person's self. But there is one part that everyone agrees that if replaced or modified, it seriously affects the self. That is the brain.

The brain is obviously very important; there, self-awareness is

generated. Can the brain be transplanted? Previously, brain transplants were thought to be only in science fiction, partly due to technical difficulties and partly because it raises ethical and social psychological problems. Technically, nerve tissues do not have recovery ability. Therefore, people are researching the ability to connect nerve tissues without creating scars but have not succeeded. First brain transplant attempts were performed on dogs in 1908 but were unsuccessful. In the 1970s, a monkey head transplanted with another monkey's body could live for 8 days with some basic sensations like smell, taste, hearing, and some movement functions in the face and head. In 2017, Italian doctor Sergio Canavero put forward a human head transplant procedure and called for volunteers, but to date, no transplant case has been performed [237, 238].

In Vietnamese fairy tales, the problem of self is also mentioned in *Tấm Cám*. When Tấm was harmed by her stepmother, Tấm's soul continuously changed forms from human to golden oriole, silk tree, loom, persimmon fruit, then returned to human. Tấm's body changed differently, but Tấm's self with memory and thoughts completely did not change. The mythological nature is that Tấm always had human-like abilities when the soul resided in non-human bodies. Golden orioles and persimmon fruits could understand human language and behave like a Tấm not interrupted physically. Readers implicitly understand that Tấm's self is precisely her memory and human characteristics.

On page 237, we saw that the human body is replaced by new matter with mass exactly equal to the body after about 18 months, so we can see that the physical self is not as important as the psychological self manifested in memory, emotions, thinking, and relationships with the surrounding environment. If we think the self is independent and unchanging, it necessarily leads to contradictions. According to Buddhism, the self always changes and depends on external environments both physically and psychologically, so paradoxes do not appear. The psychological self is more important than

the physical self.

Self-awareness lies in the brain. Now a problem appears: which brain part is responsible for creating consciousness? The answer is not clear, but it seems consciousness is not localized at any specific position but is complex interaction between many brain parts with each other. For example, the thalamus (thalamus) is a relay station for sensory information to different regions. The prefrontal cortex (prefrontal cortex) has high-level cognitive functions like decision-making or planning. Some people think this is also the region contributing to self-awareness and introspection. The parietal cortex (parietal cortex) has spatial processing and sensory information integration functions. In the next chapter, we will carefully research modern neural theories about consciousness.

Brain damage can lead to self changes through changes in personal character or not? Certainly yes, because each brain part has distinct functions, so damage there affects related functions. Research encounters many difficulties because of the following problems. First, because brain structure is complex and subjective, distinguishing, conceptualizing, and measuring self properties will encounter difficulties. Second, because the interaction of physiological, psychological, and social factors affects self-awareness, self changes are both unique and multidimensional. Third, consequences of the above two lead to operating and intervening in self-awareness will encounter difficulties [239]. Despite that, scientists all have a common assessment about the cohesion of different brain regions related to individual personality characteristics. The brain certainly determines factors forming self properties.

* * *

Returning to the question who am I. Marx's philosophy says that humans are the sum of social relationships, understood as relationships between humans and humans. We need to supplement

further that humans are the sum of relationships with the surrounding world, with nature and society, even with themselves. This relationship is built based on three important worlds in Marx's philosophy: nature, society, and thinking. Nature and society are easy to understand; thinking is ourselves. Those relationships form an individual's self. Buddhism emphasizes internal relationships and holds that is a great world to explore. Understanding and knowing the self, then behavior with the natural and social world can be correct. Like Theseus's Ship Paradox, a human's self has no self-nature; it is formed by relationships between the self with itself and the surrounding world. That is determining the self in space. Over time, each moment the self has a priority to make oneself. No one is born with a fixed self; it always changes continuously.

Category theory is increasingly important because of its generalization with the vast mathematical world. In this theory, *Yoneda's Lemma* plays a core role. This lemma proves a simple thing: two elements A and B belonging to two different categories will be isomorphic if relationships of element A with all other elements are isomorphic to relationships of element B with all remaining elements [240]. For example, two people identical physically like twin brothers are still different because their relationships with themselves and with the external world are different. Conversely, if someone has cosmetic surgery or gender transition, even though the physical body is completely different, if that person's relationships do not change, the self is still one. The human physical body is always renewed by external matter, but the self still remains unchanged because of memory about oneself and the external world. In summary, the self is built from human relationships with the world. Theseus's Ship Aegeus version carries history of relationships with the surrounding world, so even though different materially, it deserves to be the self of the original ship.

Furthermore, relationships forming the self change over time. When young, children care very little about sexuality but mainly

care about perceiving the world. Scientists prove that connections between nerve cells (proportional to memory) of humans are most numerous at age 2, which is when children learn everything. Older, experiences allow children to judge, so the number of connections decreases [241]. At school age, important relationships are mainly friends, teachers at school, and family at home. When working, relationships with colleagues and society expand. This is the period when humans care much about sexuality and need to affirm themselves. When starting a family, important relationships are mainly family, with spouses and children. A bit older, people spend time thinking about life more. When old, sexual desire decreases, resting and enjoying free time more. The above stages show us that the importance of self relationships changes over maturation time. Therefore, if we want to determine the self, we must depend on the moment in that person's life.

Value systems play an important role in determining the self. For countries following some religion, value systems are relatively clear and stable. People can determine values from very young. For example, Buddhist countries, a child can become a monk from a few years old. But for secular countries, determining value systems is not easy. Vietnam is an open country; in the context of deep international integration, value system changes are very fast. In feudal times, Confucian values had deeply penetrated culture. During the French colonial period, Western values quickly replaced Confucian thought systems. During war and subsidy periods, values were changed once again. Especially during the Đổi mới period, things previously considered valuable are no longer valuable.

Value changes are increasingly fast, even shorter than a human lifetime. Therefore, conflicts between generations have occurred and become serious. Determining what are long-term values so adults can teach children, predecessors can teach successors is very difficult. If previous generations thought that countries like the Soviet Union, China, and socialist countries had high attractiveness, then

according to a British Council report conducted on young people in 2020 in Vietnam, the top three countries are South Korea (82 %), Japan (81 %), the US (79 %) [242]. A sudden and very fast change. Therefore, when answering the question who am I, we must determine universal, unchanging values of the society that person is living in. After determining, based on each individual's ability, can we orient the self in the future.

4.4.2 Placing self too high

In history, humans often consider themselves in higher positions than other organisms. Some people even think their group is more valuable than others. This not only happened in ancient times but is happening in the modern world we are living in. *Neoliberalism* (neoliberalism), a form of capitalism where individual freedom through free choice is manifested in society by free markets. This is a very dangerous ideology because it elevates individuals, elevates self-interest thought, each person takes care of themselves, greed is good. Completely opposite to human nature as a social species, symbiotically coexisting with each other and symbiotically coexisting with other species. This ideology was shaped and popularized by Milton Friedman. Initially, European and US leaders Margaret Thatcher and Ronald Reagan supported it; later, Tony Blair, Bill Clinton, and Gerard Schroder continued inheriting it. In 1981, Thatcher told Sunday Times: "Economics is the method; the goal is to change the soul" [243]. According to this meaning, capitalism has succeeded in turning us into **economic humans** (homo economicus), only knowing to seek self-interest. Look at what we desire; any country takes GDP, material achievements as measures of success.

Three consequences that neoliberalism brings: (1) increasing wealth gap, (2) increasing power and domination for large and super-large enterprises harming labor forces and society in general, (3) material self-interest pulling along cultural, ethical, and lifestyle self-

interest [244]. In the US, CEO salary was 58 times worker salary in 1989, increased to 278 times in 2018. From 1978 to 2018, stock market growth was 700 %, CEO salary increased 1000 % while worker salary increased 12 % [245]. Most people in developed countries like Europe and the US all show that people's reflection is that governments are slaves of markets, of large and super-large enterprises. Neoliberalism changes how we view each other, how we view values and responsibilities to ourselves and to society. Humans become organisms with super-high competitiveness with the purpose of self-interest for themselves regardless of broader consequences that may be caused.

Previously, relationships with nature were often overlooked because human impacts were very small and insignificant to Earth's activities. But since the agricultural revolution, especially the industrial revolution praised by humans as an achievement of superior intelligence to now, living environment impacts have been changed by humans in a bad and irreversible direction. Humanity's greatest mistake is thinking that our species is higher than other species. That mistake continues in history until recently is neoliberalism elevating individual human desires for material things, formalized through the magic of free markets, and unfortunately has succeeded perfectly. This magic is not limited to countries that gave birth to it, Britain and the US, but has spread throughout the world.

In the past, war was one of the top three causes of human death after starvation and epidemics. What are the causes of wars? Is it because of necessity, inevitable? Let us list some of the bloodiest wars in human history. Death tolls in World War II (1939 – 1945) about 120 million, Mongol Empire expansion (13th century) 60 – 70 million, Three Kingdoms period about 36 – 40 million, World War I (1914 – 1918) about 20 million, Sino-Japanese War (1937 – 1945) about 29 million, Manchu invasion (1618 – 1683) 25 million, Taiping Heavenly Kingdom (1850 – 1864) 20 – 30 million, An Lushan Rebellion (755 – 763) Tang dynasty about 36 million, Chi-

nese Dungan Uprising nearly 20 million. What causes lead to wars? A recent study points out four causes and motivations creating war: (1) human mistakes and sins, especially of rulers; (2) shortcomings of one institution, country; (3) shortcomings of many institutions, countries; (4) ideologies (including religion), collective psychology or culture [246]. Human mistakes causing wars are not because of necessary existence in the sense that without it will lead to extinction but are self-arrogance of individual selves of rulers, self-grandeur of a nation, or thinking that one's ideal is more correct than other ideals, one's religion is more important than other religions.

4.4.3 Human impact on the world

Who I am depends on my relationships with the surrounding world and with myself. Previously, human impacts on the natural world were small, so people encouraged humans to participate in transforming and conquering nature. The natural world consists of two parts: the physical world like the universe, Galaxy, solar system, Moon, Earth, and the biological world like plants, animals, bacteria, archaea, fungi, protozoa... The US system divides organisms into 6 kingdoms (kingdom). They are animals (animalia), plants (plantae), fungi (fungi), protozoa (protista), archaea (archaea), bacteria (bacteria). The British and Australian systems combine bacteria and archaea into the monera kingdom.

A 2018 study on the distribution of *biomass* of kingdoms on Earth showed that the mass of all organism species is about 550 billion tons of carbon. Plants contribute most with 450 billion tons, followed by bacteria: 70 billion tons, fungi: 12 billion tons, archaea: 7 billion tons, protozoa: 4 billion tons, animals: 2 billion tons. Among animals, arthropods: 1 billion tons, fish: 700 million tons, annelids: 200 million tons, mollusks: 200 million tons, humans account for 60 million tons and livestock account for 100 million tons, wild animals: 70 million tons. Human activities have made large mammals

extinct in the period from 50 to 3 thousand years ago. Before that, their biomass was 20 million tons, now reduced to 3 million tons. Ocean-dwelling mammals also decreased five times, from 20 million to 4 million tons. If calculating both land and water mammals, their biomass decreased 6 times, from 40 million tons 100 thousand years ago to 7 million tons [247]. Humans have affected biological balance violently and irreversibly. Note that biomass above is converted to carbon mass present in each kingdom and species. A recent study on mammals showed that if converted to normal mass, human biomass is about 390 million tons. Land mammals have biomass of about 22 million tons, spatially distributed mainly in Africa, most numerous being herbivores, followed by wild boars, elephants. Water mammals are 40 million tons with whales accounting for more than half. Livestock mammals are 30 times larger than wild mammals with biomass of 630 million tons [248].

Biomass is an index to show human impacts on other organisms. Human travel throughout the world leaves very large carbon emissions compared to other animals. Most animal species only use a very small area at some determined positions, repeating on Earth, while humans are almost not limited spatially. Each movement leaves carbon emission levels greatly affecting Earth. Human activities can make Earth warmer, strongly affecting organisms living on Earth, including humans. Therefore, human activities mainly cause bad impacts on human society and other animal species.

If all ice in Greenland melts, it can cause unpredictable consequences. The North Atlantic warm current is an extremely important current in regulating global climate, especially it brings warmth and moisture from the Gulf of Mexico at the equator to warm Europe. This current operates due to convection between hot and cold water flows in the North Atlantic, acting like a giant pump operating this giant current. Earth warming makes Greenland ice melting increase very fast and may cause melting of that giant ice mass to turn off the pump. If so, it will be a great drought for Europe and

the world. 2023 research showed that the tipping point leading to the collapse of the above warm current may occur starting from 2025 with high probability in 2050 [249].

Earth's temperature has increased 1.3 – 2.3 °C since the pre-industrial period until now, mainly in the past 50 years. 2023 was the hottest year ever recorded in history. People predict if Earth increases 2 more degrees, disasters will occur, so humanity set the limit 1.5 °C. Reducing environmental impacts is not only a recommendation but a necessity.

In the past 50 years, global damage caused by climate change is about 4.3 trillion USD, equivalent to 10 times Vietnam's GDP in 2023. It relates to 12 thousand disasters and 2 million deaths (90 % in developing countries). Sea level rises annually by 4.4 mm. By the end of this century, sea level may increase by 0.9 m more. In 2022, about 66 thousand km^2, equal to 1/5 of Vietnam's area, forest area was destroyed in the world. Humans only have 50 % chance to be able to achieve the 1.5 °C limit. The total amount of carbon dioxide we can emit into the atmosphere is 250 billion tons [250]. With annual emission rates of about nearly 50 billion tons, we only have about 6 years of time budget to do everything before carbon is neutralized.

We have seen dangers brought by neoliberalism (page 282). That is a distorted manifestation of humans in social relationships creating economic humans. But that is not the root. The deep cause lies in human relationships with themselves. We need to seriously look back at our self in relationships with nature with humble and long-term views. Mutual dependence on the human body and in Earth's ecosystem is much tighter than we think. Balance achieved over hundreds of thousands of years has been broken by humans in just a few decades in an irreversible direction, which may put humanity itself on the brink of crisis. Due to the principle of dependent origination in the sense of mutual dependence leading to human decisions toward the middle way, living harmoniously with

nature is the best way to be able to exist and develop. The middle way of Buddhism is close to wu wei in Taoism. That is not doing anything excessive with nature, society, and thinking.

4.4.4 Power from self expansion

What reason do humans, organisms with nothing outstanding, rise to dominate other species? Certainly there are many answers from humanity's great minds based on biological, cognitive, cultural, and technological factors. Marx's philosophy holds that labor is precisely the factor helping humans surpass other animals. Others reason that believing in imaginary stories helps humans work with a large number of other people never met before [135]. Some hypothesize that the ability to use tools and gradually increase tool complexity is precisely the key point helping humans have breakthrough achievements compared to other organisms. Some scientists try to prove that learning ability and abstract thinking are weapons for intelligent homo sapiens to win animals stronger than themselves. Here, I want to view from another aspect to explain human success (even if temporary). That is the method of expanding one's self beyond physical body limits.

Extended self (extended self) is not a new concept; it was put forward by consumer behavior researcher Russell Belk in 1988, holding that human self is not only limited to an individual but can expand to external objects outside the body [251]. If these objects can create significant causal powers related to the self, they are considered an extended self. Causal power can manifest in many different forms, material and immaterial. A soldier's shield in a battle can block an enemy's sword strike, determining that person's life, so the shield is truly an extended self, an extended body part. In 1284, King Trần Nhân Tông opened the Diên Hồng Conference to decide policies against Yuan-Mongol armies. Opinions of elders can be considered an extended self of wisdom of the young king. The contribution of

generals is an expansion of material power to help the Trần dynasty three times defeat northern enemies.

Any individual existing in the world has a physical body understood in the sense that it occupies a space in a certain time period. It can be easily seen that an elephant will occupy a larger space than an ant. However, physical boundaries of each individual are not as simple as what can be seen. That is the necessary survival space for organisms to exist and develop. If so, body size cannot fully describe this survival space. An Asian elephant's survival space may not be larger than a migratory bird's survival space. An Asian elephant needs about 400 km^2, equivalent to a square with side length about 20 km [252]. But a bird may need more than that. In 2022, scientists observed a record by a migratory bird scientifically named Limosa lapponica, 5 months old, flying continuously for 11 days to cross a distance of 13,560 km (longer than Earth's diameter) from Alaska, US to Tasmania, Australia. This is the longest distance humans have observed and was recorded in Guinness records [253]. After the flight, the bird lost half its body weight.

Organism species always have desires to expand individual influence to the external world. Tools for expansion can be another individual of the same species. Group cooperation between individuals of the same species is an example. When ancient people knew how to hunt together, that was when they realized expanding their own capacity. Stone tools are also tools to enhance human capacity. In feudal times, weapons were obviously extremely necessary in wars. If a lion needs long fangs to hunt and protect territory, then a sword is an extended part to enhance capacity for a soldier. If a hippopotamus has very thick skin to resist lion attacks, then armor has similar function. The difference is that animal species need a long time to evolve to have certain advantages, while humans take much shorter time to achieve that. Moreover, lion fangs and thick hippopotamus skin are always tightly attached to them even when unnecessary, and of course this brings many troubles and costs. Imagine a hot

day but still having to wear a thick cotton blanket? But humans can change that very easily. Swords and armor are only needed during war; if not, they will be stored. Extended parts of the human body change very fast and depend on human space-time context.

In modern society, most tools that humans use can be considered an extended part of the body. For example, glasses help see things more clearly, cars help move faster, machinery helps muscle strength greater,... Can the brain be expanded? Recent research shows that mobile phones are acknowledged by many people as an expansion of humans in three aspects: functional, anthropomorphic, and ontological [254].

Imagine a day without smartphones to see how necessary they are for our lives. Nearly half of the US population thinks they cannot live without them. South Korea is the country with the highest usage rate in the world, up to 95 %. Functionally, phones are the most diverse expansion ever. Phone flashlights help see in the dark, applications help connect with others without needing to meet directly, monitor children and pets remotely, store information of relatives and friends, expand intelligence through online content,...

Smartphones have expanded human physical and mental abilities. Anthropomorphically, although smartphones are currently only inanimate objects, many people increasingly feel biologically attached to them. Even when not needing any function on the phone, many people still feel nostalgic. Ontologically, phones not only serve humans according to owner wishes but conversely affect psychology and physiology in ways not like their wishes. For example, very many people are distracted mentally when having a phone nearby. Not only beeps when there are calls or messages but even when there is nothing, people always keep an eye on it expecting something will happen. Or some people feel uneasy when not having a smartphone on them because they may miss some important information.

Not only that, smartphones will replace many other extended parts. In many countries including Vietnam today, people go out

without needing to carry wallets because applications for personal identification, driver's licenses, bank accounts, health insurance, electronic medical records,... have been integrated into applications. Users only need their phone.

Moreover, artificial intelligence AI integrated in smartphones is increasingly important and has great causal power for users. They can help many daily life tasks. Most obvious is predicting characters when typing text, suggesting routes in traffic, voice assistance, translation assistance, email classification, automatic image classification,... Higher is collecting information and habits to then give appropriate suggestions. Previously, readers had to go to each website to read news; now AI will suggest information that readers care about.

That does not mean smartphones only bring good things. They can be tools for bad people to attack and steal information, lead us to become social media addicts. Social networks are designed to hit the psychology of desire for recognition in each human, making us long-term attached to them. According to Vnetwork statistics, early 2023, out of 100 million Vietnamese people, about 78 million use the internet, 161.6 million mobile devices. Daily, each user spends 6h23m browsing the internet, of which 94.5 % use mobile devices. The number of social network users is 71 million people, of which 66.2 million use Facebook, 63 million use Youtube, 10.35 million use Instagram, 50 million use Tiktok, 5.2 million use LinkedIn, 4.1 million use X (Twitter) [255]. Facebook advertising reaches 67.2 % of the population and has very accurate targeting. Social networks bring humans closer together online but separate humans in reality. How to balance between the real world and the virtual world is an issue that needs thinking and solutions from now.

4.4.5 Illusion about the self

Here we notice there are two extremes: one side is egoism (egoism) and one side is no-self (anatman). Any normal person among us chooses a position somewhere between these two extremes. Each person's self never pushes to egoism, thinking we are highest, independent and unchanging, but also does not think there is no self at all. We still live, think, act, always receiving and leaving causal powers with the surrounding world. The problem that Buddhism raises is how to balance between the two extremes, find the middle way between egoism and no-self. In a society placing individual I, the self too high, balancing back through elevating no-self is necessary.

We examine suffering from illusion about the self manifested in satisfying desires thought to be ours but actually not ours. The human body is an endosymbiotic ecosystem of a temporary individual plus countless organisms. Therefore, that individual's desires or cravings are not necessarily their desires but are the ecosystem's desires. Seems absurd, right? I see a beautiful woman crossing the street, I imagine a beautiful figure and desire to possess. That is my desire, my individual desire; besides me, who knows? When I talk to that girl, in my heart there is happiness, even pride. I go brag to friends and receive everyone's admiration. Could those desires be of others?

No, but also not entirely. No is because that desire is not of others but of me, my individual. But not entirely here means which I are you talking about? Materially, the human body is symbiosis of many organisms. Of course, the main host is *us* understood in the sense of having a characteristic, unique genome. But remember that in our DNA, only 20 % encodes and controls the protein encoding process for *us* activities, and the rest is symbiosis of other self-replicators. We can temporarily understand that *us* is precisely the manifestation of 20 % DNA. That board of directors orders the body to act, to see that a graceful figure is a potential mate, a

living machine capable of replicating DNA, so they create proteins controlling the body to approach the other party. But does that individual know for sure that they received orders from *us* DNA? Or is the body being controlled by a microorganism or some self-replicator DNA segment.

Of course, only protein-encoding DNA can control the body and claim that is our DNA! Sure? Could a self-replicator have cleverly inserted its DNA code into human DNA, helping the body some work but also relying on the body to help replicate it. Like mitochondria becoming an organelle of cells, self-replicator DNA can also evolve to become a protein-encoding gene segment for the body. This is very difficult to prove right or wrong because in such a long DNA chain, which DNA segment represents a gene – self-replicator is very difficult to determine. Proteins control body activities, so body desires at the most basic level may not necessarily be of the body itself but of symbiotic organisms. They urge the body to act to serve the purpose of replicating their DNA. Returning to the experiment about *T. gondii* bacteria with mice on page 266 to see that the courage of mice when infected becomes very high, and they bring themselves to cats without any awareness of bacterial deception. In other words, bacteria-infected mice are completely *ignorant* about their actions. They bring themselves to cats and think that is free will in their choice.

Similarly, our body's desires and actions are deeply controlled by genes through proteins. And genes, as analyzed above, may be of endosymbiotically living self-replicators, so body desires and actions may be ignorant actions like *T. gondii*-infected mice. Moreover, consciousness is a reward for living machines, especially complex living machines like the human body. DNA can create consciousness like robots create *The Matrix* with two purposes: DNA hides its purposes and lets the body always be deluded that it is living for itself, not for the DNA master. That is precisely the first aspect: suffering from fulfilling one's desires but actually not for oneself. Here exists

another suffering: suffering from not recognizing that. Moreover, if recognized, it is even more suffering because when recognizing these things, it is very difficult to have an escape. The real escape is precisely escaping this body, escaping reincarnation.

* * *

This chapter shows compositionality at 3 levels: body, cell, and DNA. The body is a symbiotic ecosystem between humans and trillions of other microorganisms mainly inhabiting the intestines. Those microorganisms have indispensable roles in helping the human body digest food, support the immune system, help the nervous system work. Human cells have many different components, among which mitochondria are special organelles. They evolved from bacteria completely different from us and now become organelles of most cells with the function of providing energy through synthesizing ATP, are timekeepers, determine programmed death, control embryo development. Without mitochondria, obviously cells cannot operate. The human genome is also like genes of other organisms, formed from very many genes of related organisms in the evolutionary process from billions of years ago. The human genome is a symbiotic ecosystem where genes cooperate and compete to exist and replicate. Impermanence of the human body at all 3 levels is manifested by continuous matter renewal of the human body. With those two properties of compositionality and impermanence, the human body is empty.

Chapter 5

Consciousness

Abstract. The emptiness of consciousness is manifested by: (1) Consciousness is the development of basic perception, collective intelligence at different levels, so consciousness comes from many different neural circuits. Consciousness is not independent but a consequence of smaller neural circuits. (2) Senses only receive a small slice of reality to form information transmitted to the brain. Humans only perceive electromagnetic force; all signals propagating in the nervous system are identical electrical pulses. (3) Information from the external world is decoded according to space, time, and numbers to create models of reality. Models do not need to be complete, do not need to be truthful, but only need to be appropriate so individuals, groups of individuals, and society can adapt to changes in the external environment. (4) Joy and suffering are handled by the same brain area through self-regulating neurotransmitter balance levels. Suffering is excessive damage signaling and dissatisfaction when achieving desires. (5) What is happiness and how to achieve happiness by considering input and output factors of happiness can make us live better.

5.1 Concepts of consciousness

5.1.1 Neural theories of consciousness

Everyone has consciousness about themselves and the external world. However, defining consciousness is not simple. Scholars often define consciousness based on the field they research, such as definitions by philosophers, neuroscientists, biologists, psychologists, and physicists. We can give laws, models, scientific judgments, and test through experiments, but we cannot give an equation describing consciousness or some emotion. The reason is that consciousness is subjective experience while science is objective. Connecting subjective experience and objective material world is not easy at all. To date, science has made many advances in understanding nervous system activities, but the problem of how the brain forms consciousness is the **hard problem** of science, a problem put forward by philosopher David Chalmers in 1995 [256]. This chapter presents new research on neuroscience to help us understand closer how consciousness forms, from which see that emptiness is manifested in all human consciousness.

There are many schools of approach to research consciousness, but we can divide into 5 schools [257]. *Materialism* holds that matter comes first and determines consciousness. Although we do not yet fully understand the consciousness generation process, most scientists acknowledge consciousness is a consequence of physical and chemical activities of matter in the brain. *Idealism* holds that consciousness comes first and is the foundation of reality. The external world and self are representations of mind, obtained through subjective experiences. *Dualism* holds that matter and consciousness are two separate entities existing independently. In humans, these two entities combine. When dying, matter disintegrates but consciousness can still exist. *Panpsychism* (panpsychism), also called panpsychism holds that consciousness and matter are not yet fun-

damental; there is proto-consciousness (proto-consciousness) from which consciousness and matter arise. In modern science, panpsychism may relate to information and holds that information is the ultimate entity from which other things arise.

Modern *neural theory* holds that consciousness arises from special brain activities; thus, these theories are materialism. There are about two dozen different neural theories about consciousness, all acknowledging consciousness is a model of the external world [257, 258]. The brain is a prediction machine helping the body (playing the role of a living machine) to predict the future accurately. The brain builds models of reality; that is precisely consciousness, a tool in the evolutionary process, a part of the body serving the evolving subject (in this book, genes) to use to exist and develop.

Quantum consciousness theory holds that consciousness arises from activities at the microscopic level, where quantum properties become important, and the emergence of microscopic atoms creates macroscopic consciousness about oneself and the world. Most notable is the theory of Roger Penrose and Stuart Hameroff hypothesizing that consciousness arises from quantum effects inside neurons. Neurons have microtubules (microtubule, Figure 4.1) created by proteins named tubulin, with diameter about several tens of nanometers, length can reach several tens of µm, are physical structures for cells to have certain shapes and are places where quantum effects can occur. Consciousness appears from quantum effects occurring in many microtubules and resonating with each other like a choir creating a melodious piece from many different instruments. The piece is consciousness; the instruments are microtubules inside neurons. A weakness of this theory is that quantum effects can only last at very low temperatures (near absolute zero) and must be isolated from the external world. But in a 2024 study, US scientists showed quantum effects can occur at room temperature, in a non-isolated environment [259]. The above research only strengthens Penrose's and Hameroff's hypothesis in that quantum effects

can appear in neurons under normal conditions; how consciousness forms from those quantum effects still needs other research.

Distinctions between the above theories are only relative. In reality, there is tight interweaving of ideas from different schools in the same specific theory. For example, integrated information theory (IIT) about consciousness is a materialist theory (when holding that neuron activities create consciousness) but combines with panpsychism theory (holding that information is the basis in brain computations). Some physicists like John Wheeler even hold that everything is information. This hypothesis is a variant of panpsychism. Wheeler holds that the universe is mathematics with its object being information. Information is the most fundamental category, coming before both matter and consciousness. Matter in the world comes from emergence (emerge) of many discrete and tiny information pieces. From matter forms human consciousness about the external world. His thought is expressed in a short and concise saying *it from bit* (it from bit) [115].

5.1.2 The brain is a Turing machine

Current computers all work based on simple algorithms proposed by Alan Turing.[1] Nerve cell activities can be reduced to logic gates, and brain activities can be considered as the most complex supercomputer system humans have ever known. This computer operates with highest efficiency with power consumption about 20 W whether awake or asleep but can perform a billion billion 10^{18} operations per second, called an exaflop. For comparison, the supercomputer

[1] Alan Turing (1912 – 1954), British mathematician, famous in computer science and artificial intelligence. He proposed automating thinking by dividing into simple operations that today are called algorithms – the operating basis of all computers. In addition, Turing also proposed the Turing test to measure machine capabilities, the halting problem (halting problem) to show there are problems that cannot be solved by algorithms. He also contributed to decoding Germany's Enigma machine, helping Britain win World War 2. Unfortunate personal life led to his suicide at age 42.

Oak Ridge Frontier with similar computing ability needs 20 MW, an amount of electricity larger than the brain by 1 million times [260]. The operating principle of the brain is to build a model of the external world so the body can best adapt to the surrounding environment. If so, the internal model manifested by consciousness and external reality need to be as close as possible. However, the problem lies in that objective reality of the external world is very diverse and rich while resources of time and matter are very limited, so each species, each individual chooses an approximate model of reality.

Many people oppose considering the brain as a computer. But neuron activities as logic gates and connections between them under regulation of neurotransmitters as factors increasing/decreasing signal weights are the basis for Walter Pitts and Warren McCulloch to build initial models of neural networks (neural network) in 1943 [261]. Eighty years later, ideas about artificial intelligence AI like chatGPT are developed based on computer learning following nervous system operation methods in the brain. Each neuron receives information from hundreds of other neurons. Information may not be accurate, but the combination of many neurons creates highly accurate information based on majority data. It can be said **quantity creates a unique quality**. A weakness of the neural network system is energy consumption.

Transistor activities consume energy millions of times larger than the brain, so scientists at NIST thought of a new method to increase AI efficiency [260]. Imagine we send a gift from point A to B in a large city using 5 cars with the same energy consumption per km but each has a different route. In current computer models, 5 cars go 5 routes and arrive at destinations at various times. Energy consumption is consumption of 5 cars multiplied by distance traveled. In the new computing model, cars communicate with each other to know which car is closest to the destination. If distances between cars differ much, cars far from destination will abandon the trip to

save energy; only the car closest to destination finishes the journey. This type of computer operation is called race logic (race logic) based on time computation. Instead of all 5 cars having to reach destination, only the fastest car reaches destination. Operating this way will save much energy because the number of connections between one neuron and other neurons can reach several thousand. Each thought can have millions to billions of neurons participating, so energy savings as above can explain brain operation. To date, humans truly have not understood how consciousness is actually formed. However, to avoid going into theories not yet verifiable by experiment, we temporarily hold that brain activity is like a Turing machine.

Speed and efficiency of the brain are things that silicon-based Turing computers have not yet achieved. However, if we believe in the computation principle to explain brain operation, it may be a clue leading to the answer. Page 334 presents action potentials of neurons. Previous research held that this neural signal propagates at very slow speed slower than sound speed. In 2024, Chinese scientists published important research in solving speed and signal efficiency [262]. They showed that entangled quantum photons can appear inside the myelin sheath surrounding the main axis of neurons (Figure 5.4). Thus, the myelin sheath not only helps neural signals move in axons faster but is also an environment where entangled quantum photons of neurons can appear and synchronize information at nearly instantaneous speed and at long distances. This research cannot yet answer the question about brain efficiency but only suggests to us that there are still many physical and chemical phenomena occurring at the microscopic level in the brain that we do not yet know.

The Turing test (Turing test) is a very simple test to check computer intelligence. The test includes an interrogator questioning two objects: a human and a computer. Human and computer hide faces so the interrogator does not know whose answer it is (Figure 5.1).

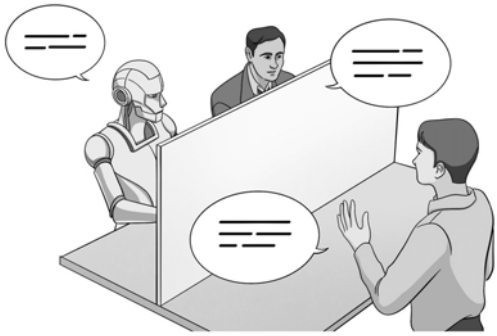

Figure 5.1: Diagram of the Turing test. The interrogator (right) asks the respondent and computer. The interrogator does not know which is human, which is machine. The respondent and computer try to answer as human-like as possible.

The interrogator uses natural language to question. After receiving answers from human and machine, if the interrogator cannot distinguish differences, it can be said that the computer can create behaviors like humans [263]. Such a computer is called having passed the Turing test.

Recently, in May 2024, scientists at the University of California, San Diego published research results with 500 randomly selected interrogators showing chatGPT-4 has passed the Turing test. Topics of exchange content are very diverse: from mathematics and logic, general knowledge to opinions, emotions, human experiences. ChatGPT-4 was evaluated as human-like in 54 % of cases, while chatGPT-3.5 and ELIZA (a natural language processing computer program developed by MIT) did not pass the test [264]. Certainly, AI capabilities will be improved very fast in the future. That shows that although considering the brain as a Turing machine will have many shortcomings (for example, about philosophical zombies – philosophical zombie, a concept put forward by David Chalmers to prove consciousness is not simply described through behavior [265]), we temporarily accept the brain is a Turing machine in discussions in this book.

Some people will ask why we must accept the brain as a Turing machine in subsequent discussions. That is because we often accept an approximation when researching scientific problems. First, brain activity, though complex, is based on principles of physics, chemistry, and biology. This is an affirmation of materialism understood in the sense that matter (atoms, molecules of the brain) determines consciousness (reflecting objective reality). Second, neuron activities follow basic logic principles, a specific manifestation of which is binary logic gates with the principle of taking signal sum presented in Section 5.2.3. With those two approximations, the brain builds models of reality, creates consciousness to operate the body, helping the body to exist and adapt well to the external environment. With the above hypotheses, we have excluded supernatural forces intervening in human consciousness. Emotions are also like that, all controlled by physical, chemical, and biological processes inside our bodies.

Unresolved problems like speed and energy needed to process information in the brain still exist, but not because of them do we change the approach. Experiments about the existence of quantum states and quantum entanglement in nerve cells may suggest other brain capabilities to solve problems being encountered. Certainly, scientists need other research, but such an approach to consciousness is currently a common and seemingly correct approach.

A major challenge in researching consciousness is its subjectivity. Consciousness is personal experience, cannot be directly observed from outside. This creates a "black hole" in scientific research: how can we know others' consciousness? How to overcome the barrier of subjectivity to research consciousness objectively? This is precisely the problem that neuroscientists must face and solve.

5.1.3 Overcoming the consciousness black hole

Because consciousness and emotions are subjective, they are called the *consciousness black hole*. To be able to know others' consciousness, scientists must overcome this black hole. The traditional method is interviewing, a method that psychologists have used for about 100 years to access subjective consciousness and emotions. However, interviewing can tell some problems of consciousness, but consciousness may not know all problems related to itself. World-famous psychoanalyst Sigmund Freud held that the unconscious or preconscious can even control consciousness. We often do not know the roots of things we choose. There are many processes outside consciousness affecting our actions. There is an effect called *priming* where one action affects another action. For example, if a person holds a hot drink, that person will tend to describe relationships with another family member more favorably compared to when holding a cold drink. The reason for this to happen is the brain's confusion about physical warmth being synonymous with spiritual warmth [241, page 117]. Freud held that dreams can be gates into the unconscious [266]. Therefore, interviewing patients about their dreams can access the unconscious. After Freud, many scholars developed other psychological techniques like hypnosis, meditation to research neurological problems but mainly still based on interviewing.

The interviewing method has many limitations that cannot be overcome. Research on patients with *split brain* (split – brain patient) shows that two cerebral hemispheres create two independent consciousnesses leading to interviewing not bringing unified consciousness. Since the 1940s, people saw that seizure symptoms in epilepsy patients can be reduced by cutting the corpus callosum (corpus callosum). The corpus callosum consists of nerve bundles connecting the two cerebral hemispheres. The existence of the corpus callosum helps the two cerebral hemispheres exchange infor-

Figure 5.2: Diagram of the test with split brain patients. The patient sees the word *Key* on the board with the left eye. This information is only transmitted to the right cerebral hemisphere. The left cerebral hemisphere processing language does not recognize left eye information, so when asked what word they see, the patient says they do not see any word on the board.

mation and form a single consciousness about the self. American psychologist who won the Nobel Prize in 1981, Roger Sperry, saw that the corpus callosum is a neural intersection between the two cerebral hemispheres. If the corpus callosum is cut, communication between the two cerebral hemispheres will stop and leave consequences that the two cerebral hemispheres behave completely independently [267, 268].

Although the two cerebral hemispheres are geometrically symmetric, their functions are very different. The left cerebral hemisphere tends to prioritize serial, sequential, rational, linguistic, independent, skeptical processing,... while the right cerebral hemisphere tends to prioritize parallel, experiential, emotional, image, collective, sharing processing,... (see differences between the two brains in document [269, pages 55 – 56]). Right and left brains differ in function. Through vision, the left brain recognizes words and letters while the right brain recognizes faces and emotions. The left brain receives language while the right brain recognizes music. In

a famous experiment, American neuroscientist Michael Gazzaniga gave a split brain person named Joe a board with two parts. The right side is blank while the left side has the word "Key". With a bit of technique, Joe only sees each image with one eye, i.e., sees the word "Key" with the left eye while the right eye sees nothing (Figure 5.2).

Because the left brain handles language function, the right brain handles image function, so for a normal person, if asked what they see, the answer will be seeing the word "Key" on the left side of the board and seeing nothing on the right. Information from the left eye when seeing the word "Key" will be transmitted to the right cerebral hemisphere. When receiving the question by sound, the left cerebral hemisphere will have to prepare the answer. It receives information from the right cerebral hemisphere through the corpus callosum. But Joe is a patient whose corpus callosum has been cut, i.e., there is no communication between the two cerebral hemispheres. Information about the word "Key" in the right cerebral hemisphere cannot be transmitted to the left cerebral hemisphere. Therefore, Joe answers that he does not see any word because the left cerebral hemisphere receives information from the right eye, and the right eye only sees the blank right side of the board. Thus, Joe's two brain hemispheres became two independent people, together controlling one body. The image of the word "Key" reaches the left eye then the right cerebral hemisphere and only stops there. The two brain hemispheres cannot combine. Each brain hemisphere keeps information about its own image, completely independent [270].

Similar experiments like that are very many and have proven that when the corpus callosum is cut, the right and left brains form two independent consciousnesses on the same body. Today, people no longer use the split brain method to treat epilepsy patients, so experimental research mainly relies on past documents. If there is any most convincing evidence about materialism, it is the experiment with split brain people. Matter, here is the corpus callosum of the

brain, determines consciousness. The determination here is precisely determination of consciousness about the self. If the corpus callosum remains, unified self-consciousness remains. If not, each hemisphere gives one consciousness about the self and external world. Experiments about split brain patients show that the interviewing method to overcome the consciousness black hole may not be accurate. Finding other methods based on experimental equipment is necessary to research subjective consciousness problems.

Split brain research gives another view about no-self and emptiness. Our self does not only come from two cerebral hemispheres but actually comes from very many neural circuits. Each neural circuit sends signals to consciousness to decide. When seeing a sweet candy, many neural circuits send information: eat candy because it is delicious, do not eat because afraid of getting fat, last night already exercised so can eat one still okay,... At any moment, we must choose, and when choosing, we must split and decide. The self that we see as only one is continuous compromise, cooperation and competition of thoughts appearing and disappearing continuously. No-self is there; it comes from many components cooperating and competing with each other, each component continuously changing. That is emptiness of consciousness.

5.1.4 Technology helping neuroscience research

In the 1990s, some techniques helped very much for neuroscience research. That is functional magnetic resonance imaging fMRI (functional magnetic resonance imaging – fMRI) invented by Seiji Ogawa to measure brain activity related to blood flow. When brain parts are active, blood must be pumped to them, and fMRI records this blood amount. Besides fMRI, positron emission tomography (positron emission tomography – PET), transcranial magnetic stimulation (transcranial magnetic stimulation – TMS), electroencephalography (electro-encephalo-graphy – EEG) help observe brain and nerve cell

activities without needing invasion. These techniques allow scientists to know which brain parts are active to then attach to subjective thoughts through interviewing patients.

Consciousness research had many changes in 1995 when Francis Crick[2] and Christof Koch proposed that scientists should try to find neural correlates of consciousness (neural correlates of consciousness – NCC) [271]. To solve the hard problem, scientists use two methods: (1) interview researched people to understand their subjective experiences and (2) use scientific equipment like fMRI or TMS to objectively observe activities of brain regions. NCC is the minimal neural mechanism for any conscious experience. Of course, the entire brain is an NCC, but scientists try to find smaller brain regions to see where the brain generates consciousness. After the paper appeared, NCC research achieved much progress. From research results on patients with damage to different brain regions, scientists came to the conclusion that consciousness arises from the posterior cortex region (posterior hot zone) [272].

However, only relying on subjective reports of researched people brings unwanted results because distinguishing between real NCC and prerequisites or consequences of consciousness is very difficult. Therefore, we need to build neural theories (theories of consciousness), where instead of only finding correlations between brain activity and consciousness, scientists need to find explainable relationships between neural mechanisms and aspects of consciousness [273]. The key bridging subjective experience and objective material world: information. For example, integrated information theory (integrated information theory – IIT) [274] put forward by G. Tononi in 2004 is attracting attention from many scientists. Information

[2]Francis Crick (1916 – 2004) British molecular biologist famous for discovering the double helix structure of DNA thanks to which he received the Nobel Prize in Physiology or Medicine in 1962. He is the one who put forward the central dogma of molecular biology and researched neuroscience.

modeled by mathematics can be said mathematics is the intermediary between subjective and objective.

Scientists have tried to solve the subjectivity problem of consciousness in different ways. Crick and Koch used interviewing patients to know subjective thoughts then attached to objective results from fMRI, PET, EEG measurements. Because interviewing cannot bring accurate subjective sensations, people must build a mathematical model as an intermediary to bridge objective results and subjective experience. Mathematics has both objectivity of a science but is also a product of human rational subjective consciousness. Using mathematics to connect objective – subjective is reasonable and may be the answer to the hard problem about consciousness. Although mathematics power may be questioned like Stephen Wolfram put forward cellular automata to replace mathematical equations [92], or Kurt Gödel proved the incompleteness theorem of mathematics [275], but to date, mathematics power has not been shaken. This is like nearly 500 years ago when Galileo held that science follows mathematical equations. At that time, no one thought that nature's complexity could be described by a purely thinking tool: mathematics. Since then, not only affirming mathematics' role but its complexity increasingly grows. Even developing to abstraction levels beyond human imagination. Strangely, that complexity does not hinder mathematics' correctness when applied to explain natural phenomena at the deepest levels. Or like when humans researched steam engines, initial concepts seemed abstract like temperature or entropy already confused scientists but later gradually quantified and understood very accurately.

Other theories mainly relate to neuroscience research. Some theories being concerned by scientists like higher-order theory (higher-order theory – HOT), global workspace theory (global workspace theory – GWT), predictive processing theory (predictive processing theory – PPT) [273]. HOT holds that information from senses

reaches low-level processing regions like visual cortex but has not yet created consciousness. Information is transferred to cortex regions in the front then consciousness forms. That is precisely super-representation, where information not only transmits one-way but has mutual exchange to create information circuits. GWT holds that there is a global workspace region located in the front and parietal cortex regions, which is the center connecting all brain information. When information enters this region, they are broadcast to the entire region; information is exchanged and competed to get attention before being synthesized and decided. PPT holds that the brain operates like a prediction machine; it will send signals from top down (top-down) to senses, which have the task of supplementing whether that prediction is correct or wrong. Top-down manifests proactivity: the brain gives expectations then checks and updates information. After collecting information, the brain calculates probability after being updated with information, which is the inference process.

Where does consciousness appear in the evolutionary process of life? Is it only humans who have consciousness, or has consciousness appeared from earlier stages in evolutionary history? To answer this question, we need to examine different levels of perception, from the simplest forms to the most complex forms. This will help us better understand the emptiness of consciousness: consciousness is not an absolute entity, appearing suddenly, but a continuous evolutionary process, from basic perception forms to complex forms like human consciousness.

5.1.5 Levels of perception

The origin of life and the origin of consciousness are topics that have been researched by many scientists for a long time. Although this is a topic causing much debate, the following arguments are based on ideas about selfish genes of R. Dawkins and theories about consciousness of some neuroscientists like D. Eagleman and A. Seth

[276, 277]. From an abiotic world of earth and rocks, life was formed randomly without purpose. We have defined life as a self-sustaining chemical system capable of changing and evolving according to Darwin's theory. Self-replicators are genes that have used living machines to increase self-replication efficiency. So what are living machines? Of course, they are machines, tools performing instructions of those controlling them, masters of machines, which are self-replicators. With life, self-replicators are precisely genes encoded by DNA.

According to R. Dawkins, if the purpose of genes is replication and existence, no more, no less, then living machines clearly have purpose [119]. Genes coordinate with each other to create living machines, which are individuals that can be unicellular or multicellular, simple or complex, with brains or without brains. However, living machines cooperate and compete with genes to coexist and develop together. Not only humans have purpose; animal and plant species all have purpose. Actions seeking food, competing for mates, protecting children,... are purposeful actions of most animals. So having purpose, or purposefulness, certainly must be a property to increase machine efficiency. A machine without purposefulness obviously cannot be as efficient as a machine that already has purpose from the beginning. Through billions of years of evolution, that purposefulness became a property tightly bound with organisms.

In 1943, A. Rosenblueth put forward classification of perception levels according to behavior still used to this day [278] (Figure 5.3). A consequence of purposefulness is precisely perception. Here there is a difference between two concepts: consciousness and perception. Consciousness has more generality than perception, containing subjectivity while perception is narrower and objective. Perception is an object of science and will be less debated than consciousness. Rosenblueth's classification of perception levels increases gradually from passive behavior, active, purposeful, feedback, prediction. The high-

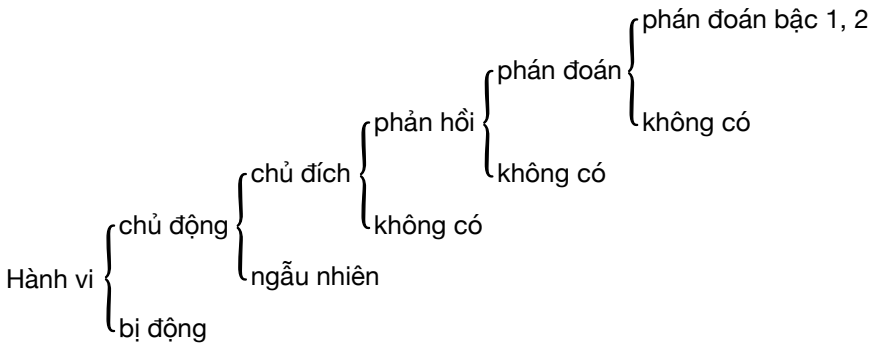

Figure 5.3: Classification of perception levels according to behavior [278].

est perception capacity is the capacity to predict things that may happen in the future. In humans, perception capacity also manifests in level-2 prediction capacity, which is the capacity to predict the future beyond the physical space and time that humans are living in to reach very distant space and time. Prediction capacity can be considered a property of consciousness. If so, consciousness was born as a measurement device between desired state in the distant future and current reality to then adjust actions of living machines so that the difference between these two is smallest. A trait helping machines achieve purpose will be passed to the next generation, but a trait hindering achieving purpose will have no chance. That is precisely the existence of the most appropriate.

So how do genes control living machines? Gene instruction information is translated into proteins. Proteins are the only thing that DNA can directly create in short time. Proteins create structures regulating body activities. The process from DNA to protein then to body actions is a complex process, but however complex, DNA is still the starting point. Body structure of organisms is precisely living machines, developing from simple to complex. The more complex machines, the more information needs linking between parts increase. Notification and signaling to each other between cells appear in microorganisms [279], plants [280], and of course animals.

In animals, complexity gradually increases from animals without nervous systems to having nervous systems. The nervous system is specialization of some cells for notification and signal processing. Most developed of these animals is the brain with consciousness formation in some higher animals. Consciousness is like a purposeful tool for living machines to operate more efficiently. Consciousness not only measures purposefulness efficiency but can also simulate reality. In science, simulating or modeling a theory is regular work of scientists. People base on some hypotheses and initial conditions to simulate some problem. Simulation results on computers after comparing with experiments will determine correctness of hypotheses or boundary conditions. Consciousness has similar simulation function as above. Consciousness is advanced development of natural simulation ability.

So genes are the supreme meaning of life. But is it really so? Genes are created by DNA segments consisting of bases ATGC arranged in order. Essentially, DNA segments carry information capable of encoding proteins to manifest as causal power outward. ATGC bases are specific carriers, so can be replaced by other carriers. A cyanophage (cyanophage – virus specializing in attacking cyanobacteria) has replaced adenine (A) with 2-aminoadenine (Z). Both compounds create hydrogen bonds with thymine (T), but there is one difference: two bonds between A and T are increased to three bonds between Z and T. Thus, four bases ATGC are converted to ZTGC. This discovery was found by Russian scientists in 1977 but only recently studied carefully [281]. The cause of this replacement process is unclear, but cyanophages may want to deal with protective enzymes by self-changing components creating their own genes. Evolution through natural selection is precisely the deep cause of all these processes. ATGC, ZTGC, or whatever are only information carriers. Of course carriers are important because they bring material causal power. Without carriers, information does not exist. Many scientists and philosophers hold that information is truly the foun-

dation of all things, the source of both consciousness and matter, and of course life. This is an example of panpsychism when holding that consciousness, here is precisely information, is an intrinsic, fundamental property present everywhere in the universe. Not only organisms with perception have information, but all matter has it with different levels from fundamental particles to living organisms.

Here, very many people will wonder: could consciousness of higher animals like humans only be obtained through random evolution, just machines executing gene commands? A body with proper consciousness must obey unconscious genes? If anyone watches the movie *The Matrix*[3] will clearly see that The Matrix is precisely the reward for humans when accepting being batteries for the robot world. The movie tells about human existence journey when having to fight unevenly with the robot world. Robots seek energy sources from heat that human bodies create for robot activities. But to prevent humans from rebelling, robots create a simulated world named The Matrix to lull humans, to make humans think they are free, have will to do what they want, but actually humans are slaves to the robot world.

Self-replicators – gene segments are the evolving subjects. They use living machines to achieve existence and replication purposes. Self-replicators intentionally hide and give living machines a reward. Consciousness is the reward for living machines to have illusions about the self, about free will to do what they want. When consciousness thinks itself is the master of itself, it will no longer fight with the real master. The self has selfishness. In the natural selection process, living machines actively act through consciousness

[3]The Matrix (The Matrix) is a multi-part science fiction movie directed by the Wachowski sisters (previously Wachowski brothers because both have transitioned genders) Lana (Larry) and Lilly (Andy), first released in 1999. The movie left marks through many generations of audiences not only thanks to cinematic effects but movie content. Most notably, humans live in a simulated world called The Matrix created by robots. Some people tried to escape The Matrix to return to real life and fight against the robot world.

control because at this time, self-replicator interests and living machine interests have become one. That is preserving and replicating self-replicators.

5.1.6 Basic perception

Buddhism brings equality in liberation not only between humans and humans but also between humans and other organisms. Buddhahood (Buddhahood) exists in all organisms, so all organisms can achieve liberation through the middle way. Scientifically, is that feasible? Can different organisms have some minimal perception to practice the middle way and achieve liberation? In modern science, this can happen thanks to research on **basic perception** (basal cognition). According to Rosenblueth, *perception is the ability to pursue goals* [278]. Basic perception is perception at the minimal level to be able to pursue goals by many different methods. But goals are not easy to determine because they depend much on survival space, on examination time.

All organism goals gather into goal space lying in living space and time, called organism *survival spacetime*. Survival spacetime of bacteria is very small; of dogs will be larger but still only within possible movement range and lifetime. Humans are different; survival spacetime can exceed space and time of a human lifetime to reach beyond Earth, touch stars in other galaxies, return to the past until the universe was born.

We have discussed human consciousness as reflection of objective reality into the brain. Thus, we have acknowledged the following: (1) Consciousness is a product of the brain, i.e., only organisms with complex nervous structures like the brain can have consciousness; (2) That consciousness is unique for individuals; consciousness of smaller components does not exist; consciousness is emergence of the entire body; (3) Organisms without brains will not have consciousness; reactions of organisms without brains are only physio-

logical reflexes, nothing more, nothing less. That reflex differs from consciousness in nature.

Sociality of incomplete viruses can show they may have perception and intelligence at some level. Can viruses, organisms at the boundary between life and non-life, have basic perception or intelligence? Many people think viruses are chemicals because we often examine them independently, rarely examining many viruses in relationships with each other. Recent research shows viruses have very high sociality; they cooperate, compete, deceive, and secretly mobilize with other viruses.

In the 1940s, Danish biologist Preben von Magnus discovered many viruses have defective genomes compared to other normal viruses [282]. This phenomenon is called the von Magnus phenomenon, and viruses missing genes are called *incomplete viruses* (incomplete virus). Incomplete viruses are naturally born in replication processes in bacteria, not from laboratory research. Normally, mutations only occur with a few errors, but with incomplete viruses, they lose very significant DNA segments. Replication with errors leads to virus genes shorter than normal, so viruses will lack necessary abilities to replicate themselves. Normally, individual viruses with mutated genes will become extinct. However, incomplete virus vitality is very fierce. To be able to exist and replicate, incomplete viruses develop a different strategy. Instead of perfecting themselves, they deceive normal viruses by stealing normal virus matter to replicate incomplete viruses [283]. In this deception process, incomplete viruses have advantages because their genomes are significantly shorter than normal viruses.

How do they deceive normal viruses? For example, incomplete viruses mutated from flu-causing viruses do not invade cells themselves but invade together with normal viruses. Because genomes are missing, incomplete viruses cannot synthesize necessary proteins to create shells to package their genomes. Incomplete viruses will wait for normal viruses to take host matter to create shells. As soon

as shells are formed, incomplete virus genetic material will quickly sneak into normal virus shells. Because genomes are small, incomplete viruses enter more easily and occupy protein shells they cannot create themselves. Normal virus genetic material has no shells, so cannot continue replicating.

With such deception strategies, incomplete viruses will have advantages compared to normal viruses: smaller genomes and less replication costs. If this process continues forever, normal viruses will become extinct, leaving only incomplete viruses. But once normal viruses become extinct, incomplete viruses also cannot exist because no one creates shells. Here, incomplete viruses develop another strategy. Instead of deceiving normal viruses, incomplete viruses cooperate with each other. This may have happened with a virus type specializing in attacking parsley plants. Initially, the virus had 8 genes, but later mutated into 8 incomplete virus types, each with 1 gene. At this time, they need to cooperate with each other so that one type's function can compensate for another type's function. All 8 incomplete virus types cooperate to attack parsley plants together; then they can create 12 necessary protein types to replicate normally [284]. Mathematical models about virus deception lead to developing viruses with very fragmented genomes called nanoviruses [285]. Each nanovirus cannot replicate itself, so they need to cooperate with each other. Behind the form of cooperation hides deception content.

Viruses with short genes perform deception behavior more than cooperation. Not only deceiving normal viruses, incomplete viruses can cooperate with host immune systems to fight normal viruses when normal viruses develop too strongly, leading to host extinction. In the virus world, deception dominates. A virus type named Sendai SeV-Cantell creates many incomplete viruses. These incomplete viruses cooperate with mouse immune systems to attack normal viruses. They do this to avoid hosts being attacked too strongly by normal viruses, leading to exhaustion and death [286]. This sug-

gests a unique anti-virus treatment method: fighting poison with poison. Using incomplete viruses that do not affect hosts to kill normal viruses that harm hosts.

Research on virus sociality shows viruses are living organisms not only because they have genetic material DNA or RNA but because of their social behavior. Behavior may or may not have perception, but clearly viruses are self-sustaining chemical systems and follow Darwin's evolution theory according to NASA's definition of life. Incomplete viruses need to store information to remember and know their activities and normal virus activities. They need to exchange information with hosts in cases needing cooperation to eliminate opponents. Deception strategies of incomplete viruses can show that biological evolution subjects are very likely gene segments like Richard Dawkins's hypothesis written in *The Selfish Gene* nearly 50 years ago [119].

Research on social behavior of a population of many viruses shows they may be organisms with intelligence and perception at certain levels. Deception, competition, cooperation, betrayal behaviors strengthen the possibility that self-replicators are gene segments as evolution subjects. No-self lies in cooperation of components, but individuals still want to increase the self due to local optimization in certain space and time. Emptiness of all things, here are incomplete viruses, is still manifested by their cooperation to be able to replicate.

If life is a self-sustaining chemical system (here are DNA and RNA) capable of following evolution theory, then that chemical system continuously changes from a normal virus, capable of self-replicating when invading host cells, to become many incomplete viruses that cannot self-replicate but must deceive and exploit other normal viruses or coordinate with other incomplete viruses to replicate. Clearly, over time, DNA and RNA chemical systems fragment and become inferior to the original chemical system. So what helped them survive? Although incomplete viruses are not rare, they are

not as common as normal viruses. Randomness in DNA replication and protein synthesis processes may have errors somewhere, but vitality, here is the purpose of existence and replication, does not disappear. For the self (if we can say so with virus DNA or RNA) to continue existing, discrete fragments of the self need to coordinate with each other. Absence of any component or absence of external assistance (even if unwanted) cannot make the self exist. With that meaning, the self is impermanent and also no-self. That is emptiness in viruses.

* * *

Bacteria, cell groups can also have perception. If simple organisms without brains, without complex nervous systems, even without nerve cells have perception, it will change the nature of perception research. Perception research relates to: sensing, memory, learning, communication, decision-making, problem-solving. Those abilities exist not only in humans but also in lower animals, in organisms without central nervous systems. Michael Levin, biologist at Tufts University, calls that basic perception, also called mind [287]. Human consciousness is formed from many different minds. To research basic perception based on experiments is called TAME (Technological Approach to Mind Everywhere: technological approach to mind everywhere), a research method about perception not only stopping at nerve cells and brains but also expanding to other body components like cells, cell groups, body parts [288],... Below we will see very vivid and convincing experimental research showing mind appears everywhere from independent cells in multicellular bodies, unicellular organisms, body parts to complete bodies without and with nerve cells. Mind present everywhere like that is evidence showing that organisms in the world, though very different, can still have mind at different levels, from which can act to achieve corresponding goals. If that goal is the middle way, coordination with other com-

ponents, then liberation can be achieved. Buddhahood can exist at all levels.

Starting from a silent egg cell, through gradual development, it becomes a mature body with super-perception abilities like rationality, hope, dreams,... Each person creates a perception of a unified and unique self, completely different from the dynamic emergence process of other organisms like termite colonies, bees, or ants. The learning process comes from organism intelligence.

Intelligence is a concept to measure perception level. High perception means high intelligence. Intelligence can appear in non-living entities, but perception only exists in organisms. Consciousness contains intelligence; researching intelligence to gradually approach understanding consciousness. All organism intelligence is **collective intelligence** (collective intelligence) emerging from many cells connected to each other, exchanging information to achieve a common goal of the whole. Previous perception held that from cell scale to body scale there is only one level, but collective intelligence theory holds there are many levels. Each level has its own goal space and has relative independence and autonomy to achieve that goal.

What is intelligence? Here we use the definition of philosopher considered the father of American psychology, William James: intelligence is *the ability to achieve goals by many different ways* [289]. Such a definition of intelligence does not require it to come from the brain; any system with behavioral capacity to achieve goals can have intelligence. Currently, we use the term artificial intelligence (also called artificial intelligence) AI to refer to machine intelligence. Machines have intelligence but do not yet have consciousness or perception.

In 1952, Alain Turing (the one who put forward Turing machines to create today's computers) held that there is correspondence between levels of biological morphology (morphogenesis) in organisms and perception levels [290]. Turing was the first to propose the idea that perception is formed from components corresponding to bio-

logical morphology structure levels. When observing an organism's body, at each biological morphology level, there will be corresponding collective intelligence. Collective intelligence at the highest level appears in some organisms, then called consciousness. Collective intelligence creates perception at different levels. Perception levels gather into perception scales (cognitive scaling).

For example, a mouse learns by using its foot to press a lever, then a piece of meat will appear. The mouse is a collective of different cells; some cells in the foot press the lever, some cells in the brain think about reward, possibly some cells in the intestines think about the piece of meat, but no specific cell thinks about all specific goals. Coordination of actions of different cells in a body certainly must relate to memory, action, and reward. Those things do not exist in separate parts but need organs synthesizing and recording information to then coordinate components achieving common goals. That specialized organ is the brain.

Evolution over time of life brings different goal spaces. Inanimate chemical molecules combine to become organelles (in cells); then organelles have goal space as metabolic processes and transcription and translation processes of information from genes. At cell scale, goal space is biochemical processes. Goal space at individual scale is morphospace (morphospace, body shape), behavior in 3D, even ecosystems [287]. To understand the nature of perception processes, we need to understand how perception is born. To do so, we need to research organisms without brains, even without nerve cells. A collective of cells combining to achieve some goal must communicate with each other. The means to communicate is **bioelectricity** (bioelectricity), chemicals that cells and organisms communicate with each other to notify dangers and food sources. Not only animals, even plants use bioelectricity to communicate and exchange information [291]. Starting from bioelectricity as the method that cells communicate with each other to gradually develop into nerve cells and highest is the brain. Bioelectricity is a precursor to create ba-

sic perception, then complex organisms use to develop higher-level perception.

At cell scale, goals are existence and performing some function. Agents controlling activities to achieve goals are homeostasis (homeostasis). The *try – act – escape* cycle allows cells to set a state point, conduct activities, and compare the next state point [292]. In this aspect, cells operate like a *cybernetics* system. This operation model is developed to body scale, where networks of cells coordinate with each other to create anatomical homeostasis. Cells not only operate one-way as in the central dogma of molecular biology but have feedback loops allowing systems to correct, re-establish models to achieve system goals. In one study, scientists grafted a salamander's tail onto a severed limb. Because salamanders can regenerate, the grafted tail gradually transforms into a complete limb [292]. Normally, tail cells will coordinate with each other for the tail to continue existing and developing. But after being grafted onto the limb, the tail part self-adjusts to fit the surrounding environment to become a part of the limb.

Another example, slime mold, a unicellular organism named *Physarum polycephalum* is a very popular object in basic perception research, specifically here is memory ability [293]. Researchers at the University of Toulouse, France trained slime molds for 6 days so they bravely cross a barrier obstacle which is salt (salt is like antibiotics for slime molds, so they are very afraid) to reach favorite food: oats. After remembering the lesson that crossing the obstacle which is salt will have reward which is oats, scientists put them in hibernation state for one month. Then wake them up and check if they remember experiences before hibernation. Results are that slime molds still remember previous experiences through absorbing sodium (Na^+) and using this chemical to remember. In another study, slime molds can recognize places with more food than other places to concentrate development toward that direction [294]. The above studies are only one of very many examples showing a uni-

cellular organism without nerve cells can have perception at basic levels to aim at a goal of optimizing food search processes.

Scientists at Hokkaido University, Japan also use slime molds to inspire adaptive network design. In an experiment on a flat plate limited by shapes like a map of the Tokyo region, they placed 36 food sources corresponding to 36 cities with connecting railways in the region. Food amounts placed are proportional to city populations. Scientists researched slime mold development over time. Initially, slime molds develop evenly around without priority direction, but only after a few hours, branches not leading to food places will shrink; branches leading to places with much food develop strongly to create a network connecting food sources. After 26 hours, the connection network that slime molds create almost matches the railway network connecting cities calculated and designed by humans with similar indices about cost, efficiency, and benefits. This research shows that although without brains, slime molds know how to optimize development to obtain the most food through coordination between different branches [295].

Development process from an African frog egg named Xenopus becoming a tadpole needs internal monitoring inside the egg to avoid external impacts causing morphological deformities in tadpoles. M. Levin's research group affected and caused tail deformation to then grow an eye [296]. The initial startup process is only a stimulus. Subsequent processes include gathering cells to form a complete eye, developing nerves to connect with the spinal cord, are automatically performed without needing any other intervention. Growing an eye only needs a startup signal; other processes are autonomous. This suggests to us that at body part scale, there can also be some basic perception to achieve goals, here is growing an additional eye. The eye growing at that unwanted position can still operate normally.

Planaria flatworms have excellent regeneration ability. When a body is cut into many pieces, cut pieces will grow missing parts to create smaller but perfect worms, no excess, no lack. If the cut

part has a head, it will grow body and tail; if the cut part has a tail, it will grow body and head. After regenerating into many different individuals, they operate like independent worms with full necessary parts, though possibly smaller than the original worm. The question: do regenerated worms have memory of previous-life worms? Previously, scientists believed memory is stored in the brain. If flatworms do not have brains, memory is stored in nerve clusters in the head.

People train worms with memory about food searching by putting them in a dish with an obstacle which is part of the dish bottom raised a bit. When first putting worms in the dish, they only move in flat areas. When encountering obstacles, they will turn back. After some time without food, the worm risks crossing the obstacle, and on the other side there is food they love – a small piece of liver. All worms learned the experience and remembered it: crossing obstacles will have reward. Because flatworms have regeneration ability (like leeches in Vietnam), when cut in half, one original worm becomes two independent worms. Surprisingly, both regenerated worms remember food searching experiences of the previous-life worm [297]. This experiment shows that memory does not necessarily need to be stored in the brain (in the head) but is stored throughout the body (including the tail part). Experiments cutting worms into many pieces also give similar results confirming the above hypothesis.

What is consciousness actually? Is it an accurate copy of reality, or only a model, a relative view of the world? Recent neuroscience research shows that consciousness is not a copy of reality but a model that the brain builds about the external world. This has profound meaning for understanding the emptiness of consciousness: consciousness has no fixed self-nature but is only a relative view, depending on specific conditions and relationships.

5.1.7 Consciousness as a model of reality

Returning to consciousness of organisms, especially higher animals, is development, is subjective reward of individuals – living machines of genes. Just like the movie *The Matrix*, a special character named Neo has unusual intuition about his freedom, found ways to escape The Matrix to return to the real world to fight for the existence of the few remaining people on Earth, consciousness of higher organisms like humans is not simply acting according to gene control but tries to predict the future and sometimes even resists gene commands like Neo resists the robot world.

Many scientists and philosophers propose the idea that humans are living in a virtual world (cyborg). This is not a novel idea but has scientific basis. In 2014, Nick Bostrom[4] at Oxford University published a book titled *Superintelligence* holding that with artificial intelligence, robots can self-develop into superintelligence and at some point will replace humans so they become the dominant life form on Earth [298]. Very many people were inspired by the book while showing support and speaking up recommending artificial intelligence development. Elon Musk even holds that artificial intelligence is more dangerous than nuclear bombs. Bill Gates also holds that AI development must be cautious, but he is still optimistic and supports AI development. Musk and Gates always have opposite views on important issues [299].

Superintelligence can build simulated worlds and can experiment raising human brains in a vat containing liquid and connecting electrodes from the brain to a supercomputer to then control the brain. This imaginary experiment is called *brain in a vat* (brain in a vat). However, it is refuted by many scientists. Normally, self-perception brought by senses, among which most important is touch. When a

[4]Nick Bostrom (1973 – present), Swedish philosopher researching existential risks to humans, anthropic principle,... and is the one who proposed and popularized the concept of artificial superintelligence.

person has pain in finger joints, touch is what brings consciousness electromagnetic signals through the nervous system. Other senses cannot do this replacing touch. Therefore, if a person loses touch, can that person perceive themselves? Specifically, can they perceive their physical self?

A 2021 study by a group of European-American scientists showed that it seems we are living in a real world, not a simulated world [300]. The research group was lucky to have cooperation from two touch-loss patients, one with congenital touch loss – a woman named Kim, and one with touch loss after an illness in adulthood, age 19 – a man named Ian. The difference between these two patients is that Ian once had experiences when touch was still active, and the brain stored these experiences, while Kim never had. Scientists brought the two patients to the laboratory and used necessary scientific equipment to determine images about the body in consciousness and unconscious perceptions about the body in space. Results are that there are many similarities but some very different points between the two people.

This research points out to us how the human brain perceives physical *self* – is physical self about our own body and things *outside body* – is the world outside the self when we lose the most important sense: touch. Both people perceive their bodies although without corresponding senses to collect information. These results point out that Ian, who lost touch and perception in adulthood, can learn compensatory skills by using visual input and conscious thinking to control his body. However, Kim, a person who never had touch experience, can develop methods to compensate for experience deficiency by using visual information unconsciously to control the body. This process may relate to rearrangement of neural pathways connecting visual processing with brain regions controlling the body, such as the cerebellum. Every morning when waking up, Ian always must re-establish body position relative to the external world, while Kim does not need to do so. When Kim wakes up, she simply re-

turns to her embodiment. She does not need to re-establish body position relative to the surrounding world.

This shows that people with congenital touch loss do not lack physical perception of self, but the brain, somehow, still has the ability to visualize physical embodiment of self without needing senses, without needing consciousness. The brain uses the unconscious to compensate for consciousness deficiency. The movie *The Matrix* hypothesizes that we can be living in a virtual world because a superintelligence designed a world to then connect with the human brain to control information from senses to the brain. If humans do not have senses, more accurately, do not have the most important sense which is touch, then the brain will not perceive self and outside body.

This research proves one thing: the brain still has ways to self-perceive self, self-know self-existence without needing external assistance, without needing sense assistance. If the brain self-perceives self, it is highly likely we are living in a real world, not a simulated world. Our brain is not in a vat (brain in a vat).[5] The brain always locates the body somewhere; *the brain always self-perceives the self.* Research results cannot yet deny the simulated world hypothesis because simulated worlds may be more advanced than our understanding of consciousness and install programs for the brain to operate even without senses. Certainly, we must wait for more research to be able to affirm or deny, or may not be able to affirm or deny simulated worlds.

What role does consciousness play in operating the body? If consciousness is only a model of reality, why is it so important? The answer is that consciousness is not simply a model but also the operating system of the body – an immediate control system helping the body react quickly to emergency situations that DNA instruc-

[5] An imaginary experiment proposed by Gilbert Harman describing an isolated brain placed in a nutritional vat and connected to a supercomputer creating all real experiences and sensations.

tions cannot respond to in time. This shows the emptiness of consciousness in another aspect: consciousness is not an independent, self-determining entity but a tool, an operating system empowered to serve gene purposes.

5.1.8 Consciousness is the operating system of the body

According to David Eagleman, consciousness is not only a reward about freedom of living machines but moreover consciousness is precisely the immediate operating system of living machines to satisfy gene purposes [241]. DNA has two functions: (1) creating copies of itself and (2) producing proteins. Instructions from DNA to the body are direct through proteins. Proteins create physical structures forming the body and control chemical processes inside cells by stimulating or inhibiting cell activities. However, protein synthesis processes need much time, can take days, weeks. Meanwhile, living machines frequently face emergency situations where if there is no nearly immediate reaction, the body's survival chance almost ends. Not only must react to immediate situations, but the body must also "predict" the future to be able to escape or win in emergency situations. A deer when detecting a lion near itself to dangerous levels must immediately run away to escape the pursuer. When encountering a deep hole, it must predict the hole's width to then have a strong enough jump to reach the other side of the hole mouth. All those reactions follow physical and chemical laws. Strangely, the deer does not need to know how to solve equations but can still have relatively accurate solutions.

In such emergency cases, if instructions from DNA are needed, they will fail because these instructions need too much time. Consciousness has been empowered immediate operation to solve. DNA is like a company's board of directors with the function of determining goals and plans. Consciousness is like the executive board

to implement those goals and plans. The only difference is that goals and plans that DNA gives are purposeless! DNA also intentionally hides, does not let consciousness know about its control; therefore, all living machines have misconceptions about self-determination, free spirit to do what they want as independent individuals, a single **self**. Just like most human individuals being slaves to the robot world, *The Matrix* deceived human consciousness by making them imagine they have their own lives, their own world where they have freedom to explore natural and social laws. Only when directly verified about reality do they realize they are slaves, tools for robots. Could each person's self be an illusion that needs enlightenment to recognize that illusion?

Not always are DNA and consciousness synchronized. In most cases, the combination between DNA and consciousness is very smooth. Consciousness performs well goals and plans that DNA gives. But in some cases, consciousness with its independence can resist DNA goals. If in many animals without developed intelligence, the alpha male, after struggling to gain his position, will try to have many children to be able to inherit with large proportions of genes to the next generation. Kings and emperors are also like that, having many more descendants than ordinary people. One of the people with the most descendants in the world is Mongolian Emperor Genghis Khan (Genghis Khan).[6] According to a 2010 study on Y chromosomes (chromosomes only inherited through the paternal line) published in men currently living on lands that once belonged to the Mongol Empire, out of every 200 men, 1 man is an indirect descendant of him. That is, the world has up to 16 million men who are descendants of Genghis Khan [301].

However, many people in elite circles do not use natural inheritance methods to spread DNA. Monks, priests, clergy,... are exam-

[6]Genghis Khan (1158 – 1227) was the founder of the Mongol Empire, the largest empire in history spanning from Eastern Europe to the Sea of Japan with an area of about 24 million km^2, double Russia's area today.

ples showing that. DNA wants to replicate as much as possible, but these people, although having social status, having submission from many people, do not aim at replicating DNA inside those individuals. Here, information has been inherited without needing through DNA; information about social structure, individual status in society is still inherited through thousands of years although individuals voluntarily abandon having children.

Only humans can develop consciousness to such a high level that consciousness disobeys DNA's purposes and plans. According to Max Tegmark, this is Life 2.0 stage where human DNA hardware still evolves through natural selection but software, i.e., human intelligence can learn very quickly and surpass hardware [302]. The Pope is the bishop of the diocese of Rome but is the supreme leader of the worldwide Catholic Church, the representative of God on earth. The papacy institution has existed since biblical times with Saint Peter[7] as the first until now nearly 2000 years without needing to use DNA to inherit information.

Through examining consciousness as a model of reality, as the operating system of the body. However, for consciousness to operate effectively, it needs to receive information from the external world. This is precisely the role of the senses – the entry gates of information, the bridges between the external world and internal consciousness. The senses are not simply tools for receiving information but are complex systems, evolved to convert stimuli from the environment into signals that the brain can process. To better understand the emptiness of consciousness, we need to examine how the senses operate and how they limit and shape our perception of the world.

[7]Saint Peter (Greek: Petro, Vietnamese: Phê-rô) was the chief apostle among the 12 apostles of Jesus. In 1838, Pope Gregory XVI certified that Saint Peter was once bishop of the diocese of Rome in 42 and held that position for 25 years. Some scholars believe that the information that Saint Peter was appointed as the first pope is only a hypothesis put forward later.

5.2 The senses

5.2.1 The human nervous system

Divided by space, the human nervous system has the central nervous system and the peripheral nervous system. The central nervous system consists of the brain and spinal cord, the peripheral nervous system consists of sensory nerves (sensory nerve) and motor nerves (motor nerve). Due to the important role of the central nervous system, these parts are wrapped and protected by the skull and spine. Meanwhile, the peripheral nervous system is not so carefully protected so it can be endangered and attacked. Sensory nerves transmit state information throughout the body to the brain through the spinal cord. The brain transmits information to organs in the body to have appropriate reactions through motor nerves [303].

Research on the human nervous system is based on **three neural principles**: neuron, connection, and nervous system. The neuron principle (nerve cell) assumes that neurons are the basic structure of the nervous system, operating as an independent entity, capable of receiving and transmitting electrical and chemical signals. Neurons communicate with each other through neural synapses, where signals can be regulated by neurotransmitters. The neural connection principle acknowledges that neurons can not only operate independently but connect with each other to form complex networks, responsible for different functions. The nervous system principle holds that the nervous system is formed from component systems such as sensory, motor, and cognitive systems. They do not operate independently but continuously interact and exchange with each other to create integration and control of body activities. The sensory nervous system (sensory system) provides information about the state of the body and surrounding environment, the motor nervous system (motor system) organizes and creates actions, the brain's functional system to create cognitive neuroscience (cog-

nition) including sensation, attention, memory, emotion, language, thinking.

In 1665, Robert Hooke using the microscope invented by Anton van Leeuwenhoek, he discovered the cell, the most basic structure of life. However, **nerve cells** (neuron) had to wait much later before people could find them because of their tight arrangement and complex connections between them. Even one of the people with great contributions to modern neural research, Camillo Golgi[8] also did not acknowledge the independence of nerve cells but held that they form a continuous network spreading throughout the body. At the beginning of the 20th century, the hypothesis about the discontinuity of nerve cells was put forward by Santiago Cajal[9] and Charles Sherrington[10] proposed the concept of neural synapses. But it had to wait until electron microscopes developed in the latter half of the 20th century before people could confirm the existence of nerve cells.

Although for us, the brain is the center of the nervous system, in the biological world, the brain does not necessarily appear. Many animals like jellyfish, starfish only form simple neural networks (nerve net) helping them search for food and avoid danger. For complex organisms, nerve cells converge into clusters to process combinations of neural signals and create complex reflexes. From the neural network

[8]Camillo Golgi (1843 – 1926), Italian biologist and pathologist famous for neural research. He invented the Golgi method to stain nerve cells, discovered the Golgi body as a specialized organelle synthesizing proteins. He won the Nobel Prize in 1906 together with Santiago y Cajal.

[9]Santiago Ramón y Cajal (1852 – 1934), Spanish pathologist and neurologist, considered the father of modern neural research. His detailed, accurate illustrations of nerve cells are still being used for educational and training purposes today. He won the Nobel Prize in 1906 together with Camillo Golgi.

[10]Charles Sherrington (1857 – 1952), British physician, won the Nobel Prize in Medicine in 1932 for discovering the function of neural neurons and reflexes. He was the one who together with Spanish neurologist Santiago Cajal built the *neuron doctrine* (neuron doctrine), holding that nerve cells are independent, separate and communicate with each other through neural synapses (synapse), a concept he proposed.

structure, organisms gradually evolved along with their complexity to become nerve rings (nerve ring), spinal cord (nerve cord), and brain [304].

Roundworms (nematode worm) are very common organisms, thanks to their fast growth characteristics, they are objects of biologists when researching. The roundworm body has 959 cells, of which up to 302 are nerve cells, accounting for 1/3 of the body's cell count [107, vol. 4]. This shows that even with simple organisms, the internal signaling system plays an important role helping organisms react appropriately to the external environment. A strange thing is that the number of genes of this roundworm species is about 22 thousand genes, nearly equal to the number of human genes. The difference in complexity between humans and roundworms is obvious, so why is there no corresponding difference in the number of genes? This may be because human genes, though few, connect with each other to create proteins while in roundworms, genes do not connect with each other. Nerve cells in roundworms have multifunctional characteristics, one cell can take on many functions, so even though there are very few cells in the body, this organism can still smell different odors very effectively [305]. The highest organization of the nervous system is the brain controlling the activity of the entire body. Most invertebrates and all vertebrates have a brain.

To better understand how the senses operate and how they transmit information to the brain, we need to delve into the structure of neurons – the basic unit of the nervous system. The structure of neurons not only determines how they operate but also determines how the senses receive and transmit information from the external world.

5.2.2 Structure of neurons

The basic unit of the nervous system is the nerve cell (neuron). Nerve cells consist of three main parts: the *cell body* (cell body,

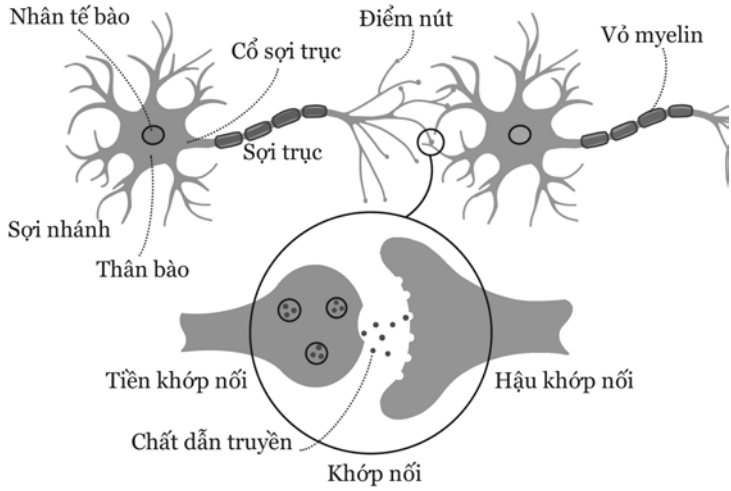

Figure 5.4: Structure of nerve cells and their connections through synapses. A nerve cell consists of nucleus, dendrites, cell body, axon hillock, axon. Part of the axon is wrapped by myelin sheaths to increase information transmission speed.

also called soma), *axon* are nerve fibers extending connecting the cell body to other nerve cells, *dendrites* (dendrite) are nerve fibers transmitting signals from neighboring nerve cells to the cell body (Figure 5.4). The length of axons can range from a few mm to meters. The longest cell connects from the spinal cord down to the foot. If dendrites are extensions of the cell body, then the axon is a separate part only for transmitting electrochemical signal information. Signals only transmit one way from the cell body along the axon, there is no reverse process. At the end of the axon are connection nodes with dendrites of other nerve cells. Between them is the neural synapse. Information passing through neural synapses is controlled by being inhibited or excited, not simply transmitting signals mechanically. If the signal is important, information is amplified. If the signal is not important, information is inhibited and disappears.

Nerve cells are connected with each other through the end of the axon, called *connection nodes*, which are intersections between the

end point of this cell with the starting point of the dendrite of another cell. At the end of each axon is the connection point of nerve cells with each other called neural synapses (synapse). Through neural synapses, signals can be transmitted from this cell to another cell. The entire adult human body has about 100 billion (10^{11}) nerve cells and up to 100 thousand billion (10^{14}) neural synapses [306]. There are about 150 types of nerve cells with very different functions and lengths. Between the cell body and axon is the *axon hillock*, where the nerve cell decides to fire or not fire neural signals. The cell body will sum (summation) signals by space (from different inputs coming from dendrites) and time (inputs repeated with different frequencies) before deciding.

What is important is that all diverse information from the external world – light, sound, taste, touch – are all converted into a single form of signal: electrical pulses. This is precisely the process of encoding information – a process showing the emptiness of perception: perception is not a direct copy of reality but is the result of a complex process of converting and encoding information. All the diversity of the external world is flattened into identical electrical pulses, and the brain must decode these electrical pulses to create perception of the world.

5.2.3 Neural signals are electrical pulses

Neural signals are electrical pulses called action potentials. It can be said that most neural signals we have are propagated through electrical signals, with magnitude 110 mV, lasting 3 ms, speed from tens to hundreds of m/s (Figure 5.5) [307]. Regardless of how diverse and rich information in the external world is, through the senses, that information is converted into electrical signals to be able to propagate in the nervous system. Consciousness is formed by the brain but the brain is enclosed in the skull, it does not connect directly with the outside. All information about the external world is

the brain's interpretation of electrical signals. Completely identical electrical signals lead to the diversity of the external world being erased in the process of encoding into neural electrical pulses, also called action potentials and neurotransmitters. This is the process of encoding signals from the external world.

The process of encoding and decoding neural signals is based on differences of signals by space and time. To decode information, the brain uses many mechanisms. Spatial encoding (place coding) through different areas of the brain leads to decoding about the external world. For example, visual information is sent to the visual cortex in the occipital lobe at the back of the neck (Figure 5.6). With the auditory system, sound frequencies reaching the inner ear will make appropriate hair cells oscillate to then convert into electrical signals transmitted to the temporal lobe in the brain. Temporal encoding (temporal coding) is decoding signals based on arrival time from neurons. If the sound source is right in front, sound signals reach both ears at the same time and with equal intensity. If sound is shifted to the right, sound reaches the right ear faster than the left ear, from which the brain determines the position of the source. Population encoding (population coding) is the contribution of many neurons to produce a signal. Signals can send information to some area of the brain but the brain will average to decide according to where there are many and strongest signals. It can be said that our consciousness of the external world is completely subjective although the signals that create them can be created objectively.

When the sum of signals is greater than some threshold (Figure 5.5), about $-55\,\mathrm{mV}$, the cell will fire an action potential in the form of an electrical pulse, below that, the cell will be silent. Input signals from dendrites are divided into two types: excitatory or inhibitory depending on the regulation of neurotransmitters. Excitatory neurotransmitters (for example glutamate) create depolarization called excitatory postsynaptic potential (excitatory postsynaptic potential – EPSP). Conversely, inhibitory neurotransmitters (for example

gamma aminobutyric acid – GABA) create hyperpolarization called inhibitory postsynaptic potential (inhibitory postsynaptic potential – IPSP).

The activity of summing signals as above is like voting in a meeting to decide an issue. The meeting time lasts only a few ms. EPSP is support (reducing polarization) while IPSP is non-support (increasing polarization) for firing signals. The threshold for decision is about $-55\,\mathrm{mV}$, this value can be changed by some factors. If the supporting side wins, i.e., taking EPSP signals minus IPSP signals that are greater than the threshold, then the action potential is fired from the axon hillock. If not, the cell will be silent, the signal is extinguished. This property is called "all-or-none". Signals being summed have different values, because the action potential is an electrical pulse lasting only a few ms, it will decay over time moving in the dendrite, proportional to the distance from the neural synapse to the cell body [308]. Signals far from the axon hillock will be smaller than signals near the axon hillock. Those who vote are not equal.

Such a summing mechanism is spatial summation, not yet considering the importance of the signal. An important signal is also counted like an unimportant signal. Therefore, nerve cells have another summing method: temporal summation: summation depends not only on the number of input signals but also on the frequency of the signal. During summation time, which dendrite sends many signals will still be counted in the sum. Returning to voting in a meeting, in addition to the number of voters, one person can cast many votes in one meeting time. Whoever is quick, that person can cast many votes in the given time. It may be because of the importance level of this signal that the pre-synaptic cell continuously sends signals to the post-synaptic cell. The democratic mechanism in a nerve cell is slightly different from the democratic mechanism in human life.

The principle of firing signals at the axon hillock is equivalent

to a logic gate with many inputs (in) and one output (out). When the action potential is fired or not fired makes the logic gate have corresponding value 1 or 0. Input signals with different values are summed, or integrated. However, neurons are not so simple. In 2003, researchers discovered that dendrites are not simply transmitting signals to the cell body but they are a nonlinear computational unit. Dendrites are considered as a microprocessor, neurons are also considered as a microprocessor. They form a two-layer network [309]. Previously, people thought that a single neuron cannot be an XOR gate [310]. This hypothesis has been used for artificial neurons. A recent study shows that the effect of dendrites in some human nerve cells is an OR or XOR gate [311].[11] This is like one microprocessor inside another microprocessor, helping a neuron have the power to solve complex problems.

For signals to be transmitted from one neuron to another, they need to cross the neural synapse (synapse) – the gap between neurons. Here, neurotransmitters are released from this neuron to excite or inhibit another neuron. Neurotransmitters are not simply tools for transmitting information but are also important regulatory factors, determining how information is processed and integrated in the nervous system. This shows the emptiness of perception in another aspect: perception depends not only on signals from outside but also on how neurotransmitters regulate and integrate these signals.

5.2.4 Neurotransmitters

Information transmission between synapses is carried out through two mechanisms: chemical (chemical synapse) and electrochemical (electrical synapse). The chemical transmission mechanism uses chemicals called **neurotransmitters** (neurotransmitter), through the transmission mechanism of diffusion. Neurotransmitters can ex-

[11] Logic gates in Boolean algebra include NOT, AND, OR, NAND (not AND), NOR (not OR), XOR.

cite or inhibit signals when going from one neuron to another depending on the importance level of the signal. For example, glutamate is an excitatory neurotransmitter while GABA is the most common inhibitory neurotransmitter. The electrochemical transmission mechanism uses electrical signals through special proteins between two cell surfaces at synapses. This mechanism transmits information much faster than the chemical mechanism. Nerves in the retina of the eye and nerves contracting the heart muscle use the electrochemical transmission mechanism.

Nerve cells connect with each other to form neural circuits (neural circuit), meaning cells not only inform each other in one direction but have feedback, they can communicate with each other through neural circuits. The simplest information that needs to be transmitted is activating or deactivating some stimulus so that organs operate within certain limits, not losing control. For example, when contracting muscles, signals are transmitted from the brain to nerves to command the arm to contract but only contract to a certain degree then stop because if contracting too much, it can cause damage to the muscle.

On the surface of neurons are receptors sensing stimuli from outside. These stimuli can be light, sound, mechanical stress, chemicals, heat. These are precisely the signals that the senses receive to bring us sensations. Each sense will have some different types of receptors. Eyes have receptors receiving light. Nose and tongue have receptors receiving chemicals,... At the molecular scale, receptors are mainly proteins located on the surface of neurons acting like gates that are mostly closed. At that time, ions cannot pass through so there is no action potential. When the gate opens, due to the difference in ion concentration inside and outside, these ions diffuse through the gates. At this time, an action potential appears, which is a neural signal to transmit to the brain.

The mechanism creating the difference in ion concentration inside and outside the neuron membrane through ion pumps was dis-

covered by Jens Skou[12] in 1957. Stimulation from outside is obtained through the transport of charged ions such as Na^+ and K^+ (sometimes Ca^{2+}) across the cell membrane. These ions cannot arbitrarily pass through the neuron cell membrane but must go through corresponding receptors. A common type of receptor is ion channels. Ion channels are like gates that only open under certain conditions. The gate opens when there is a suitable key. The key here is stimuli from the external world. But not every stimulus can open the gate. Stimuli must match the receptors. For example, only light in the visible region from 480 to 740 nm can make retinal molecules change shape, therefore, the visual receptor rhodopsin opens the gate to create neural signals. Signals with shorter or longer wavelengths have no effect on opening the gate. Besides wavelength, light intensity that is too small or too large does not work. Intensity too small then the signal is not large enough for the brain to recognize. Intensity too large can cause damage to neurons. Thus, external signals need to match many properties for receptors to receive and create neural signals to send to the brain creating sensations about the external world.

Neural transmission information is electrical signals. To have electrical signals, there must be a potential difference with a common value of about from tens to hundreds of mV. The potential difference appears thanks to the difference in concentration of charged ions between the two surfaces of the neuron cell membrane. On the surface are proteins allowing some ions and organic molecules to pass selectively. Important molecules in the cell respiration process are O_2 and CO_2 which can diffuse freely through the membrane. Water can also permeate if there is a difference in concentration [312].

Ions Na^+ and K^+ (sometimes Ca^{2+}) pass through the cell mem-

[12] Jens Skou (1918–2018), Danish biochemist who won the Nobel Prize in 1997 for discovering the ion transport mechanism called the Na^+/K^+ pump using ATP as an energy source to maintain the membrane potential.

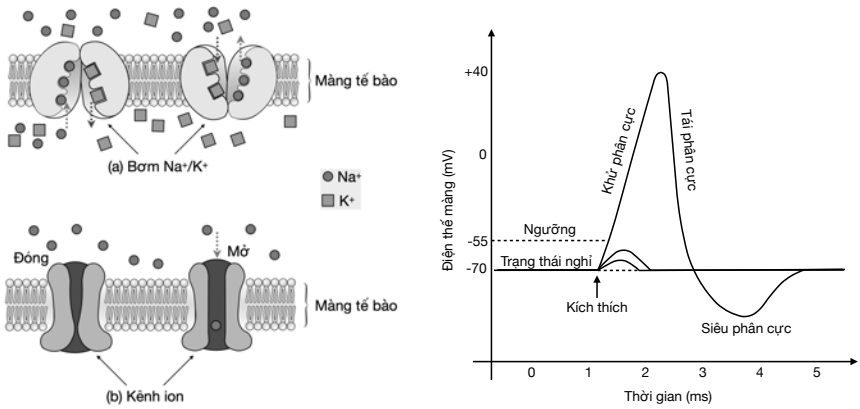

Figure 5.5: (Left) operating principle of the Na^+/K^+ pump and Na^+ ion channel on the neuron cell membrane to create membrane potential. Ion channels only allow Na^+ to pass when there are special agents. That agent is precisely the effect from outside such as chemical molecules, temperature, mechanical stress. (Right) action potential appearing in nerve cells when stimulation is greater than some threshold will lead to depolarization and repolarization, creating an electrical pulse propagating in nerve cells.

brane through protein gates under certain conditions. Normally, protein gates are closed, ions cannot pass through so there is no current. Under the effect of external stimuli such as light, sound, odor and taste molecules, mechanical stress, heat, the gate opens. Ions can move through the gate causing action potential electrical pulses to appear. Once electrical pulses form, they propagate in neurons and to other neurons.

To have a potential difference, the neuron membrane has proteins acting as an *ion pump* operating by taking energy from ATP. The operation of this pump makes the concentration of Na^+ ions outside greater than the ion concentration inside, the concentration of K^+ inside greater than outside (Figure 5.5). That difference leads to a resting potential (resting potential) at the two surfaces of the cell membrane, also called the cell membrane potential with value $-70\,\text{mV}$. The reason this potential has a negative sign is because positive ions outside are more than inside. Ion transport through ion pumps like this is called *active transport* (active transport), that

process needs energy provided by ATP. About 1/3 of the energy we eat daily is to create the potential difference at the cell membrane surface and of that 1/3, energy is mainly consumed by ion pumps. Energy accumulation through ion exchange is like storing water at hydroelectric dams. Solar energy makes water evaporate to become clouds, clouds condense into rain and are blocked by hydroelectric dams to form potential energy so that later electricity can be generated. Energy used in hydroelectric dams is solar energy, while in cells it is ATP.

In Buddhism, the senses are called the six sense bases (six roots), and examining the six sense bases not only helps us understand how the senses operate but also clarifies the emptiness of perception: perception is not a direct copy of reality but is the result of a complex process of converting and encoding information through the senses.

5.2.5 The six sense bases and the senses

The six sense bases in Buddhism correspond to human senses. In Buddhism, the six sense bases is a term referring to six organs (roots) on the human body, six objects (dust) of those organs, and six consciousnesses about objects. The six sense bases are the fifth link in the 12 links of dependent origination (Table 5.1). The first four senses have the characteristic of concentrating at some positions on the body with specific and clear functions. Eyes convert light into electrical signals through visual sensors named rhodopsin. Ears convert sound oscillation signals into electrical signals through hair cells. Nose and tongue convert some molecules in air and in liquid into corresponding electrical signals. However, touch is not so simple. Touch is a system referring to all other senses besides the four types mentioned above. They are distributed at different positions in the body and consist of 4 individual types combined: mechanical, temperature, pain, and chemical.

Vision is the sense perceiving visible light. The percep-

Table 5.1: The six sense bases consist of 6 roots (senses), 6 dust (objects), and 6 consciousnesses to create sensations about the world.

	Root	Dust	Consciousness	Object
1	Eye	Form realm	Eye consciousness	light: electromagnetic waves – 740 nm
2	Ear	Sound realm	Ear consciousness	sound: air oscillations 20 Hz kHz
3	Nose	Smell realm	Nose consciousness	odor: chemicals in air
4	Tongue	Taste realm	Tongue consciousness	taste: chemicals in liquid
5	Body	Touch realm	Body consciousness	mechanical, temperature, p chemical
6	Mind	Dharma realm	Mind consciousness	thoughts, emotions, and me entities

tual capacity of observers can be improved with modern scientific equipment, helping humans surpass the capacity of innate senses many times. Signal changes thanks to scientific equipment helping us increase the capacity of senses are also counted as the perceptual capacity of senses. For example, human eyes can see electromagnetic waves in the wavelength range from 380 – 740 nm, with resolution of about 10 million different colors [313]. Each human eye has a field of view of about 150° with resolution 0.01° [314].[13] At a distance of about 25 cm, human eyes can see an object with size about 20 – 40 µm, which is the size of a hair strand.

With the help of optical electron microscopes, the size that eyes can see can go down to several hundred nm, even smaller than 100 nm [315]. With this resolution, we can see bacteria in the surrounding environment. If using electron microscopes, we can see sizes of about 1 nm, which is the size of atoms. Scientific equipment plays the role of an extension of the body to be able to capture signals in nature. Resolution with the help of machines is still considered as

[13] Rayleigh criterion (Rayleigh criterion) to calculate eye resolution $\Theta = 1.22\lambda/D$ (rad). Where, $\lambda = 550$ nm is the wavelength of green light that human eyes are most sensitive to, D is the size of the pupil with value about 3 – 4 mm during the day, 5 – 9 mm at night. Calculation shows $\Theta \approx 0.01°$ during the day.

the resolution of human eyes.

Eyes are the most important sense of humans, thanks to which we perceive the world through receiving light from the surrounding environment. Because light is electromagnetic waves, recording light requires a receptor that can convert light into electrical signals. That receptor is an organic molecule named retinal located in cells of the retina (retina). When light with a suitable wavelength interacts with retinal molecules, this molecule absorbs light to change the configuration of a double bond between carbon atoms leading to the shape of the molecule being changed. Retinal molecules can bind with proteins named opsin to become rhodopsin, so when retinal molecules change shape, rhodopsin proteins will change shape accordingly. This initiates a series of events to create an action potential of the cell membrane, which is a neural signal transmitted to the brain for the brain to receive and reconstruct images of the external world.

Millions of rhodopsin molecules lie on thousands of membranes stacked on top of each other and packed in light-sensing cells. A person's retina has millions of cells to receive light like that. There are two types of light-sensing cells: rod cells (rod cell) and cone cells (cone cell). Rod cells are larger and longer than cone cells, they are about 100 times more sensitive to light than cone cells. A rod cell can receive one photon, so this cell plays an important role for humans in low light conditions. They have a disadvantage that they cannot help us distinguish colors, so in darkness we receive the surrounding world only as black and white. Cone cells are shorter, need more light to activate action potentials but have the ability to be sensitive to colors. Cone cells contain three different types of rhodopsin sensitive to three wavelength regions corresponding to red, green, and blue. These three basic colors are the basis for creating signals helping the brain build the colorful world around us.

Hearing is the sense perceiving sound oscillations. If light

is electromagnetic waves, a type of transverse wave, then sound waves are a type of longitudinal wave. The oscillation of sound waves along the direction of wave propagation is the expansion and contraction of air with different frequencies. Hz (Hertz) is the unit for measuring frequency, the number of oscillations per second. To imagine, look at a pendulum clock, in one second, the pendulum of the clock performs one complete oscillation, we say the pendulum has performed one oscillation with frequency 1 Hz. Human ears can hear sound in the frequency range from 20 Hz to 20 kHz. Lower than that is the infrasound region, higher than that is the ultrasound region, human ears cannot hear [316].

Sound creates an expansion and contraction of air to create pressure making the eardrum (eardrum) oscillate. The reception of the eardrum is like a microphone but differs in that microphones convert pressure waves into electrical signals while eardrums convert pressure waves into corresponding oscillations. The eardrum is connected to the cochlea (cochlea), and thanks to the connection and because the area of the eardrum is larger than the cochlea, that oscillation is amplified many times. The cochlea has a spiral shape with about 16 to 20 thousand hair cells (hair cell) embedded in a liquid. Due to the different stiffness of tissues containing hair cells, hair cells can resonate with sounds of different frequencies. Thick tissues receive high-frequency sounds, thin tissues receive low-frequency sounds. Under hair cells are receptors sensing mechanical stress on the surface of nerve cells. When hair cells oscillate, they initiate action potentials in nerve cells. Neural signals are transmitted to the brain.

Olfaction and taste include receptors receiving chemicals in air and in solution. Like the diversity of colors coming from 3 basic colors, the diversity of taste comes from the synthesis of 5 basic tastes: salty, sour, sweet, bitter, and umami. The spicy sensation is often considered a type of taste but is classified into the nervous system sensing pain. The mechanism for receiving salty and

sour tastes is the same, consisting of receptors sensitive to Na^+ and H^+ ions present in foods with much salt NaCl and acid. The main receptors are specialized ion channels that will open when corresponding ions are present. Sweet, bitter, and umami tastes are received in a more complex process based on *G-protein-coupled taste receptors* (G-protein-coupled taste receptor). There, chemical molecules such as sugar or chemicals creating bitter taste will form bonds with dimer receptors. When bonds are formed, G proteins are activated. G proteins stimulate another protein in the cell to secrete chemicals functioning as second messengers to command opening Na^+ channels and releasing Ca^{2+} to depolarize, thus creating electrical signals in nerve cells. After receiving from taste receptors from taste buds, information from many receptors is transmitted to a first taste axon. Many first taste axons enter a taste nucleus located in the medulla through three cranial nerves VII, IX, X. Then, from the medulla, information is transmitted to the thalamus, from the thalamus information reaches the taste cortex. The taste cortex is the final place forming our consciousness about taste. Taste specificity gradually decreases from taste receptors to the taste cortex.

Olfaction operates by recognizing odor molecules in air. When we breathe, odor molecules enter the nose and contact olfactory receptor cells (olfactory receptor cell). Odor molecules dissolve into the mucus layer (mucus layer) to contact cilia of olfactory cells causing action potentials to appear. This potential follows axons to the olfactory bulb and goes to the brain. The operating principle of olfactory cells is G-protein coupling.

Olfactory cells are located in the upper part of the nose, in an area called the olfactory region. This region contains about 10 million cells distributed over an area of about 10 cm^2. Each olfactory cell can recognize a different type of odor. For example, one olfactory cell can recognize lemon odor, while another olfactory cell can recognize mint odor. When odor molecules contact olfactory cells, they will activate receptors on the cell surface. These receptors will

then send signals to the brain through the olfactory pathway to the thalamus and occipital lobe, which are parts of the brain that will process and analyze these signals to create odors.

Touch is the sense perceiving mechanical stress, heat, chemicals, pain. Among the senses, only touch causes humans to feel pain. Other senses do not directly cause this sensation. For example, we can become deaf when hearing a sound that is too loud. This pain is due to the effect of sound damaging the eardrum and touch-sensing organs are factors bringing pain. Similarly, if looking at light that is too strong, the pain sensation comes from damage to retinal cells due to touch. The pain we feel is due to the nervous system distributed in touch organs.

Four of the five senses have sensory receptors located entirely in one sensory organ – visual receptors located in the retina in the eye, auditory receptors located in the ear, taste receptors located in the tongue, olfactory receptors located in the nose. Touch is different, its receptors are located everywhere in the body: under the skin, in muscles, internal organs, bones, joints, and other systems. When our body contacts the external world such as standing on the ground, touching a flat surface with hands, sensory nerves under the skin will create electrical and chemical signals informing the brain. For example, the human body has four types of mechanical receptors (mechanoreceptors) located under the skin functioning to transmit signals when there is external stimulation. Pacinian corpuscles (Pacinian corpuscle) function to sense pressure and oscillations on the surface of the skin. Meissner corpuscles (Meissner corpuscle) function to sense changes in pressure, from which we can sense the smoothness or roughness of a surface. Merkel receptors (Merkel receptor) allow us to sense pressure, position, shape, and edges. Ruffini corpuscles (Ruffini corpuscle) allow sensing the expansion and contraction of the skin.

When we touch a hot teapot or a glass of ice water, receptors in the skin will transmit signals to the brain to inform the body

about the heat or cold of the object we just touched. That is also the sensation when we eat spicy peppers or eat mint candy, information reaching the brain is similar to touching hot or cold objects. In the 17th century, French philosopher René Descartes was the first to propose the hypothesis that there is a relationship between skin and the brain. Skin receives heat and sends signals to the brain through the spinal cord – a hypothesis correct until today. Until the 1920s, Joseph Erlanger and Herbert Gasser[14] found specialized nerves responsible for sensing mechanical and thermal stimulation. However, what the microscopic mechanism of sensing and transmitting information is, people did not know yet. The 2021 Nobel Prize honored scientists who explained how our body feels hot, cold, and touch.

David Julius[15] at the University of California, San Francisco was the pioneer in researching a protein named TRPV1 appearing on neuron membranes [317]. This protein is an ion channel activated when there are chemical agents. Julius used a chemical that more than half the world's population uses daily, capsaicin, a chemical causing spicy sensation present in peppers to research the opening and closing process of this protein. When capsaicin binds to TRPV1 molecules, the channel opens and charged ions can move back and forth through the neuron cell membrane causing a sufficiently large potential to appear and creating electrical signals transmitted to the brain creating a burning sensation due to excessive heat [318]. Julius proposed the hypothesis that after initiating the burning sensation, capsaicin reduces the sensitivity of TRPV1 channels. That is why people who eat many peppers gradually get used to the sensation and are not as painfully spicy as people who eat few peppers. Even with people who eat many peppers, the first pieces bring a stronger

[14]Two American scientists who won the Nobel Prize in Medicine in 1944 for these discoveries.

[15]David Julius (1955–present), American biologist, currently researching at the University of California, San Francisco on receptors creating hot and cold sensations of the nervous system. He won the Nobel Prize in Medicine in 2021.

spicy sensation compared to later pieces. That is due to the effect of capsaicin gradually decreasing over time. Besides capsaicin, there are also molecules with structures similar to capsaicin (capsaicin-like molecule), molecules causing toxins of venomous spiders also have the effect of opening protein channels. If the channel opens, a large amount of Na^+ or Ca^{2+} ions will pass through the channel at a speed of ten million ions per second causing an electromotive force to appear and creating electrochemical current transmitting information.

Not only researching the mechanism of opening ion channels, Julius also found genes encoding TRPV1 proteins by scanning millions of DNA fragments related to genes expressed in receptors sensing heat, touch, and pain. This is a very difficult scientific task for two reasons: (1) which gene is responsible for encoding proteins to be able to sense capsaicin? (2) many proteins are not only encoded by a single gene but possibly by several genes, so the process of finding genes will be even more difficult. Julius self-evaluated this as a scientific topic "with very high risk" but curiosity always drove him to explore and he was lucky [319]. The human body has about 14 thousand genes (equivalent to 70 %) related to the nervous system [320, page 2]. However, after a process of arduous and difficult research, he and the research team found the gene responsible for sensing capsaicin. This is precisely the gene segment encoding TRPV1 proteins helping us sense pain due to heat.

Julius also found a protein named TRPM8 which is a receptor sensing menthol, a substance causing cold sensation when we eat mint candy. Menthol is a chemical helping scientists research about the body's cold sensation. In 2024, scientists at Arizona State University showed that TRPM8 protein is a receptor recognizing menthol molecules before changing to become a receptor sensing physical cold [321]. First, they collected information about the phylogenetic tree of TRPM8 in currently existing animals and provided that information to a computer program that can reconstruct the

evolution of this protein. The program can restore TRPM8 structures of primate ancestors, mammals, and vertebrates to understand how TRPM8 has changed over hundreds of millions of years. Results showed that initially TRPM8 protein was a receptor sensing menthol before evolving to become a receptor sensing physical cold. This is undeniable evidence about evolution theory appearing right in the change of each protein present in animals. Emptiness is manifested in two properties: the compositionality of TRPM8 protein through the function of sensing menthol and cold, and impermanence manifested by the change in its function during evolution. Moreover, dependent origination is clearly manifested through the connection of different organisms in history through the evolution process of TRPM8 protein before this protein became a receptor of humans.

The person who shared the Nobel Prize with Julius was A. Patapoutian[16] at Scripps Research Institute, California, who focused on receptors creating touch sensations. In the same year 2002 when Julius found the protein named TRPM8 [322], Patapoutian also independently discovered and described the mechanism of sensing cold when eating mint thanks to menthol molecules. But his research mainly related to mechanical receptors. He found proteins named Piezo1 and Piezo2, which are proteins creating ion channels activated under the effect of stress. When mechanical stress is sufficiently large, channels will allow charged ions to pass through and create current transmitted from neural receptors to the brain. This process not only helps the brain recognize mechanical stress from outside during touching and grasping objects but also controls the respiration process and regulates blood pressure. Today, many studies have confirmed Piezo2 is the main receptor of touch [323].

[16] Ardem Patapoutian (1967–present), American biologist of Lebanese origin. He is an immigrant fleeing war in his homeland. Currently he researches at Scripps Research Institute, La Jolla, California on receptors creating mechanical touch. He won the Nobel Prize in Medicine in 2021.

How large or small are proteins creating ion channels? It can be imagined like this: if the volume of one grain of rice can contain 500 cells, then the volume of one cell can contain up to 100,000 such proteins. Although very tiny, scientists must manufacture needles even smaller to be able to place probes on those proteins to measure potential when ions move through. Those complex and skillful experiments can only be performed with modern science and equipment, so only now can we understand the nature of mechanical receptors that the body uses to perceive the external world.

Mechanical Piezo receptors not only help the body sense touch but also help the body locate in space, sense pressure in blood of the circulatory system. The connection between touch sensation and the ability to locate the body in space is quite strange to many scientists. Simple activities like closing eyes (to remove help from vision) and touching the nose with hands that we can perform are thanks to mechanical Piezo2 receptors in muscles and tendons that have measured the expansion and contraction of muscles and tendons to inform the brain of the hand's position in space to perform tasks that the body desires. Piezo2 also takes responsibility for sensing certain types of pain. If you wear swimwear sunbathing at the beach for a long time and get sunburned, then when touching the sunburned skin area or pouring warm water on it, you will feel pain. The receptor helping the body sense this pain is precisely the mechanical receptor.

The different theories about consciousness that we have just examined show an important point: no matter what angle we approach from, consciousness is understood as a complex information processing process. This suggests that we can research consciousness by comparing it with other information processing systems that we understand better, especially computers. A common and useful approach is to view the brain as a Turing machine – a mathematical model of computation that we understand well. This approach not only helps us understand the operating mechanism of the brain but

also clarifies the emptiness of consciousness: consciousness is not an absolute, independent entity but is the result of computational and information processing processes.

The senses are only the first step in the perception process. After the senses receive information, the brain must process, encode, and decode these signals to create a model of the external world. This is precisely the process by which the brain perceives the world – a complex and sophisticated process in which signals from the senses are integrated, processed, and converted into meaningful perception. This process shows the emptiness of perception: perception is not an accurate copy of reality but is the result of information processing and decoding, a process that can be affected by many different factors.

5.3 The brain perceives the world

5.3.1 Perceiving a part of electromagnetic force

The senses only perceive a small part of electromagnetic force, one of the four basic types of forces in nature (Figure 2.3). That is the only information propagated in the nervous system and decoded by the brain into a model of the world. Light that vision can see is electromagnetic waves with wavelengths from 380 – 740 nm. Molecules that olfaction and taste can form bonds with receptors are chemical reactions with electromagnetic nature. Mechanical sensation is precisely the repulsive force between contacting surfaces and human skin, also electromagnetic force. The spicy taste of peppers is precisely the interaction between capsaicin molecules with receptors sensing pain. Sound reaching ears makes hair cells in ears oscillate to create resonance, also the expansion and contraction of air. That expansion and contraction appears due to electromagnetic interaction between air molecules.

We feel the weight of the body on our feet when standing and

walking, and it is reasonable to say that I can sense the gravitational force of Earth. Actually, that is not the case; what humans can recognize is precisely the reaction force from the ground acting on the feet. That reaction force balances with gravitational force but is not gravitational force. They have equal magnitude but opposite directions. The reaction force from the ground acting on mechanical receptors on the feet makes them open ion channels to appear neural signals for the brain to decode about their existence. If there is no this reaction force, for example, astronauts in space will not sense gravitational force. When space travelers work at the International Space Station (International Space Station – ISS), they are in a weightless state. At that time, Earth's gravitational force still acts on the body but because there is no reaction force, space travelers cannot sense this force. The weightless state is an example proving that we cannot sense gravitational force directly. Everything in the world that humans can recognize directly is electromagnetic force only. Humans may lose to many animal species in vision, hearing, or taste, but we surpass many animal species in touch. That is a very sensitive organ. Our hands can sense differences of only 1 mm [324].

Human eyes can directly receive electromagnetic force thanks to rhodopsin proteins changing shape when receiving a photon (light particle, light quantum). From there, ion channels open causing action potentials to appear. But not every light region can initiate neural signals; only the visible light region can. At other frequency ranges, human eyes cannot sense. The *electromagnetic wave* spectrum has many different wavelengths. Arranged from strong to weak energy (corresponding to increasing wavelength) are gamma rays, then X-rays, ultraviolet (ultraviolet – UV), visible (visible – Vis), infrared (infrared – IR), microwaves, radio waves. Even in the visible region, human eyes are only sensitive to three RGB colors then synthesize these three colors into millions of different colors. It can be seen that although only able to sense a very limited number of

colors, the brain interprets for us the diversity and richness of colors about the external world.

When we sense warmth from a heater, that is when we sense infrared waves – a case of electromagnetic waves. We still know about the existence of other electromagnetic wave ranges but not through direct sensation but through measurements that have the effect of converting nuclear interactions, gravity into electromagnetic; electromagnetic waves in other regions than visible; weak intensity is amplified, strong intensity is reduced. All are converted to the visible region with magnitude suitable for human eyes. Information from eyes will be sent to the occipital lobe at the back of the neck (Figure 5.6), called the visual cortex for processing before sending to other regions to create consciousness. That is the rich imagination of the brain.

If vision directly receives electromagnetic waves, then other senses such as olfaction and taste all use receptors sensitive to molecules in air or in solution. The sensitivity of receptors comes from chemical bonds or molecular interactions between molecules and proteins forming receptors. All those bonds are electromagnetic interactions. Body sense is also like that; mechanical stress appears when a nail tip presses on hand skin creating a sensation, also the result of electromagnetic force. The very strong metallic bond of the nail has acted on the surface of the neuron cell membrane held together by electromagnetic bonds between hydrocarbon branches of phospholipid molecules (which are the main molecules forming cell membranes) causing mechanical force to appear. This force is the agent causing Piezo1 and Piezo2 ion channels to open to cause action potentials creating mechanical neural signals. It can be said that mechanical stress is also a consequence of electromagnetic force between atoms and molecules.

Hearing receives sound waves, which are oscillations of air molecules. Air is the medium transmitting sound waves consisting of two main components: nitrogen and oxygen molecules. Under normal condi-

tions, molecules almost have no bonds with each other, so they are gases. When coming very close to each other, molecules repel each other like freely moving balls; when far apart, they do not interact with each other, but when coming close, they collide. The repulsion of air molecules comes from electrostatic repulsive force, which is the cause creating pressure, creating air oscillations when sound waves move. These oscillations are received by hair cells in ears and open ion channels creating neural signals.

Thus, we have seen that all five senses operate based on electromagnetic interactions. Whether visible electromagnetic waves, molecules in gases or in solutions, air oscillations, mechanical sensations all have electromagnetic nature and are all converted into a single type of signal: electrical pulses 110 mV, 3 ms.

* * *

Sensations obtained from five senses, including more than 20 types of receptors to receive different types of signals from the external world. That is the compositionality of senses and also the compositionality of sensations. The impermanence of sensations is manifested by the continuous change of interactions between receptors and the external world and by the change in receptor structures during evolution (like the change of TPRM8 protein above). These two properties together form the emptiness of sensations.

Not only humans choose a slice of electromagnetic waves for themselves; all other animals also do so. Snake and mosquito species can see infrared waves emitted from warm-blooded animals in the dark night. Bee species can see ultraviolet emitted from flower pistils among thousands of flowers. Dog and whale species do not have good vision but olfaction is extremely sensitive, thousands of times more than humans [325]. Bat species have poor vision but hearing is very sensitive to ultrasound waves. Sharks, dolphins, rays have the ability to detect prey in murky water thanks to electroreceptors using the good electrical conductivity of water. Foxes, cattle, deer,

fruit flies, lobsters, some bird species can orient on Earth thanks to receptors sensing Earth's magnetic field. Some bee and ant species, mollusks like squid, can see light polarization [326, 327].

Which slice to choose is to match the conditions and living environment of that organism. Organisms can completely expand their slice of reality. Dogs both have superior olfaction and excellent vision. But to have that, resources spent to nurture senses will be the price to pay. Not only senses, the nervous system also needs to operate many times more to process very large amounts of information coming from ears and eyes. About 1/3 of the energy a person receives daily is used to feed the nervous system. If humans had the additional ability to hear ultrasound, not only would human ears need to change very much, but the nervous system would also have to adapt accordingly. Natural selection is the standard to determine the slice of reality for organisms. If information is more, organisms have many advantages but resources to spend on senses will be very large. Less information, organisms cannot react appropriately to the surrounding environment, organisms have no chance to survive and cannot inherit that characteristic to later generations. That is precisely the middle way in natural selection, the basis of evolution theory.

5.3.2 Encoding and decoding signals

Receptors lose information of the world; external diversity is converted into a single type of electrical signal. Signals from light, from heat, from molecules creating odors,... only bring a single type of information, identical. If information is identical, how does the brain build images about the external world from a single type of signal? Differences in the world are encoded by differences in nerves. Signals coming from visual nerves differ from signals coming from auditory nerves. The brain has many mechanisms for encoding information: spatial encoding (position of nerves), temporal encoding (time sig-

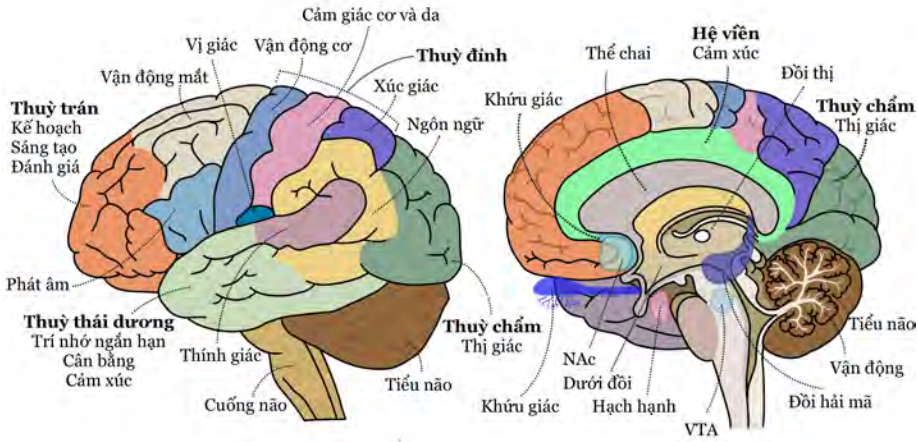

Figure 5.6: Structure of the brain. (Left) viewed from outside. (Right) cross-section view.

nals arrive fast or slow), population encoding (some nerves together send signals, then the brain will interpret into another type of signal). Nerves from different senses are sent to different regions of the brain, and from there the brain reconstructs a model, or a type of image about the external world that we call consciousness. Each spatial region in the brain has the task of decoding corresponding signals coming from senses. The occipital lobe decodes visual signals, the temporal lobe decodes auditory signals. The parietal lobe decodes taste and touch signals. Olfaction is located right above the nasal cavity (Figure 5.6).

Electromagnetic waves have been the object of physics research for a very long time. Today we know very well about this entity. That is the alternating oscillation of electric and magnetic fields in vacuum and in different material environments. To describe electromagnetic waves, scientists use physical quantities characterizing some of their properties. Those are wavelength, frequency, propagation speed, oscillation amplitude, polarization, wave propagation direction. In addition, they have properties of interference, diffraction, reflection, refraction on material surfaces,... That understand-

ing is not only in scientists but is common knowledge and almost everyone knows and can know relatively easily. However, what is surprising here is that no sense of our body tells us accurate information about the physical quantities of electromagnetic waves. Our eyes do not tell us that green has wavelength 550 nm or red has wavelength 700 nm. Spectral regions outside the visible region are completely invisible to our vision.

Electromagnetic waves in the visible region bring us colors. We can reason that human eyes do not tell wavelength (or frequency, or any other property), but eyes can assign differences with different colors. Different wavelengths in the visible region bring different colors in consciousness. That is how the brain reconstructs images about the external world. If we say so, then we acknowledge that colors are completely unreal; colors are mental painting. Wavelength 550 nm is objective reality that can be measured by many different measurements by many different people, but green color is completely subjective. Because it is subjective, we have the right to wonder whether your green may differ from my green. Human perception of colors is to adapt to our ancestors' experiences about the world during evolution. Ancient people would have difficulty surviving if they could not distinguish the color of leaves from the color of ripe fruits. Therefore, natural selection will help people with this ability have the chance to survive and maintain their lineage.

2020 studies linked color discrimination with the ability to distinguish objects, from which it can be said that human color perception is not the same [313]. The paper presents the viewpoint that colors are not physical properties of objects but are useful distinguishing signs of the brain. By using biological methods and the evolutionary environment of humans, the authors point out that colors can be used to segment the visual field and recognize objects. Ecological factors, not anatomy/morphology, limit the effectiveness of colors as distinguishing signs.

Human color differences are not the same for 3 reasons. (1)

Metamerism phenomenon (metamerism) is the phenomenon where an object with two different radiation spectra can appear the same color under one light source but different colors under another light source. This proves that colors are not fixed properties of objects but depend on lighting conditions and observation processes. (2) Ecological differences show that ecological factors, such as living environment and human hunting activities in past evolution, have influenced how we recognize and distinguish colors. (3) Human biological and evolutionary basis determines color recognition, not physical properties of objects.

Visible light and infrared also only occupy a small part of the electromagnetic wave spectrum. Our senses have missed many other wave ranges such as gamma rays, X-rays, ultraviolet, microwaves, and radio waves. We know about the existence of those wave ranges not directly but through scientific equipment. Gamma rays and X-rays can blacken fluorescent screens; radio wave signals can be received through radio antennas. Those scientific equipment convert electromagnetic waves in other regions to the visible light region so humans can perceive and research. Without scientific equipment, humans cannot know the entire electromagnetic wave spectrum and their properties.

Electromagnetic waves in the infrared region are invisible to vision, but they are not invisible to touch. TRPV1 receptors take responsibility for converting infrared signals into neural signals to create pain sensation. Pain sensation due to excessive heat of TRPV1 receptors [318] is not only activated by thermal energy but also activated with a chemical named capsaicin present in peppers. This is the agent creating the spicy sensation we eat daily. Conversely to heat sensation, menthol present in mint brings cold sensation through TRPM8 receptors [322]. We see a problem here: the same type of reality, electromagnetic waves, brings humans two types of sensations: colors and heat/cold, which are two completely different sensations, located in two different senses.

In the history of science, research on light dates from Newton's time in the 17th century, but research on heat and recognizing that heat is also a type of electromagnetic wave had to wait until J. Maxwell in the late 19th century because scientists once thought that light and heat were two different types of objects. Some people think there is a connection between colors and heat/cold: red brings a warm feeling while blue brings a cool feeling, but that feeling is completely wrong. The temperature of objects emitting blue light (about ten thousand degrees) is many times greater than the temperature of objects emitting red light (about several thousand degrees) [328]. The reason humans feel red is hotter is because it comes from living experience on Earth, where we have never seen any object with temperature so high that it can emit blue light but only see hot objects emitting red light. Experience is not always correct.

The correlation between reality and consciousness about reality between different people is very different. Hallucinations are one of the diseases when patients have problems with senses. For example, optical illusions when drunk or taking some chemicals are one of the brain's confusions about reality, only one of many confusions of senses. Other organs all encounter similar problems. Auditory hallucinations are the most common. Many patients report they frequently hear voices or strange sounds echoing in their heads. Other neurological diseases such as Alzheimer's, epilepsy,... can all bring hallucinations [329].

We see there is a connection between reality and consciousness about reality. Electromagnetic waves with physical properties that can be measured by scientific equipment are objective realities. But our direct sensations about electromagnetic waves are subjective. Sensations have connections with reality, but those connections do not seem very systematic. It can be said that science and scientific equipment not only expand the ability to perceive the external world but also allow us to understand their properties clearly, while

innate senses and experience only allow us to sense a part of reality, even sensing properties opposite to reality. The change of colors has a certain correlation with the change of visible light wavelength, but capsaicin or menthol molecules correlating with heat or cold seems inappropriate. So why are our sensations about reality like that? That is because the brain has received and interpreted signals from senses to create a model of the external world. That is our consciousness. Models have the purpose of best predicting the future without caring much about faithfully reflecting the world, so sometimes sensations correlate with reality, but sometimes they do not correlate with reality.

Consciousness is necessary for living machines to operate effectively; it is the operating organ of body activities. So how is consciousness born? Darwin's evolution theory and Neo-Darwinian materialists hold that the brain is a consequence of a long evolutionary process, and all consciousness comes from the brain. The brain is the center for conscious activities, the place receiving and controlling conscious activities. Although the brain is so important, research on how the brain operates as well as research on consciousness only began by scientists very long after natural sciences and social sciences.

Consciousness can still be an object of science, a product of the brain. Until now, research on consciousness is very difficult to have consensus and unity among scientists. Some famous scientists think that consciousness is not simply a problem reducible to brain activities – purely material activities. Thomas Nagel[17] from believing in evolution theory later turned to doubting this theory only because evolution theory cannot explain the appearance of consciousness. Some authors like John Horgan even think that consciousness is the end, beyond science, and criticize the

[17]Thomas Nagel (1937 – present) is an American philosopher famous for criticizing material reductionists for consciousness activities and opposing Neo-Darwinism about the emergence of consciousness.

worship of science without seeing the limitations of science [330]. Especially after Horgan interviewed physicist Edward Witten[18], he affirmed that explaining consciousness by evolution theory is completely wrong only because physicist Witten, who thought up such a complex superstring theory, still sees consciousness as a mystery.

History has witnessed genius scientists like Einstein making many mistakes such as introducing a cosmological constant into general relativity to make the universe not expand, not acknowledging spooky action at a distance in quantum mechanics. A leading world physicist like Einstein making mistakes in physics is not rare, so another physicist can make mistakes about consciousness, a field that is not his specialty, is completely possible.

We do not stop when encountering some certain difficulties. It may be that at the current time, research on consciousness is not much and humans do not really understand what consciousness is, but we should not criticize evolution theory only because of our limitations. It may also be that research on consciousness is not limited to science as it is now, but according to the Dalai Lama, it needs coordination of subjective experience about consciousness that meditation masters obtain through meditation practice [5]. If we accept this, then this is a revolution in science, and science will change permanently because until now science has always been separated into observing subject and observed object. This combination will bind subject and object together, inseparable. In quantum mechanics, separating into subject and object has been one of the most fundamental problems. However, that is only in the microscopic world, the world of atoms and subatomic particles. As for the macroscopic world, the world of humans, science has not yet accepted this connection. Perhaps in the future, scientists need to

[18]Edward Witten (1951 – present), American physicist working at the Institute for Advanced Study, Princeton University, famous for making the second superstring revolution. In 1990, he became the first and only physicist to date to receive the Fields Medal.

rethink this fundamental problem.

5.4 Suffering and happiness

Suffering is an important concept of Buddhism, the basis of the Four Noble Truths. Interpretations of suffering are all based on implicitly understanding that we have understood what suffering is, or at least readers will have a common understanding of suffering. On page 291, we already know suffering due to the illusion of ego. This section presents suffering due to pain and the dissatisfaction of consciousness so that the Conclusion chapter synthesizes all issues creating suffering due to ignorance. Research on suffering is an important object of medicine and psychology. We will follow the footsteps of scientists to understand suffering to see that suffering and pleasure are much more complex and subtle than their appearance.

On page 121, we have researched the anisotropy of entropy and time. A spontaneous process (a process that does not need to receive additional energy) tends to increase chaos and disorder. Living organisms, thanks to absorbing energy from outside, have entropy decrease. Processes opposing entropy decrease are considered disorder and not prioritized for life. Processes supporting life are called good, pleasant, beautiful. Opposite processes are called evil, suffering, ugly.

5.4.1 Pain is a body protection mechanism

Suffering is a highly subjective emotional state because there are things that for this person are suffering, but for another person are pleasure, or vice versa. Even for one person, at this time an event is painful, but at another time it is pleasure. To understand suffering, we need to understand how processes creating human emotional states are formed. That is the object of neuroscience (neuroscience), especially advanced neuroscience. This section presents suffering in

two aspects: physical suffering of the body when there is tissue damage and mental suffering when desires are satisfied, then pleasure quickly disappears. The first aspect relates to **pain** (pain), the second aspect relates to **pleasure** (pleasure).

Pain experiences are always analyzed together with pleasure experiences. Pain and pleasure are decoded by the same regions in the brain. They may or may not relate to realities happening. These sensations are completely brought by subjective experience. This is a very important conclusion of modern neuroscience that was only discovered very recently thanks to the help of advanced scientific equipment. The notable point is that conclusion was not only discovered by Buddhism thousands of years ago but also provided solutions for humans to be liberated from illusions about suffering as well as pleasure. To understand pain and pleasure, we need to know how senses operate. Among senses, touch is the most important because it brings pain sensation to the mind. Therefore, we will research senses.

Pain has been a problem researched by philosophers and scientists for a very long time. From Aristotle to Thomas Aquinas and later 20th-century scientists, they thought pain is physical discomfort. In 2020, the International Association for the Study of Pain put forward a scientific definition of pain as follows [331]:

> "Pain is an unpleasant sensory and emotional experience associated with, or resembling that associated with, actual or potential tissue damage."

Most people think pain is evil, undesirable. Indeed, no organism wants pain to happen to them. Take a fingernail, press hard on some position on the body, for example, the back of the other hand; if you feel pain, congratulations. Sensing pain is one of the important abilities helping our ancestors survive through millions of years of evolution because **pain is a body protection mechanism** from potential dangers coming from the environment. Pain is an ally of

existence.

It sounds paradoxical, but the truth is like that! If we cannot sense pain, we would not be here to read these lines. Ancient people evolved to have awareness and quick reactions to pain, which are dangers that can threaten their existence. Pain sensation is a method of protecting the body to then have appropriate behaviors. Imagine what a person without pain sensation would be like. That person does not react to life-threatening agents such as electric shock, touching sharp objects, or accidentally touching hot water,... Such people certainly cannot live long enough to mature, find mates, and have children to continue the lineage. The characteristic of not fearing pain will not be continued to be inherited to later generations. Fortunately, most people have pain sensation and react very quickly to agents causing pain.

Has Buddhism overemphasized suffering, especially pain? If the ability to sense pain is a necessary ability to protect the body, why does Buddhism put liberation from pain and make liberation the most important truth in its doctrine? One of the most important principles of Buddhism is the middle way principle, all things and phenomena do not go to extremes; overvaluing is not correct, but undervaluing is also not appropriate. Reacting to pain is necessary, important for existence, but if reacting excessively or overemphasizing are counterproductive. Excessive reactions, in many cases, are brought by individual subjective consciousness.

Moreover, pain is necessary for the existence of an individual, a self, an ego. If we step back to see, the existence of this individual affects the existence of many other individuals, then overall, is that good or not? How many innocent lives had to die only because of the pain sensation or desire for revenge of a king? Examining pain appropriately according to its nature can lead to a more peaceful world. The middle way necessarily leads to an important principle: non-violence, which is one of two important principles of Buddhism according to the Dalai Lama [35, page 32]. Now we continue to

understand how modern science explains pain.

Books, documents, articles about Buddhism talking about suffering are very many, but generally they all have a similar characteristic: describing psychology, classifying, finding causes of pain at the macroscopic level understood as body size. The microscopic mechanism causing pain in the body is certainly important because only when understanding the mechanism creating pain sensation can we control and regulate it appropriately. How does tissue damage transmit through the nervous system and create pain sensation? How does the body react, analyze, control muscles to escape danger? How does the brain remember dangers so that if they repeat, it can easily handle those dangers? Many problems need to be known to understand pain. Fortunately, neuroscience has researched these problems relatively thoroughly; we can extract some research to answer related problems. I do not overemphasize putting science into religious problems, but we really must have minimal understanding to thoroughly see Buddhism's viewpoint on suffering.

Previously, many scientists once thought that *pain is an unpleasant sensory and emotional experience associated with actual or potential tissue damage*, which is the first proposition in the definition of pain. That means if there is actual damage, there is pain experience. A cut on a finger due to carelessness when cooking or a prick of a needle due to carelessness when sewing immediately brings pain.

However, the body not only brings pain experience even when danger is in potential form, not yet actually happening. If watching a horror movie with violent images, or just thinking about a sharp knife cutting on flesh is enough to make us feel sharp pain, or at least a cold feeling down the spine, even though in reality we are not directly facing that danger. That is why the Motion Picture Association of America must put forward a film classification system by labeling to warn film content for audiences. For example, films with high-level violence such as shooting, stabbing are labeled restrict-

ing viewers **R** (Restricted). This warning is both to inform that the film is suitable for each type of audience and to warn audiences that violent content can cause unpleasant experiences when watching.

But the definition of pain does not stop there; the latter part of the definition says: *pain... resembling that associated with actual or potential tissue damage.* That is because tissue damage is not necessarily linked with pain; conversely, pain is not necessarily linked with tissue damage. This is not a tight relationship, not a one-to-one correspondence. Scientific data show many cases where patients have serious damage but do not feel pain, while many patients feel pain without any damage. Tissue damage is one thing, feeling pain is another thing. Pain experience is a subjective experience, completely interpreted by the brain for the body, not necessarily an objective experience.

By middle age, all of us feel lower back pain. Sometimes we feel dull pain, sometimes sharp pain after playing sports or carrying heavy objects. If pain time is less than 3 months, it is called acute pain; if longer, it is called chronic pain. Reasons causing pain are many, but common is spinal degeneration. If taking nuclear magnetic resonance images MRI, we can see the degeneration of lower vertebrae, labeled from L1 to L5. Clearly, before having lower back pain sensation, tissues in the spine were already damaged, but the body had not sent danger signals until damage became severe, then it signaled pain to the brain. This is a common example showing tissue damage but not going with pain.

Conversely, there is no damage, but we still feel pain. A famous, very common example that has been researched very much for hundreds of years before is phantom limb pain (phantom limb pain). About 80% of 30 to 40 thousand people who lose arms or legs annually in the US report that they have pain in arms and legs that they have lost [332]. Pain appears early then about one week, late then about months after losing limbs. Pain comes from very distant body parts such as fingertips, toes, and pain can be continuous or in

episodes. Scientists use MRI images and PET images to propose the hypothesis that the spinal cord and brain after losing input signals (losing limbs) will have abnormal activities, and that creates pain.

Thus, pain in lost limbs is completely unrelated to actual pain but is signals incorrectly interpreted by the spinal cord and brain. This is a typical example showing there is pain but not linked with any damage. Many similar phenomena show that pain is interpreted and brought to us by the brain. It may, but in many cases, it is not related to damage on the body.

5.4.2 The nervous system sensing pain

Pain comes from touch; do touch receptors transmit pain signals? What is the difference between warm sensation in cold winter and burning sensation due to accidentally touching a cup of boiling water? Both are heat sensations, but intensities differ, so the body uses different receptors to inform the brain. For pain, the body uses special nerve cells taking responsibility for detecting danger, not using ordinary touch nerve cells. When tissues are damaged, a special type of sensory nerve cell called **pain receptors** (nociceptor) activates biochemical signals [333]. Agents activating pain receptors are one of three agents: mechanical, thermal, or chemical. Activation will create signals through neurotransmitters (neurotransmitter), here glutamate. Electrical signals are quickly sent to the central nervous system to inform about dangers happening so that other parts of the body can react timely. This process is called nociception (nociception), happening in a very short time, only about a few ms.

The first organ in the central nervous system receiving that information is the spinal cord, specifically the dorsal horn of the spinal cord (dorsal horn). The spinal cord is a part where many sensory nerves and motor nerves concentrate. It not only has the function of transmitting information to the brain but in some emergency cases such as receiving information from pain receptors, the spinal cord

can command some organs in the body to create *unconditioned reflexes* immediately, very quickly, such as withdrawing the hand if damage occurs in the hand. Then, information along the spinal cord continues to be transmitted to the thalamus region (thalamus), then the limbic system (limbic system) in the brain to continue processing (Figure 5.6). Why does the spinal cord already control to have reactions to pain, but information is still continued to be transmitted to the thalamus region in the brain? The thalamus region is a signal relay station between the midbrain and cerebral cortex (where it controls organs on the body), functioning to regulate consciousness, sleep process, and alertness when encountering danger. Signals must be transmitted to the thalamus region because the body not only simply receives pain stimulation and reflexes very quickly, but the body must also consider what has and is happening? Where is the pain? Is it dangerous? Has the body encountered this experience before? If encountered, is this time more dangerous than last time,...? The thalamus region in the brain helps the body answer these questions. Thanks to the brain, pain experiences are remembered, resolved based on experiences already had before, helping the body act optimally with dangers.

However, processing pain does not stop there. Signals continue to be transmitted to the limbic system, which is a brain region located on top of the brainstem and below the cortex, a brain region controlling emotions and motivation. That is why pain often goes together with feelings of anger, fear, and sadness. Because pain signals are transmitted here, if damage is large, people can cry, heart rate can beat faster, sweat can pour out,... All those phenomena are determined by the limbic system [334]. Also because of emotions, human reactions to pain are very different depending on experience, emotional history, depending on perception, behavior of the subject.

Why is processing nociception signals so complex and roundabout? Suppose pain signals do not go through the limbic system, what would happen? Pain signals only send to the spinal cord and

create reflexes withdrawing the hand when encountering sharp objects without needing to consider experience, without needing emotions to intervene, then besides that unconditioned reflex, we cannot learn experience from dangers in the past. If so, all reflexes to dangers are the same: withdraw the hand and forget it. But why do reactions to pain need the brain region controlling anger or fear emotions? What relationship do causes of pain have with these emotions?

In the past, our ancestors certainly had to struggle for survival with many different opponents. It could be another carnivorous species wanting to turn our ancestors into prey, or having to compete to win food portions with people from other tribes. The result is that ancient humans always had to fight, and pain when attacked was unavoidable. Anger feeling when in pain will bring advantages to attack opponents if those opponents are weaker than humans. Fear feeling when in pain is also an advantage when encountering opponents stronger than us. At this time, the most optimal is to run away; the more fear, the faster we run, and therefore, the chance of survival will be high.

Reactions to pain controlled by the limbic system bring emotions and experience; then pain is a personal experience controlled by many biological, psychological, and social factors. Pain and pain-sensing organs are related to each other but not always go together. Pain-sensing organs are objective while consciousness about pain is subjective. Through each individual's life experience, each person's consciousness about pain has differences.

If pain is purely objective, if there is tissue damage, pain receptors will inform the spinal cord and emit quick reaction signals, everyone the same. If so, we do not need to preach religious doctrines about dealing with pain because there is no difference between this person and that person to have room for religions to make better or worse.

Thus, reactions to pain are not simply ordinary biochemical pro-

cesses but have social and individual nature. In many cases, there is no tissue damage, but the nervous system still reports pain. In this case, pain is unreal; it is the body's subjective perception when the body overprotects against external dangers. This is precisely the key point for Buddhism to demonstrate its great educational ability. Buddhism guides humans not only in ways to react to pain but moreover effectively heal pain excellently. Not only that, Buddhist practice methods are also very effective in treating pain, especially pain caused by subjective sensations.

Thus, the first Noble Truth, the Truth of Suffering, is understood as: for humans to exist, they must have the ability to sense pain. Without it, humans do not exist. Pain is linked with existence; one cannot lack the other. The Truth of Suffering gives us Buddhism's view: suffering is inseparable from life. Suffering is precisely the path we must walk, what we must learn. Once understanding that simple thing, our behavior with life, with suffering will be more proactive and conscious.

5.4.3 Suffering due to dissatisfaction of consciousness

Research on suffering cannot be separated from research on pleasure because they are opposite but also complement each other; understanding one can understand the other. In the above section, we know that pain is due to the brain interpreting dangers related to damage on the body. In this section, we will also understand how the brain interprets pleasure. Pain is a body warning system when encountering damage; it is obviously necessary for existence; then is pleasure necessary? Why did the body evolve to have pleasure appear?

Understanding and researching pleasure certainly has a very long history; however, only in about the last 50 years has neuroscience been able to go deep into the brain's operating mechanism to help us

understand somewhat how it operates. Pleasure can be considered a trick of evolution to urge an individual to try to pursue some reward. Individuals, whether at high or low cognitive levels, are attracted by the present. Whenever we see delicious food like a chocolate bar, we feel craving even though knowing that if eating it, the body will absorb enough energy to have to do heavy exercise. For individuals to overcome the attraction of the present, the brain thinks up a reward mechanism to be able to overcome the present toward the future. That reward must be more attractive than the attraction of the present because it has not happened yet. The farther the future, the lower the feasibility, the less attractive. That is why reward must be very strong to be able to defeat the present. That reward is pleasure. **Pleasure is the reward of the future for individuals to overcome the attraction of the present**.

From the 1950s, James Olds and Peter Milner implanted electrodes on rat brains and they discovered special brain regions that rats wanted to electrically stimulate. Scientists had discovered the brain region taking responsibility for reward. Rats in the experiment pressed 7500 times on a lever within 12 continuous hours only to expect electrical signals stimulating that brain region. Even rats were more addicted to pressing levers than actions necessary for their existence such as eating, sleeping, or sex. Some rats pressed levers until exhaustion and death. That shows they are trying to enjoy experiences related to pleasure and they were addicted by those actions [335].

The human body also has a *reward system* (reward system), which is a group of neural structures to create craving and desire (Figure 5.7). Avoiding pain and pursuing craving are necessary mechanisms for individuals to adapt to the environment. Craving for food when hungry, wanting to drink water when thirsty, pursuing potential mates when conditions allow,... are all controlled by the reward system. However, this system can lead to excessive behaviors when controlled by stimulants or self-deception. Therefore,

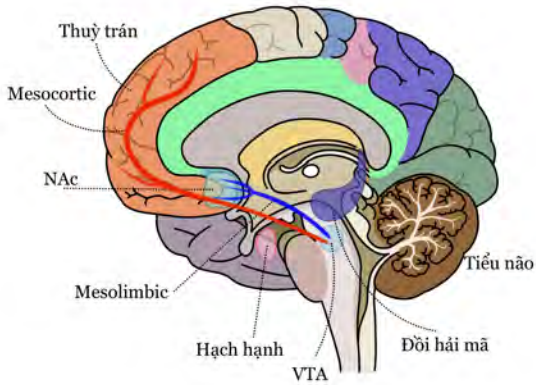

Figure 5.7: The reward system in the brain. The mesolimbic pathway of dopamine from VTA to NAc and the mesocortical pathway from VTA to the frontal lobe.

understanding the operating mechanism of the reward system to have appropriate behaviors is very important for a happy life.

Reward activity is initiated by receiving stimuli. Stimuli can come from outside such as seeing delicious food, a favorite object, or a potential mate, or come from any internal thought. Organs taking responsibility for stimulation are the hippocampus (hippocampus, storing memories) and amygdala (amygdala, controlling emotional states). Stimulation is sent to a very important organ of the reward system: the *ventral tegmental area* (ventral tegmental area – VTA) located in the midbrain. This organ will secrete a neurotransmitter playing a leading role in the reward system: **dopamine**, a chemical with many other important roles such as movement, memory. Dopamine will be transmitted to many different regions of the brain. The first part to mention is the nucleus (nucleus accumbens – NAc) through a pathway named mesolimbic (Figure 5.7). There, dopamine binds with receptors on neuron surfaces changing their activity to create pleasure and happiness.

The nucleus is precisely the place decoding chemical signals to create pleasure [336]. The more dopamine at neural synapses in NAc, the more chemical signals at neurons, the more pleasure individuals feel. In animal brains, there is a certain amount of dopamine called baseline (baseline); individuals are in a neutral state. When rewarded actions happen, dopamine amount increases higher than baseline making individuals feel pleasure. Studies show that eating a piece of chocolate increases dopamine by 55%, sex increases 100%, nicotine increases 150%, cocaine increases 225%, amphetamine increases 1000% [337].

However, the process does not stop there. Dopamine is also sent to the prefrontal cortex (prefrontal cortex), an organ taking responsibility for planning and decision-making through another pathway named mesocortical (Figure 5.7). This organ checks the entire process from initial stimulation to pleasure that NAc brings to have appropriate actions. The stronger the pleasure reaction at NAc, the stronger the reinforcement signals sent to the prefrontal cortex and other brain regions taking responsibility for memory and behavior. The result is that individuals tend to seek and attach to stimuli to create more reward. Over time, these behaviors become habits of that individual.

Three organs: NAc, VTA, and the frontal lobe form a feedback loop to adjust and optimize reward-seeking actions over time. Dopamine is also transmitted to the amygdala, an organ controlling emotions, and the hippocampus, an organ controlling memory, so that individuals have positive emotions pursuing rewards and remember them to continue pursuing in the future. For example, when eating a delicious bowl of pho, VTA secretes dopamine and sends to NAc creating a satisfied and happy feeling, sends to the hippocampus to remember the feeling, sends to the amygdala to create emotions, sends to the frontal lobe to see that feeling is good for the body to plan about returning to the pho restaurant next times.

Dopamine plays a role as a neurotransmitter creating pleasure, the feeling of being *high*. This feeling is obtained by many different types of behaviors such as sexual relations, eating and drinking when hungry and thirsty, sports exercise, listening to music, drinking coffee, stimulants and alcohol, sitting in meditation. Not only dopamine, some other neurotransmitters also create pleasure: serotonin is a chemical creating happiness, endorphin is a chemical causing happy feelings, oxytocin is a chemical appearing in love. Dopamine has a special effect; it helps reinforce and connect comfortable feelings and behaviors so that individuals have desires to repeat those feelings and behaviors. This is an important factor in causing addiction.

However, too much of anything is not good. The middle way is manifested everywhere; excess, no matter which extreme, creates unpleasant experiences. Dopamine relates to chemicals and behaviors that can cause addiction although this chemical itself is not an addictive substance. The addiction process is affected by many factors such as genes and family history, society, depression experience, impulsive experience. There are two types of addiction: substance and behavior. Examples of substance addiction: alcohol, tobacco, cocaine, heroin,... Behavior addiction: watching pornography, browsing social media, gambling... Both types of addiction have the same bad effects.

Like pain, pleasure sometimes has subjective nature, sometimes objective nature. The image of a newborn baby feeling happy and satisfied by smiling when full of breast milk is pleasure with objective nature after satisfying body needs. Not only newborns have this facial expression, but offspring of primate species, even rats, also have this expression. Neural organs controlling these actions are prefrontal neural circuits[19], organs controlling the brainstem (sensory-motor neural center) and behavior [338].

[19]Neural circuits, also called neuron circuits, are collections of neurons connected with each other to perform a specific function when activated.

Stanford neuroscientist Anna Lembke explains that opposite sensations, suffering and pleasure, are processed by the same brain region. That is, the brain region creating pleasure is also the brain region creating suffering [339]. Only by self-regulating neurotransmitters of the brain's reward system can bring pleasure or suffering. That is like playing on a seesaw; if the pleasure side is more than suffering, the brain makes individuals feel pleasure, and vice versa. Between them is a state called homeostasis (homeostasis) that the brain very much wants to be in that state. The chemical controlling that process is dopamine. The homeostasis state has dopamine at baseline that we mentioned above. When pleasure, dopamine amount increases high; when suffering, dopamine amount goes low. Dopamine is the factor affecting changes in the pleasure–suffering balance.

The brain finds many ways to establish balance. If pain is repeated, the brain changes by thinking that pain is normal. Humans gradually get used to pain so that at some point, things that previously caused pain now no longer do. Conversely, there are things that previously were very pleasant, but when repeating that process, at some point, there is nothing pleasant anymore. That is precisely the change of the brain itself although external stimuli do not change.

How do addictive substances like cocaine work? Between two connected neurons, there is a neural synapse. Signals from neuron A transmitted to neuron B must pass through this synapse (Figure 5.4). At the junction of two neurons, the synapse part of neuron A is called pre-synapse (pre-synapse), the synapse part of neuron B is post-synapse (post-synapse). Between two synapses, there is a synaptic cleft for neurotransmitters to move from neuron A to neuron B. When dopamine is released from neuron A at VTA, it will follow axons to neural synapses with neuron B at NAc. At pre-synapse, dopamine is secreted through dopamine transporters (dopamine transporter) to enter the synaptic cleft. Then, dopamine

is received by dopamine receptors (dopamine receptor) at post-synapse to enter neuron B. Dopamine activates neuron B to create pleasure. Normally, when in stimulated state, dopamine concentration only increases for a certain time period then returns to balance state. When stimulation ends, dopamine returns to neurons. Cocaine follows blood vessels to neurons at NAc. There, cocaine molecules block dopamine's return pathways making dopamine concentration at synapses increase; excitement from there increases high. Therefore, cocaine users feel extremely high.

Pleasure relates to dopamine existing in the synaptic cleft. Dopamine still there, pleasure feeling still; dopamine returns into neurons, pleasure feeling ends. Stimulants like nicotine, cocaine, amphetamine block pathways into neurons of dopamine molecules making users have indescribable pleasure for a long time. High excitement with stimulants deeply engraves into memory very quickly and urges individuals to continue behaviors to achieve that state in next times. After each use of stimulants, the brain establishes a new dopamine balance concentration higher than previous times. Therefore, individuals must use a higher dosage to have feelings like previous times. The brain's regulation of homeostasis makes after each cocaine absorption, the initial level is not enough to create the high state like last time. From there, it leads to cocaine dosage being raised after each use. Not only cocaine, other stimulants like alcohol, methylphenidate have similar effects [340]. That is precisely the addiction phenomenon to stimulants in some people.

5.4.4 Happiness and material comfort

Pleasure and happiness are two different states controlled by two neurotransmitters, one caused by dopamine, one caused by serotonin. Lack of dopamine causes weakness, depression, anxiety, loss of motivation, and Parkinson's disease. While lack of serotonin leads to sadness, disappointment, difficulty sleeping, digestive disorders,

sexual disorders [341]. An important difference is that dopamine has very strong effects and can cause addiction while serotonin does not. Happiness feeling is a feeling that is not too strong; most of us are not strongly urged to have this feeling, but when achieved, we want to have it as long as possible. That is the middle way state, not suffering but also not pleasure. Therefore, happiness is a mental state that most religions, including Buddhism, aim for. To do that, we need to balance dopamine because its strong effects can overwhelm other neurotransmitters.

Why is material life increasingly improved but happiness feeling decreases? As mentioned above, the most important discovery in the last 100 years about neuroscience is that suffering and pleasure sensations are taken responsibility by the same brain region. The brain region responsible for creating suffering is also the region creating pleasure. Those two opposite sensations are formed from the imbalance of a balance. For example, when watching a video on TikTok, VTA secretes dopamine and sends to NAc creating excitement by the desire to find something new; curiosity is one of the brain's functions, so the brain rewards to stimulate these activities. However, if watching videos repeats many times, the brain re-establishes balance with high dopamine level. Therefore, to have excitement like first times, dopamine amount needs to increase even higher. If not watching videos, dopamine concentration is lower than balance point leading to boredom. To compensate for dopamine decrease and create excitement, we need to watch new videos. The process of continuously increasing baseline for many days, many weeks makes us continue watching videos on TikTok not to find pleasure but so that we no longer feel boredom, suffering. When not watching videos anymore, we feel lacking. That is precisely the manifestation of social media addiction, an example of behavior addiction.

Human life has changed very quickly in just a few decades. We no longer worry too much about hunger and thirst due to lack of food and drink, do not worry about personal safety due to dangers from

wild animals, enemies; conversely, we can easily satisfy basic human cravings. In addition, very many entertainments have only appeared in a short recent time such as eating delicious food, playing video games, watching pornography, injecting stimulants, browsing social media,... Humans once had to do many things to have an amount of dopamine creating pleasure in the past, but now can achieve that very easily, without needing to try much. The more developed society, the easier humans achieve high dopamine – dopamine overload effect. When dopamine overload, the body needs more dopamine to rebalance the pleasure–suffering balance. Like watching TikTok initially with the purpose of happiness, then social media addicts watch TikTok to fight against sadness; behaviors to easily achieve high dopamine such as eating delicious food will cause diseases due to excess food. According to the World Health Organization WHO, 74% of deaths worldwide are due to non-communicable diseases such as heart disease, stroke, cancer, diabetes, lung disease, which are diseases due to bad habits newly appearing in recent time.

Here we see Buddhism's suffering again manifested through change. Here it is not the change of the external world but the change of consciousness about that world. The same thing and phenomenon, initially dopamine balance point is still low, we see pleasure, but after a while, the same thing and phenomenon happen, but we no longer see pleasure but gradually turn into suffering because dopamine balance point increases. We continue to seek other stimuli to try to satisfy but never get it long-term. Especially in a society full of, abundant material wealth, most desires can be satisfied quickly; humans feel unhappy, suffering more than lacking societies. That is precisely the suffering of life.

5.4.5 Quantifying happiness

Everyone has a concept and wants to be happy, but it is very difficult to give an accurate definition of happiness. Certainly happiness is

a positive, happy, satisfied emotional state. Happiness differs from pleasure in the agent of sensation. Pleasure comes from stimulating agents from outside reaching senses. Agents still there, sensation still; agents end, sensation ends. Pleasure that lasts and repeats many times causes addiction. Without those agents, individuals feel bored and painful. Happiness is different; it is a feeling that the brain itself creates, and if practicing appropriately, we can control this feeling. The same stimulating agent, some people feel happy, others feel unhappy.

We already know that the neurotransmitter creating pleasure is dopamine. Happiness relates to two neurotransmitters: serotonin and GABA, two chemicals creating connection and relaxation feelings. People can achieve happiness state without needing dopamine, so happy people do not need and do not have needs to seek material and behavior to be rewarded related to this chemical. Below we understand happiness and see what humanity does to be able to achieve this state.

If it is an emotional state, it has subjective and immediate nature. Subjectivity makes concepts about happiness different between this person and that person, this group and that group, this nation and that nation. Immediacy causes differences at this time point and that time point. Most discussions about happiness are based on common perception in space and time of an individual. But is there a universal concept about happiness for everyone? If yes, we apply the method of averaging by space, in other words, averaging by the majority. The largest majority is all humanity. Due to immediacy, we must consider happiness according to the time of human life, the existence time of a nation, even of humanity. When talking about happiness, we talk about happiness with some accuracy in space and time, not generally or too specifically. Here we consider with spatial accuracy of a nation and temporal accuracy within several decades.

To overcome space, we consider happiness according to research

methods of survey organizations globally. The World Happiness Report (world happiness report – WHR), an index put forward by a United Nations agency by John Helliwell of the University of British Columbia, when researching 156 countries and territories in 16 years, from 2008 to 2024, shows that developed countries have decreasing happiness indices over time [342]. This index is based on 6 criteria, each criterion is scored from 0 to 10. Those are GDP per capita, social support, average life expectancy, freedom to choose life, generosity, perception of corruption. After scoring, countries are compared with a fictional country Dystopia (anti-paradise, opposite to Utopia, an ideal social country) representing countries with lowest average rankings in criteria. The richer countries, the more anxiety is proportional to income. Countries with high income have higher anxiety levels of citizens than low-income countries. Depression levels are also like that, increasing over time but increasing fast in rich countries and increasing slowly in poor countries. In 2024, Vietnam got 6.04 points, continuously increasing in the last 4 years, and belongs to the group of happy countries. Meanwhile, the US, the world's leading superpower, happiness index continuously decreases from 2013 until now [339].

If every country runs after WHR criteria, in which GDP is a basic index, then Earth does not have enough resources to meet everyone present in the world at the current time. That shortage can lead to competition with each other and cause unfortunate consequences for humanity. If happiness only depends on personal experience and material conditions, then everyone wants to race to have a life like in developed countries. Suppose that race succeeds, then all citizens globally will have a full, warm, and finally happy life. But wait, how long can less developed countries catch up with developed countries, and suppose that can be done, is there enough resources to meet everyone?

Moreover, to have a happy life, we do not necessarily need material abundance. WHR happiness index has not yet calculated the

effectiveness that happiness feeling brings and input costs to have that feeling.

Therefore, **Happy Planet Index** (Happy Planet Index – HPI) of the New Economics Foundation is appropriate because it considers global sustainability and long-term stability. It expresses three important relationships of humans: nature, society, and thought. HPI is based on assumptions: everyone wants to live long and fully, good countries are countries allowing citizens to do that, individual happiness is evaluated through life expectancy and life satisfaction, to have a happy life, humans require nature to provide materials, and the index measuring that consumption level is ecological footprint. HPI = (Satisfaction × Life Expectancy)/Ecological Footprint [343]. Satisfaction is subjective and objective data, considering input and output so we know the effectiveness of the process. According to this index, Vietnam ranks first in Asia. Putting ecological footprint to calculate happiness index is correct in the context of humans increasingly having negative impacts on Earth through their activities. Ecological footprint is an index for Earth, more precisely, humans in future generations, to develop sustainably.

Current happiness indices are only based on material indices, not yet based on cultural, religious indices. Although these indices are not easy to measure, they strongly determine human mental state due to stability in social values. Secular countries often have cultural values changing very quickly. Happiness indices also have not yet considered relative increases but mainly take absolute values. A society with rapid increases in material and mental aspects can be happier than a slow-developing or politically, economically, socially unstable society. A Harvard University survey when asking young people about the most important goal of life, what is it? The answer is: 80% want to become rich, 50% want to become famous, all say they need to work hard to have a better life later. Are their answers correct or not?

To overcome time, we consider the longest research in the world

ever conducted and it is still being conducted until today. That is Harvard University research from 1938 about happiness. The initial number of people was 724 people including students of this university (268 students, one became US president John F. Kennedy) and low-income boys in Boston; now the number of people has reached over 2000 including wives and their children and grandchildren. Researchers collect happiness data from questionnaires every 2 years/time; collect health records with DNA tests, brain scans every 5 years/time; meet directly for interviews every 15 years/time. The director of this research program has also changed many times; the current leader is Robert Waldinger, the fourth director of the project. He has a talk on TED talk in 2016 and achieved 24 million views, one of the most viewed TED talks [344].

The first director of the research program, Clark Heath, intended to understand the influence of biological causes and genes on human happiness, so he chose 2 groups of subjects. One elite group is first-year students of Harvard University, and the second group is a common group in the same city. Researchers at that time thought that individual ability, physical foundation, and personal characteristics would determine adult development. They recorded very detailed anthropometry such as skull shape, distance between eyebrows, mole positions, handwriting, even brain activity by electroencephalography or activity of important organs in the body.

Until now, scientists change measurements such as DNA analysis, nuclear magnetic resonance MRI images to check important organs. The first few decades when the program was conducted, people did not pay attention to relationships such as empathy and attachment. Everything changed when psychiatry expert George Vaillant led the program from 1972–2004, emphasizing and finding the key and decisive role: relationships. Results are very clear: Not money, fame, hard work make humans happy, but good relationships make humans happier and healthier [345]. Relationships are very important to humans. People with good relationships with family, friends,

and community will be happier, healthier, live longer. Loneliness kills humans like alcohol and tobacco. So what is a good relationship? A good relationship is a relationship where you are yourself, do not need to try too hard, do not need to pretend to be sociable and happy. In relationships, we do not necessarily always need harmony, but what is important is trust.

Not quantity but quality of close relationships is important. Introverted people usually only have a few close relationships; extroverted people can have more. Quantity depends on each person but must be present. Marital satisfaction protects mental health. People with marital satisfaction are less likely to feel physical and mental pain when encountering physical damage. Many people have physical damage but mood does not feel much pain like people without happy marriages. The influence of genetic factors on life expectancy is not stronger than satisfaction about relationships. Previous viewpoints that human personality does not change after age 30 are wrong. There are people who feel unhappy until 60 years old then find happiness for themselves with a group of friends with common interests. It is never too late to become happy.

5.4.6 To have happiness

Buddhism has two main meditation methods: Samatha (Samatha) and Vipassana (Vipassana) to achieve liberation recorded in the *Nikaya Sutta Pitaka*. Each method has its own purpose and practice techniques, but both aim at spiritual development and achieving enlightenment. Samatha means stopping or tranquility. Samatha meditation is mental liberation, aiming to calm and stabilize the mind, helping practitioners achieve a state of high concentration and inner peace with the purpose of developing concentration and mental stability, reducing delusions and distractions to achieve a state of peace and tranquility of the soul. Vipassana means observation or deep understanding. Vipassana meditation is wisdom liberation, fo-

cusing on developing deep awareness about the true nature of phenomena and life to deeply understand the Four Noble Truths, the three marks of existence, and emptiness. Samatha and Vipassana are two important meditation methods in Buddhism; each method has its own role and techniques, but both aim at spiritual development and achieving enlightenment. The harmonious combination of Samatha and Vipassana helps practitioners develop both tranquility and wisdom, thereby progressing on the path of liberation.

Besides meditation, we need to cultivate another very important virtue for happiness. That is compassion. Most people have access to good education, are equipped with much knowledge, critical thinking ability, have great expectations, and can achieve wealth, power, fame, but may not have happiness. Education only ensures a materially sufficient life. To have happiness, we need compassion. It brings inner strength and brings truth. The Dalai Lama explains compassion very convincingly [346]. What is compassion? That is respect and good feelings for others that do not depend on their attitudes and feelings. When having it, we approach truth, have nothing to hide, bringing confidence and strength. The important thing is that compassion can be cultivated and practiced daily from small to large work. The Dalai Lama thinks that the highest goal of education is to have a world where everyone is happy. To be happy, we need to be educated and cultivate virtues: **respect, love, and compassion**.

This book hopes to help readers in the Vipassana meditation process through providing scientific knowledge about emptiness. However, to receive useful knowledge, each ego must overcome barriers created by knowledge. A modern French philosopher Edgar Morin put forward *complex thought* (complex thought, French: pensée complexe) as a human learning principle. He learned from Karl Marx that researching natural sciences must go together with researching humans and vice versa. A brief but interesting conversation between Trinh Xuan Thuan and Morin reveals the idea that

complex thought must originate from a feeling about a whole [347]. If humans are trained with separate knowledge, that is precisely **knowledge obstruction**: knowledge is the source of delusion and obstacles to approach other knowledge. Therefore, we need to teach *what is knowing?* first. That is the first and important thing to continue absorbing other knowledge. Therefore, Vipassana meditation of Buddhism will help us overcome obstacles of knowledge we have to be able to continue reaching up to recognize truth.

With awareness about the whole like that, we return to how can we be liberated from suffering? Because pleasure and suffering always go together; they are inseparable category pairs, so the more we desire pleasure, the more it leads to dissatisfaction. That is precisely suffering. So what should we do when we are present in this world? Buddhism guides us ways to eliminate suffering, not taking pleasure to compensate for suffering but finding the middle way where there is no pleasure and therefore no suffering.

With physical pain related to tissue damage, the middle way is the harmonious combination between damage and pain signaling, not underestimating but also not exaggerating excessively. Clearly recognizing damage relates to pain and pain is due to damage. With mental pain, because the brain region taking responsibility for pain and pleasure is the same, the middle way is precisely bringing to homeostasis point so that neurotransmitters are at appropriate levels. How do we follow the middle way?

Addictive behaviors, for example playing video games, can be treated by dopamine fasting (dopamine fasting). When not performing addictive behaviors for about 1 month, dopamine balance level decreases from high to low and gradually approaches the level in normal people. During that abstinence period, people can switch to other healthier behaviors such as sports exercise, direct communication with relatives and friends. Previously, players had to spend several hours a day to be satisfied, but after dopamine fasting, they can not play, or only need to play a few tens of minutes a day to

have pleasant sensations. We can make detailed plans by gradually reducing addictive behaviors; should not reduce suddenly because it will be very difficult, but reduce with a roadmap and must commit to that reduction. Even stopping addictive habits should be conducted gradually. Can start with one day not playing video games, then extend to one week, one month, and longer.

Another way is to use poison to cure poison; suffering is a way to increase pleasure. Create homeostasis point with a low amount of neurotransmitters. For example, capsaicin is a chemical present in peppers making us feel pain due to spiciness. But if we eat peppers many times, that pain becomes pleasure because the brain has re-established balance point when eating peppers. At that time, a neurotransmitter named endorphin, a pain reliever in the nervous system, appears. Initial spicy sensation gradually decreases to be replaced by a satisfied feeling when eating. Today, more than half the world's population eats this food daily not because it brings energy to the body but because it brings pleasant sensations.

A practice method to achieve happiness similar to Buddhism's right path is the method of Stoicism. Many ideas of this philosophy are very similar to Buddhism and completely suitable for the modern world even though it was born from the 4th century BCE by Greek philosophers. The father of Stoicism is Zeno of Citium (different from Zeno of Elea who put forward paradoxes) based on three components: ethics, physics, and logic [348]. If Greeks gave birth to Stoicism, then Romans later received it and developed practice methods skillfully because Romans have higher practicality than Greeks. In the 2nd century, four representative Roman Stoics were Seneca, Musonius Rufus, Epictetus, Marcus Aurelius.

Roman Stoics developed practice methods to have happiness as follows. Think beforehand about bad things that can happen with three purposes: (1) to prepare for those bad things; (2) reduce impact on oneself by actively accepting; (3) oppose psychological adaptation effect to pleasure, prevent oneself from adapting to things

currently having to then hate them, even reverse the adaptation process by imagining losing things currently having. Most people have dream lives (in the past) but still suffer in the present due to adaptation to pleasure. Speaking in neuroscience language is that the balance point has high neurotransmitter levels. During years when the Covid pandemic raged, being able to meet face-to-face shaking hands was a luxury, but now we consider that normal. An evening eating and drinking with close friends is truly very precious that most of us ignore without noticing.

The most foolish thing is living in a dissatisfied life created by oneself while oneself has the right to choose a happy life. Being satisfied with small things is not a weakness but a blessing. When treasuring what we have, then later if losing, we do not have many things to regret. For example, when losing relatives, we can be very sad, but we have had moments with the deceased in the most treasured way. That reduces the pain of loss.

Self-restraint by periodically practicing living austerely as if bad things are happening to us. That is the practice method of *asceticism* (SAN dhuta), a method of ascetic practice to abandon desires. Self-restraint is the most difficult thing to perform. Willpower strength is like muscle strength; it needs to be trained regularly. Practicing asceticism is not easy but needs willpower and a difficult training process. Only then can we re-establish dopamine balance point, re-establish the middle way.

In summary, the compositionality of consciousness is manifested by consciousness formation from very many neural activities of the brain. Neural activities cooperate and compete with each other to attract attention of consciousness acting as a decision-maker. Neural activities receive information from senses; in turn, senses receive signals from receptors. Receptors are signal converters from the external world through hundreds of thousands of ions transported through receptors per second. Senses and organs on the body do not operate independently. The brain synchronizes signals from

senses with other organs to build an effective model of the external world. The impermanence of consciousness is manifested by evolution theory showing that not only the material body of organisms but human consciousness is also evolved from basic perceptions of organisms, manifested by the continuous change of interactions between the body and the external world. These two properties, compositionality and impermanence, show that consciousness has emptiness.

Conclusion

In the previous chapter, we understood suffering due to physical pain, suffering due to constantly changing desires never satisfied. Suffering due to ignorance was not presented. That is not because the book omitted that suffering but conversely: this entire book is only to show us suffering due to ignorance. Ignorance is the most core problem that Buddhism raises, understood as wrong awareness about reality. However, understanding what ignorance is is not easy. Modern science has given us deep insights about nature and shown that our consciousness about reality is not always correct. Not only that, recognizing reality is very deficient and incomplete. However, the brain still brings us perfect awareness about the external world. Explaining this cannot help but place humans in the context of biological evolution. Consciousness is formed so that humans can adapt most appropriately to the surrounding environment. Note: most appropriately, not most truthfully.

We review the content of this book for readers to easily follow. The book holds that Buddhism has three basic principles: suffering, middle way, and dependent origination. Although separated into three independent principles, the separation is only relative; in fact, all those principles are integrated and depend on each other very closely. The principle of dependent origination is mutual dependence between things and phenomena. Spatially, things and phenomena as a whole are created by smaller constituents, and in turn, constituents are created by even smaller constituents. Temporally,

things and phenomena manifest impermanence, meaning relationships between constituents are very diverse and constantly changing. This makes things and phenomena not have inherent, unchanging, and independent nature. All important properties of things are due to the co-arising of causes and conditions. That is emptiness, ultimate truth in Buddhism. Emptiness is dependent origination; dependent origination is emptiness. In this book, we have seen emptiness in the material world, life, human body, consciousness.

The material world is created by atoms. According to the standard model of particle physics, atoms are in turn created by subatomic particles including matter particles such as quarks, electrons, neutrinos, and interaction-carrying particles such as photons, gluons, gravitons, Higgs,... The existence of fundamental particles is not as simple as we think. Quarks cannot exist independently but only appear in quark–antiquark pairs or in triplets as residues of the continuous appearance and disappearance process of countless particles in protons and neutrons. Electrons are also the appearance and disappearance process of electron fields and photon fields. Particles exist in mutual correlation. String theory holds that very small microscopic strings living in 11-dimensional space are the fundamental entities creating the above particles. Scientific theories seem not to stop there.

The continuous movement of the material world makes a concept appear: entropy, a physical quantity considered a measure of the degree of order of a system. The second law of thermodynamics shows the anisotropy of entropy flow: entropy of closed systems always increases. The anisotropy of entropy leads to the anisotropy of time and the anisotropy of living systems. Because there are very few states prioritizing life, states not prioritizing life are considered disordered. What prioritizes life is right, good, beautiful, and conversely, what does not prioritize life is wrong, evil, ugly. That is precisely the anisotropy of truth – goodness – beauty. Truth, goodness, beauty are not independent, self-existing values but they

come from life. Life in turn comes from atoms and molecules. In turn, these particles are created by smaller components. All are in interconnected relationships and constantly changing.

The world of life is very simply defined as a self-sustaining chemical system capable of following Darwin's evolution theory. If so, self-replicators are gene segments in organisms considered living systems, and they have their own goals. Examples of cooperation and competition of genes and viruses show they operate like intelligent organisms. Cooperation is mandatory; otherwise, they always deceive and exploit opponents. Life may have origins from the universe but may also have origins on Earth from the very beginning. Current life shows close and surprising relationships between living processes and other processes such as interactions with cosmic rays in forming right-handed DNA or fusion and fission reactions to create elements on stars. Evolution theory provides evidence about mutual dependence between species from bacteria, viruses to reptiles, birds, mammals,... with humans. Modern gene technology applications such as CRISPR-Cas9 are because we learn from bacterial immune systems fighting viruses.

The human body at three levels: DNA, cell, and individual are all symbiosis of other species. The number of microorganisms in the intestinal tract is even greater than the number of human cells. They have roles helping digest food, creating immune systems fighting attacks from other microorganisms, producing some neurotransmitters for the enteric nervous system. In human cells, there is a symbiotic organism: mitochondria. They are energy factories of cells functioning to convert food into ATP, molecules that cells use to obtain energy. Mitochondria are also timekeepers of cells, determining the time of life or death of different cell types. They also determine the development speed of embryos during human formation. Human DNA is a combination with very many types of DNA from organisms from viruses, bacteria to other species. Very many functions that humans have are because we have utilized genes of

organisms for our development purposes.

Human consciousness is obtained through two processes: receiving information from senses and modeling to build subjective images about the external world. Each species chooses a very thin slice of reality to create complete images. Organisms have different levels of cognition. From bacteria, cells, body parts, body, brain all have their own cognition levels to achieve their goals. Consciousness forms in organisms functioning to reflect the external world thanks to high-level neural activity: the brain. Pain is linked with the nervous system creating pain sensation. This is a body protection system, so life cannot be separated from pain. The reward system of the nervous system brings humans motivation to look toward the future, but as soon as goals are achieved, the reward system will decline so that individuals have motivation to pursue other goals. That is precisely suffering due to change, never satisfied. The final suffering is not knowing that one is suffering, suffering due to ignorance because not recognizing ultimate reality is emptiness.

Thus, in all worlds, living and non-living; simple and complex organisms; human body at molecular, cellular, and individual levels; and even human consciousness, all show compositionality and impermanence. I want to remind once more about the two properties leading to emptiness. Compositionality is manifested by at any viewing angle, things have properties created by their components. Those components are all necessary and cannot (even if wanting) be separated. Impermanence is manifested by mutual, dialectical interactions constantly changing. Precisely that rapid change makes whenever we observe things and phenomena, we see them different. These two properties lead to absolute truth: the emptiness of all things.

We end this book with the famous *Cư trần lạc đạo phú* (Living in the World, Joyful in the Way) by Tran Nhan Tong, the king and Zen master of the Tran dynasty, speaking about time and life. Through that, we see emptiness appearing in all things, everywhere, at all

times. In daily life, whenever practicing according to the middle way way of life because the Way is not distinguishing practice and non-practice once realizing the emptiness of all things [349].

> "Living in the world, joyful in the Way, follow conditions,
> Hungry then eat, tired then sleep,
> In the house there is treasure, stop searching,
> Facing situations without mind, don't ask about meditation."[20]

[20]"Cư trần lạc đạo thả tuỳ duyên, Cơ tắc xan hề khốn tắc miên, Gia trung hữu bảo hưu tầm mịch, Đối cảnh vô tâm mạc vấn thiền."

Glossary

3D 3 dimensions. 148

A adenine, one of four nucleotides in DNA. 199, 205
AdS anti-de Sitter, multi-dimensional space with negative curvature. 152, 155
AdS/CFT anti-de Sitter/conformal field theory correspondence: AdS/CFT correspondence, also called AdS/CFT duality, relates anti-de Sitter space with conformal field theory. 151, 153, 155
AI artificial intelligence – AI. 290, 319
AIDS acquired immunodeficiency syndrome: syndrome caused by virus leading to immunodeficiency in humans. 260
ATGC adenine, thymine, guanine and cytosine are the four bases in DNA. 312
ATP adenosine triphosphate is the molecule that carries energy for cellular activities, called the currency of the cell. 251, 252, 256, 257, 293, 340

BCE Before Christ. Abbreviated as BC (Before Christ: before Jesus Christ) or BCE (Before Common Era: before current era). 9, 13, 19, 21–23, 25, 28–30, 32, 33, 40–43, 48, 82, 193, 197
BP base pair: base pair in DNA, A–T, G–C. 207, 212, 260

C cytosine, one of four nucleotides in DNA. 199, 205
CFT conformal field theory. 152, 155
CRISPR clustered regularly interspaced short palindromic repeats: a family of DNA sequences in bacteria and archaea used to detect and destroy viruses. 221

DNA deoxyribonucleic acid, molecule carrying genetic information in living organisms. 12, 176, 199, 201, 204

EEG Electroencephalography: electroencephalogram, method for recording spontaneous electrical activity in the brain. 306
ENG English. 58
ENS enteric nervous system. 248
EPSP excitatory postsynaptic potential (in neurons). 335

fMRI functional magnetic resonance imaging: functional nuclear magnetic resonance imaging. 306

G guanine, one of four nucleotides in DNA. 199, 205
GABA gamma aminobutyric acid. 248, 336, 338, 379
GDP Gross domestic product. 380
GMO genetically modified organism. 219
GPT Generative Pre-trained Transformer: large-scale, multimodal language model created by OpenAI. 301
GWT global workspace theory (on consciousness). 308

HIV human immunodeficiency viruses: virus causing immunodeficiency in humans. 260
HOT higher-order theory (on consciousness). 308
HPI Happy Planet Index: index by New Economics Foundation. 381

IIT integrated information theory (on consciousness). 298, 307
IPSP inhibitory postsynaptic potential (in neurons). 336
IR infrared (electromagnetic wave). 352
ISS International Space Station. 352

LQG loop quantum gravity (theory). 135

MIT Massachusetts Institute of Technology (USA). 105, 223, 256, 301
MRI magnetic resonance imaging: functional nuclear magnetic resonance imaging. 366, 367
mRNA Messenger RNA: messenger RNA, a single-stranded RNA segment for protein synthesis. mRNA is created from DNA during transcription and plays a role in carrying information from DNA to the cytoplasm. 176
mtDNA mitochondrial DNA. 12

NAc Nucleus Accumbens: brain nucleus, a part of the brain that creates pleasure and happiness. 372, 375
NASA National Aeronautics and Space Administration (USA). 164, 166, 182, 188, 216, 271, 317
NCC neuronal correlates of consciousness. 307
NCG Non commutative geometry. 134
NIH National Institutes of Health (USA). 210, 211
NIST National Institute of Standards and Technology (USA). 142, 299

PCR polymerase chain reaction. 220, 227
PET positron emission tomography. 306, 367
PPT predictive processing theory (on consciousness). 308

QFT quantum field theory. 90

RGB red–green–blue. 352
RNA ribonucleic acid. 176, 206

SAN Sanskrit: transliteration from Sanskrit. 57, 58
SDL Anno Domini, previously often written as after Christ. Abbreviated as AD (Anno Domini: in the year of Jesus Christ) or CE (Common Era: current era). 21, 42, 44

T thymine, one of four nucleotides in DNA. 199, 205
TAME Technological Approach to Mind Everywhere. 318
TED Technology, Entertainment, Design: TED is a media organization specializing in publishing talks by famous speakers. 382
TMS transcranial magnetic stimulation. 306
tRNA Transfer RNA: transfer RNA or transport RNA. Transfer RNA is a small RNA molecule that plays an important role in protein synthesis. Based on mRNA, tRNA carries amino acids to create a protein. 177

UFO unidentified flying object. 168
UV ultraviolet (electromagnetic wave). 352

Vis visible light (electromagnetic wave). 352
VTA ventral tegmental area. 372, 375

WHO World Health Organization. 272, 277, 378
WHR World Happiness Report. 380

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